

Integration Routines

5.4

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jeod	
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jeod::GaussJacksonSimpleSecondOrderODEIntegrator	140
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SimpleCheckpointable	
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jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< 4, 3, er7_utils::Left← QuaternionGeneralizedPositionFunctions >	243
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jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >	243
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jeod::RestartableSimpleSecondOrderODEIntegrator< size >	256
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jeod::RestartableScalarFirstOrderODEIntegrator	250
jeod::RestartableStateIntegrator< IntegratorType >	264
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er7_utils::StandardIntegrationControls	
jeod::LsodeIntegrationControls	226
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er7_utils::TimeInterface	
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er7_utils::TwoDArray< T >	278
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er7_utils::DoubleTwoDArray	59

Chapter 4

Data Structure Index

4.1 Data Structures

Here are the data structures with brief descriptions:

er7_utils::DoubleTwoDArray	2D array, specialized for doubles	59
jeod::GaussJacksonCoefficientsPair	Contains a summed Adams and Gauss-Jackson coefficient pair	60
jeod::GaussJacksonCoeffs	Contains the Gauss-Jackson predictor and corrector coefficients	65
jeod::GaussJacksonConfig	Contains Gauss-Jackson configuration data	70
jeod::GaussJacksonFirstOrderODEIntegrator	Integrates a first order ODE using the summed Adams technique that is embedded within the Gauss-Jackson technique	75
jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator	Integrates a generalized derivative second order ODE using Gauss-Jackson	78
jeod::GaussJacksonIntegrationControls	IntegrationControls specialized for Gauss-Jackson integration	84
jeod::GaussJacksonIntegratorBase< State, Primer >	Base template class for integrating state via the Gauss-Jackson technique	95
jeod::GaussJacksonIntegratorConstructor	Create state and time integrators that propagate using Gauss-Jackson	126
jeod::GaussJacksonOneState	Essentially just a double*	134
jeod::GaussJacksonRationalCoefficients	Contains a set of Adams or Stormer-Cowell coefficients	136
jeod::GaussJacksonSimpleSecondOrderODEIntegrator	Integrates a simple second order ODE using the Gauss-Jackson technique	140
jeod::GaussJacksonStateMachine	Guides the behavior of the Gauss-Jackson integration process via a finite state machine	146
jeod::GaussJacksonTwoState	Essentially just std::pair<double*>	160
jeod::GeneralizedSecondOrderODETechnique	Enumerates the integration schemes, generalized Cartesian or Lie group integration, and provides simple utilities that work with this	162
jeod::IntegrationMessages	Declares messages associated with the integration test model	166
jeod::JeodIntegrationGroup	A JeodIntegrationGroup integrates the state of a set of objects over time	169

jeod::JeodIntegrationGroupOwner	
The abstract class <code>IntegrationGroupOwner</code> contains an <code>IntegrationGroup</code>	178
jeod::JeodIntegrationTime	
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jeod::LsodeControlDataInterface	
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jeod::LsodeFirstOrderODEIntegrator	
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jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator	
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs	224
jeod::LsodeIntegrationControls	
Contains controls for an LSODE integrator	226
jeod::LsodeIntegratorConstructor	
Create state and time integrators that propagate using standard Lsode	228
jeod::LsodeSecondOrderODEIntegrator	
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs	231
jeod::LsodeSimpleSecondOrderODEIntegrator	
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs	235
jeod::Restartable2DSecondOrderIntegrator	
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jeod::RestartableFirstOrderODEIntegrator< size >	
A <code>RestartableFirstOrderODEIntegrator</code> is-a <code>RestartableStateIntegrator</code> that manages an <code>er7_↔</code> <code>utils::FirstOrderODEIntegrator</code>	241
jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >	
A <code>RestartableGeneralizedDerivSecondOrderODEIntegrator</code> is-a <code>RestartableStateIntegrator</code> that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity	243
jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >	
A <code>RestartableGeneralizedStepSecondOrderODEIntegrator</code> is-a <code>RestartableStateIntegrator</code> that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity	247
jeod::RestartableScalarFirstOrderODEIntegrator	
A <code>RestartableScalarFirstOrderODEIntegrator</code> integrates a first order ODE, $dx/dt = v(x,t)$, where x is a scalar	250
jeod::RestartableSecondOrderODEIntegrator	
A <code>RestartableSecondOrderODEIntegrator</code> is-a <code>RestartableStateIntegrator</code> that manages the integrator for a second order ODE problem	255
jeod::RestartableSimpleSecondOrderODEIntegrator< size >	
A <code>RestartableSimpleSecondOrderODEIntegrator</code> is-a <code>RestartableSecondOrderODEIntegrator</code> that manages the integrator for a simple second order ODE problem, one in which the time derivative of the generalized position is the generalized velocity	256
jeod::RestartableSO3SecondOrderODEIntegrator	
A <code>RestartableSO3SecondOrderODEIntegrator</code> integrates a generalized second order ODE that describes rotation in three space	259

jeod::RestartableStateIntegrator< IntegratorType >	
A RestartableStateIntegrator establishes the basic capabilities needed to make a state integrator a managed resource	264
jeod::RestartableT3SecondOrderODEIntegrator	
A RestartableT3SecondOrderODEIntegrator integrates a second order ODE in three space, $d^2x/dt^2 = a(x,t)$, where x is a three-vector	271
jeod::TimeChangeSubscriber	
A TimeChangeSubscriber is some object that wants to be notified of changes in the nature of time	276
er7_utils::TwoDArray< T >	
RAII template class that implements a rectangular two dimensional array	278

Chapter 5

File Index

5.1 File List

Here is a list of all files with brief descriptions:

gauss_jackson_coefficients_pair.cc	Defines member functions for the class GaussJacksonCoefficientsPair	291
gauss_jackson_coefficients_pair.hh	Defines the class GaussJacksonCoefficientsPair, which contains summed Adams and Gauss-Jackson coefficient pair	291
gauss_jackson_coefs.cc	Defines member functions for the class GaussJacksonCoeffs	292
gauss_jackson_coefs.hh	Defines the class GaussJacksonCoeffs, which contains the Gauss-Jackson predictor and corrector coefficients	293
gauss_jackson_config.cc	Defines member functions for the class GaussJacksonIntegratorConstructor	293
gauss_jackson_config.hh	Defines the class GaussJacksonConfig, which specifies Gauss-Jackson configuration data . .	294
gauss_jackson_first_order_ode_integrator.hh	Defines the class GaussJacksonFirstOrderODEIntegrator, which integrates a first order ODE using the summed Adams technique	294
gauss_jackson_generalized_second_order_ode_integrator.cc	Defines member functions for the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator	295
gauss_jackson_generalized_second_order_ode_integrator.hh	Defines the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator, which integrates a generalized derivative second order ODE using Gauss-Jackson	295
gauss_jackson_integration_controls.cc	Defines member functions for the class GaussJacksonIntegrationControls	296
gauss_jackson_integration_controls.hh	Defines the class GaussJacksonIntegrationControls, which controls Gauss-Jackson integration process	296
gauss_jackson_integrator_base.hh	Defines the template class GaussJacksonIntegratorBase, which provides the basis for Gauss-Jackson integration	297
gauss_jackson_integrator_base_first.hh	Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE . . .	298
gauss_jackson_integrator_base_second.hh	Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE . . .	298

gauss_jackson_integrator_constructor.cc	Defines member functions for the class GaussJacksonIntegratorConstructor	299
gauss_jackson_integrator_constructor.hh	Defines the class GaussJacksonIntegratorConstructor, which constructs integrators that use Gauss-Jackson integration	300
gauss_jackson_one_state.hh	Defines the class GaussJacksonOneState, which contains a double* pointer	300
gauss_jackson_rational_coeffs.cc	Defines member functions for the class GaussJacksonRationalCoefficients	301
gauss_jackson_rational_coeffs.hh	Defines the class GaussJacksonRationalCoefficients, which contains a set of Adams or Stormer-Cowell coefficients	301
gauss_jackson_simple_second_order_ode_integrator.hh	Defines the class GaussJacksonSimpleSecondOrderODEIntegrator, which integrates a simple second order ODE using Gauss-Jackson	302
gauss_jackson_state_machine.cc	Defines member functions for the class GaussJacksonStateMachine	302
gauss_jackson_state_machine.hh	Defines the class GaussJacksonStateMachine, which guides the Gauss-Jackson integration process	303
gauss_jackson_two_state.hh	Defines the class GaussJacksonTwoState, which contains a pair of double* pointers	303
generalized_second_order_ode_technique.cc	Define class GeneralizedSecondOrderODETechnique methods	304
generalized_second_order_ode_technique.hh	Define the static class GeneralizedSecondOrderODETechnique	304
integration_messages.cc	Implement the class IntegrationMessages	305
integration_messages.hh	Define the class IntegrationMessages, the class that specifies the message IDs used in the integration model	305
jeod_integration_group.cc	Define JeodIntegrationGroup methods	306
jeod_integration_group.hh	Define the extensible class IntegrationGroup, an instance of which is responsible for integrating the states of a set of DynBody objects	306
jeod_integration_time.cc	Define JeodIntegrationTime methods	307
jeod_integration_time.hh	Define the class JeodIntegrationTime	307
lode_control_data_interface.cc	Define member functions for the class LodeControlDataInterface	308
lode_control_data_interface.hh	Define the class LodeControlDataInterface	308
lode_data_classes.cc	Define member functions for the data-grouping classes specified in lode_data_classes	309
lode_data_classes.hh	Define LSODE classes that contain just data members	309
lode_first_order_ode_integrator.hh	Define the class LodeFirstOrderODEIntegrator, the Jeod-compatible version of the Livermore ODE solver, LSODE	310
lode_first_order_ode_integrator_integrator.cc	Define member functions for the class LodeFirstOrderODEIntegrator	310
lode_first_order_ode_integrator_manager.cc	Define member functions for the class LodeFirstOrderODEIntegrator	311
lode_first_order_ode_integrator_support.cc	Define member functions for the class LodeFirstOrderODEIntegrator	311

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lsode_integrator_constructor.cc	
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Define the class LsodeIntegratorConstructor, the class that constructs the integration controls and the integrators for the LSODE method	314
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Defines the class Restartable2DSecondOrderODEIntegrator	317
restartable_state_integrator.hh	
Define classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators	317
restartable_state_integrator_templates.hh	
Define template classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators	318
time_change_subscriber.hh	
Define the class TimeChangeSubscriber	319
two_d_array.hh	
Defines the template class er7_utils::TwoDArray , which implements an RAII rectangular 2D array	320

Chapter 6

Module Documentation

6.1 Models

Modules

- [Utils](#)

6.1.1 Detailed Description

6.2 Utils

Modules

- [Integration](#)

6.2.1 Detailed Description

6.3 Integration

Modules

- [GaussJackson](#)
- [Lsode](#)

Files

- file [generalized_second_order_ode_technique.hh](#)
Define the static class GeneralizedSecondOrderODETechnique.
- file [integration_messages.hh](#)
Define the class IntegrationMessages, the class that specifies the message IDs used in the integration model.
- file [jeod_integration_group.hh](#)
Define the extensible class IntegrationGroup, an instance of which is responsible for integrating the states of a set of DynBody objects.
- file [jeod_integration_time.hh](#)
Define the class JeodIntegrationTime.
- file [restartable_2d_second_order_integrator.hh](#)
Defines the class Restartable2DSecondOrderODEIntegrator.
- file [restartable_state_integrator.hh](#)
Define classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators.
- file [restartable_state_integrator_templates.hh](#)
Define template classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators.
- file [time_change_subscriber.hh](#)
Define the class TimeChangeSubscriber.
- file [generalized_second_order_ode_technique.cc](#)
Define class GeneralizedSecondOrderODETechnique methods.
- file [integration_messages.cc](#)
Implement the class IntegrationMessages.
- file [jeod_integration_group.cc](#)
Define JeodIntegrationGroup methods.
- file [jeod_integration_time.cc](#)
Define JeodIntegrationTime methods.

6.3.1 Detailed Description

6.4 GaussJackson

Files

- file [gauss_jackson_coefficients_pair.hh](#)
Defines the class GaussJacksonCoefficientsPair, which contains summed Adams and Gauss-Jackson coefficient pair.
- file [gauss_jackson_coeffs.hh](#)
Defines the class GaussJacksonCoeffs, which contains the Gauss-Jackson predictor and corrector coefficients.
- file [gauss_jackson_config.hh](#)
Defines the class GaussJacksonConfig, which specifies Gauss-Jackson configuration data.
- file [gauss_jackson_first_order_ode_integrator.hh](#)
Defines the class GaussJacksonFirstOrderODEIntegrator, which integrates a first order ODE using the summed Adams technique.
- file [gauss_jackson_generalized_second_order_ode_integrator.hh](#)
Defines the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator, which integrates a generalized derivative second order ODE using Gauss-Jackson.
- file [gauss_jackson_integration_controls.hh](#)
Defines the class GaussJacksonIntegrationControls, which controls Gauss-Jackson integration process.
- file [gauss_jackson_integrator_base.hh](#)

- Defines the template class GaussJacksonIntegratorBase, which provides the basis for Gauss-Jackson integration.*

 - file [gauss_jackson_integrator_base_first.hh](#)

Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE.

 - file [gauss_jackson_integrator_base_second.hh](#)
- Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE.*
- file [gauss_jackson_integrator_constructor.hh](#)
- Defines the class GaussJacksonIntegratorConstructor, which constructs integrators that use Gauss-Jackson integration.*
- file [gauss_jackson_one_state.hh](#)
- Defines the class GaussJacksonOneState, which contains a double* pointer.*
- file [gauss_jackson_rational_coeffs.hh](#)
- Defines the class GaussJacksonRationalCoefficients, which contains a set of Adams or Stormer-Cowell coefficients.*
- file [gauss_jackson_simple_second_order_ode_integrator.hh](#)
- Defines the class GaussJacksonSimpleSecondOrderODEIntegrator, which integrates a simple second order ODE using Gauss-Jackson.*
- file [gauss_jackson_state_machine.hh](#)
- Defines the class GaussJacksonStateMachine, which guides the Gauss-Jackson integration process.*
- file [gauss_jackson_two_state.hh](#)
- Defines the class GaussJacksonTwoState, which contains a pair of double* pointers.*
- file [two_d_array.hh](#)
- Defines the template class `er7_utils::TwoDArray`, which implements an RAll rectangular 2D array.*
- file [gauss_jackson_coefficients_pair.cc](#)
- Defines member functions for the class GaussJacksonCoefficientsPair.*
- file [gauss_jackson_coeffs.cc](#)
- Defines member functions for the class GaussJacksonCoeffs.*
- file [gauss_jackson_config.cc](#)
- Defines member functions for the class GaussJacksonIntegratorConstructor.*
- file [gauss_jackson_generalized_second_order_ode_integrator.cc](#)
- Defines member functions for the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator.*
- file [gauss_jackson_integration_controls.cc](#)
- Defines member functions for the class GaussJacksonIntegrationControls.*
- file [gauss_jackson_integrator_constructor.cc](#)
- Defines member functions for the class GaussJacksonIntegratorConstructor.*
- file [gauss_jackson_rational_coeffs.cc](#)
- Defines member functions for the class GaussJacksonRationalCoefficients.*
- file [gauss_jackson_state_machine.cc](#)
- Defines member functions for the class GaussJacksonStateMachine.*

6.4.1 Detailed Description

6.5 Lsode

Files

- file [lsode_control_data_interface.hh](#)
- Define the class LsodeControlDataInterface.*
- file [lsode_data_classes.hh](#)
- Define LSODE classes that contain just data members.*
- file [lsode_first_order_ode_integrator.hh](#)

- Define the class *LsodeFirstOrderODEIntegrator*, the Jeod-compatible version of the Livermore ODE solver, LSODE.
- file [lsode_generalized_second_order_ode_integrator.hh](#)

Define the class *LsodeGeneralizedDerivSecondOrderODEIntegrator*.
- file [lsode_integration_controls.hh](#)

Define the class *LsodeIntegrationControls*.
- file [lsode_integrator_constructor.hh](#)

Define the class *LsodeIntegratorConstructor*, the class that constructs the integration controls and the integrators for the LSODE method.
- file [lsode_second_order_ode_integrator.hh](#)

Define the class *LsodeSecondOrderODEIntegrator*.
- file [lsode_simple_second_order_ode_integrator.hh](#)

Define the class *LsodeSimpleSecondOrderODEIntegrator*.
- file [lsode_control_data_interface.cc](#)

Define member functions for the class *LsodeControlDataInterface*.
- file [lsode_data_classes.cc](#)

Define member functions for the data-grouping classes specified in *lsode_data_classes*.
- file [lsode_first_order_ode_integrator_integrator.cc](#)

Define member functions for the class *LsodeFirstOrderODEIntegrator*.
- file [lsode_first_order_ode_integrator_manager.cc](#)

Define member functions for the class *LsodeFirstOrderODEIntegrator*.
- file [lsode_first_order_ode_integrator_support.cc](#)

Define member functions for the class *LsodeFirstOrderODEIntegrator*.
- file [lsode_first_order_ode_integrator_utility.cc](#)

Define member functions for the class *LsodeFirstOrderODEIntegrator*.
- file [lsode_generalized_second_order_ode_integrator.cc](#)

Define member functions for the class *LsodeGeneralizedDerivSecondOrderODEIntegrator*.
- file [lsode_integration_controls.cc](#)

Define the methods for the class *LsodeIntegrationControls*.
- file [lsode_integrator_constructor.cc](#)

Define the methods in the class *LsodeIntegratorConstructor*.
- file [lsode_second_order_ode_integrator.cc](#)

Define member functions for the class *LsodeSecondOrderODEIntegrator*.
- file [lsode_simple_second_order_ode_integrator.cc](#)

Define member functions for the class *LsodeSimpleSecondOrderODEIntegrator*.

Functions

- [jeod::LsodeControlDataInterface::LsodeControlDataInterface \(\)](#)

constructor
- void [jeod::LsodeControlDataInterface::check_interface_data \(\)](#)

verifies that the input data has legal values.
- void [jeod::LsodeControlDataInterface::allocate_arrays \(\)](#)

allocates space for vector-populated data to allow for restart
- void [jeod::LsodeControlDataInterface::destroy_allocated_arrays \(\)](#)

De-allocates allocated array.
- void [jeod::LsodeControlDataInterface::set_rel_tol \(int index, double value\)](#)

set values from external
- void [jeod::LsodeControlDataInterface::set_abs_tol \(int index, double value\)](#)
- void [jeod::LsodeDataArrays::allocate_arrays \(unsigned int num_odes, LsodeControlDataInterface::CorrectorMethod corrector_method\)](#)

- Allocates memory for the variable size arrays.*

 - void `jeod::LsodeDataArrays::destroy_allocated_arrays ()`
- Allows for refactoring and reallocation of newly sized arrays.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_core ()`

*integrator_core provides the front-end to all of the integrator_*i methods, which together perform one step of the integration of an initial value problem for a system of ordinary differential equations.*
- Sets/resets the method_coeffs_current array.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs ()`
- Tests h against old h.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change ()`
- Resets history arrays and time-step.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh ()`
- This section computes the predicted values by multiplying the history array by the Pascal Triangle matrix.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_predict ()`
- This method resets the iteration loop to the values generated by the integrator_predict method, which populated history[*][0].*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part1 ()`
- This code follows part 1 after the break-out to get to the external calls, and completes the reset of the integration iteration loop.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part2 ()`
- Keeps looping through the iterations until convergence or failure.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration ()`
- The corrector iteration failed to converge.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part1 ()`
- Retracts the history array in the case that the correction iteration failed to converge with either functional iteration or with an up-to-date Jacobian.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2 ()`
- Starts the processing of a converged iteration.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged ()`
- Restores the history array following the failure of the corrector for exceeding local error bounds.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed ()`
- The first steps in computing whether the order of the integrator should be changed.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep ()`
- Computes the step-size scaling factors that will result once it is determined what happens to the order at the next step.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order ()`
- Sets the new order and the step-ratio for the next step - or the current step if redoing it.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_check_step_error ()`
- Sets the new order and the step-ratio for the next step - or the current step if redoing it.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order ()`
- Repopulates the y-array from history, recomputes the original first derivatives, sets the order back to 1, and reduces the step size by a factor of 10.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1 ()`
- Continue reset, with derivatives now at hand.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2 ()`
- Wraps up the completion of the integrator.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup ()`
- this is the only succesful path back from integrator to manager.*

 - void `jeod::LsodeFirstOrderODEIntegrator::integrator_terminate ()`
- Propagate state via Lsode's method.*

 - er7_utils::IntegratorResult `jeod::LsodeFirstOrderODEIntegrator::integrate (double dyn_dt, unsigned int target_stage, const double *y_dot, double *y) override`
- Propagate state via Lsode's method.*

 - void `jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start ()`

- The code block from the main integrate routine for re_entry_point=CycleStartFinish.*
- void `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1 ()`
Sets the values for the case with calculation_phase = 1.
 - void `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2 ()`
 - int `jeod::LsodeFirstOrderODEIntegrator::manager_check_stop_conditions ()`
verifies whether the convergence conditions have been met to end the cycle.
 - void `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1 ()`
The iteration loop for the integration process.
 - void `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part2 ()`
 - void `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part3 ()`
 - void `jeod::LsodeFirstOrderODEIntegrator::reset_integrator ()` override
Resets the integrator when the timestep changes or when identified as needing a reset.
 - void `jeod::LsodeFirstOrderODEIntegrator::manager_set_calculation_phase_eq_2_reload ()`
 - void `jeod::LsodeFirstOrderODEIntegrator::calculate_epsilon ()`
Identify the smallest double precision value, epsilon, such that the computer can distinguish (1+epsilon) from 1.
 - void `jeod::LsodeFirstOrderODEIntegrator::calculate_integration_coefficients ()`
Modified from original DCFODE subroutine.
 - void `jeod::LsodeFirstOrderODEIntegrator::interpolate_y ()`
Interpolates the zeroth-derivative of y Adapted from subroutine DINTDY, which was a general method for interpolating the K-th derivative of the dependent variable vector, y.
 - void `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init ()`
Modified from DPREPJ.
 - bool `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop ()`
 - bool `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up ()`
 - void `jeod::LsodeFirstOrderODEIntegrator::linear_chord_iteration ()`
Modified from DSOLSY.
 - void `jeod::LsodeFirstOrderODEIntegrator::load_ew_values ()`
 - `jeod::LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator ()`
LsodeFirstOrderODEIntegrator default constructor.
 - `jeod::LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator (const LsodeControlDataInterface &data_in, er7_utils::IntegrationControls &controls, unsigned int size)`
LsodeFirstOrderODEIntegrator non-default constructor.
 - `jeod::LsodeFirstOrderODEIntegrator::~~LsodeFirstOrderODEIntegrator ()` override
LsodeFirstOrderODEIntegrator destructor.
 - void `jeod::LsodeFirstOrderODEIntegrator::update_control_data ()`
Gets the control data from where it can be populated in the constructor and verifies that the input control parameters are not out of sensible range.
 - `LsodeFirstOrderODEIntegrator * jeod::LsodeFirstOrderODEIntegrator::create_copy ()` const override
Create a copy of 'this' LsodeFirstOrderODEIntegrator object.
 - double `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array (const double *v)`
returns the RMS value of {V dot W}, where V and W are N-vectors.
 - double `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array (unsigned int ix, double **v)`
returns RMS value of v[][index]*
 - int `jeod::LsodeFirstOrderODEIntegrator::gauss_elim_factor ()`
Factors a double array (arrays.lin_alg) by Gaussian elimination.
 - void `jeod::LsodeFirstOrderODEIntegrator::linear_solver ()`
Solves the equation Y' = A Y, with A = arrays.lin_alg.
 - unsigned int `jeod::LsodeFirstOrderODEIntegrator::index_of_max_magnitude (unsigned int num_points, double **mx, int starting_ix)`
Modified version of IDAMAX.
 - void `jeod::LsodeFirstOrderODEIntegrator::load_derivatives (double *derivs)`
Load the externally generated derivative values (incoming as y_dot)i into the array derivs.

- `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::LsodeGeneralizedDerivSecondOrderODEIntegrator` (const `LsodeControlDataInterface` &data_in, `er7_utils::IntegrationControls` &controls, const `er7_utils::GeneralizedPositionDerivativeFunctions` &deriv_funs, unsigned int position_size, unsigned int velocity_size)
non-default constructor
- `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::LsodeGeneralizedDerivSecondOrderODEIntegrator` (const `LsodeGeneralizedDerivSecondOrderODEIntegrator` &src)
LsodeGeneralizedDerivSecondOrderODEIntegrator copy constructor.
- `LsodeGeneralizedDerivSecondOrderODEIntegrator * jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::create_copy` () const override
Clone a LsodeGeneralizedDerivSecondOrderODEIntegrator.
- `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::~~LsodeGeneralizedDerivSecondOrderODEIntegrator` () override
LsodeGeneralizedDerivSecondOrderODEIntegrator destructor.
- `er7_utils::IntegratorResult jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate` (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position) override
Propagate state via Lsode's method.
- `jeod::LsodeIntegrationControls::LsodeIntegrationControls` (unsigned int num_stages)
- `LsodeIntegrationControls * jeod::LsodeIntegrationControls::create_copy` () const override
Create a copy of 'this' StandardIntegrationControls object.
- unsigned int `jeod::LsodeIntegrationControls::integrate` (double start_time, double sim_dt, `er7_utils::TimeInterface` &time_interface, `er7_utils::IntegratorInterface` &integ_interface, `er7_utils::BaseIntegrationGroup` &integ_group) override
Perform one step of the integration process.
- `jeod::LsodeIntegratorConstructor::LsodeIntegratorConstructor` (const `LsodeIntegratorConstructor` &src)
- static `er7_utils::IntegratorConstructor * jeod::LsodeIntegratorConstructor::create_constructor` ()
Named constructor; create an LsodeIntegratorConstructor instance.
- `er7_utils::IntegratorConstructor * jeod::LsodeIntegratorConstructor::create_copy` () const override
Create a duplicate of the constructor.
- `er7_utils::IntegrationControls * jeod::LsodeIntegratorConstructor::create_integration_controls` () const override
Create an integration controls that guides the Lsode integration process.
- `er7_utils::FirstOrderODEIntegrator * jeod::LsodeIntegratorConstructor::create_first_order_ode_integrator` (unsigned int size, `er7_utils::IntegrationControls` &controls) const override
Create an Lsode state integrator for a first order ODE.
- `er7_utils::SecondOrderODEIntegrator * jeod::LsodeIntegratorConstructor::create_second_order_ode_integrator` (unsigned int size, `er7_utils::IntegrationControls` &controls) const override
Create an Lsode state integrator for a simple second order ODE.
- `er7_utils::SecondOrderODEIntegrator * jeod::LsodeIntegratorConstructor::create_generalized_deriv_second_order_ode_integrator` (unsigned int position_size, unsigned int velocity_size, const `er7_utils::GeneralizedPositionDerivativeFunctions` &deriv_funs, `er7_utils::IntegrationControls` &controls) const override
Create an Lsode state integrator for a generalized second order ODE where generalized position is advanced with the use of the position derivative function.
- `jeod::LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator` (const `LsodeControlDataInterface` &data_in, `er7_utils::IntegrationControls` &controls, unsigned int size)
LsodeSecondOrderODEIntegrator non-default constructor.
- `jeod::LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator` (const `LsodeControlDataInterface` &data_in, `er7_utils::IntegrationControls` &controls, const `er7_utils::GeneralizedPositionDerivativeFunctions` &deriv_funs, unsigned int position_size, unsigned int velocity_size)
- `jeod::LsodeSecondOrderODEIntegrator::~~LsodeSecondOrderODEIntegrator` () override
LsodeSecondOrderODEIntegrator destructor.
- `jeod::LsodeSimpleSecondOrderODEIntegrator::LsodeSimpleSecondOrderODEIntegrator` (const `LsodeControlDataInterface` &data_in, `er7_utils::IntegrationControls` &controls, unsigned int size)
LsodeSimpleSecondOrderODEIntegrator non-default constructor.

- `LsodeSimpleSecondOrderODEIntegrator * jeod::LsodeSimpleSecondOrderODEIntegrator::create_copy ()` const override
- `er7_utils::IntegratorResult jeod::LsodeSimpleSecondOrderODEIntegrator::integrate (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position)` override

Propagate state via Lsode's method.

6.5.1 Detailed Description

6.5.2 Function Documentation

6.5.2.1 `allocate_arrays()` [1/2]

```
void LsodeControlDataInterface::allocate_arrays ( )
```

allocates space for vector-populated data to allow for restart

Definition at line 260 of file `lsode_control_data_interface.cc`.

References `jeod::LsodeControlDataInterface::abs_tolerance_error_control`, `jeod::LsodeControlDataInterface::abs_tolerance_error_control_vec`, `jeod::LsodeControlDataInterface::error_control_vector_copied_over`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeControlDataInterface::num_odes_at_alloc`, `jeod::LsodeControlDataInterface::rel_tolerance_error_control`, and `jeod::LsodeControlDataInterface::rel_tolerance_error_control_vec`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1()`.

6.5.2.2 `allocate_arrays()` [2/2]

```
void LsodeDataArrays::allocate_arrays (
    unsigned int num_odes,
    LsodeControlDataInterface::CorrectorMethod corrector_method )
```

Allocates memory for the variable size arrays.

Definition at line 52 of file `lsode_data_classes.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeDataArrays::allocated`, `jeod::LsodeDataArrays::error_weight`, `jeod::LsodeControlDataInterface::FunctionalIteration`, `jeod::LsodeDataArrays::history`, `jeod::LsodeControlDataInterface::JacobiNewtonInternalJac`, `jeod::LsodeDataArrays::lin_alg`, `jeod::LsodeDataArrays::lin_alg_index1`, `jeod::LsodeControlDataInterface::NewtonIterInternalJac`, `jeod::LsodeControlDataInterface::NewtonIterUserJac`, `jeod::LsodeDataArrays::num_odes`, `jeod::LsodeDataArrays::pivots`, and `jeod::LsodeDataArrays::save`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1()`.

6.5.2.3 calculate_epsilon()

```
void LsodeFirstOrderODEIntegrator::calculate_epsilon ( ) [protected]
```

Identify the smallest double precision value, epsilon, such that the computer can distinguish (1+epsilon) from 1.

Definition at line 57 of file lsode_first_order_ode_integrator__support.cc.

References jeod::LsodeFirstOrderODEIntegrator::epsilon, and jeod::LsodeFirstOrderODEIntegrator::sqrt_epsilon.

Referenced by jeod::LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator().

6.5.2.4 calculate_integration_coefficients()

```
void LsodeFirstOrderODEIntegrator::calculate_integration_coefficients ( ) [protected]
```

Modified from original DCFODE subroutine.

calculate_integration_coefficients is called by dstode to set coefficients needed there. The coefficients for the current method, as given by the value of integration_method, are set for all orders and saved. The maximum order assumed here is 12 if integration_method = ImplicitAdamsNonStiff and 5 if integration_method = ImplicitBackDiffStiff.

NOTE - A smaller value of the maximum order is also allowed and may be set by the user with the value control_data.max_order, which gets copied to the protected value max_order_internal.

calculate_integration_coefficients is called once at the beginning of the problem, and again only if integration_method is changed.

The coefficients are stored in two arrays: method_coeffs_complete is a 13x12 array that contains a complete set of coefficients for the method test_coeffs_complete is a 3x12 array that contains the coefficients for local error tests and selection of the step size and/or order. The 1st set of 12 coeffs is for order method_order_current - 1 The 2nd set of 12 coeffs is for order method_order_current The 3rd set of 12 coeffs is for order method_order_current + 1

The coefficients in method_coeffs_complete are computed by a generating polynomial. For a given order (note that order changes during the integration process up to the maximum allowable, and is identified in the integrator as the variable method_order_current), abbreviate method_coeffs_complete[i][order-1] to mcco[i]

Then $l(x) = mcco[0] + (mcco[1] * x) + (mcco[2] * x^2) + \dots + mcco[order] * (x^{order})$ For the implicit Adams methods, $l(x)$ is given by $dl/dx = (x+1)*(x+2)*\dots*(x+order-1)/(order-1)!$, $l(-1) = 0$.; For the BDF methods, $l(x)$ is given by $l(x) = (x+1)*(x+2)*\dots*(x+order) / ((order)! * (1 + 1/2 + \dots + 1/order))$

(! represents factorial)

Note that while method_coeffs_compelte is a rectangular array for convenience, it is effectively a triangular array since method_coeffs_complete[i][order-1] has no meaning for $i \geq (order + 2)$ order=1: method_coeffs_complete[0-1][0] order=2: method_coeffs_complete[0-2][1] ... order=12: method_coeffs_complete[0-12][11] Hence a 13x12 array.

Definition at line 118 of file lsode_first_order_ode_integrator__support.cc.

References jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeControlDataInterface::ImplicitAdamsNonStiff, jeod::LsodeControlDataInterface::ImplicitBackDiffStiff, jeod::LsodeControlDataInterface::integration_method, jeod::LsodeFirstOrderODEIntegrator::method_coeffs_complete, and jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_core().

6.5.2.5 check_interface_data()

```
void LsodeControlDataInterface::check_interface_data ( )
```

verifies that the input data has legal values.

Definition at line 59 of file lsode_control_data_interface.cc.

References `jeod::LsodeControlDataInterface::abs_tolerance_error_control_vec`, `jeod::LsodeControlDataInterface::CommonAbsSpecificRel`, `jeod::LsodeControlDataInterface::corrector_method`, `jeod::LsodeControlDataInterface::error_control_indicator`, `jeod::LsodeControlDataInterface::error_control_vector_copied_over`, `jeod::LsodeControlDataInterface::integration_method`, `jeod::LsodeControlDataInterface::max_num_small_step_warnings`, `jeod::LsodeControlDataInterface::max_num_steps`, `jeod::LsodeControlDataInterface::max_order`, `jeod::LsodeControlDataInterface::max_step_size`, `jeod::LsodeControlDataInterface::min_step_size`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeControlDataInterface::rel_tolerance_error_control_vec`, `jeod::LsodeControlDataInterface::SpecificAbsCommonRel`, and `jeod::LsodeControlDataInterface::SpecificAbsSpecificRel`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::update_control_data()`.

6.5.2.6 create_constructor()

```
er7_utils::IntegratorConstructor * LsodeIntegratorConstructor::create_constructor ( ) [static]
```

Named constructor; create an [LsodeIntegratorConstructor](#) instance.

The caller is responsible for deleting the returned object.

Returns

Newly created [LsodeIntegratorConstructor](#) instance.

Definition at line 64 of file lsode_integrator_constructor.cc.

6.5.2.7 create_copy() [1/5]

```
LsodeFirstOrderODEIntegrator * LsodeFirstOrderODEIntegrator::create_copy ( ) const [override]
```

Create a copy of 'this' [LsodeFirstOrderODEIntegrator](#) object.

Returns

Clone of 'this'.

Definition at line 118 of file lsode_first_order_ode_integrator__utility.cc.

6.5.2.8 create_copy() [2/5]

```
LsodeGeneralizedDerivSecondOrderODEIntegrator * LsodeGeneralizedDerivSecondOrderODEIntegrator↔
::create_copy ( ) const [override]
```

Clone a [LsodeGeneralizedDerivSecondOrderODEIntegrator](#).

Definition at line 81 of file `lsode_generalized_second_order_ode_integrator.cc`.

6.5.2.9 create_copy() [3/5]

```
LsodeIntegrationControls * LsodeIntegrationControls::create_copy ( ) const [override]
```

Create a copy of 'this' `StandardIntegrationControls` object.

Returns

Clone of 'this'.

Definition at line 56 of file `lsode_integration_controls.cc`.

6.5.2.10 create_copy() [4/5]

```
er7_utils::IntegratorConstructor * LsodeIntegratorConstructor::create_copy ( ) const [override]
```

Create a duplicate of the constructor.

The caller is responsible for deleting the returned object.

Returns

Duplicated constructor.

Definition at line 70 of file `lsode_integrator_constructor.cc`.

6.5.2.11 create_copy() [5/5]

```
LsodeSimpleSecondOrderODEIntegrator * LsodeSimpleSecondOrderODEIntegrator::create_copy ( )
const [override]
```

Definition at line 57 of file `lsode_simple_second_order_ode_integrator.cc`.

6.5.2.12 create_first_order_ode_integrator()

```
er7_utils::FirstOrderODEIntegrator * LsodeIntegratorConstructor::create_first_order_ode↔
integrator (
    unsigned int size,
    er7_utils::IntegrationControls & controls ) const [override]
```

Create an Lsode state integrator for a first order ODE.

The caller is responsible for deleting the created object.

Returns

State integrator

Parameters

<i>in</i>	<i>size</i>	State size
<i>in, out</i>	<i>controls</i>	Integration controls

Definition at line 82 of file `lsode_integrator_constructor.cc`.

References `jeod::LsodeIntegratorConstructor::data_interface`.

6.5.2.13 create_generalized_deriv_second_order_ode_integrator()

```
er7_utils::SecondOrderODEIntegrator * LsodeIntegratorConstructor::create_generalized_deriv_↵
second_order_ode_integrator (
    unsigned int position_size,
    unsigned int velocity_size,
    const er7_utils::GeneralizedPositionDerivativeFunctions & deriv_funs,
    er7_utils::IntegrationControls & controls ) const [override]
```

Create an Lsode state integrator for a generalized second order ODE where generalized position is advanced with the use of the position derivative function.

The caller is responsible for deleting the created object.

Returns

State integrator

Parameters

<i>in</i>	<i>position_size</i>	Size of the generalized position
<i>in</i>	<i>velocity_size</i>	Size of the generalized velocity
<i>in</i>	<i>deriv_funs</i>	Position derivative functions container
<i>in, out</i>	<i>controls</i>	Integration controls

Definition at line 102 of file `lsode_integrator_constructor.cc`.

References `jeod::LsodeIntegratorConstructor::data_interface`.

6.5.2.14 create_integration_controls()

```
er7_utils::IntegrationControls * LsodeIntegratorConstructor::create_integration_controls ( )
const [override]
```

Create an integration controls that guides the Lsode integration process.

The caller is responsible for deleting the created object.

Returns

Integration controls object

Definition at line 76 of file `lsode_integrator_constructor.cc`.

6.5.2.15 create_second_order_ode_integrator()

```
er7_utils::SecondOrderODEIntegrator * LsodeIntegratorConstructor::create_second_order_ode_↵
integrator (
    unsigned int size,
    er7_utils::IntegrationControls & controls ) const [override]
```

Create an Lsode state integrator for a simple second order ODE.

The caller is responsible for deleting the created object.

Returns

State integrator

Parameters

<i>in</i>	<i>size</i>	State size
<i>in, out</i>	<i>controls</i>	Integration controls

Definition at line 92 of file `lsode_integrator_constructor.cc`.

References `jeod::LsodeIntegratorConstructor::data_interface`.

6.5.2.16 destroy_allocated_arrays() [1/2]

```
void LsodeControlDataInterface::destroy_allocated_arrays ( )
```

De-allocates allocated array.

Definition at line 312 of file `lsode_control_data_interface.cc`.

References `jeod::LsodeControlDataInterface::abs_tolerance_error_control`, `jeod::LsodeControlDataInterface::error_control_vector_copied_over`, and `jeod::LsodeControlDataInterface::rel_tolerance_error_control`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::~LsodeFirstOrderODEIntegrator()`.

6.5.2.17 `destroy_allocated_arrays()` [2/2]

```
void LsodeDataArrays::destroy_allocated_arrays ( )
```

Allows for refactoring and reallocation of newly sized arrays.

Definition at line 148 of file `lsode_data_classes.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeDataArrays::allocated`, `jeod::LsodeDataArrays::error_weight`, `jeod::LsodeDataArrays::history`, `jeod::LsodeDataArrays::lin_alg`, `jeod::LsodeDataArrays::lin_alg_index1`, `jeod::LsodeDataArrays::num_odes`, `jeod::LsodeDataArrays::pivots`, and `jeod::LsodeDataArrays::save`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::~LsodeFirstOrderODEIntegrator()`.

6.5.2.18 `gauss_elim_factor()`

```
int LsodeFirstOrderODEIntegrator::gauss_elim_factor ( ) [protected]
```

Factors a double array (`arrays.lin_alg`) by Gaussian elimination.

Modified version of DGEFA.

Definition at line 174 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::index_of_max_magnitude()`, `jeod::LsodeDataArrays::lin_alg`, `jeod::LsodeControlDataInterface::num_odes`, and `jeod::LsodeDataArrays::pivots`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up()`.

6.5.2.19 `index_of_max_magnitude()`

```
unsigned int LsodeFirstOrderODEIntegrator::index_of_max_magnitude (
    unsigned int num_points,
    double ** array,
    int start_ix ) [protected]
```

Modified version of IDAMAX.

IDAMAX has 2 operations, one for situations in which the index increments by 1, and another for the converse. Since all instances in LSODE use the unit-increment method, that is the only one represented here. Search through matrix "array", starting at `array[start_ix_1][start_ix_2]` for the next "num_points" elements. The boolean `search_ix_1` controls whether to increment index#1 (true), or index#2 (false). Method returns the searched index that corresponds to the largest magnitude.

Note

The only call to this method passed "k" in for both indices, so I stripped the second argument. If DGBFA gets implemented, it will have to be added back in; the call from DGBFA is for array starting at (M,K)

Definition at line 337 of file `lsode_first_order_ode_integrator__utility.cc`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::gauss_elim_factor()`.

6.5.2.20 integrate() [1/4]

```
er7_utils::IntegratorResult LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * accel,
    double * velocity,
    double * position ) [override]
```

Propagate state via Lsode's method.

Parameters

in	<i>dyn_dt</i>	Integration interval step, dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>accel</i>	Generalized acceleration vector.
in, out	<i>velocity</i>	Generalized velocity vector.
in, out	<i>position</i>	Generalized position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 101 of file `Lsode_generalized_second_order_ode_integrator.cc`.

References `jeod::LsodeSecondOrderODEIntegrator::first_derivative_size`, `jeod::LsodeSecondOrderODEIntegrator::first_order_integrator`, `jeod::LsodeFirstOrderODEIntegrator::integrate()`, `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::posdot`, `jeod::LsodeSecondOrderODEIntegrator::y`, `jeod::LsodeSecondOrderODEIntegrator::y_dot`, and `jeod::LsodeSecondOrderODEIntegrator::zeroth_derivative_size`.

6.5.2.21 integrate() [2/4]

```
er7_utils::IntegratorResult LsodeSimpleSecondOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * accel,
    double * velocity,
    double * position ) [override]
```

Propagate state via Lsode's method.

Parameters

in	<i>dyn_dt</i>	Integration interval step, dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>accel</i>	Generalized acceleration vector.
in, out	<i>velocity</i>	Generalized velocity vector.
in, out	<i>position</i>	Generalized position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 67 of file `lsode_simple_second_order_ode_integrator.cc`.

References `jeod::LsodeSecondOrderODEIntegrator::first_order_integrator`, `jeod::LsodeFirstOrderODEIntegrator::integrate()`, `jeod::LsodeSecondOrderODEIntegrator::y`, `jeod::LsodeSecondOrderODEIntegrator::y_dot`, and `jeod::LsodeSecondOrderODEIntegrator::zeroth_derivative_size`.

6.5.2.22 integrate() [3/4]

```
er7_utils::IntegratorResult LsodeFirstOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * y_dot,
    double * y ) [override]
```

Propagate state via Lsode's method.

Propagate state via the LSODE method.

Parameters

in	<i>dyn_dt</i>	Integration interval step, dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in, out	<i>y_dot</i>	Generalized velocity vector.
in, out	<i>y</i>	Generalized position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 60 of file `lsode_first_order_ode_integrator__manager.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::cycle_target_time`, `jeod::LsodeFirstOrderODEIntegrator::CycleStartFinish`, `jeod::LsodeFirstOrderODEIntegrator::DstodeResetStep`, `jeod::LsodeFirstOrderODEIntegrator::InitCalc`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part2()`, `jeod::LsodeFirstOrderODEIntegrator::IterationLoop`, `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init()`, `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop()`, `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up()`, `jeod::LsodeFirstOrderODEIntegrator::JacobianPrep`, `jeod::LsodeFirstOrderODEIntegrator::load_derivatives()`, `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2()`, `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part2()`, `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part3()`, `jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start()`, `jeod::LsodeFirstOrderODEIntegrator::re_entry_point`, `jeod::LsodeFirstOrderODEIntegrator::ResetIterLoop`, `jeod::LsodeDataArrays::save`, `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`, `jeod::LsodeFirstOrderODEIntegrator::step_error`, `jeod::LsodeFirstOrderODEIntegrator::update_jacobian`, `jeod::LsodeFirstOrderODEIntegrator::y`, and `jeod::LsodeFirstOrderODEIntegrator::y_dot`.

Referenced by `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate()`, and `jeod::LsodeSimpleSecondOrderODEIntegrator::integrate()`.

6.5.2.23 integrate() [4/4]

```
unsigned int LsodeIntegrationControls::integrate (
    double start_time,
    double sim_dt,
    er7_utils::TimeInterface & time_interface,
    er7_utils::IntegratorInterface & integ_interface,
    er7_utils::BaseIntegrationGroup & integ_group ) [override]
```

Perform one step of the integration process.

Definition at line 69 of file `lsode_integration_controls.cc`.

6.5.2.24 integrator_compute_new_order()

```
void LsodeFirstOrderODEIntegrator::integrator_compute_new_order ( ) [protected]
```

Computes the step-size scaling factors that will result once it is determined what happens to the order at the next step.

Definition at line 746 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeDataStode::dsm`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_check_step_error()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup()`, `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array()`, `jeod::LsodeFirstOrderODEIntegrator::method_coeffs_current`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeDataStode::new_method_order`, `jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::order_select_para`, `jeod::LsodeDataStode::step_ratio`, `jeod::LsodeDataStode::step_ratio_order_inc`, and `jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed()`.

6.5.2.25 integrator_compute_new_order_check_step_error()

```
void LsodeFirstOrderODEIntegrator::integrator_compute_new_order_check_step_error ( ) [protected]
```

Definition at line 816 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup()`, `jeod::LsodeFirstOrderODEIntegrator::order_select_para`, `jeod::LsodeFirstOrderODEIntegrator::step_error`, and `jeod::LsodeDataStode::step_ratio`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`.

6.5.2.26 integrator_compute_new_order_prep()

```
void LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep ( ) [protected]
```

The first steps in computing whether the order of the integrator should be changed.

Regardless of the success or failure of the step, the step-ratio factors for an increase, decrease, or retention of the integrator order are computed. In the case of failure, the increase ratio (`data_stode.step_ratio_order_inc`) has already been set to 0.0 to prevent an order increase. The largest of these factors is determined and the new order chosen accordingly. In the unusual case of equality, the priority is given to:

1. retain the order
2. increase the order (if `inc = dec > same`)

If the order is to be increased, we compute one additional scaled derivative.

This process is spread over four methods - `integrator_compute_new_order_prep` `integrator_compute_new_order` `integrator_compute_new_order_check_step_error` `integrator_set_new_order`

extracted from lines 520-540

Definition at line 724 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array()`, `jeod::LsodeFirstOrderODEIntegrator::max_history_size`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeDataArrays::save`, `jeod::LsodeDataStode::step_ratio_order_inc`, and `jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged()`.

6.5.2.27 integrator_core()

```
void LsodeFirstOrderODEIntegrator::integrator_core ( ) [protected]
```

`integrator_core` provides the front-end to all of the `integrator_*i` methods, which together perform one step of the integration of an initial value problem for a system of ordinary differential equations.

Modified from DSTODE

NOTES: The entire `integrator_*` suite is independent of the value of the iteration method indicator, `corrector_method`, when said is `!= 0`, and hence is independent of the type of chord method used, or the Jacobian structure.

The value `internal_state` (JSTART) controls the direction that this method takes.

By commenting out substantial parts of the package that are not useful to the ER7 / JEOD / Trick implementation, the only viable values for `internal_state` are now 0 or 1. `internal_state = 0` : take the first step. `internal_state = 1` : take another step, continuing from the last. `internal_state = -1` was associated with externally driven changes to the input aparameters, something we do not allow. `internal_state = -2` is associated with the critical / singularity time avoidance, something we have not implemented.

Definition at line 77 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::calculate_↵
integration_coefficients()`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator↵
::convergence_factor`, `jeod::LsodeFirstOrderODEIntegrator::convergence_jacobian_flag`, `jeod::LsodeFirstOrder↵
ODEIntegrator::convergence_rate`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeControlData↵
Interface::integration_method`, `jeod::LsodeFirstOrderODEIntegrator::integrator_predict()`, `jeod::LsodeFirstOrder↵
ODEIntegrator::integrator_reset_method_coeffs()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh()`,
`jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change()`, `jeod::LsodeFirstOrderODEIntegrator↵
::internal_state`, `jeod::LsodeDataStode::iredo`, `jeod::LsodeDataStode::iret`, `jeod::LsodeControlDataInterface↵
::is_corrector_method_functional_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::iteration_matrix_singular`,
`jeod::LsodeFirstOrderODEIntegrator::jacobian_current`, `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_↵
weighted_array()`, `jeod::LsodeFirstOrderODEIntegrator::max_history_size`, `jeod::LsodeFirstOrderODEIntegrator↵
::max_order_internal`, `jeod::LsodeFirstOrderODEIntegrator::max_step_increase_ratio`, `jeod::LsodeFirstOrder↵
ODEIntegrator::method_coeff_first`, `jeod::LsodeFirstOrderODEIntegrator::method_coeffs_complete`, `jeod::Lsode↵
FirstOrderODEIntegrator::method_coeffs_current`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`,
`jeod::LsodeControlDataInterface::min_step_size`, `jeod::LsodeFirstOrderODEIntegrator::modified_iteration_↵
matrix_singular`, `jeod::LsodeDataStode::ncf`, `jeod::LsodeFirstOrderODEIntegrator::num_equations`, `jeod::Lsode↵
FirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::LsodeFirstOrderODEIntegrator::num_predictor_elements`,
`jeod::LsodeFirstOrderODEIntegrator::order_select_para`, `jeod::LsodeFirstOrderODEIntegrator::prev_integration_↵
method`, `jeod::LsodeFirstOrderODEIntegrator::prev_step_size`, `jeod::LsodeFirstOrderODEIntegrator::rel_change↵
_since_jacobian`, `jeod::LsodeDataArrays::save`, `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`, `jeod↵
::LsodeFirstOrderODEIntegrator::step_at_last_jacobian_update`, `jeod::LsodeFirstOrderODEIntegrator::step_↵
error`, `jeod::LsodeDataStode::step_ratio`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, `jeod::LsodeFirstOrder↵
ODEIntegrator::test_coeffs_complete`, `jeod::LsodeDataStode::told`, and `jeod::LsodeFirstOrderODEIntegrator↵
::update_jacobian`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part2()`.

6.5.2.28 integrator_corrector_converged()

```
void LsodeFirstOrderODEIntegrator::integrator_corrector_converged ( ) [protected]
```

Starts the processing of a converged iteration.

Definition at line 587 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::↵
LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::Lsode↵
DataStode::dsm`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_↵
new_order_prep()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed()`, `jeod::LsodeFirstOrder↵
ODEIntegrator::integrator_wrapup()`, `jeod::LsodeDataStode::iredo`, `jeod::LsodeFirstOrderODEIntegrator::iter_delta`,
`jeod::LsodeFirstOrderODEIntegrator::iteration_count`, `jeod::LsodeFirstOrderODEIntegrator::jacobian_current`,
`jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array()`, `jeod::LsodeFirstOrderODEIntegrator↵
::max_history_size`, `jeod::LsodeFirstOrderODEIntegrator::method_coeffs_current`, `jeod::LsodeFirstOrder↵
ODEIntegrator::method_order_current`, `jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::Lsode↵
ControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::num_steps_taken`, `jeod::LsodeFirst↵
OrderODEIntegrator::order_select_para`, `jeod::LsodeFirstOrderODEIntegrator::prev_good_step_size`, `jeod::↵
LsodeFirstOrderODEIntegrator::prev_method_order`, `jeod::LsodeFirstOrderODEIntegrator::step_error`, `jeod::↵
LsodeFirstOrderODEIntegrator::step_size`, and `jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration()`.

6.5.2.29 integrator_corrector_failed_part1()

```
void LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part1 ( ) [protected]
```

The corrector iteration failed to converge.

If corrector_method != FunctionalIteration and the Jacobian is out of date, exit so that the Jacobian method can be called (externally) for the next try. Otherwise, try changing the step-size in part 2 of the failure recovery.

Extracted from DSTODE lines 410-430

Definition at line 515 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::convergence↵_jacobian_flag, jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2(), jeod::LsodeFirstOrder↵ODEIntegrator::integrator_reset_iteration_loop_part1(), jeod::LsodeControlDataInterface::is_corrector_method↵_functional_iteration(), jeod::LsodeFirstOrderODEIntegrator::jacobian_current, and jeod::LsodeFirstOrder↵ODEIntegrator::update_jacobian.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration().

6.5.2.30 integrator_corrector_failed_part2()

```
void LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2 ( ) [protected]
```

Retracts the history array in the case that the correction iteration failed to converge with either functional iteration or with an up-to-date Jacobian.

If there are problems, the associated flags are set.

Definition at line 541 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::convergence_jacobian_flag, jeod::LsodeFirstOrderODEIntegrator::data_↵stode, jeod::LsodeDataArrays::history, jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh(), jeod::Lsode↵FirstOrderODEIntegrator::integrator_terminate(), jeod::LsodeDataStode::iredo, jeod::LsodeControlDataInterface↵::is_corrector_method_functional_iteration(), jeod::LsodeControlDataInterface::max_num_conv_failure, jeod::↵LsodeFirstOrderODEIntegrator::max_step_increase_ratio, jeod::LsodeFirstOrderODEIntegrator::method_order↵_current, jeod::LsodeControlDataInterface::min_step_size, jeod::LsodeDataStode::ncf, jeod::LsodeFirstOrder↵ODEIntegrator::num_equations, jeod::LsodeFirstOrderODEIntegrator::stage_target_time, jeod::LsodeFirstOrder↵ODEIntegrator::step_error, jeod::LsodeDataStode::step_ratio, jeod::LsodeFirstOrderODEIntegrator::step_size, jeod::LsodeDataStode::told, and jeod::LsodeFirstOrderODEIntegrator::update_jacobian.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part1(), and jeod::LsodeFirst↵OrderODEIntegrator::jacobian_prep_wrap_up().

6.5.2.31 integrator_corrector_iteration()

```
void LsodeFirstOrderODEIntegrator::integrator_corrector_iteration ( ) [protected]
```

Keeps looping through the iterations until convergence or failure.

Definition at line 410 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::convergence_factor`, `jeod::LsodeFirstOrderODEIntegrator::convergence_rate`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part1()`, `jeod::LsodeControlDataInterface::is_corrector_method_functional_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::iter_delta`, `jeod::LsodeFirstOrderODEIntegrator::iteration_count`, `jeod::LsodeFirstOrderODEIntegrator::iterationLoop`, `jeod::LsodeFirstOrderODEIntegrator::linear_chord_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array()`, `jeod::LsodeControlDataInterface::max_correction_iters`, `jeod::LsodeFirstOrderODEIntegrator::method_coeffs_current`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeFirstOrderODEIntegrator::modified_iteration_matrix_singular`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::prev_iter_delta`, `jeod::LsodeFirstOrderODEIntegrator::re_entry_point`, `jeod::LsodeDataArrays::save`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, `jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete`, and `jeod::LsodeFirstOrderODEIntegrator::y`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrate()`.

6.5.2.32 integrator_error_test_failed()

```
void LsodeFirstOrderODEIntegrator::integrator_error_test_failed ( ) [protected]
```

Restores the history array following the failure of the corrector for exceeding local error bounds.

Definition at line 662 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_terminate()`, `jeod::LsodeDataStode::iredo`, `jeod::LsodeFirstOrderODEIntegrator::max_step_increase_ratio`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeControlDataInterface::min_step_size`, `jeod::LsodeFirstOrderODEIntegrator::num_equations`, `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`, `jeod::LsodeFirstOrderODEIntegrator::step_error`, `jeod::LsodeDataStode::step_ratio_order_inc`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, and `jeod::LsodeDataStode::told`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged()`.

6.5.2.33 integrator_fail_reset_order_1_part1()

```
void LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1 ( ) [protected]
```

Repopulates the y-array from history, recomputes the original first derivatives, sets the order back to 1, and reduces the step size by a factor of 10.

Called when 3 or more failures have occurred. It is assumed that the derivatives that have accumulated in the history array have errors of the wrong order. Hence the first derivative is recomputed, and the order is set to 1. Then the step-size is reduced by a factor of 10, and the step is retried. Repeat until successful, or the step reaches the minimum step-size.

If 10 failures occur, exit with `step_error = -1`.

This method is divided in two by a call to calculate the derivatives. Part1 precedes that call, the execution exits from the integrator back to the sim control engine; then on return to the integrator, execution immediately proceeds with part2.

extracted from lines 640-

Definition at line 877 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeFirstOrderODEIntegrator::DstodeResetStep`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_terminate()`, `jeod::LsodeControlDataInterface::min_step_size`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::re_entry_point`, `jeod::LsodeFirstOrderODEIntegrator::step_error`, `jeod::LsodeDataStode::step_ratio`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, and `jeod::LsodeFirstOrderODEIntegrator::y`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed()`.

6.5.2.34 integrator_fail_reset_order_1_part2()

```
void LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2 ( ) [protected]
```

Continue reset, with derivatives now at hand.

See `integrator_fail_reset_order_1_part1` for details.

Definition at line 901 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_predict()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs()`, `jeod::LsodeDataStode::iret`, `jeod::LsodeControlDataInterface::is_corrector_method_functional_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::order_select_para`, `jeod::LsodeDataArrays::save`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, and `jeod::LsodeFirstOrderODEIntegrator::update_jacobian`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrate()`.

6.5.2.35 integrator_predict()

```
void LsodeFirstOrderODEIntegrator::integrator_predict ( ) [protected]
```

This section computes the predicted values by multiplying the history array by the Pascal Triangle matrix.

Extracted from DSTODE lines 200-215

Definition at line 342 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeDataArrays::history, jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part1(), jeod::LsodeControlDataInterface::is_corrector_method_functional_iteration(), jeod::LsodeControlDataInterface::max_num_steps_jacobian, jeod::LsodeFirstOrderODEIntegrator::max_rel_change_without_jacobian, jeod::LsodeFirstOrderODEIntegrator::method_order_current, jeod::LsodeFirstOrderODEIntegrator::num_equations, jeod::LsodeFirstOrderODEIntegrator::num_steps_taken, jeod::LsodeFirstOrderODEIntegrator::rel_change_since_jacobian, jeod::LsodeFirstOrderODEIntegrator::stage_target_time, jeod::LsodeFirstOrderODEIntegrator::step_at_last_jacobian_update, jeod::LsodeFirstOrderODEIntegrator::step_size, and jeod::LsodeFirstOrderODEIntegrator::update_jacobian.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh(), and jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change().

6.5.2.36 integrator_reset_iteration_loop_part1()

```
void LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part1 ( ) [protected]
```

This method resets the iteration loop to the values generated by the integrator_predict method, which populated history[*][0].

Definition at line 377 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeDataArrays::history, jeod::LsodeFirstOrderODEIntegrator::iteration_count, jeod::LsodeControlDataInterface::num_odes, jeod::LsodeFirstOrderODEIntegrator::re_entry_point, jeod::LsodeFirstOrderODEIntegrator::ResetIterLoop, and jeod::LsodeFirstOrderODEIntegrator::y.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part1(), and jeod::LsodeFirstOrderODEIntegrator::integrator_predict().

6.5.2.37 integrator_reset_iteration_loop_part2()

```
void LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part2 ( ) [protected]
```

This code follows part 1 after the break-out to get to the external calls, and completes the reset of the integration iteration loop.

Definition at line 394 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeDataArrays::accum_correction, jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::iter_delta, jeod::LsodeControlDataInterface::num_odes, and jeod::LsodeFirstOrderODEIntegrator::prev_iter_delta.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrate().

6.5.2.38 integrator_reset_method_coeffs()

```
void LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs ( ) [protected]
```

Sets/resets the method_coeffs_current array.

Definition at line 237 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::convergence_factor, jeod::LsodeFirstOrderODEIntegrator::data_stode, jeod::LsodeFirstOrderODEIntegrator::integrator_predict(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh(), jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change(), jeod::LsodeDataStode::iret, jeod::LsodeFirstOrderODEIntegrator::method_coeff_first, jeod::LsodeFirstOrderODEIntegrator::method_coeffs_complete, jeod::LsodeFirstOrderODEIntegrator::method_coeffs_current, jeod::LsodeFirstOrderODEIntegrator::method_order_current, jeod::LsodeControlDataInterface::min_step_size, jeod::LsodeFirstOrderODEIntegrator::num_equations, jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols, jeod::LsodeFirstOrderODEIntegrator::num_predictor_elements, jeod::LsodeFirstOrderODEIntegrator::rel_change_since_jacobian, jeod::LsodeDataStode::step_ratio, and jeod::LsodeFirstOrderODEIntegrator::step_size.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2(), and jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order().

6.5.2.39 integrator_reset_yh()

```
void LsodeFirstOrderODEIntegrator::integrator_reset_yh ( ) [protected]
```

Resets history arrays and time-step.

Definition at line 302 of file lsode_first_order_ode_integrator__integrator.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::data_stode, jeod::LsodeDataArrays::history, jeod::LsodeFirstOrderODEIntegrator::integrator_predict(), jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup(), jeod::LsodeDataStode::iredo, jeod::LsodeFirstOrderODEIntegrator::max_step_increase_ratio, jeod::LsodeFirstOrderODEIntegrator::max_step_size_inv, jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols, jeod::LsodeControlDataInterface::num_odes, jeod::LsodeFirstOrderODEIntegrator::order_select_para, jeod::LsodeFirstOrderODEIntegrator::rel_change_since_jacobian, jeod::LsodeDataStode::step_ratio, and jeod::LsodeFirstOrderODEIntegrator::step_size.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs(), jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order(), and jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change().

6.5.2.40 integrator_set_new_order()

```
void LsodeFirstOrderODEIntegrator::integrator_set_new_order ( ) [protected]
```

Sets the new order and the step-ratio for the next step - or the current step if redoing it.

Definition at line 839 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh()`, `jeod::LsodeDataStode::iret`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeControlDataInterface::min_step_size`, `jeod::LsodeDataStode::new_method_order`, `jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::LsodeDataStode::step_ratio`, and `jeod::LsodeFirstOrderODEIntegrator::step_size`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_check_step_error()`.

6.5.2.41 integrator_terminate()

```
void LsodeFirstOrderODEIntegrator::integrator_terminate ( ) [protected]
```

this is the only succesful path back from integrator to manager.

All other returns from `integrator_*` back to `manager_*` are in response to a need for new derivatives and carry with them a modified `re_entry_point` to provide access back to the integrator on reentry. All returns with `re_entry_point = CycleStartFinish` should go through this method.

Definition at line 944 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::internal_state`, `jeod::LsodeFirstOrderODEIntegrator::prev_step_size`, and `jeod::LsodeFirstOrderODEIntegrator::step_size`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup()`.

6.5.2.42 integrator_test_stepsize_change()

```
void LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change ( ) [protected]
```

Tests `h` against old `h`.

Definition at line 279 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::data_stode`, `jeod::LsodeFirstOrderODEIntegrator::integrator_predict()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh()`, `jeod::LsodeDataStode::iredo`, `jeod::LsodeFirstOrderODEIntegrator::prev_step_size`, `jeod::LsodeDataStode::step_ratio`, and `jeod::LsodeFirstOrderODEIntegrator::step_size`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs()`.

6.5.2.43 integrator_wrapup()

```
void LsodeFirstOrderODEIntegrator::integrator_wrapup ( ) [protected]
```

Wraps up the completion of the integrator.

Definition at line 924 of file `lsode_first_order_ode_integrator__integrator.cc`.

References `jeod::LsodeDataArrays::accum_correction`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::integrator_terminate()`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::prev_method_order`, and `jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_check_step_error()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh()`.

6.5.2.44 interpolate_y()

```
void LsodeFirstOrderODEIntegrator::interpolate_y ( ) [protected]
```

Interpolates the zeroth-derivative of y. Adapted from subroutine DINTDY, which was a general method for interpolating the K-th derivative of the dependent variable vector, y.

Implementation notes - DINTDY was called exclusively with three input values that always matched.

1. Argument #1 T (time). Calls to DINTDY always passed TOUT (now `cycle_target_time`) in for T
2. Argument #2 K (order). Calls to DINTDY always passed 0 in for K
3. Argument #5 DKY (value). DKY is the value that DINTDY interpolates. Calls to DINTDY always passed Y in for DKY. This routine uses y directly in place of DKY, so can only interpolate the 0-th derivative (since that is y), and always evaluates at `cycle_target_time`. replaced accordingly.

The computed values are gotten by interpolation using the Nordsieck history array, `arrays.history`. The formula for Y is:

$$Y[i] = \sum_{j=0 \text{ to } \text{method_order_current}} \{ (\text{cycle_target_time} - \text{stage_target_time})^j * \text{arrays.history}[i-1][j] / h^j \}$$

Definition at line 262 of file `lsode_first_order_ode_integrator__support.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::cycle_target_time`, `jeod::LsodeFirstOrderODEIntegrator::epsilon`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::prev_good_step_size`, `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, and `jeod::LsodeFirstOrderODEIntegrator::y`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::manager_check_stop_conditions()`, and `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part3()`.

6.5.2.45 jacobian_prep_init()

```
void LsodeFirstOrderODEIntegrator::jacobian_prep_init ( ) [protected]
```

Modified from DPREPJ.

DPREPJ was called by DSTODE to compute and process the matrix $P = I - h * el(1) * J$, where J is an approximation to the Jacobian.

NOTES

DPREPJ has been split into 3 parts, book-ended by the external-calls. jacobian_prep_init contains the code that precedes the first external-call jacobian_prep_loop contains the code that continues to loop according to the limits as written is DPREPJ. jacobian_prep_wrap_up contains the code that follows successful completion of the looping section of DPREPJ.

Note that the division in this implementation is not linear with that in the original Fortran. The external calls within the original fortran are embedded within switch-blocks and for loops; the return points - to go to the next routine will pick up from one of several locations in the Fortran code, depending on the configuration at the time the external call was made.

For the ER7_Utils / JEOD / Trick implementation, the derivative/jac calls must be external to the integrate call, so we must fully back out and then reenter. Rentry goes to jacobian_prep_loop.

Some variables that were local have been moved to the class so that their value is not lost in going from jacobian_↵_prep_init to jacobian_prep_loop. These are typically identified with ***_dprepj to indicate that their sole purpose is within dprepj (the original name of the jacobian_prep_* routines).

Here the jacobian is computed by the user-supplied routine JAC if corrector_method = NewtonIterUserJac or NewtonIterUserBandJac, or by finite differencing if corrector_method = NewtonIterInternalJac, JacobiNewton↵InternalJac, or NewtonIterInternalBandJac.

If corrector_method = JacobiNewtonInternalJac, a diagonal approximation to the Jacobian is used. The Jacobian is stored in arrays lin_alg

If corrector_method != JacobiNewtonInternalJac, P is subjected to LU decomposition in preparation for later solution of linear systems with P as coefficient matrix. This is done by gauss_elim_factor (DGEFA) if corrector_method = NewtonIterUserJac or NewtonIterInternalJac, and by linear_solver (DGBFA) if corrector_method = NewtonIter↵UserBandJac or NewtonIterInternalBandJac.

Note that the corrector_method using the banded Jacobians is not supported in this release, so linear_solver is not used.

FTEM and ACOR were effectively the same, now arrays.accum_correction. SAVF is now arrays.save. WM is now arrays.lin_alg

Definition at line 373 of file lsode_first_order_ode_integrator__support.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeControlDataInterface::corrector_method, jeod::LsodeFirstOrderODEIntegrator::data_prepj, jeod::↵LsodeFirstOrderODEIntegrator::epsilon, jeod::LsodeDataArrays::error_weight, jeod::LsodeDataJacobianPrep::fac, jeod::LsodeControlDataInterface::FunctionalIteration, jeod::LsodeDataArrays::history, jeod::LsodeDataJacobian↵Prep::hl0, jeod::LsodeDataJacobianPrep::index, jeod::LsodeDataJacobianPrep::index_max, jeod::LsodeFirst↵OrderODEIntegrator::iteration_matrix_singular, jeod::LsodeFirstOrderODEIntegrator::jacobian_current, jeod::↵LsodeFirstOrderODEIntegrator::JacobianPrep, jeod::LsodeControlDataInterface::JacobiNewtonInternalJac, jeod↵::LsodeDataArrays::lin_alg, jeod::LsodeDataArrays::lin_alg_1, jeod::LsodeDataArrays::lin_alg_2, jeod::Lsode↵FirstOrderODEIntegrator::magnitude_of_weighted_array(), jeod::LsodeFirstOrderODEIntegrator::method_coeff↵_first, jeod::LsodeControlDataInterface::NewtonIterInternalBandJac, jeod::LsodeControlDataInterface::Newton↵IterInternalJac, jeod::LsodeControlDataInterface::NewtonIterUserBandJac, jeod::LsodeControlDataInterface::↵NewtonIterUserJac, jeod::LsodeFirstOrderODEIntegrator::num_jacobian_evals, jeod::LsodeControlDataInterface↵::num_odes, jeod::LsodeDataJacobianPrep::r0, jeod::LsodeFirstOrderODEIntegrator::re_entry_point, jeod::↵LsodeDataArrays::save, jeod::LsodeFirstOrderODEIntegrator::step_size, jeod::LsodeFirstOrderODEIntegrator::y, and jeod::LsodeDataJacobianPrep::yj.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrate().

6.5.2.46 jacobian_prep_loop()

```
bool LsodeFirstOrderODEIntegrator::jacobian_prep_loop ( ) [protected]
```

Definition at line 514 of file lsode_first_order_ode_integrator__support.cc.

References jeod::LsodeDataArrays::accum_correction, jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeControlDataInterface::corrector_method, jeod::LsodeFirstOrderODEIntegrator::data_prepj, jeod::LsodeDataArrays::error_weight, jeod::LsodeDataJacobianPrep::fac, jeod::LsodeControlDataInterface::FunctionalIteration, jeod::LsodeDataJacobianPrep::hl0, jeod::LsodeDataJacobianPrep::index, jeod::LsodeDataJacobianPrep::index_max, jeod::LsodeControlDataInterface::JacobiNewtonInternalJac, jeod::LsodeDataArrays::lin_alg, jeod::LsodeDataArrays::lin_alg_1, jeod::LsodeFirstOrderODEIntegrator::load_derivatives(), jeod::LsodeControlDataInterface::NewtonIterInternalBandJac, jeod::LsodeControlDataInterface::NewtonIterInternalJac, jeod::LsodeControlDataInterface::NewtonIterUserBandJac, jeod::LsodeControlDataInterface::NewtonIterUserJac, jeod::LsodeControlDataInterface::num_odes, jeod::LsodeDataJacobianPrep::r0, jeod::LsodeDataArrays::save, jeod::LsodeFirstOrderODEIntegrator::y, and jeod::LsodeDataJacobianPrep::yj.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrate().

6.5.2.47 jacobian_prep_wrap_up()

```
bool LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up ( ) [protected]
```

Definition at line 611 of file lsode_first_order_ode_integrator__support.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::convergence_rate, jeod::LsodeControlDataInterface::corrector_method, jeod::LsodeFirstOrderODEIntegrator::data_prepj, jeod::LsodeFirstOrderODEIntegrator::epsilon, jeod::LsodeDataArrays::error_weight, jeod::LsodeControlDataInterface::FunctionalIteration, jeod::LsodeFirstOrderODEIntegrator::gauss_elim_factor(), jeod::LsodeDataArrays::history, jeod::LsodeDataJacobianPrep::hl0, jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2(), jeod::LsodeFirstOrderODEIntegrator::iteration_matrix_singular, jeod::LsodeControlDataInterface::JacobiNewtonInternalJac, jeod::LsodeDataArrays::lin_alg, jeod::LsodeFirstOrderODEIntegrator::load_derivatives(), jeod::LsodeControlDataInterface::NewtonIterInternalBandJac, jeod::LsodeControlDataInterface::NewtonIterInternalJac, jeod::LsodeControlDataInterface::NewtonIterUserBandJac, jeod::LsodeControlDataInterface::NewtonIterUserJac, jeod::LsodeControlDataInterface::num_odes, jeod::LsodeFirstOrderODEIntegrator::num_steps_taken, jeod::LsodeFirstOrderODEIntegrator::rel_change_since_jacobian, jeod::LsodeDataArrays::save, jeod::LsodeFirstOrderODEIntegrator::step_at_last_jacobian_update, jeod::LsodeFirstOrderODEIntegrator::step_size, and jeod::LsodeFirstOrderODEIntegrator::update_jacobian.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrate().

6.5.2.48 linear_chord_iteration()

```
void LsodeFirstOrderODEIntegrator::linear_chord_iteration ( ) [protected]
```

Modified from DSOLSY.

This routine manages the solution of the linear system arising from a chord iteration. It is called if `corrector_method != FunctionalIteration`.

If `corrector_method == NewtonIterUserJac || NewtonIterInternalJac`, it calls `linear_solver` (was DGESE). If `corrector_method = JacobiNewtonInternalJac` it updates the coefficient `h10 = step_size * method_coeff_first` (previously $H \cdot EL0$) in the diagonal matrix, and then computes the solution.

Definition at line 735 of file `Lsode_first_order_ode_integrator__support.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::corrector_method`, `jeod::LsodeControlDataInterface::FunctionalIteration`, `jeod::LsodeControlDataInterface::JacobiNewtonInternalJac`, `jeod::LsodeDataArrays::lin_alg`, `jeod::LsodeDataArrays::lin_alg_2`, `jeod::LsodeFirstOrderODEIntegrator::linear_solver()`, `jeod::LsodeFirstOrderODEIntegrator::method_coeff_first`, `jeod::LsodeFirstOrderODEIntegrator::modified_iteration_matrix_singular`, `jeod::LsodeControlDataInterface::NewtonIterInternalBandJac`, `jeod::LsodeControlDataInterface::NewtonIterInternalJac`, `jeod::LsodeControlDataInterface::NewtonIterUserBandJac`, `jeod::LsodeControlDataInterface::NewtonIterUserJac`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, and `jeod::LsodeFirstOrderODEIntegrator::y`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration()`.

6.5.2.49 linear_solver()

```
void LsodeFirstOrderODEIntegrator::linear_solver ( ) [protected]
```

Solves the equation $Y' = A Y$, with $A = \text{arrays.lin_alg}$.

Definition at line 253 of file `Lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeDataArrays::lin_alg`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeDataArrays::pivots`, and `jeod::LsodeFirstOrderODEIntegrator::y`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::linear_chord_iteration()`.

6.5.2.50 load_derivatives()

```
void LsodeFirstOrderODEIntegrator::load_derivatives (
    double * derivs ) [protected]
```

Load the externally generated derivative values (incoming as `y_dot`) into the array `derivs`.

Definition at line 360 of file `Lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::num_odes`, and `jeod::LsodeFirstOrderODEIntegrator::y_dot`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrate()`, `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop()`, and `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up()`.

6.5.2.51 load_ew_values()

```
void LsodeFirstOrderODEIntegrator::load_ew_values ( ) [protected]
```

Definition at line 799 of file `lsode_first_order_ode_integrator__support.cc`.

References `jeod::LsodeControlDataInterface::abs_tolerance_error_control`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeControlDataInterface::CommonAbsCommonRel`, `jeod::LsodeControlDataInterface::CommonAbsSpecificRel`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::error_control_indicator`, `jeod::LsodeDataArrays::error_weight`, `jeod::LsodeDataArrays::history`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeControlDataInterface::rel_tolerance_error_control`, `jeod::LsodeControlDataInterface::SpecificAbsCommonRel`, and `jeod::LsodeControlDataInterface::SpecificAbsSpecificRel`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2()`, and `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1()`.

6.5.2.52 LsodeControlDataInterface()

```
LsodeControlDataInterface::LsodeControlDataInterface ( )
```

constructor

Definition at line 50 of file `lsode_control_data_interface.cc`.

References `jeod::LsodeControlDataInterface::abs_tolerance_error_control_vec`, and `jeod::LsodeControlDataInterface::rel_tolerance_error_control_vec`.

6.5.2.53 LsodeFirstOrderODEIntegrator() [1/2]

```
LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator ( )
```

[LsodeFirstOrderODEIntegrator](#) default constructor.

Definition at line 51 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::calculate_epsilon()`.

6.5.2.54 LsodeFirstOrderODEIntegrator() [2/2]

```
LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator (
    const LsodeControlDataInterface & data_in,
    er7_utils::IntegrationControls & controls,
    unsigned int size )
```

[LsodeFirstOrderODEIntegrator](#) non-default constructor.

Parameters

in	<i>data_in</i>	state variable data
in	<i>size</i>	State size
in, out	<i>controls</i>	Integration controls

Definition at line 58 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::calculate_epsilon()`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::num_odes`, and `jeod::LsodeFirstOrderODEIntegrator::update_control_data()`.

6.5.2.55 LsodeGeneralizedDerivSecondOrderODEIntegrator() [1/2]

```
LsodeGeneralizedDerivSecondOrderODEIntegrator::LsodeGeneralizedDerivSecondOrderODEIntegrator (
    const LsodeControlDataInterface & data_in,
    er7_utils::IntegrationControls & controls,
    const er7_utils::GeneralizedPositionDerivativeFunctions & deriv_funs,
    unsigned int position_size,
    unsigned int velocity_size )
```

non-default constructor

Definition at line 53 of file `lsode_generalized_second_order_ode_integrator.cc`.

References `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::posdot`.

6.5.2.56 LsodeGeneralizedDerivSecondOrderODEIntegrator() [2/2]

```
LsodeGeneralizedDerivSecondOrderODEIntegrator::LsodeGeneralizedDerivSecondOrderODEIntegrator (
    const LsodeGeneralizedDerivSecondOrderODEIntegrator & src )
```

[LsodeGeneralizedDerivSecondOrderODEIntegrator](#) copy constructor.

Copy Constructor.

Parameters

in	<i>src</i>	Item to be copied.
----	------------	--------------------

Definition at line 67 of file `lsode_generalized_second_order_ode_integrator.cc`.

6.5.2.57 LsodeIntegrationControls()

```
LsodeIntegrationControls::LsodeIntegrationControls (
    unsigned int num_stages ) [explicit]
```

Definition at line 51 of file `lsode_integration_controls.cc`.

6.5.2.58 `LsodeIntegratorConstructor()`

```
LsodeIntegratorConstructor::LsodeIntegratorConstructor (
    const LsodeIntegratorConstructor & src )
```

Definition at line 57 of file `lsode_integrator_constructor.cc`.

6.5.2.59 `LsodeSecondOrderODEIntegrator()` [1/2]

```
LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator (
    const LsodeControlDataInterface & data_in,
    er7_utils::IntegrationControls & controls,
    const er7_utils::GeneralizedPositionDerivativeFunctions & deriv_funs,
    unsigned int position_size,
    unsigned int velocity_size ) [protected]
```

Definition at line 68 of file `lsode_second_order_ode_integrator.cc`.

References `jeod::LsodeSecondOrderODEIntegrator::arrays_allocated`, `jeod::LsodeSecondOrderODEIntegrator::y`, and `jeod::LsodeSecondOrderODEIntegrator::y_dot`.

6.5.2.60 `LsodeSecondOrderODEIntegrator()` [2/2]

```
LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator (
    const LsodeControlDataInterface & data_in,
    er7_utils::IntegrationControls & controls,
    unsigned int size ) [protected]
```

[LsodeSecondOrderODEIntegrator](#) non-default constructor.

Parameters

in	<i>data_in</i>	LSODE-specific control data.
in, out	<i>controls</i>	Integration controls.
in	<i>size</i>	State size.

Definition at line 52 of file `lsode_second_order_ode_integrator.cc`.

References `jeod::LsodeSecondOrderODEIntegrator::arrays_allocated`, `jeod::LsodeSecondOrderODEIntegrator::y`, and `jeod::LsodeSecondOrderODEIntegrator::y_dot`.

6.5.2.61 LsodeSimpleSecondOrderODEIntegrator()

```
LsodeSimpleSecondOrderODEIntegrator::LsodeSimpleSecondOrderODEIntegrator (
    const LsodeControlDataInterface & data_in,
    er7_utils::IntegrationControls & controls,
    unsigned int size )
```

[LsodeSimpleSecondOrderODEIntegrator](#) non-default constructor.

Parameters

in	<i>data_in</i>	State Variable Data
in	<i>size</i>	State size
in, out	<i>controls</i>	Integration controls

Definition at line 49 of file `lsode_simple_second_order_ode_integrator.cc`.

6.5.2.62 magnitude_of_weighted_array() [1/2]

```
double LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array (
    const double * v ) [protected]
```

returns the RMS value of $\{V \text{ dot } W\}$, where V and W are N-vectors.

Modified version of DVNORM

The only places DVNORM is used, it is multiplying some array by the error_weight array (`arrays.error_weight`) across `control_data.num_odes` terms. These values are fixed for our application, and do not need to be passed in.

We provide two implementations, one for a one-dimensional array, and one for a two-dimensional array in which the first index is the variable.

Parameters

v	array
---	-------

Definition at line 140 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeDataArrays::error_weight`, and `jeod::LsodeControlDataInterface::num_odes`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init()`, `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2()`, and `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part2()`.

6.5.2.63 magnitude_of_weighted_array() [2/2]

```
double LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array (
    unsigned int index,
    double ** v ) [protected]
```

returns RMS value of $v[*][index]$

Modified version of DVNORM, second implementation.

Parameters

<i>index</i>	use this index
<i>v</i>	array

Definition at line 158 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeDataArrays::error_weight`, and `jeod::LsodeControlDataInterface::num_odes`.

6.5.2.64 manager_check_stop_conditions()

```
int LsodeFirstOrderODEIntegrator::manager_check_stop_conditions ( ) [protected]
```

verifies whether the convergence conditions have been met to end the cycle.

Definition at line 528 of file `lsode_first_order_ode_integrator__manager.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::calculation_task`, `jeod::LsodeFirstOrderODEIntegrator::CompleteCycle`, `jeod::LsodeFirstOrderODEIntegrator::cycle_target_time`, `jeod::LsodeFirstOrderODEIntegrator::interpolate_y()`, `jeod::LsodeFirstOrderODEIntegrator::Normal`, `jeod::LsodeFirstOrderODEIntegrator::NormalWithSingularity`, `jeod::LsodeFirstOrderODEIntegrator::num_steps_taken`, `jeod::LsodeFirstOrderODEIntegrator::OneStep`, `jeod::LsodeFirstOrderODEIntegrator::OneStepWithSingularity`, `jeod::LsodeFirstOrderODEIntegrator::prior_num_steps`, `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`, and `jeod::LsodeFirstOrderODEIntegrator::step_size`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start()`.

6.5.2.65 manager_initialize_calculation_part1()

```
void LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1 ( ) [protected]
```

Sets the values for the case with `calculation_phase = 1`.

Definition at line 342 of file `lsode_first_order_ode_integrator__manager.cc`.

References `jeod::LsodeControlDataInterface::allocate_arrays()`, `jeod::LsodeDataArrays::allocate_arrays()`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::corrector_method`, `jeod::LsodeFirstOrderODEIntegrator::InitCalc`, `jeod::LsodeFirstOrderODEIntegrator::internal_state`, `jeod::LsodeControlDataInterface::is_corrector_method_functional_iteration()`, `jeod::LsodeDataArrays::lin_alg_1`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::re_entry_point`, and `jeod::LsodeFirstOrderODEIntegrator::sqrt_epsilon`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start()`.

6.5.2.66 manager_initialize_calculation_part2()

```
void LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2 ( ) [protected]
```

Definition at line 389 of file `Lsode_first_order_ode_integrator__manager.cc`.

References `jeod::LsodeControlDataInterface::abs_tolerance_error_control`, `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeControlDataInterface::CommonAbsSpecificRel`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::cycle_target_time`, `jeod::LsodeFirstOrderODEIntegrator::epsilon`, `jeod::LsodeControlDataInterface::error_control_indicator`, `jeod::LsodeDataArrays::error_weight`, `jeod::LsodeDataArrays::history`, `jeod::LsodeControlDataInterface::initial_step_size`, `jeod::LsodeFirstOrderODEIntegrator::load_ew_values()`, `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array()`, `jeod::LsodeFirstOrderODEIntegrator::max_step_size_inv`, `jeod::LsodeFirstOrderODEIntegrator::method_order_current`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeControlDataInterface::rel_tolerance_error_control`, `jeod::LsodeControlDataInterface::SpecificAbsCommonRel`, `jeod::LsodeControlDataInterface::SpecificAbsSpecificRel`, `jeod::LsodeFirstOrderODEIntegrator::step_size`, `jeod::LsodeFirstOrderODEIntegrator::y`, and `jeod::LsodeFirstOrderODEIntegrator::y_dot`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrate()`.

6.5.2.67 manager_integration_loop_part1()

```
void LsodeFirstOrderODEIntegrator::manager_integration_loop_part1 ( ) [protected]
```

The iteration loop for the integration process.

Definition at line 645 of file `Lsode_first_order_ode_integrator__manager.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeDataArrays::error_weight`, `jeod::LsodeFirstOrderODEIntegrator::load_ew_values()`, `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part2()`, `jeod::LsodeControlDataInterface::max_num_steps`, `jeod::LsodeControlDataInterface::num_odes`, `jeod::LsodeFirstOrderODEIntegrator::num_steps_taken`, `jeod::LsodeFirstOrderODEIntegrator::prior_num_steps`, and `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part3()`, and `jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start()`.

6.5.2.68 manager_integration_loop_part2()

```
void LsodeFirstOrderODEIntegrator::manager_integration_loop_part2 ( ) [protected]
```

Definition at line 701 of file `Lsode_first_order_ode_integrator__manager.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeFirstOrderODEIntegrator::CycleStartFinish`, `jeod::LsodeFirstOrderODEIntegrator::epsilon`, `jeod::LsodeDataArrays::history`, `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, `jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array()`, `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part3()`, `jeod::LsodeControlDataInterface::max_num_small_step_warnings`, `jeod::LsodeFirstOrderODEIntegrator::num_small_step_warnings`, `jeod::LsodeFirstOrderODEIntegrator::re_entry_point`, `jeod::LsodeFirstOrderODEIntegrator::stage_target_time`, and `jeod::LsodeFirstOrderODEIntegrator::step_size`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrate()`, and `jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1()`.

6.5.2.69 manager_integration_loop_part3()

```
void LsodeFirstOrderODEIntegrator::manager_integration_loop_part3 ( ) [protected]
```

Definition at line 766 of file lsode_first_order_ode_integrator__manager.cc.

References jeod::LsodeFirstOrderODEIntegrator::calculation_task, jeod::LsodeFirstOrderODEIntegrator::CompleteCycle, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::cycle_target_time, jeod::LsodeFirstOrderODEIntegrator::initialized, jeod::LsodeFirstOrderODEIntegrator::interpolate_y(), jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1(), jeod::LsodeControlDataInterface::min_step_size, jeod::LsodeFirstOrderODEIntegrator::Normal, jeod::LsodeFirstOrderODEIntegrator::NormalWithSingularity, jeod::LsodeFirstOrderODEIntegrator::OneStep, jeod::LsodeFirstOrderODEIntegrator::OneStepWithSingularity, jeod::LsodeFirstOrderODEIntegrator::stage_target_time, jeod::LsodeFirstOrderODEIntegrator::step_error, and jeod::LsodeFirstOrderODEIntegrator::step_size.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrate(), and jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part2().

6.5.2.70 manager_set_calculation_phase_eq_2_reload()

```
void LsodeFirstOrderODEIntegrator::manager_set_calculation_phase_eq_2_reload ( ) [protected]
```

Definition at line 920 of file lsode_first_order_ode_integrator__manager.cc.

6.5.2.71 process_entry_point_cycle_start()

```
void LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start ( ) [protected]
```

The code block from the main integrate routine for re_entry_point=CycleStartFinish.

Definition at line 269 of file lsode_first_order_ode_integrator__manager.cc.

References jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::cycle_target_time, jeod::LsodeFirstOrderODEIntegrator::first_pass, jeod::LsodeControlDataInterface::initial_step_size, jeod::LsodeFirstOrderODEIntegrator::initialized, jeod::LsodeFirstOrderODEIntegrator::manager_check_stop_conditions(), jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1(), jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1(), jeod::LsodeFirstOrderODEIntegrator::num_equations, jeod::LsodeControlDataInterface::num_odes, and jeod::LsodeFirstOrderODEIntegrator::stage_target_time.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrate().

6.5.2.72 reset_integrator()

```
void LsodeFirstOrderODEIntegrator::reset_integrator ( ) [override]
```

Resets the integrator when the timestep changes or when identified as needing a reset.

Definition at line 896 of file lsode_first_order_ode_integrator__manager.cc.

References jeod::LsodeFirstOrderODEIntegrator::arrays, jeod::LsodeFirstOrderODEIntegrator::control_data, jeod::LsodeFirstOrderODEIntegrator::cycle_target_time, jeod::LsodeFirstOrderODEIntegrator::first_pass, jeod::LsodeDataArrays::history, jeod::LsodeFirstOrderODEIntegrator::InitCalc, jeod::LsodeControlDataInterface::initial_step_size, jeod::LsodeFirstOrderODEIntegrator::method_order_current, jeod::LsodeControlDataInterface::num_odes, jeod::LsodeFirstOrderODEIntegrator::prev_good_step_size, jeod::LsodeFirstOrderODEIntegrator::reentry_point, and jeod::LsodeFirstOrderODEIntegrator::stage_target_time.

6.5.2.73 set_abs_tol()

```
void LsodeControlDataInterface::set_abs_tol (
    int index,
    double value )
```

Definition at line 382 of file lsode_control_data_interface.cc.

References jeod::LsodeControlDataInterface::abs_tolerance_error_control, jeod::LsodeControlDataInterface::abs_tolerance_error_control_vec, jeod::LsodeControlDataInterface::error_control_vector_copied_over, and jeod::LsodeControlDataInterface::num_odes_at_alloc.

6.5.2.74 set_rel_tol()

```
void LsodeControlDataInterface::set_rel_tol (
    int index,
    double value )
```

set values from external

Definition at line 325 of file lsode_control_data_interface.cc.

References jeod::LsodeControlDataInterface::error_control_vector_copied_over, jeod::LsodeControlDataInterface::num_odes_at_alloc, jeod::LsodeControlDataInterface::rel_tolerance_error_control, and jeod::LsodeControlDataInterface::rel_tolerance_error_control_vec.

6.5.2.75 `update_control_data()`

```
void LsodeFirstOrderODEIntegrator::update_control_data ( )
```

Gets the control data from where it can be populated in the constructor and verifies that the input control parameters are not out of sensible range.

Definition at line 87 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeControlDataInterface::check_interface_data()`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::ImplicitAdamsNonStiff`, `jeod::LsodeControlDataInterface::integration_method`, `jeod::LsodeControlDataInterface::max_order`, `jeod::LsodeFirstOrderODEIntegrator::max_order_internal`, `jeod::LsodeControlDataInterface::max_step_size`, and `jeod::LsodeFirstOrderODEIntegrator::max_step_size_inv`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator()`.

6.5.2.76 `~LsodeFirstOrderODEIntegrator()`

```
LsodeFirstOrderODEIntegrator::~LsodeFirstOrderODEIntegrator ( ) [override]
```

[LsodeFirstOrderODEIntegrator](#) destructor.

Definition at line 73 of file `lsode_first_order_ode_integrator__utility.cc`.

References `jeod::LsodeFirstOrderODEIntegrator::arrays`, `jeod::LsodeFirstOrderODEIntegrator::control_data`, `jeod::LsodeControlDataInterface::destroy_allocated_arrays()`, `jeod::LsodeDataArrays::destroy_allocated_arrays()`, and `jeod::LsodeFirstOrderODEIntegrator::first_pass`.

6.5.2.77 `~LsodeGeneralizedDerivSecondOrderODEIntegrator()`

```
LsodeGeneralizedDerivSecondOrderODEIntegrator::~LsodeGeneralizedDerivSecondOrderODEIntegrator ( ) [override]
```

[LsodeGeneralizedDerivSecondOrderODEIntegrator](#) destructor.

Destructor.

Definition at line 95 of file `lsode_generalized_second_order_ode_integrator.cc`.

References `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::posdot`.

6.5.2.78 `~LsodeSecondOrderODEIntegrator()`

```
LsodeSecondOrderODEIntegrator::~LsodeSecondOrderODEIntegrator ( ) [override]
```

[LsodeSecondOrderODEIntegrator](#) destructor.

Definition at line 86 of file `lsode_second_order_ode_integrator.cc`.

References `jeod::LsodeSecondOrderODEIntegrator::arrays_allocated`, `jeod::LsodeSecondOrderODEIntegrator::y`, and `jeod::LsodeSecondOrderODEIntegrator::y_dot`.

Chapter 7

Namespace Documentation

7.1 er7_utils Namespace Reference

Data Structures

- class [TwoDArray](#)
RAII template class that implements a rectangular two dimensional array.
- class [DoubleTwoDArray](#)
2D array, specialized for doubles.

7.2 jeod Namespace Reference

Namespace jeod.

Data Structures

- class [GaussJacksonCoefficientsPair](#)
Contains a summed Adams and Gauss-Jackson coefficient pair.
- class [GaussJacksonCoeffs](#)
Contains the Gauss-Jackson predictor and corrector coefficients.
- class [GaussJacksonConfig](#)
Contains Gauss-Jackson configuration data.
- class [GaussJacksonFirstOrderODEIntegrator](#)
Integrates a first order ODE using the summed Adams technique that is embedded within the Gauss-Jackson technique.
- class [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#)
Integrates a generalized derivative second order ODE using Gauss-Jackson.
- class [GaussJacksonIntegrationControls](#)
IntegrationControls specialized for Gauss-Jackson integration.
- class [GaussJacksonIntegratorBase](#)
Base template class for integrating state via the Gauss-Jackson technique.
- class [GaussJacksonIntegratorConstructor](#)
Create state and time integrators that propagate using Gauss-Jackson.

- class [GaussJacksonOneState](#)
Essentially just a double.*
- class [GaussJacksonRationalCoefficients](#)
Contains a set of Adams or Stormer-Cowell coefficients.
- class [GaussJacksonSimpleSecondOrderODEIntegrator](#)
Integrates a simple second order ODE using the Gauss-Jackson technique.
- class [GaussJacksonStateMachine](#)
Guides the behavior of the Gauss-Jackson integration process via a finite state machine.
- class [GaussJacksonTwoState](#)
Essentially just `std::pair<double*>`.
- class [GeneralizedSecondOrderODETechnique](#)
Enumerates the integration schemes, generalized Cartesian or Lie group integration, and provides simple utilities that work with this.
- class [IntegrationMessages](#)
Declares messages associated with the integration test model.
- class [JeodIntegrationGroupOwner](#)
The abstract class `IntegrationGroupOwner` contains an `IntegrationGroup`.
- class [JeodIntegrationGroup](#)
A `JeodIntegrationGroup` integrates the state of a set of objects over time.
- class [JeodIntegrationTime](#)
The class `JeodIntegrationTime` adds the concepts of timestamping and notification of changes in the nature of time to the ER7 numerical utilities `TimeInterface` class.
- class [Restartable2DSecondOrderIntegrator](#)
Integrates a second order ODE in two dimensional space, $d^2x/dt^2 = a(x,t)$, where x is a two-vector.
- class [RestartableScalarFirstOrderODEIntegrator](#)
A `RestartableScalarFirstOrderODEIntegrator` integrates a first order ODE, $dx/dt = v(x,t)$, where x is a scalar.
- class [RestartableT3SecondOrderODEIntegrator](#)
A `RestartableT3SecondOrderODEIntegrator` integrates a second order ODE in three space, $d^2x/dt^2 = a(x,t)$, where x is a three-vector.
- class [RestartableSO3SecondOrderODEIntegrator](#)
A `RestartableSO3SecondOrderODEIntegrator` integrates a generalized second order ODE that describes rotation in three space.
- class [RestartableStateIntegrator](#)
A `RestartableStateIntegrator` establishes the basic capabilities needed to make a state integrator a managed resource.
- class [RestartableFirstOrderODEIntegrator](#)
A `RestartableFirstOrderODEIntegrator` is-a `RestartableStateIntegrator` that manages an `er7_utils::FirstOrderODEIntegrator`.
- class [RestartableSecondOrderODEIntegrator](#)
A `RestartableSecondOrderODEIntegrator` is-a `RestartableStateIntegrator` that manages the integrator for a second order ODE problem.
- class [RestartableSimpleSecondOrderODEIntegrator](#)
A `RestartableSimpleSecondOrderODEIntegrator` is-a `RestartableSecondOrderODEIntegrator` that manages the integrator for a simple second order ODE problem, one in which the time derivative of the generalized position is the generalized velocity.
- class [RestartableGeneralizedDerivSecondOrderODEIntegrator](#)
A `RestartableGeneralizedDerivSecondOrderODEIntegrator` is-a `RestartableStateIntegrator` that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.
- class [RestartableGeneralizedStepSecondOrderODEIntegrator](#)
A `RestartableGeneralizedStepSecondOrderODEIntegrator` is-a `RestartableStateIntegrator` that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

- class [TimeChangeSubscriber](#)
A *TimeChangeSubscriber* is some object that wants to be notified of changes in the nature of time.
- class [LsodeControlDataInterface](#)
Specifies controls for an LSODE integrator.
- class [LsodeDataJacobianPrep](#)
Data associated with the method DPREPJ.
- class [LsodeDataArrays](#)
The data arrays.
- class [LsodeDataStode](#)
The data associated with method Dstode.
- class [LsodeFirstOrderODEIntegrator](#)
Jeod-compatible version of the Livermore ODE solver, LSODE.
- class [LsodeGeneralizedDerivSecondOrderODEIntegrator](#)
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.
- class [LsodeIntegrationControls](#)
Contains controls for an LSODE integrator.
- class [LsodeIntegratorConstructor](#)
Create state and time integrators that propagate using standard Lsode.
- class [LsodeSecondOrderODEIntegrator](#)
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.
- class [LsodeSimpleSecondOrderODEIntegrator](#)
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Typedefs

- using [GaussJacksonIntegratorBaseFirst](#) = [GaussJacksonIntegratorBase](#)< [GaussJacksonOneState](#), [er7_utils::FirstOrderODEIntegrator](#) >
Alias for a first order Gauss Jackson integrator.
- using [GaussJacksonIntegratorBaseSecond](#) = [GaussJacksonIntegratorBase](#)< [GaussJacksonTwoState](#), [er7_utils::SecondOrderODEIntegrator](#) >
Alias for a second order Gauss Jackson integrator.

Functions

- [std::ostream & operator<<](#) ([std::ostream &stream](#), const [GaussJacksonCoeffs](#) &coeff)
- static [GaussJacksonConfig set_default_config_values](#) (const [GaussJacksonConfig](#) &config)
Swap the negative ones in the supplied config with the default values, some of which are computed.
- static unsigned int [validate_config](#) (const [GaussJacksonConfig](#) &config)
Check for invalid values in the supplied config.
- static [GaussJacksonIntegrationControls](#) * [cast_to_gj_controls](#) ([er7_utils::IntegrationControls](#) &controls)
Cast the provided integration controls to a *GaussJacksonIntegrationControls*.

7.2.1 Detailed Description

Namespace jeod.

7.2.2 Typedef Documentation

7.2.2.1 GaussJacksonIntegratorBaseFirst

```
using jeod::GaussJacksonIntegratorBaseFirst = typedef GaussJacksonIntegratorBase<GaussJacksonOneState,
er7_utils::FirstOrderODEIntegrator>
```

Alias for a first order Gauss Jackson integrator.

Definition at line 80 of file gauss_jackson_integrator_base_first.hh.

7.2.2.2 GaussJacksonIntegratorBaseSecond

```
using jeod::GaussJacksonIntegratorBaseSecond = typedef GaussJacksonIntegratorBase<GaussJacksonTwoState,
er7_utils::SecondOrderODEIntegrator>
```

Alias for a second order Gauss Jackson integrator.

Definition at line 80 of file gauss_jackson_integrator_base_second.hh.

7.2.3 Function Documentation

7.2.3.1 cast_to_gj_controls()

```
static GaussJacksonIntegrationControls* jeod::cast_to_gj_controls (
    er7_utils::IntegrationControls & controls ) [static]
```

Cast the provided integration controls to a [GaussJacksonIntegrationControls](#).

Parameters

<i>controls</i>	Generic controls to be cast.
-----------------	------------------------------

Returns

[GaussJacksonIntegrationControls](#) pointer, guaranteed to be non-null.

Definition at line 47 of file gauss_jackson_integrator_constructor.cc.

Referenced by `jeod::GaussJacksonIntegratorConstructor::create_first_order_ode_integrator()`, `jeod::GaussJacksonIntegratorConstructor::create_generalized_deriv_second_order_ode_integrator()`, and `jeod::GaussJacksonIntegratorConstructor::create_second_order_ode_integrator()`.

7.2.3.2 operator<<()

```
std::ostream& jeod::operator<< (
    std::ostream & stream,
    const GaussJacksonCoeffs & coeff )
```

Parameters

<i>stream</i>	The stream to be printed to.
<i>coeff</i>	The coefficients to be printed.

Definition at line 130 of file gauss_jackson_coeffs.cc.

7.2.3.3 set_default_config_values()

```
static GaussJacksonConfig jeod::set_default_config_values (
    const GaussJacksonConfig & config ) [static]
```

Swap the negative ones in the supplied config with the default values, some of which are computed.

Definition at line 75 of file gauss_jackson_config.cc.

References `jeod::GaussJacksonConfig::absolute_tolerance`, `jeod::GaussJacksonConfig::final_order`, `jeod::GaussJacksonConfig::initial_order`, `jeod::GaussJacksonConfig::max_correction_iterations`, `jeod::GaussJacksonConfig::ndoubling_steps`, `jeod::GaussJacksonConfig::priming_technique`, and `jeod::GaussJacksonConfig::relative_tolerance`.

Referenced by `jeod::GaussJacksonConfig::validate_configuration()`.

7.2.3.4 validate_config()

```
static unsigned int jeod::validate_config (
    const GaussJacksonConfig & config ) [static]
```

Check for invalid values in the supplied config.

Definition at line 167 of file gauss_jackson_config.cc.

References `jeod::GaussJacksonConfig::absolute_tolerance`, `jeod::GaussJacksonConfig::final_order`, `jeod::GaussJacksonConfig::initial_order`, `jeod::GaussJacksonConfig::ndoubling_steps`, and `jeod::GaussJacksonConfig::relative_tolerance`.

Referenced by `jeod::GaussJacksonConfig::validate_configuration()`.

Chapter 8

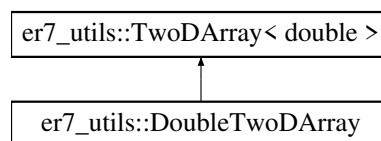
Data Structure Documentation

8.1 er7_utils::DoubleTwoDArray Class Reference

2D array, specialized for doubles.

```
#include <two_d_array.hh>
```

Inheritance diagram for er7_utils::DoubleTwoDArray:



Friends

- class [InputProcessor](#)
- void [init_attrer7_utils__DoubleTwoDArray](#) ()

Additional Inherited Members

8.1.1 Detailed Description

2D array, specialized for doubles.

Definition at line 403 of file two_d_array.hh.

8.1.2 Friends And Related Function Documentation

8.1.2.1 `init_attrer7_utils__DoubleTwoDArray`

```
void init_attrer7_utils__DoubleTwoDArray ( ) [friend]
```

8.1.2.2 `InputProcessor`

```
friend class InputProcessor [friend]
```

Definition at line 405 of file `two_d_array.hh`.

The documentation for this class was generated from the following file:

- [two_d_array.hh](#)

8.2 `jeod::GaussJacksonCoefficientsPair` Class Reference

Contains a summed Adams and Gauss-Jackson coefficient pair.

```
#include <gauss_jackson_coefficients_pair.hh>
```

Public Member Functions

- [GaussJacksonCoefficientsPair](#) ()=default
Default constructor.
- [~GaussJacksonCoefficientsPair](#) ()
Destructor.
- void [configure](#) (int max_order)
Allocate (re-allocate) memory for the coefficients.
- void [swap](#) ([GaussJacksonCoefficientsPair](#) &other)
Non-throwing swap.
- void [allocate_arrays](#) (int size)
Allocate space for the coefficients.
- void [deallocate_arrays](#) ()
Release allocated memory.
- void [apply](#) (int nelelem, int ncoeff, const double *const *acc_hist, const [GaussJacksonTwoState](#) &state_sum) const
Apply both sets of coefficients to the supplied history data.
- void [apply](#) (int nelelem, int ncoeff, const double *const *acc_hist, const [GaussJacksonOneState](#) &state_sum) const
Apply just the Adams coefficients to the supplied history data.
- void [printOut](#) (int order, std::ostream &stream) const
Print the coefficients.
- [GaussJacksonCoefficientsPair](#) (const [GaussJacksonCoefficientsPair](#) &)=delete
- [GaussJacksonCoefficientsPair](#) & operator= (const [GaussJacksonCoefficientsPair](#) &)=delete

Data Fields

- double * [sa_coefs](#) {}
Summed Adams coefficients, in ordinate form.
- double * [gj_coefs](#) {}
Gauss Jackson coefficients, in ordinate form.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [GaussJacksonCoefficientsPair](#))

8.2.1 Detailed Description

Contains a summed Adams and Gauss-Jackson coefficient pair.

Definition at line 82 of file gauss_jackson_coefficients_pair.hh.

8.2.2 Constructor & Destructor Documentation

8.2.2.1 GaussJacksonCoefficientsPair() [1/2]

```
jeod::GaussJacksonCoefficientsPair::GaussJacksonCoefficientsPair ( ) [default]
```

Default constructor.

8.2.2.2 ~GaussJacksonCoefficientsPair()

```
jeod::GaussJacksonCoefficientsPair::~~GaussJacksonCoefficientsPair ( ) [inline]
```

Destructor.

Definition at line 107 of file gauss_jackson_coefficients_pair.hh.

8.2.2.3 GaussJacksonCoefficientsPair() [2/2]

```
jeod::GaussJacksonCoefficientsPair::GaussJacksonCoefficientsPair (
    const GaussJacksonCoefficientsPair & ) [delete]
```

8.2.3 Member Function Documentation

8.2.3.1 allocate_arrays()

```
void jeod::GaussJacksonCoefficientsPair::allocate_arrays (
    int size )
```

Allocate space for the coefficients.

Parameters

<i>size</i>	Array size.
-------------	-------------

Definition at line 34 of file gauss_jackson_coefficients_pair.cc.

References `gj_coefs`, and `sa_coefs`.

8.2.3.2 apply() [1/2]

```
void jeod::GaussJacksonCoefficientsPair::apply (
    int nelem,
    int ncoeff,
    const double *const * acc_hist,
    const GaussJacksonOneState & state_sum ) const
```

Apply just the Adams coefficients to the supplied history data.

Parameters

<i>nelem</i>	Dimensionality of each acceleration history element
<i>ncoeff</i>	Number of elements in the acceleration history
<i>acc_hist</i>	Acceleration history
<i>state_sum</i>	Output inner products

Definition at line 84 of file gauss_jackson_coefficients_pair.cc.

8.2.3.3 apply() [2/2]

```
void jeod::GaussJacksonCoefficientsPair::apply (
    int nelem,
    int ncoeff,
    const double *const * acc_hist,
    const GaussJacksonTwoState & state_sum ) const
```

Apply both sets of coefficients to the supplied history data.

The first element of the output `state_sum` is calculated as the inner products of the acceleration history with the summed Adams coefficients. The second element is calculated as the inner product with the Gauss-Jackson coefficients. (First = first integral; second = second integral.)

Parameters

<i>nelem</i>	Dimensionality of each acceleration history element
<i>ncoeff</i>	Number of elements in the acceleration history
<i>acc_hist</i>	Acceleration history
<i>state_sum</i>	Output inner products

Definition at line 52 of file gauss_jackson_coefficients_pair.cc.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj().

8.2.3.4 configure()

```
void jeod::GaussJacksonCoefficientsPair::configure (
    int max_order ) [inline]
```

Allocate (re-allocate) memory for the coefficients.

Arrays are size & to contain max_order+1 elements.

Parameters

<i>max_order</i>	Maximum order that will be used.
------------------	----------------------------------

Definition at line 117 of file gauss_jackson_coefficients_pair.hh.

Referenced by jeod::GaussJacksonCoeffs::configure().

8.2.3.5 deallocate_arrays()

```
void jeod::GaussJacksonCoefficientsPair::deallocate_arrays ( )
```

Release allocated memory.

Definition at line 40 of file gauss_jackson_coefficients_pair.cc.

References gj_coefs, and sa_coefs.

8.2.3.6 operator=()

```
GaussJacksonCoefficientsPair& jeod::GaussJacksonCoefficientsPair::operator= (
    const GaussJacksonCoefficientsPair & ) [delete]
```

8.2.3.7 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonCoefficientsPair::p (
    jeod ,
    GaussJacksonCoefficientsPair ) [private]
```

8.2.3.8 printOut()

```
void jeod::GaussJacksonCoefficientsPair::printOut (
    int order,
    std::ostream & stream ) const
```

Print the coefficients.

Parameters

<i>order</i>	Coefficients order
<i>stream</i>	Output stream

Definition at line 110 of file gauss_jackson_coefficients_pair.cc.

References `gj_coefs`, and `sa_coefs`.

8.2.3.9 swap()

```
void jeod::GaussJacksonCoefficientsPair::swap (
    GaussJacksonCoefficientsPair & other )
```

Non-throwing swap.

Parameters

<i>other</i>	Coeffs pair with which contents are to be swapped.
--------------	--

Definition at line 46 of file gauss_jackson_coefficients_pair.cc.

References `gj_coefs`, and `sa_coefs`.

Referenced by `jeod::GaussJacksonCoeffs::swap()`.

8.2.4 Field Documentation**8.2.4.1 gj_coefs**

```
double* jeod::GaussJacksonCoefficientsPair::gj_coefs {}
```

Gauss Jackson coefficients, in ordinate form.

`trick_units(-)`

Definition at line 95 of file gauss_jackson_coefficients_pair.hh.

Referenced by `allocate_arrays()`, `jeod::GaussJacksonCoeffs::compute_coefs()`, `deallocate_arrays()`, `printOut()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`, and `swap()`.

8.2.4.2 sa_coefs

```
double* jeod::GaussJacksonCoefficientsPair::sa_coefs {}
```

Summed Adams coefficients, in ordinate form.

trick_units(—)

Definition at line 90 of file gauss_jackson_coefficients_pair.hh.

Referenced by `allocate_arrays()`, `jeod::GaussJacksonCoeffs::compute_coefs()`, `deallocate_arrays()`, `printOut()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`, and `swap()`.

The documentation for this class was generated from the following files:

- [gauss_jackson_coefficients_pair.hh](#)
- [gauss_jackson_coefficients_pair.cc](#)

8.3 jeod::GaussJacksonCoeffs Class Reference

Contains the Gauss-Jackson predictor and corrector coefficients.

```
#include <gauss_jackson_coefs.hh>
```

Public Member Functions

- [GaussJacksonCoeffs](#) ()=default
Default constructor.
- [GaussJacksonCoeffs](#) (const [GaussJacksonCoeffs](#) &src)
Copy constructor.
- [~GaussJacksonCoeffs](#) ()
Destructor.
- [GaussJacksonCoeffs](#) & operator= ([GaussJacksonCoeffs](#) src)
Copy-and-swap assignment operator.
- void [swap](#) ([GaussJacksonCoeffs](#) &src)
Non-throwing swap.
- void [configure](#) (unsigned int max_order_in)
Configure to enable coefficients up to the specified maximum order.
- void [compute_coefs](#) (unsigned int order_in)
Compute coefficients for the specified order.

Data Fields

- [GaussJacksonCoefficientsPair](#) predictor
Summed Adams and Gauss-Jackson predictor coefficients.
- [GaussJacksonCoefficientsPair](#) * corrector {}
Summed Adams and Gauss-Jackson corrector coefficients.
- unsigned int [max_order](#) {}
Maximum order; used for sizing.
- unsigned int [order](#) {}
Current order; dictates the coefficient values.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonCoeffs](#))

Friends

- std::ostream & [operator<<](#) (std::ostream &stream, const [GaussJacksonCoeffs](#) &coeff)
Print the coefficients.

8.3.1 Detailed Description

Contains the Gauss-Jackson predictor and corrector coefficients.

Definition at line 81 of file gauss_jackson_coeffs.hh.

8.3.2 Constructor & Destructor Documentation

8.3.2.1 GaussJacksonCoeffs() [1/2]

```
jeod::GaussJacksonCoeffs::GaussJacksonCoeffs ( ) [default]
```

Default constructor.

8.3.2.2 GaussJacksonCoeffs() [2/2]

```
jeod::GaussJacksonCoeffs::GaussJacksonCoeffs (
    const GaussJacksonCoeffs & src ) [inline]
```

Copy constructor.

Note that this doesn't copy; it recomputes. The end result is as if a copy had been made.

Parameters

<i>src</i>	Object to be copied.
------------	----------------------

Definition at line 119 of file gauss_jackson_coeffs.hh.

8.3.2.3 ~GaussJacksonCoeffs()

```
jeod::GaussJacksonCoeffs::~GaussJacksonCoeffs ( )
```

Destructor.

Definition at line 43 of file gauss_jackson_coeffs.cc.

References corrector.

8.3.3 Member Function Documentation

8.3.3.1 compute_coeffs()

```
void jeod::GaussJacksonCoeffs::compute_coeffs (
    unsigned int order_in )
```

Compute coefficients for the specified order.

Parameters

<i>order_in</i>	The current order.
-----------------	--------------------

Definition at line 75 of file gauss_jackson_coeffs.cc.

References jeod::GaussJacksonRationalCoefficients::configure_adams_corrector(), jeod::GaussJacksonRationalCoefficients::construct_predictor(), jeod::GaussJacksonRationalCoefficients::construct_stormer_cowell_corrector(), jeod::GaussJacksonRationalCoefficients::convert_to_ordinate_form(), corrector, jeod::GaussJacksonRationalCoefficients::discard_extra_terms(), jeod::GaussJacksonRationalCoefficients::displace_back(), jeod::GaussJacksonCoefficientsPair::gj_coefs, max_order, order, predictor, and jeod::GaussJacksonCoefficientsPair::sa_coefs.

Referenced by jeod::GaussJacksonIntegrationControls::GaussJacksonIntegrationControls(), and jeod::GaussJacksonIntegrationControls::start_cycle().

8.3.3.2 configure()

```
void jeod::GaussJacksonCoeffs::configure (
    unsigned int max_order_in )
```

Configure to enable coefficients up to the specified maximum order.

Parameters

<i>max_order_in</i>	The maximum order to be used.
---------------------	-------------------------------

Definition at line 58 of file gauss_jackson_coeffs.cc.

References `jeod::GaussJacksonCoefficientsPair::configure()`, `corrector`, `max_order`, `order`, and `predictor`.

Referenced by `jeod::GaussJacksonIntegrationControls::GaussJacksonIntegrationControls()`.

8.3.3.3 operator=()

```
GaussJacksonCoeffs& jeod::GaussJacksonCoeffs::operator= (
    GaussJacksonCoeffs src ) [inline]
```

Copy-and-swap assignment operator.

Parameters

<code>src</code>	Object to be copied.
------------------	----------------------

Definition at line 141 of file `gauss_jackson_coeffs.hh`.

8.3.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonCoeffs::p (
    jeod ,
    GaussJacksonCoeffs ) [private]
```

8.3.3.5 swap()

```
void jeod::GaussJacksonCoeffs::swap (
    GaussJacksonCoeffs & src )
```

Non-throwing swap.

Parameters

<code>src</code>	Object to swap contents with.
------------------	-------------------------------

Definition at line 49 of file `gauss_jackson_coeffs.cc`.

References `corrector`, `max_order`, `order`, `predictor`, and `jeod::GaussJacksonCoefficientsPair::swap()`.

8.3.4 Friends And Related Function Documentation

8.3.4.1 operator<<

```
std::ostream& operator<< (
    std::ostream & stream,
    const GaussJacksonCoeffs & coeff ) [friend]
```

Print the coefficients.

Parameters

<i>stream</i>	The stream to be printed to.
<i>coeff</i>	The coefficients to be printed.

Definition at line 130 of file gauss_jackson_coeffs.cc.

8.3.5 Field Documentation

8.3.5.1 corrector

```
GaussJacksonCoefficientsPair* jeod::GaussJacksonCoeffs::corrector {}
```

Summed Adams and Gauss-Jackson corrector coefficients.

trick_units(—)

Definition at line 94 of file gauss_jackson_coeffs.hh.

Referenced by compute_coeffs(), configure(), jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj(), jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), swap(), and ~GaussJacksonCoeffs().

8.3.5.2 max_order

```
unsigned int jeod::GaussJacksonCoeffs::max_order {}
```

Maximum order; used for sizing.

trick_units(—)

Definition at line 99 of file gauss_jackson_coeffs.hh.

Referenced by compute_coeffs(), configure(), and swap().

8.3.5.3 order

```
unsigned int jeod::GaussJacksonCoeffs::order {}
```

Current order; dictates the coefficient values.

trick_units(-)

Definition at line 104 of file gauss_jackson_coeffs.hh.

Referenced by compute_coeffs(), configure(), and swap().

8.3.5.4 predictor

```
GaussJacksonCoefficientsPair jeod::GaussJacksonCoeffs::predictor
```

Summed Adams and Gauss-Jackson predictor coefficients.

trick_units(-)

Definition at line 89 of file gauss_jackson_coeffs.hh.

Referenced by compute_coeffs(), configure(), and swap().

The documentation for this class was generated from the following files:

- [gauss_jackson_coeffs.hh](#)
- [gauss_jackson_coeffs.cc](#)

8.4 jeod::GaussJacksonConfig Class Reference

Contains Gauss-Jackson configuration data.

```
#include <gauss_jackson_config.hh>
```

Public Member Functions

- [GaussJacksonConfig](#) ()

Static Public Member Functions

- static [GaussJacksonConfig default_configuration](#) ()
Creates a [GaussJacksonConfig](#) with all members set to -1.
- static [GaussJacksonConfig standard_configuration](#) ()
Creates a [GaussJacksonConfig](#) with all members set to their defaults.
- static [GaussJacksonConfig validate_configuration](#) (const [GaussJacksonConfig](#) &config)
Creates a [GaussJacksonConfig](#) based on the supplied configuration.

Data Fields

- er7_utils::Integration::Technique [priming_technique](#) {er7_utils::Integration::Unspecified}
The integration technique to be used to prime the Gauss-Jackson process.
- unsigned int [initial_order](#) {4}
The order of the Gauss Jackson integrator immediately after priming.
- unsigned int [final_order](#) {12}
The order of the Gauss Jackson integrator once it's operational.
- unsigned int [ndoubling_steps](#)
The number of time doubling steps involved in the bootstrap operation.
- unsigned int [max_correction_iterations](#) {10}
Maximum number of correction steps allowed before the integrator is deemed to be not converging.
- double [relative_tolerance](#) {1E-14}
Number that indicates the allowable relative difference for two states to be considered converged.
- double [absolute_tolerance](#) {1E-10}
Number that indicates the allowable absolute difference for two states to be considered converged.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [GaussJacksonConfig](#))

8.4.1 Detailed Description

Contains Gauss-Jackson configuration data.

All member data are public; this is essentially a struct.

Definition at line 75 of file gauss_jackson_config.hh.

8.4.2 Constructor & Destructor Documentation

8.4.2.1 GaussJacksonConfig()

```
jeod::GaussJacksonConfig::GaussJacksonConfig ( ) [inline]
```

Definition at line 78 of file gauss_jackson_config.hh.

8.4.3 Member Function Documentation

8.4.3.1 default_configuration()

```
GaussJacksonConfig jeod::GaussJacksonConfig::default_configuration ( ) [static]
```

Creates a [GaussJacksonConfig](#) with all members set to -1.

This otherwise invalid value has a special meaning to the validation function. When encountered, the item is silently replaced with the default for that item.

Definition at line 39 of file gauss_jackson_config.cc.

References [priming_technique](#).

8.4.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonConfig::p (
    jeod ,
    GaussJacksonConfig ) [private]
```

8.4.3.3 standard_configuration()

```
GaussJacksonConfig jeod::GaussJacksonConfig::standard_configuration ( ) [static]
```

Creates a [GaussJacksonConfig](#) with all members set to their defaults.

Definition at line 55 of file gauss_jackson_config.cc.

References [priming_technique](#).

Referenced by [jeod::GaussJacksonIntegratorConstructor::create_integration_controls\(\)](#), and [validate_configuration\(\)](#).

8.4.3.4 validate_configuration()

```
GaussJacksonConfig jeod::GaussJacksonConfig::validate_configuration (
    const GaussJacksonConfig & config ) [static]
```

Creates a [GaussJacksonConfig](#) based on the supplied configuration.

Values of -1 are replaced with their defaults. The standard configuration is used if any invalid item is invalid.

Definition at line 236 of file gauss_jackson_config.cc.

References [jeod::set_default_config_values\(\)](#), [standard_configuration\(\)](#), and [jeod::validate_config\(\)](#).

Referenced by [jeod::GaussJacksonIntegratorConstructor::configure\(\)](#).

8.4.4 Field Documentation

8.4.4.1 absolute_tolerance

```
double jeod::GaussJacksonConfig::absolute_tolerance {1E-10}
```

Number that indicates the allowable absolute difference for two states to be considered converged.

Defaults to 1e-10. trick_units(-)

Definition at line 157 of file gauss_jackson_config.hh.

Referenced by jeod::set_default_config_values(), and jeod::validate_config().

8.4.4.2 final_order

```
unsigned int jeod::GaussJacksonConfig::final_order {12}
```

The order of the Gauss Jackson integrator once it's operational.

This must be an even number between initial_order and 14, inclusive. Defaults to 12. trick_units(-)

Definition at line 126 of file gauss_jackson_config.hh.

Referenced by jeod::GaussJacksonStateMachine::configure(), jeod::GaussJacksonIntegrationControls::Gauss↵
JacksonIntegrationControls(), jeod::set_default_config_values(), and jeod::validate_config().

8.4.4.3 initial_order

```
unsigned int jeod::GaussJacksonConfig::initial_order {4}
```

The order of the Gauss Jackson integrator immediately after priming.

This must be an even number and must be 14 or less. Defaults to 4. trick_units(-)

Definition at line 119 of file gauss_jackson_config.hh.

Referenced by jeod::GaussJacksonStateMachine::configure(), jeod::set_default_config_values(), and jeod↵
::validate_config().

8.4.4.4 max_correction_iterations

```
unsigned int jeod::GaussJacksonConfig::max_correction_iterations {10}
```

Maximum number of correction steps allowed before the integrator is deemed to be not converging.

The algorithm is run in predict-only mode if this limit is zero. The corrector is applied but once with the limit is one. A one-time warning is issued if the limit is 2 or more and if the the algorithm would make more corrections were it not for this limit. Defaults to 10. `trick_units(-)`

Definition at line 143 of file `gauss_jackson_config.hh`.

Referenced by `jeod::GaussJacksonStateMachine::configure()`, and `jeod::set_default_config_values()`.

8.4.4.5 ndoubling_steps

```
unsigned int jeod::GaussJacksonConfig::ndoubling_steps
```

The number of time doubling steps involved in the bootstrap operation.

Defaults to $(\text{final_order} - \text{initial_order})/2$. `trick_units(-)`

Definition at line 132 of file `gauss_jackson_config.hh`.

Referenced by `jeod::GaussJacksonStateMachine::configure()`, `jeod::set_default_config_values()`, and `jeod::validate_config()`.

8.4.4.6 priming_technique

```
er7_utils::Integration::Technique jeod::GaussJacksonConfig::priming_technique {er7_utils::↔  
Integration::Unspecified}
```

The integration technique to be used to prime the Gauss-Jackson process.

Defaults to `er7_utils::Integration::Unspecified`, the interpretation of which depends on the initial order. `trick_units(-)`

Definition at line 112 of file `gauss_jackson_config.hh`.

Referenced by `jeod::GaussJacksonIntegratorConstructor::configure()`, `default_configuration()`, `jeod::set_default_↔
config_values()`, and `standard_configuration()`.

8.4.4.7 relative_tolerance

```
double jeod::GaussJacksonConfig::relative_tolerance {1E-14}
```

Number that indicates the allowable relative difference for two states to be considered converged.

Defaults to 1e-14. `trick_units(-)`

Definition at line 150 of file `gauss_jackson_config.hh`.

Referenced by `jeod::set_default_config_values()`, and `jeod::validate_config()`.

The documentation for this class was generated from the following files:

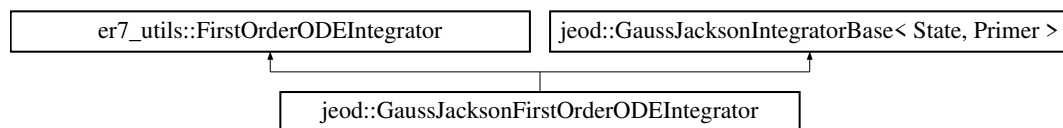
- [gauss_jackson_config.hh](#)
- [gauss_jackson_config.cc](#)

8.5 jeod::GaussJacksonFirstOrderODEIntegrator Class Reference

Integrates a first order ODE using the summed Adams technique that is embedded within the Gauss-Jackson technique.

```
#include <gauss_jackson_first_order_ode_integrator.hh>
```

Inheritance diagram for `jeod::GaussJacksonFirstOrderODEIntegrator`:



Private Member Functions

- `JEOD_MAKE_SIM_INTERFACES(jeod, GaussJacksonFirstOrderODEIntegrator)` public `~GaussJacksonFirstOrderODEIntegrator()` override=default
- `GaussJacksonFirstOrderODEIntegrator` (const `er7_utils::IntegratorConstructor` &priming_constructor, `GaussJacksonIntegrationControls` &controls, unsigned int size_in, `er7_utils::IntegrationControls` &priming_controls)
 Non-default constructor.
- `GaussJacksonFirstOrderODEIntegrator` (const `GaussJacksonFirstOrderODEIntegrator` &src)
 Copy constructor.
- `GaussJacksonFirstOrderODEIntegrator` & operator= (`GaussJacksonFirstOrderODEIntegrator` src)
 Assignment operator.
- void `swap` (`GaussJacksonFirstOrderODEIntegrator` &other)
 Non-throwing swap.
- `er7_utils::FirstOrderODEIntegrator` * `create_copy` () const override
 Replicate this.
- void `reset_integrator` () override
 Reset the integrator.
- `er7_utils::IntegratorResult` `integrate` (double dyn_dt, unsigned int target_stage, const double *ER7_UTILS_RESTRICT deriv, double *ER7_UTILS_RESTRICT state) override
 Integrate.
- void `swap` (`GaussJacksonIntegratorBase` &other)
 Non-throwing swap.

Additional Inherited Members

8.5.1 Detailed Description

Integrates a first order ODE using the summed Adams technique that is embedded within the Gauss-Jackson technique.

Definition at line 79 of file `gauss_jackson_first_order_ode_integrator.hh`.

8.5.2 Constructor & Destructor Documentation

8.5.2.1 ~GaussJacksonFirstOrderODEIntegrator()

```
JEOD_MAKE_SIM_INTERFACES (jeod, GaussJacksonFirstOrderODEIntegrator) public jeod::Gauss←
JacksonFirstOrderODEIntegrator::~~GaussJacksonFirstOrderODEIntegrator ( ) [override], [private],
[default]
```

8.5.2.2 GaussJacksonFirstOrderODEIntegrator() [1/2]

```
jeod::GaussJacksonFirstOrderODEIntegrator::GaussJacksonFirstOrderODEIntegrator (
    const er7_utils::IntegratorConstructor & priming_constructor,
    GaussJacksonIntegrationControls & controls,
    unsigned int size_in,
    er7_utils::IntegrationControls & priming_controls ) [inline], [private]
```

Non-default constructor.

Parameters

<i>priming_constructor</i>	Integrator constructor for the technique used during priming.
<i>controls</i>	The Gauss-Jackson integration controls that drives this state integrator.
<i>size_in</i>	State size.
<i>priming_controls</i>	Integration controls used during priming.

Definition at line 98 of file `gauss_jackson_first_order_ode_integrator.hh`.

8.5.2.3 GaussJacksonFirstOrderODEIntegrator() [2/2]

```
jeod::GaussJacksonFirstOrderODEIntegrator::GaussJacksonFirstOrderODEIntegrator (
    const GaussJacksonFirstOrderODEIntegrator & src ) [inline], [private]
```

Copy constructor.

Definition at line 110 of file `gauss_jackson_first_order_ode_integrator.hh`.

8.5.3 Member Function Documentation

8.5.3.1 create_copy()

```
er7_utils::FirstOrderODEIntegrator* jeod::GaussJacksonFirstOrderODEIntegrator::create_copy ( )
const [inline], [override], [private]
```

Replicate this.

Definition at line 137 of file gauss_jackson_first_order_ode_integrator.hh.

8.5.3.2 integrate()

```
er7_utils::IntegratorResult jeod::GaussJacksonFirstOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double *ER7_UTILS_RESTRICT deriv,
    double *ER7_UTILS_RESTRICT state ) [inline], [override], [private]
```

Integrate.

Definition at line 153 of file gauss_jackson_first_order_ode_integrator.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate().

8.5.3.3 operator=()

```
GaussJacksonFirstOrderODEIntegrator& jeod::GaussJacksonFirstOrderODEIntegrator::operator= (
    GaussJacksonFirstOrderODEIntegrator src ) [inline], [private]
```

Assignment operator.

Definition at line 119 of file gauss_jackson_first_order_ode_integrator.hh.

References swap().

8.5.3.4 reset_integrator()

```
void jeod::GaussJacksonFirstOrderODEIntegrator::reset_integrator ( ) [inline], [override],
[private]
```

Reset the integrator.

Definition at line 145 of file gauss_jackson_first_order_ode_integrator.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::base_reset().

8.5.3.5 swap() [1/2]

```
void jeod::GaussJacksonFirstOrderODEIntegrator::swap (
    GaussJacksonFirstOrderODEIntegrator & other ) [inline], [private]
```

Non-throwing swap.

Definition at line 128 of file gauss_jackson_first_order_ode_integrator.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

Referenced by jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::swap().

8.5.3.6 swap() [2/2]

```
void jeod::GaussJacksonIntegratorBase< State, Primer >::swap [inline], [private]
```

Non-throwing swap.

Parameters

<i>other</i>	Item whose contents are to be swapped with this.
--------------	--

Definition at line 397 of file gauss_jackson_integrator_base.hh.

Referenced by operator=().

The documentation for this class was generated from the following file:

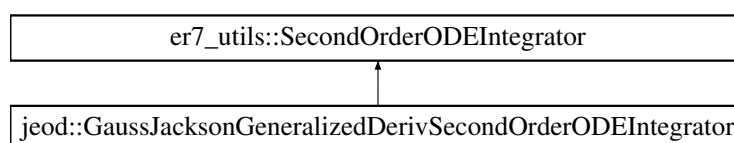
- [gauss_jackson_first_order_ode_integrator.hh](#)

8.6 jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator Class Reference

Integrates a generalized derivative second order ODE using Gauss-Jackson.

```
#include <gauss_jackson_generalized_second_order_ode_integrator.hh>
```

Inheritance diagram for jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator:



Public Member Functions

- [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) ()=default
Default constructor.
- [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) (const er7_utils::IntegratorConstructor &priming_constructor, [GaussJacksonIntegrationControls](#) &controls, unsigned int position_size, unsigned int velocity_size, const er7_utils::GeneralizedPositionDerivativeFunctions &deriv_funs, er7_utils::IntegrationControls &priming_controls)
Non-default constructor.
- [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) (const [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) &src)
Copy constructor.
- [~GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) () override
Destructor.
- [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) & operator= ([GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) &src)
Assignment operator.
- void [swap](#) ([GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#) &other)
Non-throwing swap.
- er7_utils::SecondOrderODEIntegrator * [create_copy](#) () const override
Replicate this.
- void [reset_integrator](#) () override
Reset the integrator.
- er7_utils::IntegratorResult [integrate](#) (double dyn_dt, unsigned int target_stage, const double *acc, double *vel, double *pos) override
Integrate state.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#))

Private Attributes

- [GaussJacksonFirstOrderODEIntegrator](#) vel_integrator
Integrator for the generalized velocity.
- [GaussJacksonSimpleSecondOrderODEIntegrator](#) pos_integrator
Integrator for the generalized position.
- double * [posdot](#) {}
Generalized position time derivative.
- double * [posdotdot](#) {}
Generalized position second time derivative.

8.6.1 Detailed Description

Integrates a generalized derivative second order ODE using Gauss-Jackson.

Generalized position is integrated via a simple second order Gauss-Jackson integrator. Generalized velocity is integrated via a first order summed Adams integrator.

Definition at line 82 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

8.6.2 Constructor & Destructor Documentation

8.6.2.1 GaussJacksonGeneralizedDerivSecondOrderODEIntegrator() [1/3]

```
jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::GaussJacksonGeneralizedDeriv↵
SecondOrderODEIntegrator ( ) [default]
```

Default constructor.

8.6.2.2 GaussJacksonGeneralizedDerivSecondOrderODEIntegrator() [2/3]

```
jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::GaussJacksonGeneralizedDeriv↵
SecondOrderODEIntegrator (
    const er7_utils::IntegratorConstructor & priming_constructor,
    GaussJacksonIntegrationControls & controls,
    unsigned int position_size,
    unsigned int velocity_size,
    const er7_utils::GeneralizedPositionDerivativeFunctions & deriv_funs,
    er7_utils::IntegrationControls & priming_controls )
```

Non-default constructor.

Parameters

<i>priming_constructor</i>	Integrator constructor for the technique used during priming.
<i>controls</i>	The Gauss-Jackson integration controls that drives this state integrator.
<i>position_size</i>	Generalized position vector size.
<i>velocity_size</i>	Generalized velocity vector size.
<i>deriv_funs</i>	Position vector time deriv functions.
<i>priming_controls</i>	Integration controls used during priming.

Definition at line 34 of file gauss_jackson_generalized_second_order_ode_integrator.cc.

References posdot, and posdotdot.

8.6.2.3 GaussJacksonGeneralizedDerivSecondOrderODEIntegrator() [3/3]

```
jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::GaussJacksonGeneralizedDeriv↵
SecondOrderODEIntegrator (
    const GaussJacksonGeneralizedDerivSecondOrderODEIntegrator & src )
```

Copy constructor.

Definition at line 50 of file gauss_jackson_generalized_second_order_ode_integrator.cc.

References posdot, and posdotdot.

8.6.2.4 ~GaussJacksonGeneralizedDerivSecondOrderODEIntegrator()

```
jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::~~GaussJacksonGeneralizedDerivSecondOrderODEIntegrator ( ) [override]
```

Destructor.

Definition at line 61 of file gauss_jackson_generalized_second_order_ode_integrator.cc.

References posdot, and posdotdot.

8.6.3 Member Function Documentation

8.6.3.1 create_copy()

```
er7_utils::SecondOrderODEIntegrator * jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::create_copy ( ) const [override]
```

Replicate this.

Definition at line 81 of file gauss_jackson_generalized_second_order_ode_integrator.cc.

8.6.3.2 integrate()

```
er7_utils::IntegratorResult jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * acc,
    double * vel,
    double * pos ) [inline], [override]
```

Integrate state.

Definition at line 154 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

8.6.3.3 operator=()

```
GaussJacksonGeneralizedDerivSecondOrderODEIntegrator& jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::operator= (
    GaussJacksonGeneralizedDerivSecondOrderODEIntegrator src ) [inline]
```

Assignment operator.

Definition at line 123 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

8.6.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::p (
    jeod ,
    GaussJacksonGeneralizedDerivSecondOrderODEIntegrator ) [private]
```

8.6.3.5 reset_integrator()

```
void jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::reset_integrator ( ) [inline],
[override]
```

Reset the integrator.

Definition at line 145 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

8.6.3.6 swap()

```
void jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::swap (
    GaussJacksonGeneralizedDerivSecondOrderODEIntegrator & other )
```

Non-throwing swap.

Definition at line 68 of file gauss_jackson_generalized_second_order_ode_integrator.cc.

References pos_integrator, posdot, posdotdot, jeod::GaussJacksonFirstOrderODEIntegrator::swap(), jeod::↔GaussJacksonSimpleSecondOrderODEIntegrator::swap(), and vel_integrator.

8.6.4 Field Documentation

8.6.4.1 pos_integrator

```
GaussJacksonSimpleSecondOrderODEIntegrator jeod::GaussJacksonGeneralizedDerivSecondOrder↔
ODEIntegrator::pos_integrator [private]
```

Integrator for the generalized position.

trick_units(−)

Definition at line 176 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

Referenced by swap().

8.6.4.2 posdot

```
double* jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::posdot {} [private]
```

Generalized position time derivative.

trick_units(-)

Definition at line 181 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

Referenced by GaussJacksonGeneralizedDerivSecondOrderODEIntegrator(), swap(), and ~GaussJacksonGeneralizedDerivSecondOrderODEIntegrator().

8.6.4.3 posdotdot

```
double* jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::posdotdot {} [private]
```

Generalized position second time derivative.

trick_units(-)

Definition at line 186 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

Referenced by GaussJacksonGeneralizedDerivSecondOrderODEIntegrator(), swap(), and ~GaussJacksonGeneralizedDerivSecondOrderODEIntegrator().

8.6.4.4 vel_integrator

```
GaussJacksonFirstOrderODEIntegrator jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::vel_integrator [private]
```

Integrator for the generalized velocity.

trick_units(-)

Definition at line 171 of file gauss_jackson_generalized_second_order_ode_integrator.hh.

Referenced by swap().

The documentation for this class was generated from the following files:

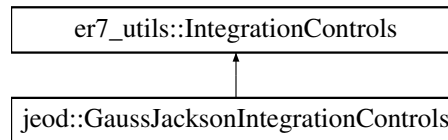
- [gauss_jackson_generalized_second_order_ode_integrator.hh](#)
- [gauss_jackson_generalized_second_order_ode_integrator.cc](#)

8.7 jeod::GaussJacksonIntegrationControls Class Reference

IntegrationControls specialized for Gauss-Jackson integration.

```
#include <gauss_jackson_integration_controls.hh>
```

Inheritance diagram for jeod::GaussJacksonIntegrationControls:



Public Member Functions

- [GaussJacksonIntegrationControls](#) ()
Default constructor.
- [GaussJacksonIntegrationControls](#) (const er7_utils::IntegratorConstructor &priming_constructor, const [GaussJacksonConfig](#) &config_in)
Non-default constructor.
- [GaussJacksonIntegrationControls](#) (const [GaussJacksonIntegrationControls](#) &src)
Copy constructor.
- [~GaussJacksonIntegrationControls](#) () override
Destructor.
- [GaussJacksonIntegrationControls](#) & operator= (GaussJacksonIntegrationControls src)
Copy and swap assignment operator.
- er7_utils::IntegrationControls * [create_copy](#) () const override
Create a duplicate of this object.
- er7_utils::IntegrationControls & [get_priming_controls](#) () const
Getter for the priming_controls data member.
- const [GaussJacksonCoeffs](#) & [get_coeff](#) () const
Getter for the coeff data member.
- const [GaussJacksonConfig](#) & [get_config](#) () const
Getter for the config data member.
- const [GaussJacksonStateMachine](#) & [get_state_machine](#) () const
Getter for the state_machine data member.
- void [reset_integrator](#) () override
Reset the integration controls object.
- unsigned int [integrate](#) (double start_time, double sim_dt, er7_utils::TimeInterface &time_interface, er7_utils::IntegratorInterface &integ_interface, er7_utils::BaseIntegrationGroup &integ_group) override
Make one step in the process that eventually integrates state from the start_time to start_time+sim_dt.

Protected Member Functions

- virtual void [swap](#) ([GaussJacksonIntegrationControls](#) &other)
Non-throwing swap function.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonIntegrationControls](#))
- void [start_cycle](#) (double sim_dt)
Perform start of integration cycle actions.
- void [integrate_edit](#) (er7_utils::TimeInterface &time_interface, er7_utils::BaseIntegrationGroup &integ_group)
Guide integration while in BootstrapEdit mode.
- void [integrate_gj](#) (er7_utils::TimeInterface &time_interface, er7_utils::BaseIntegrationGroup &integ_group)
Guide integration while in BootstrapStep or Operational mode.

Private Attributes

- er7_utils::IntegrationControls * [priming_controls](#) {}
The integration controls object used to prime the Gauss-Jackson integration process.
- double [cycle_starttime](#) {}
The simulation time of the start of the current integration cycle.
- double [cycle_simdt](#) {}
The simulation time span of the current integration cycle.
- double [cycle_dyndt](#) {}
The dynamic time span corresponding to cycle_simdt.
- double [reset_time](#) {}
The simulation time of the most recent reset.
- [GaussJacksonCoeffs](#) [coeff](#)
The Gauss-Jackson corrector and predictor coefficients.
- [GaussJacksonConfig](#) [config](#)
The Gauss-Jackson configuration data.
- [GaussJacksonStateMachine](#) [state_machine](#)
The Gauss-Jackson state machine.
- [GaussJacksonStateMachine::FsmState](#) [fsm_state](#) {[GaussJacksonStateMachine::Reset](#)}
The state machine's finite state.
- unsigned int [max_correction_iterations](#) {}
Maximum number of correction iterations allowed.
- unsigned int [initial_order](#) {}
The order of the Gauss Jackson integrator immediately after priming.
- unsigned int [order](#) {}
The current order of the Gauss Jackson integrator.
- unsigned int [edit_count](#) {}
Number of times that the current set of history have been edited.
- bool [at_end_of_tour](#) {}
Flag indicating that the current integration cycle is the last one in an integration tour (i.e., that a major time step will be completed).

8.7.1 Detailed Description

IntegrationControls specialized for Gauss-Jackson integration.

Definition at line 80 of file gauss_jackson_integration_controls.hh.

8.7.2 Constructor & Destructor Documentation

8.7.2.1 GaussJacksonIntegrationControls() [1/3]

```
jeod::GaussJacksonIntegrationControls::GaussJacksonIntegrationControls ( )
```

Default constructor.

Definition at line 40 of file gauss_jackson_integration_controls.cc.

8.7.2.2 GaussJacksonIntegrationControls() [2/3]

```
jeod::GaussJacksonIntegrationControls::GaussJacksonIntegrationControls (
    const er7_utils::IntegratorConstructor & priming_constructor,
    const GaussJacksonConfig & config_in )
```

Non-default constructor.

This is the constructor invoked by the [GaussJacksonIntegratorConstructor](#).

Parameters

<i>priming_constructor</i>	Integrator constructor for the technique used during priming.
<i>config_in</i>	Gauss-Jackson configuration data.

Definition at line 46 of file gauss_jackson_integration_controls.cc.

References [coeff](#), [jeod::GaussJacksonCoeffs::compute_coeffs\(\)](#), [config](#), [jeod::GaussJacksonStateMachine::configure\(\)](#), [jeod::GaussJacksonCoeffs::configure\(\)](#), [jeod::GaussJacksonConfig::final_order](#), [initial_order](#), [priming_controls](#), and [state_machine](#).

8.7.2.3 GaussJacksonIntegrationControls() [3/3]

```
jeod::GaussJacksonIntegrationControls::GaussJacksonIntegrationControls (
    const GaussJacksonIntegrationControls & src )
```

Copy constructor.

Parameters

<i>src</i>	Object to be copied.
------------	----------------------

Definition at line 62 of file gauss_jackson_integration_controls.cc.

References [priming_controls](#).

8.7.2.4 ~GaussJacksonIntegrationControls()

```
jeod::GaussJacksonIntegrationControls::~~GaussJacksonIntegrationControls ( ) [override]
```

Destructor.

Definition at line 85 of file [gauss_jackson_integration_controls.cc](#).

References [priming_controls](#).

8.7.3 Member Function Documentation

8.7.3.1 create_copy()

```
er7_utils::IntegrationControls * jeod::GaussJacksonIntegrationControls::create_copy ( ) const  
[override]
```

Create a duplicate of this object.

Returns

Replicated [GaussJacksonIntegrationControls](#).

Definition at line 90 of file [gauss_jackson_integration_controls.cc](#).

8.7.3.2 get_coeff()

```
const GaussJacksonCoeffs& jeod::GaussJacksonIntegrationControls::get_coeff ( ) const [inline]
```

Getter for the `coeff` data member.

Returns

Reference to the Gauss-Jackson coefficients object.

Definition at line 138 of file [gauss_jackson_integration_controls.hh](#).

8.7.3.3 get_config()

```
const GaussJacksonConfig& jeod::GaussJacksonIntegrationControls::get_config ( ) const [inline]
```

Getter for the config data member.

Returns

Reference to the Gauss-Jackson configuration object.

Definition at line 147 of file gauss_jackson_integration_controls.hh.

8.7.3.4 get_priming_controls()

```
er7_utils::IntegrationControls& jeod::GaussJacksonIntegrationControls::get_priming_controls (
) const [inline]
```

Getter for the priming_controls data member.

Returns

Reference to the integration controls used during priming.

Definition at line 129 of file gauss_jackson_integration_controls.hh.

Referenced by jeod::GaussJacksonIntegratorConstructor::create_first_order_ode_integrator(), jeod::GaussJacksonIntegratorConstructor::create_generalized_deriv_second_order_ode_integrator(), and jeod::GaussJacksonIntegratorConstructor::create_second_order_ode_integrator().

8.7.3.5 get_state_machine()

```
const GaussJacksonStateMachine& jeod::GaussJacksonIntegrationControls::get_state_machine ( )
const [inline]
```

Getter for the state_machine data member.

Returns

Reference to the Gauss-Jackson state_machine object.

Definition at line 156 of file gauss_jackson_integration_controls.hh.

8.7.3.6 integrate()

```
unsigned int jeod::GaussJacksonIntegrationControls::integrate (
    double start_time,
    double sim_dt,
    er7_utils::TimeInterface & time_interface,
    er7_utils::IntegratorInterface & integ_interface,
    er7_utils::BaseIntegrationGroup & integ_group ) [override]
```

Make one step in the process that eventually integrates state from the start_time to start_time+sim_dt.

Returns

Step number; zero when finished.

Parameters

in	<i>start_time</i>	The simulation engine time at the start of the integration tour.
in	<i>sim_dt</i>	The difference between the simulation time at the end and start of the integration tour.
in, out	<i>time_interface</i>	Object external to the ER7 utilities suite that represents time.
in, out	<i>integ_interface</i>	Interface with the simulation engine for this integration controls.
in, out	<i>integ_group</i>	The integration group that contains this integration controls.

Definition at line 126 of file gauss_jackson_integration_controls.cc.

References `at_end_of_tour`, `jeod::GaussJacksonStateMachine::BootstrapEdit`, `jeod::GaussJacksonState↵Machine::BootstrapStep`, `cycle_dyndt`, `cycle_simdt`, `cycle_starttime`, `fsm_state`, `jeod::GaussJacksonState↵Machine::get_cycle_scale()`, `integrate_edit()`, `integrate_gj()`, `jeod::GaussJacksonStateMachine::Operational`, `jeod↵::GaussJacksonStateMachine::Priming`, `priming_controls`, `jeod::GaussJacksonStateMachine::Reset`, `reset↵integrator()`, `reset_time`, `start_cycle()`, and `state_machine`.

8.7.3.7 integrate_edit()

```
void jeod::GaussJacksonIntegrationControls::integrate_edit (
    er7_utils::TimeInterface & time_interface,
    er7_utils::BaseIntegrationGroup & integ_group ) [private]
```

Guide integration while in BootstrapEdit mode.

Definition at line 222 of file gauss_jackson_integration_controls.cc.

References `cycle_dyndt`, `cycle_starttime`, `edit_count`, `jeod::GaussJacksonStateMachine::get_history_length()`, `max_correction_iterations`, `order`, `jeod::GaussJacksonStateMachine::set_bootstrap_edit_redo_needed()`, and `state_machine`.

Referenced by `integrate()`.

8.7.3.8 integrate_gj()

```
void jeod::GaussJacksonIntegrationControls::integrate_gj (
    er7_utils::TimeInterface & time_interface,
    er7_utils::BaseIntegrationGroup & integ_group ) [private]
```

Guide integration while in BootstrapStep or Operational mode.

Definition at line 255 of file gauss_jackson_integration_controls.cc.

References `cycle_dyndt`, `cycle_simdt`, `cycle_starttime`, `edit_count`, and `max_correction_iterations`.

Referenced by `integrate()`.

8.7.3.9 operator=()

```
GaussJacksonIntegrationControls& jeod::GaussJacksonIntegrationControls::operator= (
    GaussJacksonIntegrationControls src ) [inline]
```

Copy and swap assignment operator.

Parameters

<i>src</i>	Object to be copied.
------------	----------------------

Definition at line 113 of file gauss_jackson_integration_controls.hh.

8.7.3.10 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonIntegrationControls::p (
    jeod ,
    GaussJacksonIntegrationControls ) [private]
```

8.7.3.11 reset_integrator()

```
void jeod::GaussJacksonIntegrationControls::reset_integrator ( ) [override]
```

Reset the integration controls object.

Definition at line 116 of file gauss_jackson_integration_controls.cc.

References `at_end_of_tour`, `edit_count`, `fsm_state`, `initial_order`, `order`, `jeod::GaussJacksonStateMachine::Reset`, `jeod::GaussJacksonStateMachine::reset()`, and `state_machine`.

Referenced by `integrate()`.

8.7.3.12 start_cycle()

```
void jeod::GaussJacksonIntegrationControls::start_cycle (
    double sim_dt ) [private]
```

Perform start of integration cycle actions.

Definition at line 288 of file gauss_jackson_integration_controls.cc.

References `at_end_of_tour`, `jeod::GaussJacksonStateMachine::BootstrapEdit`, `coeff`, `jeod::GaussJacksonCoeffs::compute_coeffs()`, `cycle_dyndt`, `cycle_simdt`, `cycle_starttime`, `edit_count`, `fsm_state`, `jeod::GaussJacksonStateMachine::get_at_downsample()`, `jeod::GaussJacksonStateMachine::get_at_end_of_tour()`, `jeod::GaussJacksonStateMachine::get_at_order_change()`, `jeod::GaussJacksonStateMachine::get_at_reinitialize()`, `jeod::GaussJacksonStateMachine::get_current_order()`, `jeod::GaussJacksonStateMachine::get_cycle_scale()`, `jeod::GaussJacksonStateMachine::get_cycle_start_time()`, `jeod::GaussJacksonStateMachine::get_fsm_state()`, `order`, `jeod::GaussJacksonStateMachine::perform_step()`, `reset_time`, and `state_machine`.

Referenced by `integrate()`.

8.7.3.13 swap()

```
void jeod::GaussJacksonIntegrationControls::swap (
    GaussJacksonIntegrationControls & other ) [protected], [virtual]
```

Non-throwing swap function.

Swap contents of 'this' with that of the other.

Parameters

<code>in</code>	<code>other</code>	Item with which contents are to be swapped.
-----------------	--------------------	---

Definition at line 96 of file gauss_jackson_integration_controls.cc.

References `at_end_of_tour`, `coeff`, `config`, `cycle_dyndt`, `cycle_simdt`, `cycle_starttime`, `edit_count`, `fsm_state`, `initial_order`, `max_correction_iterations`, `order`, `priming_controls`, `reset_time`, and `state_machine`.

8.7.4 Field Documentation

8.7.4.1 at_end_of_tour

```
bool jeod::GaussJacksonIntegrationControls::at_end_of_tour {} [private]
```

Flag indicating that the current integration cycle is the last one in an integration tour (i.e., that a major time step will be completed).

trick_units(-)

Definition at line 271 of file gauss_jackson_integration_controls.hh.

Referenced by `integrate()`, `reset_integrator()`, `start_cycle()`, and `swap()`.

8.7.4.2 coeff

```
GaussJacksonCoeffs jeod::GaussJacksonIntegrationControls::coeff [private]
```

The Gauss-Jackson corrector and predictor coefficients.

trick_units(-)

Definition at line 230 of file gauss_jackson_integration_controls.hh.

Referenced by `GaussJacksonIntegrationControls()`, `start_cycle()`, and `swap()`.

8.7.4.3 config

```
GaussJacksonConfig jeod::GaussJacksonIntegrationControls::config [private]
```

The Gauss-Jackson configuration data.

trick_units(-)

Definition at line 235 of file gauss_jackson_integration_controls.hh.

Referenced by `GaussJacksonIntegrationControls()`, and `swap()`.

8.7.4.4 cycle_dyndt

```
double jeod::GaussJacksonIntegrationControls::cycle_dyndt {} [private]
```

The dynamic time span corresponding to cycle_simdt.

trick_units(s)

Definition at line 220 of file gauss_jackson_integration_controls.hh.

Referenced by integrate(), integrate_edit(), integrate_gj(), start_cycle(), and swap().

8.7.4.5 cycle_simdt

```
double jeod::GaussJacksonIntegrationControls::cycle_simdt {} [private]
```

The simulation time span of the current integration cycle.

trick_units(-)

Definition at line 215 of file gauss_jackson_integration_controls.hh.

Referenced by integrate(), integrate_gj(), start_cycle(), and swap().

8.7.4.6 cycle_starttime

```
double jeod::GaussJacksonIntegrationControls::cycle_starttime {} [private]
```

The simulation time of the start of the current integration cycle.

An integration cycle starts when cycle_stage is zero and ends when it reaches zero once again. trick_units(-)

Definition at line 210 of file gauss_jackson_integration_controls.hh.

Referenced by integrate(), integrate_edit(), integrate_gj(), start_cycle(), and swap().

8.7.4.7 edit_count

```
unsigned int jeod::GaussJacksonIntegrationControls::edit_count {} [private]
```

Number of times that the current set of history have been edited.

trick_units(-)

Definition at line 265 of file gauss_jackson_integration_controls.hh.

Referenced by integrate_edit(), integrate_gj(), reset_integrator(), start_cycle(), and swap().

8.7.4.8 fsm_state

```
GaussJacksonStateMachine::FsmState jeod::GaussJacksonIntegrationControls::fsm_state {GaussJacksonStateMachine::FsmState} [private]
```

The state machine's finite state.

trick_units(–)

Definition at line 245 of file gauss_jackson_integration_controls.hh.

Referenced by integrate(), reset_integrator(), start_cycle(), and swap().

8.7.4.9 initial_order

```
unsigned int jeod::GaussJacksonIntegrationControls::initial_order {} [private]
```

The order of the Gauss Jackson integrator immediately after priming.

trick_units(–)

Definition at line 255 of file gauss_jackson_integration_controls.hh.

Referenced by GaussJacksonIntegrationControls(), reset_integrator(), and swap().

8.7.4.10 max_correction_iterations

```
unsigned int jeod::GaussJacksonIntegrationControls::max_correction_iterations {} [private]
```

Maximum number of correction iterations allowed.

trick_units(–)

Definition at line 250 of file gauss_jackson_integration_controls.hh.

Referenced by integrate_edit(), integrate_gj(), and swap().

8.7.4.11 order

```
unsigned int jeod::GaussJacksonIntegrationControls::order {} [private]
```

The current order of the Gauss Jackson integrator.

trick_units(–)

Definition at line 260 of file gauss_jackson_integration_controls.hh.

Referenced by integrate_edit(), reset_integrator(), start_cycle(), and swap().

8.7.4.12 priming_controls

```
er7_utils::IntegrationControls* jeod::GaussJacksonIntegrationControls::priming_controls {}  
[private]
```

The integration controls object used to prime the Gauss-Jackson integration process.

trick_units(-)

Definition at line 203 of file gauss_jackson_integration_controls.hh.

Referenced by GaussJacksonIntegrationControls(), integrate(), swap(), and ~GaussJacksonIntegrationControls().

8.7.4.13 reset_time

```
double jeod::GaussJacksonIntegrationControls::reset_time {} [private]
```

The simulation time of the most recent reset.

trick_units(-)

Definition at line 225 of file gauss_jackson_integration_controls.hh.

Referenced by integrate(), start_cycle(), and swap().

8.7.4.14 state_machine

```
GaussJacksonStateMachine jeod::GaussJacksonIntegrationControls::state_machine [private]
```

The Gauss-Jackson state machine.

trick_units(-)

Definition at line 240 of file gauss_jackson_integration_controls.hh.

Referenced by GaussJacksonIntegrationControls(), integrate(), integrate_edit(), reset_integrator(), start_cycle(), and swap().

The documentation for this class was generated from the following files:

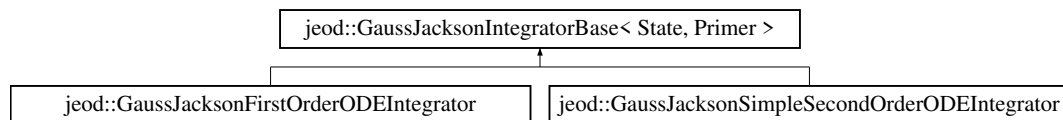
- [gauss_jackson_integration_controls.hh](#)
- [gauss_jackson_integration_controls.cc](#)

8.8 jeod::GaussJacksonIntegratorBase< State, Primer > Class Template Reference

Base template class for integrating state via the Gauss-Jackson technique.

```
#include <gauss_jackson_integrator_base.hh>
```

Inheritance diagram for jeod::GaussJacksonIntegratorBase< State, Primer >:



Public Member Functions

- [GaussJacksonIntegratorBase](#) ()=default
Default constructor.
- [GaussJacksonIntegratorBase](#) (const er7_utils::IntegratorConstructor &priming_constructor, const [GaussJacksonIntegrationCon](#) &controls, unsigned int size_in, er7_utils::IntegrationControls &priming_controls)
Non-default constructor.
- [GaussJacksonIntegratorBase](#) (const [GaussJacksonIntegratorBase](#) &src)
Copy constructor.
- [~GaussJacksonIntegratorBase](#) ()
Destructor.
- [GaussJacksonIntegratorBase](#) & operator= (const [GaussJacksonIntegratorBase](#) &)=delete

Data Fields

- const [GaussJacksonCoeffs](#) * coeff {}
The summed Adams and Gauss-Jackson coefficients, in ordinate form.
- const [GaussJacksonStateMachine](#) * state_machine {}
The Gauss-Jackson state machine.
- Primer * primer {}
The integrator used to prime the Gauss-Jackson integration process.
- State [init_state](#)
The state at the time of the last reset.
- State [delinv](#)
Inverse backward differences.
- State [corrector_sum](#)
Speed hack for the corrector.
- er7_utils::DoubleTwoDArray [acc_hist](#)
Acceleration history.
- er7_utils::DoubleTwoDArray [pos_hist](#)
Position history (or velocity history in case of a first order ODE).
- double [relative_tolerance](#) {}
Number that indicates the allowable relative difference for two states to be considered converged.
- double [absolute_tolerance](#) {}
Number that indicates the allowable absolute difference for two states to be considered converged.

- double `velocity_corrector` {}
Correction coefficient for the first integral (velocity).
- double `position_corrector` {}
Correction coefficient for the second integral (position).
- `GaussJacksonStateMachine::FsmState fsm_state` {`GaussJacksonStateMachine::Reset`}
Finite state machine state.
- unsigned int `max_history_size` {}
Maximum history size.
- unsigned int `initial_order` {}
Initial order.
- unsigned int `order` {}
Current order.
- unsigned int `size` {}
State size.
- unsigned int `history_length` {}
Current history length.

Protected Member Functions

- void `base_reset` ()
Reset the integrator.
- `er7_utils::IntegratorResult base_integrate` (double `dyn_dt`, unsigned int `target_stage`, const double `*deriv`, State `state`)
Propagate state to the specified target_stage.
- void `swap` (`GaussJacksonIntegratorBase` &`other`)
Non-throwing swap.

Private Member Functions

- void `start_cycle` (double `dt`, const double `*acc`, const State &`state`)
Start an integration cycle.
- bool `edit_point` (double `dt`, const double `*acc`, State &`state`)
Edit the specified point using the mid-corrector that pertains to the point being edited, which is that at history_length.
- bool `integrate_gj` (double `dt`, unsigned int `target_stage`, int `advance_index`, int `target_index`, const double `*acc`, const double `*const` `*ahist`, State &`state`)
Integrate using the Gauss-Jackson predictor and corrector.
- void `downsample_hist` ()
Downsample the acceleration and position histories.
- void `rotate_acc_hist` ()
Rotate the acceleration history.
- `er7_utils::IntegratorResult integrate_primer` (double `dyn_dt`, unsigned int `target_stage`, const double `*deriv`, State &`state`)
Integrate state using the primer.
- void `save_epoch_data` (const double `*acc`, const State &`state`)
Save epoch data.
- void `save_comparison_data` (const State &`state`, double `*pos_hist_elem`)
Save comparison data.
- void `initialize_edit_integration_constants` (double `dt`)
Initialize the integration constants (i.e., delinv).
- void `initialize_predictor_integration_constants` (double `dt`)

- Initialize the integration constants (i.e., delinv).*
- void [advance_edit_integration_constants](#) (unsigned int index)
 - Advance the integration constants by one cycle.*
- void [advance_predictor_integration_constants](#) (unsigned int index)
 - Advance the integration constants by one cycle.*
- void [mid_correct](#) (unsigned int coeff_idx, double dt, State &state)
 - Apply a mid-corrector.*
- void [predict](#) (double dt, const double *const *ahist, State &state)
 - Apply the predictor.*
- void [correct](#) (double dt, const double *acc, State &state)
 - Apply the corrector.*
- bool [test_for_convergence](#) (const State &state, double *hist_data)
 - Test for convergence.*
- void [swap_state](#) (State &item, State &other_item)
 - Swap state data with another of the same.*
- void [replicate_state](#) (const State &source, State &target)
 - Replicate state data.*
- void [allocate_state_contents](#) (State &item)
 - Allocate memory for a state item.*
- void [deallocate_state_contents](#) (State &item)
 - Deallocate state item memory.*
- er7_utils::FirstOrderODEIntegrator * [create_primer](#) (const er7_utils::IntegratorConstructor &priming_constructor, unsigned int size, er7_utils::IntegrationControls &priming_controls)
 - Create the priming integrator.*
- er7_utils::FirstOrderODEIntegrator * [replicate_primer](#) (const er7_utils::FirstOrderODEIntegrator *src)
 - Replicate the priming integrator.*
- er7_utils::IntegratorResult [integrate_primer](#) (double dyn_dt, unsigned int target_stage, const double *deriv, GaussJacksonOneState &state)
 - Integrate with the primer.*
- void [save_epoch_data](#) (const double *acc, const GaussJacksonOneState &state)
 - Save epoch data.*
- void [save_comparison_data](#) (const GaussJacksonOneState &state, double *pos_hist_elem)
 - Save comparison data.*
- void [initialize_edit_integration_constants](#) (double dt)
 - Initialize the integration constants (i.e., delinv).*
- void [advance_edit_integration_constants](#) (unsigned int index)
 - Advance the integration constants by one cycle.*
- void [initialize_predictor_integration_constants](#) (double dt)
 - Initialize the integration constants (i.e., delinv).*
- void [advance_predictor_integration_constants](#) (unsigned int index)
 - Advance the integration constants by one cycle.*
- void [mid_correct](#) (unsigned int coeff_idx, double dt, GaussJacksonOneState &state)
 - Apply a mid-corrector.*
- void [predict](#) (double dt, const double *const *ahist, GaussJacksonOneState &state)
 - Apply the predictor.*
- void [correct](#) (double dt, const double *acc, GaussJacksonOneState &state)
 - Apply the corrector.*
- bool [test_for_convergence](#) (const GaussJacksonOneState &state, double *hist_data)
 - Test for convergence.*
- void [swap_state](#) (GaussJacksonOneState &item, GaussJacksonOneState &other_item)
 - Swap state data with another of the same.*

- void `replicate_state` (const `GaussJacksonOneState` &source, `GaussJacksonOneState` &target)
Replicate state data.
- void `allocate_state_contents` (`GaussJacksonOneState` &item)
Allocate memory for a state item.
- void `deallocate_state_contents` (`GaussJacksonOneState` &item)
Deallocate state item memory.
- `er7_utils::SecondOrderODEIntegrator` * `create_primer` (const `er7_utils::IntegratorConstructor` &priming_constructor, unsigned int size, `er7_utils::IntegrationControls` &priming_controls)
- `er7_utils::SecondOrderODEIntegrator` * `replicate_primer` (const `er7_utils::SecondOrderODEIntegrator` *src)
- `er7_utils::IntegratorResult` `integrate_primer` (double dyn_dt, unsigned int target_stage, const double *deriv, `GaussJacksonTwoState` &state)
Integrate with the primer.
- void `save_epoch_data` (const double *acc, const `GaussJacksonTwoState` &state)
Save epoch data.
- void `save_comparison_data` (const `GaussJacksonTwoState` &state, double *pos_hist_elem)
Save comparison data.
- void `initialize_edit_integration_constants` (double dt)
Initialize the integration constants (i.e., delinv).
- void `advance_edit_integration_constants` (unsigned int index)
Advance the integration constants by one cycle.
- void `initialize_predictor_integration_constants` (double dt)
Initialize the integration constants (i.e., delinv).
- void `advance_predictor_integration_constants` (unsigned int index)
Advance the integration constants by one cycle.
- void `mid_correct` (unsigned int coeff_idx, double dt, `GaussJacksonTwoState` &state)
Apply a mid-corrector.
- void `predict` (double dt, const double *const *ahist, `GaussJacksonTwoState` &state)
Apply the predictor.
- void `correct` (double dt, const double *acc, `GaussJacksonTwoState` &state)
Apply the corrector.
- bool `test_for_convergence` (const `GaussJacksonTwoState` &state, double *hist_data)
Test for convergence.
- void `swap_state` (`GaussJacksonTwoState` &item, `GaussJacksonTwoState` &other_item)
Swap state data with another of the same.
- void `replicate_state` (const `GaussJacksonTwoState` &source, `GaussJacksonTwoState` &target)
Replicate state data.
- void `allocate_state_contents` (`GaussJacksonTwoState` &item)
Allocate memory for a state item.
- void `deallocate_state_contents` (`GaussJacksonTwoState` &item)
Deallocate state item memory.

Static Private Member Functions

- static Primer * `create_primer` (const `er7_utils::IntegratorConstructor` &priming_constructor, unsigned int size_in, `er7_utils::IntegrationControls` &priming_controls)
Create the integrator to be used during priming.
- static Primer * `replicate_primer` (const Primer *src)
Create a replica of the provided primer.

8.8.1 Detailed Description

```
template<typename State, typename Primer>
class jeod::GaussJacksonIntegratorBase< State, Primer >
```

Base template class for integrating state via the Gauss-Jackson technique.

Template Parameters

<i>State</i>	Structure that contains the state.
<i>Primer</i>	Class for priming the Gauss-Jackson integrator.

Definition at line 87 of file gauss_jackson_integrator_base.hh.

8.8.2 Constructor & Destructor Documentation

8.8.2.1 GaussJacksonIntegratorBase() [1/3]

```
template<typename State , typename Primer >
jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase ( ) [default]
```

Default constructor.

8.8.2.2 GaussJacksonIntegratorBase() [2/3]

```
template<typename State , typename Primer >
jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase (
    const er7_utils::IntegratorConstructor & priming_constructor,
    const GaussJacksonIntegrationControls & controls,
    unsigned int size_in,
    er7_utils::IntegrationControls & priming_controls ) [inline]
```

Non-default constructor.

Parameters

<i>priming_constructor</i>	Integrator constructor for the technique used during priming.
<i>controls</i>	The Gauss-Jackson integration controls that drives this state integrator.
<i>size_in</i>	State size.
<i>priming_controls</i>	Integration controls used during priming.

Definition at line 201 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist, er7_utils::TwoDArray< T >::allocate(),

jeod::GaussJacksonIntegratorBase< State, Primer >::allocate_state_contents(), jeod::GaussJacksonIntegratorBase< State, Primer >::corrector_sum, jeod::GaussJacksonIntegratorBase< State, Primer >::create_primer(), jeod::GaussJacksonIntegratorBase< State, Primer >::delinv, jeod::GaussJacksonIntegratorBase< State, Primer >::init_state, jeod::GaussJacksonIntegratorBase< State, Primer >::max_history_size, jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist, jeod::GaussJacksonIntegratorBase< State, Primer >::primer, and jeod::GaussJacksonIntegratorBase< State, Primer >::size.

8.8.2.3 GaussJacksonIntegratorBase() [3/3]

```
template<typename State , typename Primer >
jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase (
    const GaussJacksonIntegratorBase< State, Primer > & src ) [inline]
```

Copy constructor.

Parameters

src	Item to be copied.
-----	--------------------

Definition at line 230 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::corrector_sum, jeod::GaussJacksonIntegratorBase< State, Primer >::delinv, jeod::GaussJacksonIntegratorBase< State, Primer >::init_state, jeod::GaussJacksonIntegratorBase< State, Primer >::primer, jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_primer(), and jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_state().

8.8.2.4 ~GaussJacksonIntegratorBase()

```
template<typename State , typename Primer >
jeod::GaussJacksonIntegratorBase< State, Primer >::~GaussJacksonIntegratorBase ( ) [inline]
```

Destructor.

Definition at line 256 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::corrector_sum, jeod::GaussJacksonIntegratorBase< State, Primer >::deallocate_state_contents(), jeod::GaussJacksonIntegratorBase< State, Primer >::delinv, jeod::GaussJacksonIntegratorBase< State, Primer >::init_state, and jeod::GaussJacksonIntegratorBase< State, Primer >::primer.

8.8.3 Member Function Documentation

8.8.3.1 advance_edit_integration_constants() [1/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::advance_edit_integration_constants (
    unsigned int index ) [private]
```

Advance the integration constants by one cycle.

Parameters

<i>index</i>	Coefficient index.
--------------	--------------------

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point().

8.8.3.2 advance_edit_integration_constants() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::advance_edit_integration_constants (
    unsigned int index ) [inline], [private]
```

Advance the integration constants by one cycle.

Definition at line 172 of file gauss_jackson_integrator_base_first.hh.

8.8.3.3 advance_edit_integration_constants() [3/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::advance_edit_integration_constants (
    unsigned int index ) [inline], [private]
```

Advance the integration constants by one cycle.

Definition at line 173 of file gauss_jackson_integrator_base_second.hh.

8.8.3.4 advance_predictor_integration_constants() [1/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::advance_predictor_integration_constants
(
    unsigned int index ) [private]
```

Advance the integration constants by one cycle.

Parameters

<i>index</i>	Coefficient index.
--------------	--------------------

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj().

8.8.3.5 advance_predictor_integration_constants() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::advance_predictor_integration_constants (
    unsigned int index ) [inline], [private]
```

Advance the integration constants by one cycle.

Definition at line 203 of file gauss_jackson_integrator_base_first.hh.

8.8.3.6 advance_predictor_integration_constants() [3/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::advance_predictor_integration_constants (
    unsigned int index ) [inline], [private]
```

Advance the integration constants by one cycle.

Definition at line 207 of file gauss_jackson_integrator_base_second.hh.

8.8.3.7 allocate_state_contents() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::allocate_state_contents (
    GaussJacksonOneState & item ) [inline], [private]
```

Allocate memory for a state item.

Definition at line 317 of file gauss_jackson_integrator_base_first.hh.

References jeod::GaussJacksonOneState::first.

8.8.3.8 allocate_state_contents() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::allocate_state_contents (
    GaussJacksonTwoState & item ) [inline], [private]
```

Allocate memory for a state item.

Definition at line 332 of file gauss_jackson_integrator_base_second.hh.

References jeod::GaussJacksonTwoState::first, and jeod::GaussJacksonTwoState::second.

8.8.3.9 allocate_state_contents() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::allocate_state_contents (
    State & item ) [private]
```

Allocate memory for a state item.

Parameters

<i>item</i>	State item to be allocated.
-------------	-----------------------------

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase()`.

8.8.3.10 base_integrate()

```
template<typename State , typename Primer >
er7_utils::IntegratorResult jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate
(
    double dyn_dt,
    unsigned int target_stage,
    const double * deriv,
    State state ) [inline], [protected]
```

Propagate state to the specified target_stage.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>deriv</i>	Acceleration vector.
in, out	<i>state</i>	State vector(s).

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 288 of file `gauss_jackson_integrator_base.hh`.

References `jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist`, `jeod::GaussJacksonStateMachine::BootstrapEdit`, `jeod::GaussJacksonStateMachine::BootstrapStep`, `jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::fsm_state`, `jeod::GaussJacksonIntegratorBase< State, Primer >::history_length`, `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_primer()`, `jeod::GaussJacksonStateMachine::Operational`, `jeod::GaussJacksonIntegratorBase< State, Primer >::order`, `jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist`, `jeod::GaussJacksonStateMachine::Priming`, `jeod::GaussJacksonStateMachine::Reset`, `jeod::GaussJacksonIntegratorBase< State, Primer >::rotate_acc_hist()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_comparison_data()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::size`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`.

Referenced by `jeod::GaussJacksonFirstOrderODEIntegrator::integrate()`.

8.8.3.11 base_reset()

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::base_reset ( ) [inline], [protected]
```

Reset the integrator.

Definition at line 271 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::fsm_state, jeod::GaussJacksonIntegratorBase< State, Primer >::history_length, jeod::GaussJacksonIntegratorBase< State, Primer >::initial_order, jeod::GaussJacksonIntegratorBase< State, Primer >::order, and jeod::GaussJacksonStateMachine::Reset.

Referenced by jeod::GaussJacksonFirstOrderODEIntegrator::reset_integrator().

8.8.3.12 correct() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator >::correct (
    double dt,
    const double * acc,
    GaussJacksonOneState & state ) [inline], [private]
```

Apply the corrector.

Definition at line 255 of file gauss_jackson_integrator_base_first.hh.

References jeod::GaussJacksonOneState::first.

8.8.3.13 correct() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator >::correct (
    double dt,
    const double * acc,
    GaussJacksonTwoState & state ) [inline], [private]
```

Apply the corrector.

Definition at line 261 of file gauss_jackson_integrator_base_second.hh.

References jeod::GaussJacksonTwoState::first, and jeod::GaussJacksonTwoState::second.

8.8.3.14 correct() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::correct (
    double dt,
    const double * acc,
    State & state ) [private]
```

Apply the corrector.

Parameters

<i>dt</i>	Dynamic time step.
<i>acc</i>	Acceleration data.
<i>state</i>	Corrected state.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj()`.

8.8.3.15 create_primer() [1/3]

```
er7_utils::FirstOrderODEIntegrator * jeod::GaussJacksonIntegratorBase< GaussJacksonOneState,
er7_utils::FirstOrderODEIntegrator >::create_primer (
    const er7_utils::IntegratorConstructor & priming_constructor,
    unsigned int size,
    er7_utils::IntegrationControls & priming_controls ) [inline], [private]
```

Create the priming integrator.

Definition at line 92 of file `gauss_jackson_integrator_base_first.hh`.

8.8.3.16 create_primer() [2/3]

```
er7_utils::SecondOrderODEIntegrator * jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState,
er7_utils::SecondOrderODEIntegrator >::create_primer (
    const er7_utils::IntegratorConstructor & priming_constructor,
    unsigned int size,
    er7_utils::IntegrationControls & priming_controls ) [inline], [private]
```

Definition at line 91 of file `gauss_jackson_integrator_base_second.hh`.

8.8.3.17 create_primer() [3/3]

```
template<typename State , typename Primer >
static Primer* jeod::GaussJacksonIntegratorBase< State, Primer >::create_primer (
    const er7_utils::IntegratorConstructor & priming_constructor,
    unsigned int size_in,
    er7_utils::IntegrationControls & priming_controls ) [static], [private]
```

Create the integrator to be used during priming.

Parameters

<i>priming_constructor</i>	Integrator constructor for the technique used during priming.
<i>size_in</i>	State size.
<i>priming_controls</i>	Integration controls used during priming.

Returns

Constructed primer.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase()`.

8.8.3.18 deallocate_state_contents() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::deallocate_state_contents (
    GaussJacksonOneState & item ) [inline], [private]
```

Deallocate state item memory.

Definition at line 328 of file `gauss_jackson_integrator_base_first.hh`.

References `jeod::GaussJacksonOneState::first`.

8.8.3.19 deallocate_state_contents() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::deallocate_state_contents (
    GaussJacksonTwoState & item ) [inline], [private]
```

Deallocate state item memory.

Definition at line 344 of file `gauss_jackson_integrator_base_second.hh`.

References `jeod::GaussJacksonTwoState::first`, and `jeod::GaussJacksonTwoState::second`.

8.8.3.20 deallocate_state_contents() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::deallocate_state_contents (
    State & item ) [private]
```

Deallocate state item memory.

Parameters

<i>item</i>	State item to be deallocated.
-------------	-------------------------------

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::~~GaussJacksonIntegratorBase()`.

8.8.3.21 downsample_hist()

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::downsample_hist ( ) [inline], [private]
```

Downsample the acceleration and position histories.

Definition at line 559 of file gauss_jackson_integrator_base.hh.

References `jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist`, `er7_utils::TwoDArray< T >::downsample()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::history_length`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`.

8.8.3.22 edit_point()

```
template<typename State , typename Primer >
bool jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point (
    double dt,
    const double * acc,
    State & state ) [inline], [private]
```

Edit the specified point using the mid-corrector that pertains to the point being edited, which is that at `history_length`.

Parameters

in	<i>dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>acc</i>	Acceleration vector.
out	<i>state</i>	State vector(s).

Definition at line 511 of file gauss_jackson_integrator_base.hh.

References `jeod::GaussJacksonIntegratorBase< State, Primer >::advance_edit_integration_constants()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::history_length`, `jeod::GaussJacksonIntegratorBase< State, Primer >::mid_correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::order`, `jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::test_for_convergence()`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate()`.

8.8.3.23 initialize_edit_integration_constants() [1/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::initialize_edit_integration_constants
(
    double dt ) [private]
```

Initialize the integration constants (i.e., `delinv`).

Parameters

dt	Dynamic time step.
------	--------------------

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`.

8.8.3.24 `initialize_edit_integration_constants()` [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::initialize_edit_integration_constants (
    double dt ) [inline], [private]
```

Initialize the integration constants (i.e., `delinv`).

Definition at line 156 of file `gauss_jackson_integrator_base_first.hh`.

8.8.3.25 `initialize_edit_integration_constants()` [3/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::initialize_edit_integration_constants (
    double dt ) [inline], [private]
```

Initialize the integration constants (i.e., `delinv`).

Definition at line 154 of file `gauss_jackson_integrator_base_second.hh`.

8.8.3.26 `initialize_predictor_integration_constants()` [1/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::initialize_predictor_integration_↵
constants (
    double dt ) [private]
```

Initialize the integration constants (i.e., `delinv`).

Parameters

dt	Dynamic time step.
------	--------------------

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`.

8.8.3.27 initialize_predictor_integration_constants() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::initialize_predictor_integration_constants (
    double dt ) [inline], [private]
```

Initialize the integration constants (i.e., delinv).

Definition at line 189 of file gauss_jackson_integrator_base_first.hh.

8.8.3.28 initialize_predictor_integration_constants() [3/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::initialize_predictor_integration_constants (
    double dt ) [inline], [private]
```

Initialize the integration constants (i.e., delinv).

Definition at line 188 of file gauss_jackson_integrator_base_second.hh.

8.8.3.29 integrate_gj()

```
template<typename State , typename Primer >
bool jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj (
    double dt,
    unsigned int target_stage,
    int advance_index,
    int target_index,
    const double * acc,
    const double *const * ahist,
    State & state ) [inline], [private]
```

Integrate using the Gauss-Jackson predictor and corrector.

Parameters

in	<i>dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>advance_index</i>	Acceleration history index.
in	<i>target_index</i>	Position history index.
in	<i>acc</i>	Acceleration vector.
in	<i>ahist</i>	Acceleration vector history.
out	<i>state</i>	State vector(s).

Returns

True if step was successful, false otherwise.

Definition at line 532 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::advance_predictor_integration_constants(), jeod::GaussJacksonCoefficientsPair::apply(), jeod::GaussJacksonIntegratorBase< State, Primer >::coeff, jeod::GaussJacksonIntegratorBase< State, Primer >::correct(), jeod::GaussJacksonCoeffs::corrector, jeod::GaussJacksonIntegratorBase< State, Primer >::corrector_sum, jeod::GaussJacksonIntegratorBase< State, Primer >::order, jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist, jeod::GaussJacksonIntegratorBase< State, Primer >::predict(), jeod::GaussJacksonIntegratorBase< State, Primer >::save_comparison_data(), jeod::GaussJacksonIntegratorBase< State, Primer >::size, and jeod::GaussJacksonIntegratorBase< State, Primer >::test_for_convergence().

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate().

8.8.3.30 integrate_primer() [1/3]

```
er7_utils::IntegratorResult jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator >::integrate_primer (
    double dyn_dt,
    unsigned int target_stage,
    const double * deriv,
    GaussJacksonOneState & state ) [inline], [private]
```

Integrate with the primer.

Definition at line 123 of file gauss_jackson_integrator_base_first.hh.

References jeod::GaussJacksonOneState::first.

8.8.3.31 integrate_primer() [2/3]

```
er7_utils::IntegratorResult jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator >::integrate_primer (
    double dyn_dt,
    unsigned int target_stage,
    const double * deriv,
    GaussJacksonTwoState & state ) [inline], [private]
```

Integrate with the primer.

Definition at line 121 of file gauss_jackson_integrator_base_second.hh.

References jeod::GaussJacksonTwoState::first, and jeod::GaussJacksonTwoState::second.

8.8.3.32 integrate_primer() [3/3]

```
template<typename State , typename Primer >
er7_utils::IntegratorResult jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_primer (
    double dyn_dt,
    unsigned int target_stage,
    const double * deriv,
    State & state ) [private]
```

Integrate state using the primer.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>deriv</i>	Acceleration vector.
in, out	<i>state</i>	State vector(s).

Returns

The status (time advance, pass/fail status) of the integration.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate()`.

8.8.3.33 mid_correct() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::mid_correct (
    unsigned int coeff_idx,
    double dt,
    GaussJacksonOneState & state ) [inline], [private]
```

Apply a mid-corrector.

Definition at line 219 of file `gauss_jackson_integrator_base_first.hh`.

References `jeod::GaussJacksonOneState::first`.

8.8.3.34 mid_correct() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::mid_correct (
    unsigned int coeff_idx,
    double dt,
    GaussJacksonTwoState & state ) [inline], [private]
```

Apply a mid-corrector.

Definition at line 221 of file `gauss_jackson_integrator_base_second.hh`.

References `jeod::GaussJacksonTwoState::first`, and `jeod::GaussJacksonTwoState::second`.

8.8.3.35 mid_correct() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::mid_correct (
    unsigned int coeff_idx,
    double dt,
    State & state ) [private]
```

Apply a mid-corrector.

Parameters

<i>coeff_idx</i>	Coefficient index; item to be corrected.
<i>dt</i>	Dynamic time step.
<i>state</i>	Corrected state.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point()`.

8.8.3.36 operator=()

```
template<typename State , typename Primer >
GaussJacksonIntegratorBase& jeod::GaussJacksonIntegratorBase< State, Primer >::operator= (
    const GaussJacksonIntegratorBase< State, Primer > & ) [delete]
```

8.8.3.37 predict() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::predict (
    double dt,
    const double *const * ahist,
    GaussJacksonOneState & state ) [inline], [private]
```

Apply the predictor.

Definition at line 237 of file `gauss_jackson_integrator_base_first.hh`.

References `jeod::GaussJacksonOneState::first`.

8.8.3.38 predict() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::predict (
    double dt,
    const double *const * ahist,
    GaussJacksonTwoState & state ) [inline], [private]
```

Apply the predictor.

Definition at line 239 of file `gauss_jackson_integrator_base_second.hh`.

References `jeod::GaussJacksonTwoState::first`, and `jeod::GaussJacksonTwoState::second`.

8.8.3.39 predict() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::predict (
    double dt,
    const double *const * ahist,
    State & state ) [private]
```

Apply the predictor.

Parameters

<i>dt</i>	Dynamic time step.
<i>ahist</i>	Acceleration history.
<i>state</i>	Corrected state.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj()`.

8.8.3.40 replicate_primer() [1/3]

```
er7_utils::FirstOrderODEIntegrator * jeod::GaussJacksonIntegratorBase< GaussJacksonOneState,
er7_utils::FirstOrderODEIntegrator >::replicate_primer (
    const er7_utils::FirstOrderODEIntegrator * src ) [inline], [private]
```

Replicate the priming integrator.

Definition at line 105 of file `gauss_jackson_integrator_base_first.hh`.

8.8.3.41 replicate_primer() [2/3]

```
er7_utils::SecondOrderODEIntegrator * jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState,
er7_utils::SecondOrderODEIntegrator >::replicate_primer (
    const er7_utils::SecondOrderODEIntegrator * src ) [inline], [private]
```

Definition at line 103 of file `gauss_jackson_integrator_base_second.hh`.

8.8.3.42 replicate_primer() [3/3]

```
template<typename State , typename Primer >
static Primer* jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_primer (
    const Primer * src ) [static], [private]
```

Create a replica of the provided primer.

Parameters

<i>src</i>	Primer to be replicated.
------------	--------------------------

Returns

Constructed primer.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase()`.

8.8.3.43 replicate_state() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::replicate_state (
    const GaussJacksonOneState & source,
    GaussJacksonOneState & target ) [inline], [private]
```

Replicate state data.

Definition at line 306 of file gauss_jackson_integrator_base_first.hh.

References jeod::GaussJacksonOneState::first.

8.8.3.44 replicate_state() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::replicate_state (
    const GaussJacksonTwoState & source,
    GaussJacksonTwoState & target ) [inline], [private]
```

Replicate state data.

Definition at line 320 of file gauss_jackson_integrator_base_second.hh.

References jeod::GaussJacksonTwoState::first, and jeod::GaussJacksonTwoState::second.

8.8.3.45 replicate_state() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_state (
    const State & source,
    State & target ) [private]
```

Replicate state data.

Parameters

<i>source</i>	State item to be copied.
<i>target</i>	Replicated state item.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase().

8.8.3.46 rotate_acc_hist()

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::rotate_acc_hist ( ) [inline], [private]
```

Rotate the acceleration history.

Definition at line 571 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist, jeod::GaussJacksonIntegratorBase< State, Primer >::order, and er7_utils::TwoDArray< T >::rotate_down().

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate().

8.8.3.47 save_comparison_data() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::save_comparison_data (
    const GaussJacksonOneState & state,
    double * pos_hist_elem ) [inline], [private]
```

Save comparison data.

Definition at line 144 of file gauss_jackson_integrator_base_first.hh.

References jeod::GaussJacksonOneState::first.

8.8.3.48 save_comparison_data() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::save_comparison_data (
    const GaussJacksonTwoState & state,
    double * pos_hist_elem ) [inline], [private]
```

Save comparison data.

Definition at line 142 of file gauss_jackson_integrator_base_second.hh.

References jeod::GaussJacksonTwoState::second.

8.8.3.49 save_comparison_data() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::save_comparison_data (
    const State & state,
    double * pos_hist_elem ) [private]
```

Save comparison data.

Parameters

<i>state</i>	State to be saved.
<i>pos_hist_elem</i>	Element of the position history to be updated.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj()`.

8.8.3.50 save_epoch_data() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::save_epoch_data (
    const double * acc,
    const GaussJacksonOneState & state ) [inline], [private]
```

Save epoch data.

Definition at line 133 of file `gauss_jackson_integrator_base_first.hh`.

References `jeod::GaussJacksonOneState::first`.

8.8.3.51 save_epoch_data() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::save_epoch_data (
    const double * acc,
    const GaussJacksonTwoState & state ) [inline], [private]
```

Save epoch data.

Definition at line 131 of file `gauss_jackson_integrator_base_second.hh`.

References `jeod::GaussJacksonTwoState::first`, and `jeod::GaussJacksonTwoState::second`.

8.8.3.52 save_epoch_data() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::save_epoch_data (
    const double * acc,
    const State & state ) [private]
```

Save epoch data.

Parameters

<i>acc</i>	Acceleration to be saved.
<i>state</i>	State to be saved.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`.

8.8.3.53 start_cycle()

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle (
    double dt,
    const double * acc,
    const State & state ) [inline], [private]
```

Start an integration cycle.

Parameters

in	<i>dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>acc</i>	Acceleration vector.
in	<i>state</i>	State vector(s).

Definition at line 456 of file `gauss_jackson_integrator_base.hh`.

References `jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist`, `jeod::GaussJacksonStateMachine::BootstrapEdit`, `jeod::GaussJacksonIntegratorBase< State, Primer >::coeff`, `jeod::GaussJacksonCoeffs::corrector`, `jeod::GaussJacksonIntegratorBase< State, Primer >::downsample_hist()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::fsm_state`, `jeod::GaussJacksonStateMachine::get_at_downsample()`, `jeod::GaussJacksonStateMachine::get_at_order_change()`, `jeod::GaussJacksonStateMachine::get_at_reinitialize()`, `jeod::GaussJacksonStateMachine::get_current_order()`, `jeod::GaussJacksonStateMachine::get_fsm_state()`, `jeod::GaussJacksonCoefficientsPair::gj_coefs`, `jeod::GaussJacksonIntegratorBase< State, Primer >::history_length`, `jeod::GaussJacksonIntegratorBase< State, Primer >::initialize_edit_integration_constants()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::initialize_predictor_integration_constants()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::order`, `jeod::GaussJacksonIntegratorBase< State, Primer >::position_corrector`, `jeod::GaussJacksonStateMachine::Reset`, `jeod::GaussJacksonCoefficientsPair::sa_coefs`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_epoch_data()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::size`, `jeod::GaussJacksonIntegratorBase< State, Primer >::state_machine`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::velocity_corrector`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate()`.

8.8.3.54 swap()

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::swap (
    GaussJacksonIntegratorBase< State, Primer > & other ) [inline], [protected]
```

Non-throwing swap.

Parameters

<i>other</i>	Item whose contents are to be swapped with this.
--------------	--

Definition at line 397 of file gauss_jackson_integrator_base.hh.

References jeod::GaussJacksonIntegratorBase< State, Primer >::absolute_tolerance, jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist, jeod::GaussJacksonIntegratorBase< State, Primer >::coeff, jeod::GaussJacksonIntegratorBase< State, Primer >::corrector_sum, jeod::GaussJacksonIntegratorBase< State, Primer >::delinv, jeod::GaussJacksonIntegratorBase< State, Primer >::fsm_state, jeod::GaussJacksonIntegratorBase< State, Primer >::history_length, jeod::GaussJacksonIntegratorBase< State, Primer >::init_state, jeod::GaussJacksonIntegratorBase< State, Primer >::initial_order, jeod::GaussJacksonIntegratorBase< State, Primer >::max_history_size, jeod::GaussJacksonIntegratorBase< State, Primer >::order, jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist, jeod::GaussJacksonIntegratorBase< State, Primer >::position_corrector, jeod::GaussJacksonIntegratorBase< State, Primer >::primer, jeod::GaussJacksonIntegratorBase< State, Primer >::relative_tolerance, jeod::GaussJacksonIntegratorBase< State, Primer >::size, jeod::GaussJacksonIntegratorBase< State, Primer >::state_machine, er7_utils::TwoDArray< T >::swap(), jeod::GaussJacksonIntegratorBase< State, Primer >::swap_state(), and jeod::GaussJacksonIntegratorBase< State, Primer >::velocity_corrector.

Referenced by jeod::GaussJacksonFirstOrderODEIntegrator::swap().

8.8.3.55 swap_state() [1/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator >::swap_state (
    GaussJacksonOneState & item,
    GaussJacksonOneState & other_item ) [inline], [private]
```

Swap state data with another of the same.

Definition at line 296 of file gauss_jackson_integrator_base_first.hh.

References jeod::GaussJacksonOneState::first.

8.8.3.56 swap_state() [2/3]

```
void jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator >::swap_state (
    GaussJacksonTwoState & item,
    GaussJacksonTwoState & other_item ) [inline], [private]
```

Swap state data with another of the same.

Definition at line 309 of file gauss_jackson_integrator_base_second.hh.

References jeod::GaussJacksonTwoState::first, and jeod::GaussJacksonTwoState::second.

8.8.3.57 swap_state() [3/3]

```
template<typename State , typename Primer >
void jeod::GaussJacksonIntegratorBase< State, Primer >::swap_state (
    State & item,
    State & other_item ) [private]
```

Swap state data with another of the same.

Parameters

<i>item</i>	State item.
<i>other_item</i>	The other state item.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::swap()`.

8.8.3.58 test_for_convergence() [1/3]

```
bool jeod::GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator
>::test_for_convergence (
    const GaussJacksonOneState & state,
    double * hist_data ) [inline], [private]
```

Test for convergence.

Definition at line 274 of file `gauss_jackson_integrator_base_first.hh`.

References `jeod::GaussJacksonOneState::first`.

8.8.3.59 test_for_convergence() [2/3]

```
bool jeod::GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator
>::test_for_convergence (
    const GaussJacksonTwoState & state,
    double * hist_data ) [inline], [private]
```

Test for convergence.

Definition at line 288 of file `gauss_jackson_integrator_base_second.hh`.

References `jeod::GaussJacksonTwoState::second`.

8.8.3.60 test_for_convergence() [3/3]

```
template<typename State , typename Primer >
bool jeod::GaussJacksonIntegratorBase< State, Primer >::test_for_convergence (
    const State & state,
    double * hist_data ) [private]
```

Test for convergence.

Parameters

<i>state</i>	Item to be compared.
<i>hist_data</i>	Previous state value.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj()`.

8.8.4 Field Documentation

8.8.4.1 absolute_tolerance

```
template<typename State , typename Primer >
double jeod::GaussJacksonIntegratorBase< State, Primer >::absolute_tolerance {}
```

Number that indicates the allowable absolute difference for two states to be considered converged.

`trick_units(-)`

Definition at line 143 of file `gauss_jackson_integrator_base.hh`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::swap()`.

8.8.4.2 acc_hist

```
template<typename State , typename Primer >
er7_utils::DoubleTwoDArray jeod::GaussJacksonIntegratorBase< State, Primer >::acc_hist
```

Acceleration history.

`trick_units(-)`

Definition at line 126 of file `gauss_jackson_integrator_base.hh`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::downsample_hist()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::rotate_acc_hist()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::swap()`.

8.8.4.3 coeff

```
template<typename State , typename Primer >
const GaussJacksonCoeffs* jeod::GaussJacksonIntegratorBase< State, Primer >::coeff {}
```

The summed Adams and Gauss-Jackson coefficients, in ordinate form.

`trick_units(-)`

Definition at line 96 of file `gauss_jackson_integrator_base.hh`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::swap()`.

8.8.4.4 corrector_sum

```
template<typename State , typename Primer >
State jeod::GaussJacksonIntegratorBase< State, Primer >::corrector_sum
```

Speed hack for the corrector.

trick_units(–)

Definition at line 121 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj(), jeod::GaussJacksonIntegratorBase< State, Primer >::swap(), and jeod::GaussJacksonIntegratorBase< State, Primer >::~GaussJacksonIntegratorBase().

8.8.4.5 delinv

```
template<typename State , typename Primer >
State jeod::GaussJacksonIntegratorBase< State, Primer >::delinv
```

Inverse backward differences.

trick_units(–)

Definition at line 116 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), jeod::GaussJacksonIntegratorBase< State, Primer >::swap(), and jeod::GaussJacksonIntegratorBase< State, Primer >::~GaussJacksonIntegratorBase().

8.8.4.6 fsm_state

```
template<typename State , typename Primer >
GaussJacksonStateMachine::FsmState jeod::GaussJacksonIntegratorBase< State, Primer >::fsm_
state {GaussJacksonStateMachine::Reset}
```

Finite state machine state.

trick_units(–)

Definition at line 158 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate(), jeod::GaussJacksonIntegratorBase< State, Primer >::base_reset(), jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.7 history_length

```
template<typename State , typename Primer >
unsigned int jeod::GaussJacksonIntegratorBase< State, Primer >::history_length {}
```

Current history length.

trick_units(-)

Definition at line 183 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate(), jeod::GaussJacksonIntegratorBase< State, Primer >::base_reset(), jeod::GaussJacksonIntegratorBase< State, Primer >::downsample_hist(), jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point(), jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.8 init_state

```
template<typename State , typename Primer >
State jeod::GaussJacksonIntegratorBase< State, Primer >::init_state
```

The state at the time of the last reset.

trick_units(-)

Definition at line 111 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), jeod::GaussJacksonIntegratorBase< State, Primer >::swap(), and jeod::GaussJacksonIntegratorBase< State, Primer >::~~GaussJacksonIntegratorBase().

8.8.4.9 initial_order

```
template<typename State , typename Primer >
unsigned int jeod::GaussJacksonIntegratorBase< State, Primer >::initial_order {}
```

Initial order.

trick_units(-)

Definition at line 168 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_reset(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.10 max_history_size

```
template<typename State , typename Primer >
unsigned int jeod::GaussJacksonIntegratorBase< State, Primer >::max_history_size {}
```

Maximum history size.

trick_units(–)

Definition at line 163 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.11 order

```
template<typename State , typename Primer >
unsigned int jeod::GaussJacksonIntegratorBase< State, Primer >::order {}
```

Current order.

trick_units(–)

Definition at line 173 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate(), jeod::GaussJacksonIntegratorBase< State, Primer >::base_reset(), jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point(), jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj(), jeod::GaussJacksonIntegratorBase< State, Primer >::rotate_acc_hist(), jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.12 pos_hist

```
template<typename State , typename Primer >
er7_utils::DoubleTwoDArray jeod::GaussJacksonIntegratorBase< State, Primer >::pos_hist
```

Position history (or velocity history in case of a first order ODE).

trick_units(–)

Definition at line 131 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate(), jeod::GaussJacksonIntegratorBase< State, Primer >::downsample_hist(), jeod::GaussJacksonIntegratorBase< State, Primer >::edit_point(), jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.13 position_corrector

```
template<typename State , typename Primer >
double jeod::GaussJacksonIntegratorBase< State, Primer >::position_corrector {}
```

Correction coefficient for the second integral (position).

trick_units(-)

Definition at line 153 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.14 primer

```
template<typename State , typename Primer >
Primer* jeod::GaussJacksonIntegratorBase< State, Primer >::primer {}
```

The integrator used to prime the Gauss-Jackson integration process.

trick_units(-)

Definition at line 106 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), jeod::GaussJacksonIntegratorBase< State, Primer >::swap(), and jeod::GaussJacksonIntegratorBase< State, Primer >::~~GaussJacksonIntegratorBase().

8.8.4.15 relative_tolerance

```
template<typename State , typename Primer >
double jeod::GaussJacksonIntegratorBase< State, Primer >::relative_tolerance {}
```

Number that indicates the allowable relative difference for two states to be considered converged.

trick_units(-)

Definition at line 137 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.16 size

```
template<typename State , typename Primer >
unsigned int jeod::GaussJacksonIntegratorBase< State, Primer >::size {}
```

State size.

trick_units(–)

Definition at line 178 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::base_integrate(), jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase(), jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_gj(), jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.17 state_machine

```
template<typename State , typename Primer >
const GaussJacksonStateMachine* jeod::GaussJacksonIntegratorBase< State, Primer >::state_machine {}
```

The Gauss-Jackson state machine.

trick_units(–)

Definition at line 101 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

8.8.4.18 velocity_corrector

```
template<typename State , typename Primer >
double jeod::GaussJacksonIntegratorBase< State, Primer >::velocity_corrector {}
```

Correction coefficient for the first integral (velocity).

trick_units(–)

Definition at line 148 of file gauss_jackson_integrator_base.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJacksonIntegratorBase< State, Primer >::swap().

The documentation for this class was generated from the following file:

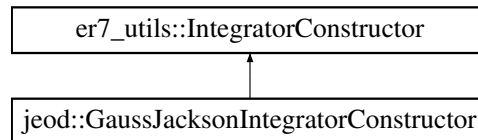
- [gauss_jackson_integrator_base.hh](#)

8.9 jeod::GaussJacksonIntegratorConstructor Class Reference

Create state and time integrators that propagate using Gauss-Jackson.

```
#include <gauss_jackson_integrator_constructor.hh>
```

Inheritance diagram for jeod::GaussJacksonIntegratorConstructor:



Public Member Functions

- [GaussJacksonIntegratorConstructor](#) ()=default
GaussJackson default constructor.
- [GaussJacksonIntegratorConstructor](#) (const [GaussJacksonIntegratorConstructor](#) &src)
GaussJacksonIntegratorConstructor copy constructor.
- [~GaussJacksonIntegratorConstructor](#) () override
GaussJacksonIntegratorConstructor destructor.
- [GaussJacksonIntegratorConstructor](#) & operator= ([GaussJacksonIntegratorConstructor](#) src)
GaussJacksonIntegratorConstructor assignment operator.
- void [configure](#) (const [GaussJacksonConfig](#) &config_in, er7_utils::Integration::Technique priming_↵ technique=er7_utils::Integration::Unspecified)
Configure the Gauss-Jackson integrator constructor.
- void [configure](#) (const [GaussJacksonConfig](#) &config_in, const er7_utils::IntegratorConstructor &priming_↵ cotr_in)
Configure the Gauss-Jackson integrator constructor.
- const char * [get_class_name](#) () const override
Return the class name.
- bool [implements](#) (er7_utils::Integration::ODEProblemType problem_type) const override
GaussJackson does not implement a 2nd order generalized step integrator.
- bool [provides](#) (er7_utils::Integration::ODEProblemType problem_type) const override
GaussJackson does not provide a 2nd order generalized step integrator.
- virtual void [swap](#) ([GaussJacksonIntegratorConstructor](#) &src)
Non-throwing swap.
- er7_utils::IntegratorConstructor * [create_copy](#) () const override
Create a duplicate of the constructor.
- er7_utils::IntegrationControls * [create_integration_controls](#) () const override
Create an integration controls that guides the GaussJackson integration process.
- er7_utils::FirstOrderODEIntegrator * [create_first_order_ode_integrator](#) (unsigned int size, er7_utils::↵ IntegrationControls &controls) const override
Create a GaussJackson state integrator for a first order ODE.
- er7_utils::SecondOrderODEIntegrator * [create_second_order_ode_integrator](#) (unsigned int size, er7_utils::↵ ::IntegrationControls &controls) const override
Create a GaussJackson state integrator for a simple second order ODE.
- er7_utils::SecondOrderODEIntegrator * [create_generalized_deriv_second_order_ode_integrator](#) (unsigned int position_size, unsigned int velocity_size, const er7_utils::GeneralizedPositionDerivativeFunctions &deriv_↵ _funcs, er7_utils::IntegrationControls &controls) const override

Create a GaussJackson state integrator for a generalized second order ODE where generalized position is advanced with the use of the position derivative function.

- unsigned int [get_buffer_size](#) () const override

GaussJackson can use a large number of steps per Trick cycle.

- unsigned int [get_transition_table_size](#) () const override

GaussJackson uses two steps per cycle once primed.

Static Public Member Functions

- static er7_utils::IntegratorConstructor * [create_constructor](#) ()

Named constructor; create an [GaussJacksonIntegratorConstructor](#) instance.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonIntegratorConstructor](#))

Private Attributes

- er7_utils::IntegratorConstructor * [priming_constructor](#) {}

The integrator constructor that creates the priming integrators.

- [GaussJacksonConfig](#) config

Data used to configure the Gauss-Jackson integration process.

8.9.1 Detailed Description

Create state and time integrators that propagate using Gauss-Jackson.

Definition at line 80 of file gauss_jackson_integrator_constructor.hh.

8.9.2 Constructor & Destructor Documentation

8.9.2.1 GaussJacksonIntegratorConstructor() [1/2]

```
jeod::GaussJacksonIntegratorConstructor::GaussJacksonIntegratorConstructor ( ) [default]
```

GaussJackson default constructor.

8.9.2.2 GaussJacksonIntegratorConstructor() [2/2]

```
jeod::GaussJacksonIntegratorConstructor::GaussJacksonIntegratorConstructor (
    const GaussJacksonIntegratorConstructor & src )
```

[GaussJacksonIntegratorConstructor](#) copy constructor.

Definition at line 69 of file gauss_jackson_integrator_constructor.cc.

References [priming_constructor](#).

8.9.2.3 ~GaussJacksonIntegratorConstructor()

```
jeod::GaussJacksonIntegratorConstructor::~~GaussJacksonIntegratorConstructor ( ) [override]
```

[GaussJacksonIntegratorConstructor](#) destructor.

Definition at line 80 of file gauss_jackson_integrator_constructor.cc.

References [priming_constructor](#).

8.9.3 Member Function Documentation

8.9.3.1 configure() [1/2]

```
void jeod::GaussJacksonIntegratorConstructor::configure (
    const GaussJacksonConfig & config_in,
    const er7_utils::IntegratorConstructor & priming_cotr_in )
```

Configure the Gauss-Jackson integrator constructor.

Definition at line 107 of file gauss_jackson_integrator_constructor.cc.

References [config](#), [priming_constructor](#), and [jeod::GaussJacksonConfig::validate_configuration\(\)](#).

8.9.3.2 configure() [2/2]

```
void jeod::GaussJacksonIntegratorConstructor::configure (
    const GaussJacksonConfig & config_in,
    er7_utils::Integration::Technique priming_technique = er7_utils::Integration::↔
:Unspecified )
```

Configure the Gauss-Jackson integrator constructor.

Definition at line 93 of file gauss_jackson_integrator_constructor.cc.

References [config](#), [priming_constructor](#), [jeod::GaussJacksonConfig::priming_technique](#), and [jeod::Gauss↔JacksonConfig::validate_configuration\(\)](#).

8.9.3.3 create_constructor()

```
er7_utils::IntegratorConstructor * jeod::GaussJacksonIntegratorConstructor::create_constructor  
( ) [static]
```

Named constructor; create an [GaussJacksonIntegratorConstructor](#) instance.

The caller is responsible for deleting the returned object.

Returns

Newly created [GaussJacksonIntegratorConstructor](#) instance.

Definition at line 63 of file gauss_jackson_integrator_constructor.cc.

8.9.3.4 create_copy()

```
er7_utils::IntegratorConstructor * jeod::GaussJacksonIntegratorConstructor::create_copy ( )  
const [override]
```

Create a duplicate of the constructor.

The caller is responsible for deleting the returned object.

Returns

Duplicated constructor.

Definition at line 115 of file gauss_jackson_integrator_constructor.cc.

8.9.3.5 create_first_order_ode_integrator()

```
er7_utils::FirstOrderODEIntegrator * jeod::GaussJacksonIntegratorConstructor::create_first_order_ode_integrator (   
    unsigned int size,  
    er7_utils::IntegrationControls & controls ) const [override]
```

Create a GaussJackson state integrator for a first order ODE.

The caller is responsible for deleting the created object.

Returns

State integrator

Parameters

<i>in</i>	<i>size</i>	State size
<i>in, out</i>	<i>controls</i>	Integration controls

Definition at line 136 of file gauss_jackson_integrator_constructor.cc.

References `jeod::cast_to_gj_controls()`, `jeod::GaussJacksonIntegrationControls::get_priming_controls()`, and `priming_constructor`.

8.9.3.6 create_generalized_deriv_second_order_ode_integrator()

```
er7_utils::SecondOrderODEIntegrator * jeod::GaussJacksonIntegratorConstructor::create_generalized←
_deriv_second_order_ode_integrator (
    unsigned int position_size,
    unsigned int velocity_size,
    const er7_utils::GeneralizedPositionDerivativeFunctions & deriv_funs,
    er7_utils::IntegrationControls & controls ) const [override]
```

Create a GaussJackson state integrator for a generalized second order ODE where generalized position is advanced with the use of the position derivative function.

The caller is responsible for deleting the created object.

Returns

State integrator

Parameters

<i>in</i>	<i>position_size</i>	Size of the generalized position
<i>in</i>	<i>velocity_size</i>	Size of the generalized velocity
<i>in</i>	<i>deriv_funs</i>	Position derivative functions container
<i>in, out</i>	<i>controls</i>	Integration controls

Definition at line 169 of file gauss_jackson_integrator_constructor.cc.

References `jeod::cast_to_gj_controls()`, `jeod::GaussJacksonIntegrationControls::get_priming_controls()`, and `priming_constructor`.

8.9.3.7 create_integration_controls()

```
er7_utils::IntegrationControls * jeod::GaussJacksonIntegratorConstructor::create_integration←
_controls ( ) const [override]
```

Create an integration controls that guides the GaussJackson integration process.

The caller is responsible for deleting the created object.

Returns

Integration controls object

Definition at line 121 of file gauss_jackson_integrator_constructor.cc.

References `config`, `priming_constructor`, and `jeod::GaussJacksonConfig::standard_configuration()`.

8.9.3.8 create_second_order_ode_integrator()

```
er7_utils::SecondOrderODEIntegrator * jeod::GaussJacksonIntegratorConstructor::create_second_order_ode_integrator (
    unsigned int size,
    er7_utils::IntegrationControls & controls ) const [override]
```

Create a GaussJackson state integrator for a simple second order ODE.

The caller is responsible for deleting the created object.

Returns

State integrator

Parameters

<code>in</code>	<i>size</i>	State size
<code>in, out</code>	<i>controls</i>	Integration controls

Definition at line 152 of file gauss_jackson_integrator_constructor.cc.

References `jeod::cast_to_gj_controls()`, `jeod::GaussJacksonIntegrationControls::get_priming_controls()`, and `priming_constructor`.

8.9.3.9 get_buffer_size()

```
unsigned int jeod::GaussJacksonIntegratorConstructor::get_buffer_size ( ) const [inline],
[override]
```

GaussJackson can use a large number of steps per Trick cycle.

The magic number 192 is for `order=16`, `ndboubling=6`.

Returns

Always returns 192.

Definition at line 217 of file gauss_jackson_integrator_constructor.hh.

8.9.3.10 `get_class_name()`

```
const char* jeod::GaussJacksonIntegratorConstructor::get_class_name ( ) const [inline], [override]
```

Return the class name.

Definition at line 134 of file `gauss_jackson_integrator_constructor.hh`.

8.9.3.11 `get_transition_table_size()`

```
unsigned int jeod::GaussJacksonIntegratorConstructor::get_transition_table_size ( ) const [inline], [override]
```

GaussJackson uses two steps per cycle once primed.

Returns

Always returns 2.

Definition at line 226 of file `gauss_jackson_integrator_constructor.hh`.

8.9.3.12 `implements()`

```
bool jeod::GaussJacksonIntegratorConstructor::implements (
    er7_utils::Integration::ODEProblemType problem_type ) const [inline], [override]
```

GaussJackson does not implement a 2nd order generalized step integrator.

Definition at line 142 of file `gauss_jackson_integrator_constructor.hh`.

8.9.3.13 `operator=()`

```
GaussJacksonIntegratorConstructor& jeod::GaussJacksonIntegratorConstructor::operator= (
    GaussJacksonIntegratorConstructor src ) [inline]
```

[GaussJacksonIntegratorConstructor](#) assignment operator.

Definition at line 112 of file `gauss_jackson_integrator_constructor.hh`.

8.9.3.14 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonIntegratorConstructor::p (
    jeod ,
    GaussJacksonIntegratorConstructor ) [private]
```

8.9.3.15 provides()

```
bool jeod::GaussJacksonIntegratorConstructor::provides (
    er7_utils::Integration::ODEProblemType problem_type ) const [inline], [override]
```

GaussJackson does not provide a 2nd order generalized step integrator.

Definition at line 150 of file gauss_jackson_integrator_constructor.hh.

8.9.3.16 swap()

```
void jeod::GaussJacksonIntegratorConstructor::swap (
    GaussJacksonIntegratorConstructor & src ) [virtual]
```

Non-throwing swap.

Parameters

<i>in, out</i>	<i>src</i>	Object with which contents are to be swapped.
----------------	------------	---

Definition at line 86 of file gauss_jackson_integrator_constructor.cc.

References `config`, and `priming_constructor`.

8.9.4 Field Documentation

8.9.4.1 config

```
GaussJacksonConfig jeod::GaussJacksonIntegratorConstructor::config [private]
```

Data used to configure the Gauss-Jackson integration process.

`trick_units(-)`

Definition at line 240 of file gauss_jackson_integrator_constructor.hh.

Referenced by `configure()`, `create_integration_controls()`, and `swap()`.

8.9.4.2 priming_constructor

```
er7_utils::IntegratorConstructor* jeod::GaussJacksonIntegratorConstructor::priming_constructor
{} [private]
```

The integrator constructor that creates the priming integrators.

trick_units(-)

Definition at line 235 of file gauss_jackson_integrator_constructor.hh.

Referenced by `configure()`, `create_first_order_ode_integrator()`, `create_generalized_deriv_second_order_ode_integrator()`, `create_integration_controls()`, `create_second_order_ode_integrator()`, `GaussJacksonIntegratorConstructor()`, `swap()`, and `~GaussJacksonIntegratorConstructor()`.

The documentation for this class was generated from the following files:

- [gauss_jackson_integrator_constructor.hh](#)
- [gauss_jackson_integrator_constructor.cc](#)

8.10 jeod::GaussJacksonOneState Class Reference

Essentially just a double*.

```
#include <gauss_jackson_one_state.hh>
```

Public Member Functions

- [GaussJacksonOneState](#) ()=default
Default constructor.
- [GaussJacksonOneState](#) (double *first_in)
Conversion constructor.

Data Fields

- double * [first](#) {}
The pointed-to data.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonOneState](#))

8.10.1 Detailed Description

Essentially just a double*.

Definition at line 71 of file gauss_jackson_one_state.hh.

8.10.2 Constructor & Destructor Documentation

8.10.2.1 GaussJacksonOneState() [1/2]

```
jeod::GaussJacksonOneState::GaussJacksonOneState ( ) [default]
```

Default constructor.

8.10.2.2 GaussJacksonOneState() [2/2]

```
jeod::GaussJacksonOneState::GaussJacksonOneState (
    double * first_in ) [inline], [explicit]
```

Conversion constructor.

Parameters

<i>first_in</i>	The pointed-to data.
-----------------	----------------------

Definition at line 88 of file gauss_jackson_one_state.hh.

8.10.3 Member Function Documentation

8.10.3.1 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonOneState::p (
    jeod ,
    GaussJacksonOneState ) [private]
```

8.10.4 Field Documentation

8.10.4.1 first

```
double* jeod::GaussJacksonOneState::first {}
```

The pointed-to data.

```
trick_units(-)
```

Definition at line 77 of file gauss_jackson_one_state.hh.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::allocate_state_contents()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::deallocate_state_contents()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_primer()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::mid_correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::predict()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_state()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_comparison_data()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_epoch_data()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::swap_state()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::test_for_convergence()`.

The documentation for this class was generated from the following file:

- [gauss_jackson_one_state.hh](#)

8.11 jeod::GaussJacksonRationalCoefficients Class Reference

Contains a set of Adams or Stormer-Cowell coefficients.

```
#include <gauss_jackson_rational_coeffs.hh>
```

Public Member Functions

- [GaussJacksonRationalCoefficients](#) ()=default
Default constructor.
- void [configure_adams_corrector](#) (unsigned int nelelem)
Configure the coefficients as an Adams corrector in difference form.
- [GaussJacksonRationalCoefficients construct_stormer_cowell_corrector](#) () const
Construct a [GaussJacksonRationalCoefficients](#) that contains the Stormer-Cowell corrector coefficients.
- [GaussJacksonRationalCoefficients construct_predictor](#) () const
Construct a [GaussJacksonRationalCoefficients](#) that contains a set of predictor coefficients.
- void [convert_to_ordinate_form](#) (er7_utils::NChooseM &n_choose_m, double *result) const
Convert the coefficients to ordinate form.
- void [discard_extra_terms](#) (unsigned int nfront, unsigned int nback)
Discard the specified number of terms from the front and back of the coefficients array.
- void [displace_back](#) ()
Displace the corrector coefficients back one time step.

Data Fields

- `std::vector< er7_utils::Ratio128 >` [coefficients](#)
The coefficients.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [GaussJacksonRationalCoefficients](#))

8.11.1 Detailed Description

Contains a set of Adams or Stormer-Cowell coefficients.

Definition at line 78 of file gauss_jackson_rational_coeffs.hh.

8.11.2 Constructor & Destructor Documentation

8.11.2.1 GaussJacksonRationalCoefficients()

```
jeod::GaussJacksonRationalCoefficients::GaussJacksonRationalCoefficients ( ) [default]
```

Default constructor.

8.11.3 Member Function Documentation

8.11.3.1 configure_adams_corrector()

```
void jeod::GaussJacksonRationalCoefficients::configure_adams_corrector (
    unsigned int nelem )
```

Configure the coefficients as an Adams corrector in difference form.

Parameters

<i>nelem</i>	The number of elements in the coefficients vector.
--------------	--

Definition at line 32 of file gauss_jackson_rational_coeffs.cc.

References coefficients.

Referenced by jeod::GaussJacksonCoeffs::compute_coeffs().

8.11.3.2 `construct_predictor()`

```
GaussJacksonRationalCoefficients jeod::GaussJacksonRationalCoefficients::construct_predictor (
) const
```

Construct a [GaussJacksonRationalCoefficients](#) that contains a set of predictor coefficients.

The coefficients are assumed to be configured as either Adams or Stormer-Cowell corrector coefficients.

Returns

A [GaussJacksonRationalCoefficients](#) object with the coefficients configured as Adams or Stormer-Cowell predictor coefficients.

Definition at line 78 of file `gauss_jackson_rational_coeffs.cc`.

References coefficients.

Referenced by `jeod::GaussJacksonCoeffs::compute_coeffs()`.

8.11.3.3 `construct_stormer_cowell_corrector()`

```
GaussJacksonRationalCoefficients jeod::GaussJacksonRationalCoefficients::construct_stormer_↵
cowell_corrector ( ) const
```

Construct a [GaussJacksonRationalCoefficients](#) that contains the Stormer-Cowell corrector coefficients.

The coefficients are assumed to be configured as Adams coefficients in difference form.

Returns

A [GaussJacksonRationalCoefficients](#) object with the coefficients configured as Stormer-Cowell corrector coefficients.

Definition at line 54 of file `gauss_jackson_rational_coeffs.cc`.

References coefficients.

Referenced by `jeod::GaussJacksonCoeffs::compute_coeffs()`.

8.11.3.4 `convert_to_ordinate_form()`

```
void jeod::GaussJacksonRationalCoefficients::convert_to_ordinate_form (
    er7_utils::NChooseM & n_choose_m,
    double * result ) const
```

Convert the coefficients to ordinate form.

Parameters

<i>n_choose_m</i>	An NChooseM object that computes N choose M.
<i>result</i>	The output ordinate form coefficients.

Definition at line 101 of file gauss_jackson_rational_coefs.cc.

References coefficients.

Referenced by jeod::GaussJacksonCoeffs::compute_coefs().

8.11.3.5 discard_extra_terms()

```
void jeod::GaussJacksonRationalCoefficients::discard_extra_terms (
    unsigned int nfront,
    unsigned int nback )
```

Discard the specified number of terms from the front and back of the coefficients array.

Parameters

<i>nfront</i>	The number of terms to be discarded from the front of the coefficients vector.
<i>nback</i>	The number of terms to be discarded from the back of the coefficients vector.

Definition at line 133 of file gauss_jackson_rational_coefs.cc.

References coefficients.

Referenced by jeod::GaussJacksonCoeffs::compute_coefs().

8.11.3.6 displace_back()

```
void jeod::GaussJacksonRationalCoefficients::displace_back ( )
```

Displace the corrector coefficients back one time step.

Definition at line 146 of file gauss_jackson_rational_coefs.cc.

References coefficients.

Referenced by jeod::GaussJacksonCoeffs::compute_coefs().

8.11.3.7 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonRationalCoefficients::p (
    jeod ,
    GaussJacksonRationalCoefficients ) [private]
```

8.11.4 Field Documentation

8.11.4.1 coefficients

```
std::vector<er7_utils::Ratio128> jeod::GaussJacksonRationalCoefficients::coefficients
```

The coefficients.

trick_units(-)

Definition at line 84 of file gauss_jackson_rational_coefs.hh.

Referenced by `configure_adams_corrector()`, `construct_predictor()`, `construct_stormer_cowell_corrector()`, `convert_to_ordinate_form()`, `discard_extra_terms()`, and `displace_back()`.

The documentation for this class was generated from the following files:

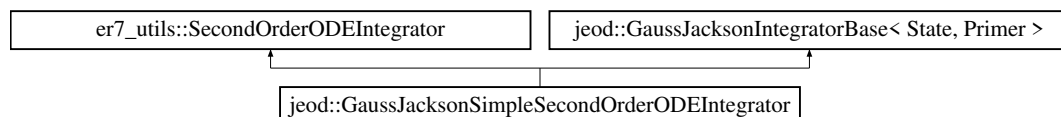
- [gauss_jackson_rational_coefs.hh](#)
- [gauss_jackson_rational_coefs.cc](#)

8.12 jeod::GaussJacksonSimpleSecondOrderODEIntegrator Class Reference

Integrates a simple second order ODE using the Gauss-Jackson technique.

```
#include <gauss_jackson_simple_second_order_ode_integrator.hh>
```

Inheritance diagram for `jeod::GaussJacksonSimpleSecondOrderODEIntegrator`:



Public Member Functions

- [GaussJacksonSimpleSecondOrderODEIntegrator](#) ()=default
- [~GaussJacksonSimpleSecondOrderODEIntegrator](#) () override=default
- [GaussJacksonSimpleSecondOrderODEIntegrator](#) (const [er7_utils::IntegratorConstructor](#) &priming_↔ constructor, [GaussJacksonIntegrationControls](#) &controls, unsigned int size_in, [er7_utils::IntegrationControls](#) &priming_controls)
Non-default constructor.
- [GaussJacksonSimpleSecondOrderODEIntegrator](#) (const [GaussJacksonSimpleSecondOrderODEIntegrator](#) &src)
Copy constructor.
- [GaussJacksonSimpleSecondOrderODEIntegrator](#) & operator= ([GaussJacksonSimpleSecondOrderODEIntegrator](#) src)
Copy and swap assignment operator.
- void [swap](#) ([GaussJacksonSimpleSecondOrderODEIntegrator](#) &other)
Non-throwing swap.
- [er7_utils::SecondOrderODEIntegrator](#) * [create_copy](#) () const override
Replicate this.
- void [reset_integrator](#) () override
Reset the integrator.
- [er7_utils::IntegratorResult](#) [integrate](#) (double dyn_dt, unsigned int target_stage, const double *acc, double *vel, double *pos) override
Propagate state using Gauss-Jackson.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonSimpleSecondOrderODEIntegrator](#))
- void [swap](#) ([GaussJacksonIntegratorBase](#) &other)
Non-throwing swap.

Additional Inherited Members

8.12.1 Detailed Description

Integrates a simple second order ODE using the Gauss-Jackson technique.

The class inherits from [er7_utils::SecondOrderODEIntegrator](#) as an is-a relationship (public inheritance) and from [GaussJacksonIntegratorBaseSecond](#) as an is-implemented-by relationship (private inheritance). Using composition instead of private inheritance would make Trick 13 checkpoint/restart a lot trickier to implement. With private inheritance, the Trick 13 io_src file contains all the necessary information.

Definition at line 83 of file [gauss_jackson_simple_second_order_ode_integrator.hh](#).

8.12.2 Constructor & Destructor Documentation

8.12.2.1 GaussJacksonSimpleSecondOrderODEIntegrator() [1/3]

```
jeod::GaussJacksonSimpleSecondOrderODEIntegrator::GaussJacksonSimpleSecondOrderODEIntegrator (
) [default]
```

8.12.2.2 ~GaussJacksonSimpleSecondOrderODEIntegrator()

```
jeod::GaussJacksonSimpleSecondOrderODEIntegrator::~~GaussJacksonSimpleSecondOrderODEIntegrator
( ) [override], [default]
```

8.12.2.3 GaussJacksonSimpleSecondOrderODEIntegrator() [2/3]

```
jeod::GaussJacksonSimpleSecondOrderODEIntegrator::GaussJacksonSimpleSecondOrderODEIntegrator (
    const er7_utils::IntegratorConstructor & priming_constructor,
    GaussJacksonIntegrationControls & controls,
    unsigned int size_in,
    er7_utils::IntegrationControls & priming_controls ) [inline]
```

Non-default constructor.

This is the constructor invoked by the [GaussJacksonIntegratorConstructor](#).

Parameters

<i>priming_constructor</i>	Integrator constructor for the technique used during priming.
<i>controls</i>	The Gauss-Jackson integration controls that drives this state integrator.
<i>size_in</i>	State size.
<i>priming_controls</i>	Integration controls used during priming.

Definition at line 100 of file `gauss_jackson_simple_second_order_ode_integrator.hh`.

8.12.2.4 GaussJacksonSimpleSecondOrderODEIntegrator() [3/3]

```
jeod::GaussJacksonSimpleSecondOrderODEIntegrator::GaussJacksonSimpleSecondOrderODEIntegrator (
    const GaussJacksonSimpleSecondOrderODEIntegrator & src ) [inline]
```

Copy constructor.

Parameters

<i>src</i>	Item to be copied.
------------	--------------------

Definition at line 113 of file `gauss_jackson_simple_second_order_ode_integrator.hh`.

8.12.3 Member Function Documentation

8.12.3.1 create_copy()

```
er7_utils::SecondOrderODEIntegrator* jeod::GaussJacksonSimpleSecondOrderODEIntegrator::create←
_copy ( ) const [inline], [override]
```

Replicate this.

Returns

Replicate of this.

Definition at line 143 of file gauss_jackson_simple_second_order_ode_integrator.hh.

8.12.3.2 integrate()

```
er7_utils::IntegratorResult jeod::GaussJacksonSimpleSecondOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * acc,
    double * vel,
    double * pos ) [inline], [override]
```

Propagate state using Gauss-Jackson.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>acc</i>	Acceleration vector.
in, out	<i>vel</i>	Velocity vector.
in, out	<i>pos</i>	Position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 167 of file gauss_jackson_simple_second_order_ode_integrator.hh.

8.12.3.3 operator=()

```
GaussJacksonSimpleSecondOrderODEIntegrator& jeod::GaussJacksonSimpleSecondOrderODEIntegrator←
::operator= (
    GaussJacksonSimpleSecondOrderODEIntegrator src ) [inline]
```

Copy and swap assignment operator.

Parameters

<i>src</i>	Item to be copied.
------------	--------------------

Definition at line 123 of file gauss_jackson_simple_second_order_ode_integrator.hh.

8.12.3.4 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonSimpleSecondOrderODEIntegrator::p (
    jeod ,
    GaussJacksonSimpleSecondOrderODEIntegrator ) [private]
```

8.12.3.5 reset_integrator()

```
void jeod::GaussJacksonSimpleSecondOrderODEIntegrator::reset_integrator ( ) [inline], [override]
```

Reset the integrator.

Definition at line 151 of file gauss_jackson_simple_second_order_ode_integrator.hh.

8.12.3.6 swap() [1/2]

```
void jeod::GaussJacksonIntegratorBase< State, Primer >::swap [inline], [private]
```

Non-throwing swap.

Parameters

<i>other</i>	Item whose contents are to be swapped with this.
--------------	--

Definition at line 397 of file gauss_jackson_integrator_base.hh.

8.12.3.7 swap() [2/2]

```
void jeod::GaussJacksonSimpleSecondOrderODEIntegrator::swap (
    GaussJacksonSimpleSecondOrderODEIntegrator & other ) [inline]
```

Non-throwing swap.

Parameters

<i>other</i>	Item whose contents are to be swapped with this.
--------------	--

Definition at line 133 of file `gauss_jackson_simple_second_order_ode_integrator.hh`.

Referenced by `jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator::swap()`.

The documentation for this class was generated from the following file:

- [gauss_jackson_simple_second_order_ode_integrator.hh](#)

8.13 jeod::GaussJacksonStateMachine Class Reference

Guides the behavior of the Gauss-Jackson integration process via a finite state machine.

```
#include <gauss_jackson_state_machine.hh>
```

Public Types

- enum `FsmState` {
[Reset](#) , [Priming](#) , [BootstrapEdit](#) , [BootstrapStep](#) ,
[Operational](#) }
Specifies the Gauss-Jackson finite state machine states.

Public Member Functions

- [GaussJacksonStateMachine](#) ()=default
Default constructor.
- [FsmState get_fsm_state](#) () const
Get the finite state machine state.
- unsigned int [get_max_history_size](#) () const
Get the maximum history size.
- unsigned int [get_current_order](#) () const
Get the current order.
- unsigned int [get_history_length](#) () const
Get the current history length.
- double [get_cycle_scale](#) () const
Get the current time scale factor.
- double [get_cycle_start_time](#) () const
Get the current cycle start time.
- bool [get_at_downsample](#) () const
Get the at_downsample flag.
- bool [get_at_reinitialize](#) () const
Get the at_reinitialize flag.
- bool [get_at_order_change](#) () const
Get the at_order_change flag.
- bool [get_at_end_of_tour](#) () const

- *Get the at_end_of_tour flag.*
- void [set_bootstrap_edit_redo_needed](#) ()
Tell the state machine that the edit did not pass a convergence test.
- void [configure](#) (const [GaussJacksonConfig](#) &config)
Configure (or reconfigure) the Gauss-Jackson state machine.
- void [reset](#) ()
Reset the Gauss-Jackson state machine.
- void [perform_step](#) ()
Advance the state machine by one step.

Static Public Member Functions

- static std::string [state_name](#) ([FsmState](#) state)
Translates a finite state machine state value to a string.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonStateMachine](#))
- void [transition_state](#) ()
Make a state transition.
- void [exit_priming](#) ()
Make the transition out of Priming.
- void [exit_bootstrap_edit](#) ()
Make a transition out of BootstrapEdit.
- void [exit_bootstrap_step](#) ()
Make a transition out of BootstrapStep.

Private Attributes

- unsigned int [initial_order](#) {}
The order to be used immediately after priming is complete.
- unsigned int [final_order](#) {}
The order to be used in operational mode.
- unsigned int [ndoubling_steps](#) {}
The number of times the time step is to be doubled between priming and operational modes.
- unsigned int [max_correction_iterations](#) {}
The maximum number of corrections to be performed.
- unsigned int [max_history_size](#) {}
The maximum history size.
- unsigned int [tour_count](#) {}
*The number of small steps that represent a step to the simulation engine, 2**n_doubling_steps.*
- [FsmState](#) [fsm_state](#) {Reset}
The finite state machine state.
- unsigned int [current_order](#) {}
The current order.
- unsigned int [history_size](#) {}
The current history size, the number of history elements that must be accumulated to transition to the next state.
- unsigned int [history_length](#) {}
The current history length, the number of history elements that have been accumulated so far.

- unsigned int `scale_factor` {}
A power of two that starts at $2^{ndoubling_steps}$ and is halved with each downsample.
- unsigned int `step_increment` {}
A power of two that starts at 1 and is doubled with each downsample.
- unsigned int `steps_since_reset` {}
The number of steps since the reset, measured in units of priming cycle steps.
- unsigned int `correction_iterations` {}
The number of correction iterations made during BootstrapEdit.
- double `cycle_scale` {}
The unitless time step size of the current integration cycle, measured in integration tour time step units.
- double `cycle_start_time` {}
The unitless start time of the current integration cycle, measured in integration tour time step units.
- bool `bootstrap_edit_redo_needed` {}
Flag indicating that the current edit sequence has failed to converge.
- bool `at_downsample` {}
Flag indicating that history data are to be downsampled and the time step is to be doubled.
- bool `at_reinitialize` {}
Flag indicating that the Gauss-Jackson integration constants are to be reinitialized.
- bool `at_order_change` {}
Flag indicating that the order is to be increased.
- bool `at_end_of_tour` {}
Flag indicating that the current integration cycle is the last one in an integration tour (i.e., that a major time step will be completed).

8.13.1 Detailed Description

Guides the behavior of the Gauss-Jackson integration process via a finite state machine.

The Gauss-Jackson integration process comprises four distinct modes:

- Priming: Using an alternate integrator, the primer, to build the requisite number of data points needed by the initial Gauss-Jackson algorithm.
- Editing: Using a Gauss-Jackson mid-corrector to make the collected data consistent with the Gauss-Jackson technique.
- Stepping: Using a Gauss-Jackson predictor/corrector to build the requisite number of data points needed by the next step of the Gauss-Jackson algorithm.
- Operational: Using the Gauss-Jackson predictor/corrector at the final user-specified time step and order.

Definition at line 89 of file `gauss_jackson_state_machine.hh`.

8.13.2 Member Enumeration Documentation

8.13.2.1 FsmState

```
enum jeod::GaussJacksonStateMachine::FsmState
```

Specifies the Gauss-Jackson finite state machine states.

Enumerator

Reset	Module was just commanded to reset itself.
Priming	Using primer to build initial set of data.
BootstrapEdit	Editing primer / lower-level Gauss-Jackson data.
BootstrapStep	Building toward downsample / change in order.
Operational	At desired rate and order.

Definition at line 95 of file gauss_jackson_state_machine.hh.

8.13.3 Constructor & Destructor Documentation

8.13.3.1 GaussJacksonStateMachine()

```
jeod::GaussJacksonStateMachine::GaussJacksonStateMachine ( ) [default]
```

Default constructor.

8.13.4 Member Function Documentation

8.13.4.1 configure()

```
void jeod::GaussJacksonStateMachine::configure (
    const GaussJacksonConfig & config )
```

Configure (or reconfigure) the Gauss-Jackson state machine.

Definition at line 55 of file gauss_jackson_state_machine.cc.

References `jeod::GaussJacksonConfig::final_order`, `final_order`, `jeod::GaussJacksonConfig::initial_order`, `initial_order`, `jeod::GaussJacksonConfig::max_correction_iterations`, `max_correction_iterations`, `max_history_size`, `jeod::GaussJacksonConfig::ndoubling_steps`, `ndoubling_steps`, and `tour_count`.

Referenced by `jeod::GaussJacksonIntegrationControls::GaussJacksonIntegrationControls()`.

8.13.4.2 `exit_bootstrap_edit()`

```
void jeod::GaussJacksonStateMachine::exit_bootstrap_edit ( ) [private]
```

Make a transition out of BootstrapEdit.

Definition at line 197 of file `gauss_jackson_state_machine.cc`.

References `at_reinitialize`, `bootstrap_edit_redo_needed`, `BootstrapEdit`, `BootstrapStep`, `correction_iterations`, `current_order`, `final_order`, `fsm_state`, `history_length`, `history_size`, `Operational`, `scale_factor`, `step_increment`, and `steps_since_reset`.

Referenced by `exit_bootstrap_step()`, `exit_priming()`, and `transition_state()`.

8.13.4.3 `exit_bootstrap_step()`

```
void jeod::GaussJacksonStateMachine::exit_bootstrap_step ( ) [private]
```

Make a transition out of BootstrapStep.

Definition at line 252 of file `gauss_jackson_state_machine.cc`.

References `at_downsample`, `at_order_change`, `at_reinitialize`, `bootstrap_edit_redo_needed`, `BootstrapEdit`, `correction_iterations`, `current_order`, `cycle_scale`, `exit_bootstrap_edit()`, `final_order`, `fsm_state`, `history_length`, `history_size`, `max_correction_iterations`, `scale_factor`, and `step_increment`.

Referenced by `transition_state()`.

8.13.4.4 `exit_priming()`

```
void jeod::GaussJacksonStateMachine::exit_priming ( ) [private]
```

Make the transition out of Priming.

Definition at line 174 of file `gauss_jackson_state_machine.cc`.

References `at_order_change`, `at_reinitialize`, `bootstrap_edit_redo_needed`, `BootstrapEdit`, `correction_iterations`, `current_order`, `exit_bootstrap_edit()`, `fsm_state`, `history_length`, `initial_order`, and `max_correction_iterations`.

Referenced by `transition_state()`.

8.13.4.5 `get_at_downsample()`

```
bool jeod::GaussJacksonStateMachine::get_at_downsample ( ) const [inline]
```

Get the `at_downsample` flag.

Definition at line 171 of file `gauss_jackson_state_machine.hh`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`, and `jeod::GaussJacksonIntegrationControls::start_cycle()`.

8.13.4.6 get_at_end_of_tour()

```
bool jeod::GaussJacksonStateMachine::get_at_end_of_tour ( ) const [inline]
```

Get the at_end_of_tour flag.

Definition at line 195 of file gauss_jackson_state_machine.hh.

Referenced by jeod::GaussJacksonIntegrationControls::start_cycle().

8.13.4.7 get_at_order_change()

```
bool jeod::GaussJacksonStateMachine::get_at_order_change ( ) const [inline]
```

Get the at_order_change flag.

Definition at line 187 of file gauss_jackson_state_machine.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJackson↵
IntegrationControls::start_cycle().

8.13.4.8 get_at_reinitialize()

```
bool jeod::GaussJacksonStateMachine::get_at_reinitialize ( ) const [inline]
```

Get the at_reinitialize flag.

Definition at line 179 of file gauss_jackson_state_machine.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJackson↵
IntegrationControls::start_cycle().

8.13.4.9 get_current_order()

```
unsigned int jeod::GaussJacksonStateMachine::get_current_order ( ) const [inline]
```

Get the current order.

Definition at line 139 of file gauss_jackson_state_machine.hh.

Referenced by jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle(), and jeod::GaussJackson↵
IntegrationControls::start_cycle().

8.13.4.10 `get_cycle_scale()`

```
double jeod::GaussJacksonStateMachine::get_cycle_scale ( ) const [inline]
```

Get the current time scale factor.

Definition at line 155 of file `gauss_jackson_state_machine.hh`.

Referenced by `jeod::GaussJacksonIntegrationControls::integrate()`, and `jeod::GaussJacksonIntegrationControls::start_cycle()`.

8.13.4.11 `get_cycle_start_time()`

```
double jeod::GaussJacksonStateMachine::get_cycle_start_time ( ) const [inline]
```

Get the current cycle start time.

Definition at line 163 of file `gauss_jackson_state_machine.hh`.

Referenced by `jeod::GaussJacksonIntegrationControls::start_cycle()`.

8.13.4.12 `get_fsm_state()`

```
FsmState jeod::GaussJacksonStateMachine::get_fsm_state ( ) const [inline]
```

Get the finite state machine state.

Definition at line 123 of file `gauss_jackson_state_machine.hh`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::start_cycle()`, and `jeod::GaussJacksonIntegrationControls::start_cycle()`.

8.13.4.13 `get_history_length()`

```
unsigned int jeod::GaussJacksonStateMachine::get_history_length ( ) const [inline]
```

Get the current history length.

Definition at line 147 of file `gauss_jackson_state_machine.hh`.

Referenced by `jeod::GaussJacksonIntegrationControls::integrate_edit()`.

8.13.4.14 get_max_history_size()

```
unsigned int jeod::GaussJacksonStateMachine::get_max_history_size ( ) const [inline]
```

Get the maximum history size.

Definition at line 131 of file gauss_jackson_state_machine.hh.

8.13.4.15 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonStateMachine::p (
    jeod ,
    GaussJacksonStateMachine ) [private]
```

8.13.4.16 perform_step()

```
void jeod::GaussJacksonStateMachine::perform_step ( )
```

Advance the state machine by one step.

Definition at line 103 of file gauss_jackson_state_machine.cc.

References `at_downsample`, `at_end_of_tour`, `at_order_change`, `at_reinitialize`, `BootstrapEdit`, `cycle_start_time`, `fsm_state`, `history_length`, `history_size`, `step_increment`, `steps_since_reset`, `tour_count`, and `transition_state()`.

Referenced by `jeod::GaussJacksonIntegrationControls::start_cycle()`.

8.13.4.17 reset()

```
void jeod::GaussJacksonStateMachine::reset ( )
```

Reset the Gauss-Jackson state machine.

Definition at line 71 of file gauss_jackson_state_machine.cc.

References `at_downsample`, `at_end_of_tour`, `at_order_change`, `at_reinitialize`, `current_order`, `cycle_scale`, `cycle_start_time`, `fsm_state`, `history_length`, `history_size`, `initial_order`, `Reset`, `scale_factor`, `step_increment`, `steps_since_reset`, and `tour_count`.

Referenced by `jeod::GaussJacksonIntegrationControls::reset_integrator()`.

8.13.4.18 `set_bootstrap_edit_redo_needed()`

```
void jeod::GaussJacksonStateMachine::set_bootstrap_edit_redo_needed ( )
```

Tell the state machine that the edit did not pass a convergence test.

Definition at line 96 of file `gauss_jackson_state_machine.cc`.

References `bootstrap_edit_redo_needed`, `BootstrapEdit`, `correction_iterations`, `fsm_state`, and `max_correction_↵ iterations`.

Referenced by `jeod::GaussJacksonIntegrationControls::integrate_edit()`.

8.13.4.19 `state_name()`

```
std::string jeod::GaussJacksonStateMachine::state_name (
    FsmState state ) [static]
```

Translates a finite state machine state value to a string.

Definition at line 30 of file `gauss_jackson_state_machine.cc`.

References `BootstrapEdit`, `BootstrapStep`, `Operational`, `Priming`, and `Reset`.

8.13.4.20 `transition_state()`

```
void jeod::GaussJacksonStateMachine::transition_state ( ) [private]
```

Make a state transition.

Definition at line 131 of file `gauss_jackson_state_machine.cc`.

References `BootstrapEdit`, `BootstrapStep`, `current_order`, `exit_bootstrap_edit()`, `exit_bootstrap_step()`, `exit_↵ priming()`, `fsm_state`, `history_size`, `initial_order`, `Operational`, `Priming`, `Reset`, and `steps_since_reset`.

Referenced by `perform_step()`.

8.13.5 Field Documentation

8.13.5.1 at_downsample

```
bool jeod::GaussJacksonStateMachine::at_downsample {} [private]
```

Flag indicating that history data are to be downsampled and the time step is to be doubled.

The flag is set on transitions from BootstrapStep to BootstrapEdit when the step size has not yet reached the desired value, clear otherwise. `trick_units(-)`

Definition at line 330 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_step()`, `perform_step()`, and `reset()`.

8.13.5.2 at_end_of_tour

```
bool jeod::GaussJacksonStateMachine::at_end_of_tour {} [private]
```

Flag indicating that the current integration cycle is the last one in an integration tour (i.e., that a major time step will be completed).

The flag is set at the start of the cycle that completes the tour, clear otherwise. This flag is never set during BootstrapEdit. `trick_units(-)`

Definition at line 352 of file `gauss_jackson_state_machine.hh`.

Referenced by `perform_step()`, and `reset()`.

8.13.5.3 at_order_change

```
bool jeod::GaussJacksonStateMachine::at_order_change {} [private]
```

Flag indicating that the order is to be increased.

The flag is set on on transitions from BootstrapStep to BootstrapEdit when the order has has not yet reached the desired value, clear otherwise. `trick_units(-)`

Definition at line 344 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_step()`, `exit_priming()`, `perform_step()`, and `reset()`.

8.13.5.4 at_reinitialize

```
bool jeod::GaussJacksonStateMachine::at_reinitialize {} [private]
```

Flag indicating that the Gauss-Jackson integration constants are to be reinitialized.

The flag is set on entry to any state except Reset and Priming, clear otherwise. `trick_units(-)`

Definition at line 337 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_edit()`, `exit_bootstrap_step()`, `exit_priming()`, `perform_step()`, and `reset()`.

8.13.5.5 bootstrap_edit_redo_needed

```
bool jeod::GaussJacksonStateMachine::bootstrap_edit_redo_needed {} [private]
```

Flag indicating that the current edit sequence has failed to converge.

This flag is set externally by the Gauss-Jackson integration controls. `trick_units(-)`

Definition at line 322 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_edit()`, `exit_bootstrap_step()`, `exit_priming()`, and `set_bootstrap_edit_redo_needed()`.

8.13.5.6 correction_iterations

```
unsigned int jeod::GaussJacksonStateMachine::correction_iterations {} [private]
```

The number of correction iterations made during `BootstrapEdit`.

`trick_units(-)`

Definition at line 303 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_edit()`, `exit_bootstrap_step()`, `exit_priming()`, and `set_bootstrap_edit_redo_needed()`.

8.13.5.7 current_order

```
unsigned int jeod::GaussJacksonStateMachine::current_order {} [private]
```

The current order.

This is incremented by two on transitions from `BootstrapStep` to `BootstrapEdit` until the `final_order` is reached.
`trick_units(-)`

Definition at line 267 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_edit()`, `exit_bootstrap_step()`, `exit_priming()`, `reset()`, and `transition_state()`.

8.13.5.8 cycle_scale

```
double jeod::GaussJacksonStateMachine::cycle_scale {} [private]
```

The unitless time step size of the current integration cycle, measured in integration tour time step units.

This starts at $2^{*(-ndoubling_steps)}$ and doubles with each downsample. `trick_units(-)`

Definition at line 310 of file `gauss_jackson_state_machine.hh`.

Referenced by `exit_bootstrap_step()`, and `reset()`.

8.13.5.9 cycle_start_time

```
double jeod::GaussJacksonStateMachine::cycle_start_time {} [private]
```

The unitless start time of the current integration cycle, measured in integration four time step units.

trick_units(-)

Definition at line 316 of file gauss_jackson_state_machine.hh.

Referenced by perform_step(), and reset().

8.13.5.10 final_order

```
unsigned int jeod::GaussJacksonStateMachine::final_order {} [private]
```

The order to be used in operational mode.

This must be an even integer and must not be less than initial_order. trick_units(-)

Definition at line 233 of file gauss_jackson_state_machine.hh.

Referenced by configure(), exit_bootstrap_edit(), and exit_bootstrap_step().

8.13.5.11 fsm_state

```
FsmState jeod::GaussJacksonStateMachine::fsm_state {Reset} [private]
```

The finite state machine state.

trick_units(-)

Definition at line 261 of file gauss_jackson_state_machine.hh.

Referenced by exit_bootstrap_edit(), exit_bootstrap_step(), exit_priming(), perform_step(), reset(), set_bootstrap↵_edit_redo_needed(), and transition_state().

8.13.5.12 history_length

```
unsigned int jeod::GaussJacksonStateMachine::history_length {} [private]
```

The current history length, the number of history elements that have been accumulated so far.

trick_units(-)

Definition at line 279 of file gauss_jackson_state_machine.hh.

Referenced by exit_bootstrap_edit(), exit_bootstrap_step(), exit_priming(), perform_step(), and reset().

8.13.5.13 history_size

```
unsigned int jeod::GaussJacksonStateMachine::history_size {} [private]
```

The current history size, the number of history elements that must be accumulated to transition to the next state.

trick_units(-)

Definition at line 273 of file gauss_jackson_state_machine.hh.

Referenced by exit_bootstrap_edit(), exit_bootstrap_step(), perform_step(), reset(), and transition_state().

8.13.5.14 initial_order

```
unsigned int jeod::GaussJacksonStateMachine::initial_order {} [private]
```

The order to be used immediately after priming is complete.

This must be an even, non-negative integer. trick_units(-)

Definition at line 227 of file gauss_jackson_state_machine.hh.

Referenced by configure(), exit_priming(), reset(), and transition_state().

8.13.5.15 max_correction_iterations

```
unsigned int jeod::GaussJacksonStateMachine::max_correction_iterations {} [private]
```

The maximum number of corrections to be performed.

trick_units(-)

Definition at line 244 of file gauss_jackson_state_machine.hh.

Referenced by configure(), exit_bootstrap_step(), exit_priming(), and set_bootstrap_edit_redo_needed().

8.13.5.16 max_history_size

```
unsigned int jeod::GaussJacksonStateMachine::max_history_size {} [private]
```

The maximum history size.

This is calculated for the benefit of state integrators. trick_units(-)

Definition at line 250 of file gauss_jackson_state_machine.hh.

Referenced by configure().

8.13.5.17 ndoubling_steps

```
unsigned int jeod::GaussJacksonStateMachine::ndoubling_steps {} [private]
```

The number of times the time step is to be doubled between priming and operational modes.

trick_units(-)

Definition at line 239 of file gauss_jackson_state_machine.hh.

Referenced by configure().

8.13.5.18 scale_factor

```
unsigned int jeod::GaussJacksonStateMachine::scale_factor {} [private]
```

A power of two that starts at $2^{ndoubling_steps}$ and is halved with each downsample.

When the scale_factor reaches 1 it is time to transition to operational mode. trick_units(-)

Definition at line 286 of file gauss_jackson_state_machine.hh.

Referenced by exit_bootstrap_edit(), exit_bootstrap_step(), and reset().

8.13.5.19 step_increment

```
unsigned int jeod::GaussJacksonStateMachine::step_increment {} [private]
```

A power of two that starts at 1 and is doubled with each downsample.

trick_units(-)

Definition at line 291 of file gauss_jackson_state_machine.hh.

Referenced by exit_bootstrap_edit(), exit_bootstrap_step(), perform_step(), and reset().

8.13.5.20 steps_since_reset

```
unsigned int jeod::GaussJacksonStateMachine::steps_since_reset {} [private]
```

The number of steps since the reset, measured in units of priming cycle steps.

The counter is incremented by the step_increment upon completion of a cycle and is reset to zero on entry into BootstrapEdit. trick_units(-)

Definition at line 298 of file gauss_jackson_state_machine.hh.

Referenced by exit_bootstrap_edit(), perform_step(), reset(), and transition_state().

8.13.5.21 tour_count

```
unsigned int jeod::GaussJacksonStateMachine::tour_count {} [private]
```

The number of small steps that represent a step to the simulation engine, $2 \times n_doubling_steps$.

trick_units(–)

Definition at line 256 of file gauss_jackson_state_machine.hh.

Referenced by configure(), perform_step(), and reset().

The documentation for this class was generated from the following files:

- [gauss_jackson_state_machine.hh](#)
- [gauss_jackson_state_machine.cc](#)

8.14 jeod::GaussJacksonTwoState Class Reference

Essentially just `std::pair<double*>`.

```
#include <gauss_jackson_two_state.hh>
```

Public Member Functions

- [GaussJacksonTwoState](#) ()=default
Default constructor.
- [GaussJacksonTwoState](#) (double *first_in, double *second_in)
Non-default constructor.

Data Fields

- double * [first](#) {}
The first element of the pair.
- double * [second](#) {}
The second element of the pair.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [GaussJacksonTwoState](#))

8.14.1 Detailed Description

Essentially just `std::pair<double*>`.

Definition at line 72 of file gauss_jackson_two_state.hh.

8.14.2 Constructor & Destructor Documentation

8.14.2.1 GaussJacksonTwoState() [1/2]

```
jeod::GaussJacksonTwoState::GaussJacksonTwoState ( ) [default]
```

Default constructor.

8.14.2.2 GaussJacksonTwoState() [2/2]

```
jeod::GaussJacksonTwoState::GaussJacksonTwoState (
    double * first_in,
    double * second_in ) [inline]
```

Non-default constructor.

Parameters

<i>first_in</i>	The first element of the pair.
<i>second_in</i>	The second element of the pair.

Definition at line 95 of file gauss_jackson_two_state.hh.

8.14.3 Member Function Documentation

8.14.3.1 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::GaussJacksonTwoState::p (
    jeod ,
    GaussJacksonTwoState ) [private]
```

8.14.4 Field Documentation

8.14.4.1 first

```
double* jeod::GaussJacksonTwoState::first {}
```

The first element of the pair.

trick_units(-)

Definition at line 78 of file gauss_jackson_two_state.hh.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::allocate_state_contents()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::deallocate_state_contents()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_primer()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::mid_correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::predict()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_state()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_epoch_data()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::swap_state()`.

8.14.4.2 second

```
double* jeod::GaussJacksonTwoState::second {}
```

The second element of the pair.

trick_units(-)

Definition at line 83 of file gauss_jackson_two_state.hh.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::allocate_state_contents()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::deallocate_state_contents()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::integrate_primer()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::mid_correct()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::predict()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::replicate_state()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_comparison_data()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::save_epoch_data()`, `jeod::GaussJacksonIntegratorBase< State, Primer >::swap_state()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::test_for_convergence()`.

The documentation for this class was generated from the following file:

- [gauss_jackson_two_state.hh](#)

8.15 jeod::GeneralizedSecondOrderODETechnique Class Reference

Enumerates the integration schemes, generalized Cartesian or Lie group integration, and provides simple utilities that work with this.

```
#include <generalized_second_order_ode_technique.hh>
```

Public Types

- enum [TechniqueType](#) { [Unspecified](#) , [Cartesian](#) , [LieGroup](#) }

Enumerates the types of second order ODE solvers that can be used to integrate a generalized second order ODE.

Public Member Functions

- [GeneralizedSecondOrderODETechnique](#) ()=delete
- [GeneralizedSecondOrderODETechnique](#) (const [GeneralizedSecondOrderODETechnique](#) &)=delete
- [GeneralizedSecondOrderODETechnique](#) & [operator=](#) (const [GeneralizedSecondOrderODETechnique](#) &)=delete

Static Public Member Functions

- static bool [is_provided_by](#) (const er7_utils::IntegratorConstructor &generator, [TechniqueType](#) technique)
Test whether an integration method provides an integrator for the specified technique.
- static [TechniqueType](#) [validate_technique](#) (const er7_utils::IntegratorConstructor &generator, [TechniqueType](#) technique, const char *file, unsigned int line, const char *requester, const char *name)
Validate the specified technique with respect to the integration method.

Private Member Functions

- [~GeneralizedSecondOrderODETechnique](#) ()

8.15.1 Detailed Description

Enumerates the integration schemes, generalized Cartesian or Lie group integration, and provides simple utilities that work with this.

Definition at line 78 of file `generalized_second_order_ode_technique.hh`.

8.15.2 Member Enumeration Documentation

8.15.2.1 TechniqueType

```
enum jeod::GeneralizedSecondOrderODETechnique::TechniqueType
```

Enumerates the types of second order ODE solvers that can be used to integrate a generalized second order ODE.

Enumerator

Unspecified	No technique specified (an error).
Cartesian	Integrate using a generalized derivative scheme. The integrator treats generalized position as if it lives in some Cartesian space.
LieGroup	Integrate using a generalized step scheme. The integrator treats generalized position as if it lives in some Lie group.

Definition at line 91 of file generalized_second_order_ode_technique.hh.

8.15.3 Constructor & Destructor Documentation

8.15.3.1 GeneralizedSecondOrderODETechnique() [1/2]

```
jeod::GeneralizedSecondOrderODETechnique::GeneralizedSecondOrderODETechnique ( ) [delete]
```

8.15.3.2 GeneralizedSecondOrderODETechnique() [2/2]

```
jeod::GeneralizedSecondOrderODETechnique::GeneralizedSecondOrderODETechnique (
    const GeneralizedSecondOrderODETechnique & ) [delete]
```

8.15.3.3 ~GeneralizedSecondOrderODETechnique()

```
jeod::GeneralizedSecondOrderODETechnique::~~GeneralizedSecondOrderODETechnique ( ) [private]
```

8.15.4 Member Function Documentation

8.15.4.1 is_provided_by()

```
bool jeod::GeneralizedSecondOrderODETechnique::is_provided_by (
    const er7_utils::IntegratorConstructor & generator,
    TechniqueType technique ) [static]
```

Test whether an integration method provides an integrator for the specified technique.

Parameters

in	<i>generator</i>	Integrator constructor for the integration technique.
in	<i>technique</i>	Technique to be queried.

Returns

True if the constructor can create an integrator for the specified technique, false otherwise.

Definition at line 41 of file generalized_second_order_ode_technique.cc.

References Cartesian, jeod::IntegrationMessages::invalid_request, LieGroup, and Unspecified.

Referenced by validate_technique().

8.15.4.2 operator=()

```
GeneralizedSecondOrderODETechnique& jeod::GeneralizedSecondOrderODETechnique::operator= (
    const GeneralizedSecondOrderODETechnique & ) [delete]
```

8.15.4.3 validate_technique()

```
GeneralizedSecondOrderODETechnique::TechniqueType jeod::GeneralizedSecondOrderODETechnique↔
::validate_technique (
    const er7_utils::IntegratorConstructor & generator,
    TechniqueType technique,
    const char * file,
    unsigned int line,
    const char * requester,
    const char * name ) [static]
```

Validate the specified technique with respect to the integration method.

Possible outcomes are:

- Failure if the generator doesn't provide either of the generalized second order ODE integrators.
- Switch to plan B if the generator doesn't provide the requested integrator but does provide the alternate technique.
- Nothing happens if the generator does provide the requested integrator.

Parameters

in	<i>generator</i>	Integrator constructor for the integration technique.
in	<i>technique</i>	Technique to be queried.
in	<i>file</i>	Typically FILE
in	<i>line</i>	Typically LINE
in	<i>requester</i>	Something to identify the caller.
in	<i>name</i>	The name of the object associated with the caller.

Returns

Input technique if supported, alternate if not. The function does not return is neither of the options is supported.

Definition at line 65 of file generalized_second_order_ode_technique.cc.

References Cartesian, jeod::IntegrationMessages::invalid_request, is_provided_by(), LieGroup, and jeod::IntegrationMessages::unsupported_option.

The documentation for this class was generated from the following files:

- [generalized_second_order_ode_technique.hh](#)
- [generalized_second_order_ode_technique.cc](#)

8.16 jeod::IntegrationMessages Class Reference

Declares messages associated with the integration test model.

```
#include <integration_messages.hh>
```

Public Member Functions

- [IntegrationMessages](#) ()=delete
- [IntegrationMessages](#) (const [IntegrationMessages](#) &)=delete
- [IntegrationMessages](#) & operator= (const [IntegrationMessages](#) &)=delete

Static Public Attributes

- static const char * [unsupported_option](#) = "utils/integration/" "unsupported_option"
Issued when some user input is invalid.
- static const char * [invalid_item](#) = "utils/integration/" "invalid_item"
Issued when an item is somehow invalid; a duplicate entry for example.
- static const char * [internal_error](#) = "utils/integration/" "internal_error"
Issued when the JEOD programmer messed up.
- static const char * [invalid_request](#) = "utils/integration/" "invalid_request"
Issued when a non-JEOD programmer messed up.
- static const char * [information](#) = "utils/integration/" "information"
Issued in non-error messages.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [IntegrationMessages](#))

8.16.1 Detailed Description

Declares messages associated with the integration test model.

Definition at line 82 of file integration_messages.hh.

8.16.2 Constructor & Destructor Documentation

8.16.2.1 IntegrationMessages() [1/2]

```
jeod::IntegrationMessages::IntegrationMessages ( ) [delete]
```

8.16.2.2 IntegrationMessages() [2/2]

```
jeod::IntegrationMessages::IntegrationMessages (
    const IntegrationMessages & ) [delete]
```

8.16.3 Member Function Documentation

8.16.3.1 operator=()

```
IntegrationMessages& jeod::IntegrationMessages::operator= (
    const IntegrationMessages & ) [delete]
```

8.16.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::IntegrationMessages::p (
    jeod ,
    IntegrationMessages ) [private]
```

8.16.4 Field Documentation

8.16.4.1 information

```
char const * jeod::IntegrationMessages::information = "utils/integration/" "information" [static]
```

Issued in non-error messages.

trick_units(-)

Definition at line 110 of file integration_messages.hh.

8.16.4.2 internal_error

```
char const * jeod::IntegrationMessages::internal_error = "utils/integration/" "internal_error"  
[static]
```

Issued when the JEOD programmer messed up.

trick_units(-)

Definition at line 100 of file integration_messages.hh.

8.16.4.3 invalid_item

```
char const * jeod::IntegrationMessages::invalid_item = "utils/integration/" "invalid_item"  
[static]
```

Issued when an item is somehow invalid; a duplicate entry for example.

trick_units(-)

Definition at line 95 of file integration_messages.hh.

Referenced by jeod::JeodIntegrationGroup::remove_integrable_object().

8.16.4.4 invalid_request

```
char const * jeod::IntegrationMessages::invalid_request = "utils/integration/" "invalid_  
request" [static]
```

Issued when a non-JEOD programmer messed up.

trick_units(-)

Definition at line 105 of file integration_messages.hh.

Referenced by jeod::JeodIntegrationTime::add_time_change_subscriber(), jeod::RestartableStateIntegrator< IntegratorType >::create_integrator(), jeod::GeneralizedSecondOrderODETechnique::is_provided_by(), jeod::JeodIntegrationTime::remove_time_change_subscriber(), and jeod::GeneralizedSecondOrderODETechnique::validate_technique().

8.16.4.5 unsupported_option

```
char const * jeod::IntegrationMessages::unsupported_option = "utils/integration/" "unsupported←
_option" [static]
```

Issued when some user input is invalid.

trick_units(−)

Definition at line 90 of file integration_messages.hh.

Referenced by jeod::GeneralizedSecondOrderODETechnique::validate_technique().

The documentation for this class was generated from the following files:

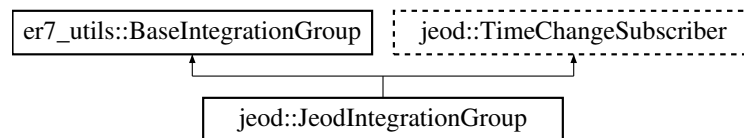
- [integration_messages.hh](#)
- [integration_messages.cc](#)

8.17 jeod::JeodIntegrationGroup Class Reference

A [JeodIntegrationGroup](#) integrates the state of a set of objects over time.

```
#include <jeod_integration_group.hh>
```

Inheritance diagram for jeod::JeodIntegrationGroup:



Public Member Functions

- [JeodIntegrationGroup](#) ()
JeodIntegrationGroup default constructor, needed for checkpoint/restart, and to support derived classes' default constructors.
- [JeodIntegrationGroup](#) ([JeodIntegrationGroupOwner](#) &owner, [er7_utils::IntegratorConstructor](#) &integ_cotr, [JeodIntegratorInterface](#) &integ_inter, [JeodIntegrationTime](#) &time_mgr)
JeodIntegrationGroup non-default constructor.
- [~JeodIntegrationGroup](#) () override
JeodIntegrationGroup destructor.
- [JeodIntegrationGroup](#) (const [JeodIntegrationGroup](#) &)=delete
- [JeodIntegrationGroup](#) & operator= (const [JeodIntegrationGroup](#) &)=delete
- bool [need_first_step_derivatives](#) () const
Indicate whether derivatives need to be calculated on the initial step of an integration cycle.
- void [update_from_owner](#) ()
Update the group via its owner.
- bool [merge_integrator_result](#) (const [er7_utils::IntegratorResult](#) &new_result, [er7_utils::IntegratorResult](#) &merged_result) const
Merge an *IntegratorResult* into another.

- void [respond_to_time_change](#) () override
Respond to a change in the nature of time.
- void [initialize_group](#) () override
Initialize the integration group.
- void [reset_body_integrators](#) () override
Reset the integrators for the integrable objects managed by this group.
- er7_utils::IntegratorResult [integrate_bodies](#) (double cycle_dyndt, unsigned int target_stage) override
Integrate the states of the integrable objects managed by this group.
- virtual void [add_integrable_object](#) (er7_utils::IntegrableObject &integrable_object)
Add an integrable object to the vector of such.
- virtual void [remove_integrable_object](#) (er7_utils::IntegrableObject &integrable_object)
Remove an integrable object from the vector of such.

Static Public Member Functions

- static void [register_classes](#) ()
Register classes associated with integration.

Protected Member Functions

- template<typename T >
void [reset_container](#) (const T &container)
Issue a reset to each member of a container.
- template<typename T >
er7_utils::IntegratorResult [integrate_container](#) (double dyn_dt, unsigned int target_stage, const T &container)
Integrate each member of a collection.

Protected Attributes

- [JeodIntegrationGroupOwner](#) *const [group_owner](#) {}
The object that owns this integration group, typically by containment.
- er7_utils::IntegratorResultMergerContainer [integ_merger](#)
The object that merges results from multiple integrators.
- JeodIntegratorInterface *const [jeod_integ_interface](#) {}
The interface between the integration module and the simulation engine's integration structure.
- [JeodIntegrationTime](#) *const [jeod_time_manager](#) {}
The interface between the integration module and the object that represents time.
- JeodPointerVector< er7_utils::IntegrableObject >::type [integrable_objects](#)
The objects whose states are integrated by this integration group.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2p](#) (jeod, [JeodIntegrationGroup](#))

8.17.1 Detailed Description

A [JeodIntegrationGroup](#) integrates the state of a set of objects over time.

This class is designed for extensibility. Authors of derived classes should follow the extension notes in the source file.

Definition at line 115 of file `jeod_integration_group.hh`.

8.17.2 Constructor & Destructor Documentation

8.17.2.1 JeodIntegrationGroup() [1/3]

```
jeod::JeodIntegrationGroup::JeodIntegrationGroup ( )
```

[JeodIntegrationGroup](#) default constructor, needed for checkpoint/restart, and to support derived classes' default constructors.

Definition at line 60 of file `jeod_integration_group.cc`.

References `integrable_objects`, and `register_classes()`.

8.17.2.2 JeodIntegrationGroup() [2/3]

```
jeod::JeodIntegrationGroup::JeodIntegrationGroup (
    JeodIntegrationGroupOwner & owner,
    er7_utils::IntegratorConstructor & integ_cotr,
    JeodIntegratorInterface & integ_inter,
    JeodIntegrationTime & time_mgr )
```

[JeodIntegrationGroup](#) non-default constructor.

Parameters

in	<i>owner</i>	The object that contains this group.
in	<i>integ_cotr</i>	Integrator constructor
in	<i>integ_inter</i>	Integrator interface
in	<i>time_mgr</i>	Time manager

Definition at line 68 of file `jeod_integration_group.cc`.

References `jeod::JeodIntegrationTime::add_time_change_subscriber()`, `integ_merger`, `integrable_objects`, and `register_classes()`.

8.17.2.3 ~JeodIntegrationGroup()

```
jeod::JeodIntegrationGroup::~~JeodIntegrationGroup ( ) [override]
```

[JeodIntegrationGroup](#) destructor.

Definition at line 86 of file `jeod_integration_group.cc`.

References `integrable_objects`, `jeod_time_manager`, and `jeod::JeodIntegrationTime::remove_time_change_`↔
`subscriber()`.

8.17.2.4 JeodIntegrationGroup() [3/3]

```
jeod::JeodIntegrationGroup::JeodIntegrationGroup (
    const JeodIntegrationGroup & ) [delete]
```

8.17.3 Member Function Documentation

8.17.3.1 add_integrable_object()

```
void jeod::JeodIntegrationGroup::add_integrable_object (
    er7_utils::IntegrableObject & integrable_object ) [virtual]
```

Add an integrable object to the vector of such.

Parameters

in	<i>integrable_object</i>	Object to be added.
----	--------------------------	---------------------

Definition at line 97 of file `jeod_integration_group.cc`.

References `integrable_objects`.

8.17.3.2 initialize_group()

```
void jeod::JeodIntegrationGroup::initialize_group ( ) [override]
```

Initialize the integration group.

Some integration techniques are configurable by user input, and thus the creation of the controls and integrators needs to be delayed a bit.

Definition at line 153 of file `jeod_integration_group.cc`.

References `integrable_objects`.

8.17.3.3 integrate_bodies()

```
er7_utils::IntegratorResult jeod::JeodIntegrationGroup::integrate_bodies (
    double cycle_dyndt,
    unsigned int target_stage ) [inline], [override]
```

Integrate the states of the integrable objects managed by this group.

This function should only be called by `IntegrationControls::integrate` or by an override of that function. Derived classes are free to override this default implementation. However, those derived class overrides either must call this method to integrate the states of the registered integrable bodies or must somehow take on the burden of integrating those states.

Parameters

in	<i>cycle_dyndt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 234 of file `jeod_integration_group.hh`.

8.17.3.4 integrate_container()

```
template<typename T >
er7_utils::IntegratorResult jeod::JeodIntegrationGroup::integrate_container (
    double dyn_dt,
    unsigned int target_stage,
    const T & container ) [inline], [protected]
```

Integrate each member of a collection.

Template Parameters

<i>T</i>	The container type.
----------	---------------------

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in, out	<i>container</i>	The container to be integrated.

Definition at line 277 of file `jeod_integration_group.hh`.

References `integ_merger`.

8.17.3.5 JEOD_CLASS_ESTABLISH_FRIENDS2p()

```
jeod::JeodIntegrationGroup::JEOD_CLASS_ESTABLISH_FRIENDS2p (
    jeod ,
    JeodIntegrationGroup ) [private]
```

8.17.3.6 merge_integrator_result()

```
bool jeod::JeodIntegrationGroup::merge_integrator_result (
    const er7_utils::IntegratorResult & new_result,
    er7_utils::IntegratorResult & merged_result ) const [inline]
```

Merge an IntegratorResult into another.

Returns

True if merger was successful, false if some error occurred.

Parameters

in	<i>new_result</i>	Size of the generalized position vector
in, out	<i>merged_result</i>	Size of the generalized position vector

Definition at line 182 of file jeod_integration_group.hh.

8.17.3.7 need_first_step_derivatives()

```
bool jeod::JeodIntegrationGroup::need_first_step_derivatives ( ) const [inline]
```

Indicate whether derivatives need to be calculated on the initial step of an integration cycle.

Returns

Desired flag.

Definition at line 163 of file jeod_integration_group.hh.

8.17.3.8 operator=()

```
JeodIntegrationGroup& jeod::JeodIntegrationGroup::operator= (
    const JeodIntegrationGroup & ) [delete]
```

8.17.3.9 register_classes()

```
void jeod::JeodIntegrationGroup::register_classes ( ) [static]
```

Register classes associated with integration.

This is a static method, and is best called prior to initialization time.

Definition at line 51 of file `jeod_integration_group.cc`.

Referenced by `JeodIntegrationGroup()`.

8.17.3.10 remove_integrable_object()

```
void jeod::JeodIntegrationGroup::remove_integrable_object (
    er7_utils::IntegrableObject & integrable_object ) [virtual]
```

Remove an integrable object from the vector of such.

Parameters

in	<i>integrable_object</i>	Object to be removed.
----	--------------------------	-----------------------

Definition at line 125 of file `jeod_integration_group.cc`.

References `integrable_objects`, and `jeod::IntegrationMessages::invalid_item`.

8.17.3.11 reset_body_integrators()

```
void jeod::JeodIntegrationGroup::reset_body_integrators ( ) [inline], [override]
```

Reset the integrators for the integrable objects managed by this group.

Resets can occur when time changes behavior (call is internal to the integration process) or when some external event would render an integrator's history invalid (call comes from outside). When either happens, integrators that depend on history need to reset their internal state to indicate that the saved data are invalid.)

Definition at line 216 of file `jeod_integration_group.hh`.

8.17.3.12 reset_container()

```
template<typename T >
void jeod::JeodIntegrationGroup::reset_container (
    const T & container ) [inline], [protected]
```

Issue a reset to each member of a container.

Template Parameters

<i>T</i>	The container type.
----------	---------------------

Parameters

<i>in, out</i>	<i>container</i>	The container to be reset.
----------------	------------------	----------------------------

Definition at line 259 of file `jeod_integration_group.hh`.

8.17.3.13 respond_to_time_change()

```
void jeod::JeodIntegrationGroup::respond_to_time_change ( ) [inline], [override], [virtual]
```

Respond to a change in the nature of time.

Implements [jeod::TimeChangeSubscriber](#).

Definition at line 193 of file `jeod_integration_group.hh`.

8.17.3.14 update_from_owner()

```
void jeod::JeodIntegrationGroup::update_from_owner ( ) [inline]
```

Update the group via its owner.

Definition at line 171 of file `jeod_integration_group.hh`.

8.17.4 Field Documentation**8.17.4.1 group_owner**

```
JeodIntegrationGroupOwner* const jeod::JeodIntegrationGroup::group_owner {} [protected]
```

The object that owns this integration group, typically by containment.

`trick_units(-)`

Definition at line 298 of file `jeod_integration_group.hh`.

8.17.4.2 integ_merger

```
er7_utils::IntegratorResultMergerContainer jeod::JeodIntegrationGroup::integ_merger [protected]
```

The object that merges results from multiple integrators.

trick_units(—)

Definition at line 303 of file jeod_integration_group.hh.

Referenced by integrate_container(), and JeodIntegrationGroup().

8.17.4.3 integrable_objects

```
JeodPointerVector<er7_utils::IntegrableObject>::type jeod::JeodIntegrationGroup::integrable_↵  
objects [protected]
```

The objects whose states are integrated by this integration group.

trick_io(**)

Definition at line 320 of file jeod_integration_group.hh.

Referenced by add_integrable_object(), initialize_group(), JeodIntegrationGroup(), remove_integrable_object(), and ~JeodIntegrationGroup().

8.17.4.4 jeod_integ_interface

```
JeodIntegratorInterface* const jeod::JeodIntegrationGroup::jeod_integ_interface {} [protected]
```

The interface between the integration module and the simulation engine's integration structure.

trick_units(—)

Definition at line 309 of file jeod_integration_group.hh.

8.17.4.5 jeod_time_manager

```
JeodIntegrationTime* const jeod::JeodIntegrationGroup::jeod_time_manager {} [protected]
```

The interface between the integration module and the object that represents time.

trick_units(—)

Definition at line 315 of file jeod_integration_group.hh.

Referenced by ~JeodIntegrationGroup().

The documentation for this class was generated from the following files:

- [jeod_integration_group.hh](#)
- [jeod_integration_group.cc](#)

8.18 jeod::JeodIntegrationGroupOwner Class Reference

The abstract class IntegrationGroupOwner contains an IntegrationGroup.

```
#include <jeod_integration_group.hh>
```

Public Member Functions

- virtual [~JeodIntegrationGroupOwner](#) ()=default
Destructor.
- virtual void [update_integration_group](#) ([JeodIntegrationGroup](#) &group)=0
Somehow update the specified integration group.

8.18.1 Detailed Description

The abstract class IntegrationGroupOwner contains an IntegrationGroup.

This simple interface class has no data members.

Definition at line 95 of file jeod_integration_group.hh.

8.18.2 Constructor & Destructor Documentation

8.18.2.1 ~JeodIntegrationGroupOwner()

```
virtual jeod::JeodIntegrationGroupOwner::~~JeodIntegrationGroupOwner ( ) [virtual], [default]
```

Destructor.

8.18.3 Member Function Documentation

8.18.3.1 update_integration_group()

```
virtual void jeod::JeodIntegrationGroupOwner::update_integration_group (
    JeodIntegrationGroup & group ) [pure virtual]
```

Somehow update the specified integration group.

Parameters

in, out	<i>group</i>	Integration group to be updated.
---------	--------------	----------------------------------

The documentation for this class was generated from the following file:

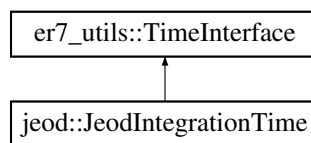
- [jeod_integration_group.hh](#)

8.19 jeod::JeodIntegrationTime Class Reference

The class [JeodIntegrationTime](#) adds the concepts of timestamping and notification of changes in the nature of time to the ER7 numerical utilities TimeInterface class.

```
#include <jeod_integration_time.hh>
```

Inheritance diagram for jeod::JeodIntegrationTime:



Public Member Functions

- [JeodIntegrationTime](#) ()
JeodIntegrationTime constructor.
- [~JeodIntegrationTime](#) () override
JeodIntegrationTime destructor.
- [JeodIntegrationTime](#) (const [JeodIntegrationTime](#) &)=delete
- [JeodIntegrationTime](#) & operator= (const [JeodIntegrationTime](#) &)=delete
- virtual double [get_timestamp_time](#) () const =0
Get the time used to timestamp some object.
- void [add_time_change_subscriber](#) ([TimeChangeSubscriber](#) &subscriber)
Add a time change subscriber.
- void [remove_time_change_subscriber](#) ([TimeChangeSubscriber](#) &subscriber)
Remove a time change subscriber.

Protected Member Functions

- void [notify_time_change_subscribers](#) ()
Notify subscribers that the nature of time has changed.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [JeodIntegrationTime](#))

Private Attributes

- JeodPointerVector< [TimeChangeSubscriber](#) >::type [time_change_subscribers](#)
List of pointers to objects that wish to be notified of a change in the nature of time.

8.19.1 Detailed Description

The class [JeodIntegrationTime](#) adds the concepts of timestamping and notification of changes in the nature of time to the ER7 numerical utilities TimeInterface class.

Definition at line 84 of file jeod_integration_time.hh.

8.19.2 Constructor & Destructor Documentation

8.19.2.1 JeodIntegrationTime() [1/2]

```
jeod::JeodIntegrationTime::JeodIntegrationTime ( )
```

[JeodIntegrationTime](#) constructor.

Definition at line 45 of file jeod_integration_time.cc.

References [time_change_subscribers](#).

8.19.2.2 ~JeodIntegrationTime()

```
jeod::JeodIntegrationTime::~~JeodIntegrationTime ( ) [override]
```

[JeodIntegrationTime](#) destructor.

Definition at line 57 of file jeod_integration_time.cc.

References [time_change_subscribers](#).

8.19.2.3 JeodIntegrationTime() [2/2]

```
jeod::JeodIntegrationTime::JeodIntegrationTime (
    const JeodIntegrationTime & ) [delete]
```

8.19.3 Member Function Documentation

8.19.3.1 add_time_change_subscriber()

```
void jeod::JeodIntegrationTime::add_time_change_subscriber (
    TimeChangeSubscriber & subscriber )
```

Add a time change subscriber.

Parameters

<i>subscriber</i>	Object to be added to list of subscribers.
-------------------	--

Definition at line 68 of file jeod_integration_time.cc.

References jeod::IntegrationMessages::invalid_request, and time_change_subscribers.

Referenced by jeod::JeodIntegrationGroup::JeodIntegrationGroup().

8.19.3.2 get_timestamp_time()

```
virtual double jeod::JeodIntegrationTime::get_timestamp_time ( ) const [pure virtual]
```

Get the time used to timestamp some object.

8.19.3.3 notify_time_change_subscribers()

```
void jeod::JeodIntegrationTime::notify_time_change_subscribers ( ) [protected]
```

Notify subscribers that the nature of time has changed.

Definition at line 108 of file jeod_integration_time.cc.

References time_change_subscribers.

8.19.3.4 operator=()

```
JeodIntegrationTime& jeod::JeodIntegrationTime::operator= (
    const JeodIntegrationTime & ) [delete]
```

8.19.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::JeodIntegrationTime::p (
    jeod ,
    JeodIntegrationTime ) [private]
```

8.19.3.6 remove_time_change_subscriber()

```
void jeod::JeodIntegrationTime::remove_time_change_subscriber (
    TimeChangeSubscriber & subscriber )
```

Remove a time change subscriber.

Parameters

<i>subscriber</i>	Object to be removed from list of subscribers.
-------------------	--

Definition at line 88 of file jeod_integration_time.cc.

References jeod::IntegrationMessages::invalid_request, and time_change_subscribers.

Referenced by jeod::JeodIntegrationGroup::~~JeodIntegrationGroup().

8.19.4 Field Documentation**8.19.4.1 time_change_subscribers**

```
JeodPointerVector<TimeChangeSubscriber>::type jeod::JeodIntegrationTime::time_change_subscribers
[private]
```

List of pointers to objects that wish to be notified of a change in the nature of time.

trick_io(**)

Definition at line 116 of file jeod_integration_time.hh.

Referenced by add_time_change_subscriber(), JeodIntegrationTime(), notify_time_change_subscribers(), remove_time_change_subscriber(), and ~JeodIntegrationTime().

The documentation for this class was generated from the following files:

- [jeod_integration_time.hh](#)
- [jeod_integration_time.cc](#)

8.20 jeod::LsodeControlDataInterface Class Reference

Specifies controls for an LSODE integrator.

```
#include <lsode_control_data_interface.hh>
```

Public Types

- enum [IntegrationMethod](#) { [ImplicitAdamsNonStiff](#) = 1 , [ImplicitBackDiffStiff](#) = 2 }
- enum [CorrectorMethod](#) { [FunctionalIteration](#) = 0 , [NewtonIterUserJac](#) = 1 , [NewtonIterInternalJac](#) = 2 , [JacobiNewtonInternalJac](#) = 3 , [NewtonIterUserBandJac](#) = 4 , [NewtonIterInternalBandJac](#) = 5 }
- enum [ErrorControlIndicator](#) { [CommonAbsCommonRel](#) = 1 , [SpecificAbsCommonRel](#) = 2 , [CommonAbsSpecificRel](#) = 3 , [SpecificAbsSpecificRel](#) = 4 }

Public Member Functions

- [LsodeControlDataInterface](#) ()
constructor
- virtual [~LsodeControlDataInterface](#) ()
- [LsodeControlDataInterface](#) (const [LsodeControlDataInterface](#) &)=default
- [LsodeControlDataInterface](#) & [operator=](#) (const [LsodeControlDataInterface](#) &)=delete
- void [check_interface_data](#) ()
verifies that the input data has legal values.
- void [set_rel_tol](#) (int index, double value)
set values from external
- void [set_abs_tol](#) (int index, double value)
- void [allocate_arrays](#) ()
allocates space for vector-populated data to allow for restart
- void [destroy_allocated_arrays](#) ()
De-allocates allocated array.
- bool [is_corrector_method_functional_iteration](#) ()
Tests whether corrector is functional iteration.

Data Fields

- [ErrorControlIndicator](#) [error_control_indicator](#) {CommonAbsCommonRel}
Was ITOL.
- std::vector< double > [abs_tolerance_error_control_vec](#)
Temporary pre-initialized place to store loaded error values.
- std::vector< double > [rel_tolerance_error_control_vec](#)
Temporary pre-initialized place to store loaded error values.
- bool [error_control_vector_copied_over](#) {}
- unsigned int [num_odes_at_alloc](#) {}
- double * [abs_tolerance_error_control](#) {}
Was ATOL.
- double * [rel_tolerance_error_control](#) {}
Was RTOL.
- unsigned int [num_odes](#) {3}
Was N, in DLS001 common block.
- [IntegrationMethod](#) [integration_method](#) {ImplicitAdamsNonStiff}
Was METH, in DLS001 common block.
- [CorrectorMethod](#) [corrector_method](#) {FunctionalIteration}
Was MITER, in DLS001 common block.
- double [min_step_size](#) {}
was HMIN, in DLS001 common block.
- double [max_step_size](#) {}
was HMAX.
- double [initial_step_size](#) {}
Was H0.
- unsigned int [max_order](#) {12}
Was MAXORD, in DLS001 common block.
- unsigned int [max_num_small_step_warnings](#) {10}
Was MXHNILI, in DLS001 common block.
- unsigned int [max_correction_iters](#) {3}
Was MAXCOR, in DLS001 common block Maximum number of corrector-iterations to attempt on any step.

- unsigned int `max_num_steps_jacobian` {20}
Was MSBP, in DLS001 common block.
- unsigned int `max_num_conv_failure` {10}
Was MXNCF, in DLS001 common block.
- unsigned int `max_num_steps` {500}
Was MXSTEP, in DLS001 common block.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 `p` (`jeod`, `LsodeControlDataInterface`)

8.20.1 Detailed Description

Specifies controls for an LSODE integrator.

Definition at line 83 of file `lsode_control_data_interface.hh`.

8.20.2 Member Enumeration Documentation

8.20.2.1 CorrectorMethod

```
enum jeod::LsodeControlDataInterface::CorrectorMethod
```

Enumerator

<code>FunctionalIteration</code>	Functional iteration.
<code>NewtonIterUserJac</code>	Modified Newton iteration with.
<code>NewtonIterInternalJac</code>	Modified Newton iteration with internally.
<code>JacobiNewtonInternalJac</code>	Modified Jacobi-Newton iteration with.
<code>NewtonIterUserBandJac</code>	Modified Newton iteration with.
<code>NewtonIterInternalBandJac</code>	Modified Newton iteration with internally.

Definition at line 101 of file `lsode_control_data_interface.hh`.

8.20.2.2 ErrorControlIndicator

```
enum jeod::LsodeControlDataInterface::ErrorControlIndicator
```

Enumerator

<code>CommonAbsCommonRel</code>	Use the same absolute and relative values.
<code>SpecificAbsCommonRel</code>	Use separate absolute values for each.
<code>CommonAbsSpecificRel</code>	Use a common absolute values and separate.
<code>SpecificAbsSpecificRel</code>	Use separate absolute and relative values.

Definition at line 121 of file lsode_control_data_interface.hh.

8.20.2.3 IntegrationMethod

```
enum jeod::LsodeControlDataInterface::IntegrationMethod
```

Enumerator

ImplicitAdamsNonStiff	Variable-step, variable-order, implicit Adams.
ImplicitBackDiffStiff	Variable-step, variable-order, implicit.

Definition at line 89 of file lsode_control_data_interface.hh.

8.20.3 Constructor & Destructor Documentation

8.20.3.1 ~LsodeControlDataInterface()

```
virtual jeod::LsodeControlDataInterface::~~LsodeControlDataInterface ( ) [inline], [virtual]
```

Definition at line 135 of file lsode_control_data_interface.hh.

8.20.3.2 LsodeControlDataInterface()

```
jeod::LsodeControlDataInterface::LsodeControlDataInterface (
    const LsodeControlDataInterface & ) [default]
```

8.20.4 Member Function Documentation

8.20.4.1 is_corrector_method_functional_iteration()

```
bool jeod::LsodeControlDataInterface::is_corrector_method_functional_iteration ( ) [inline]
```

Tests whether corrector is functional iteration.

Definition at line 152 of file lsode_control_data_interface.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part1(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_predict(), and jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1().

8.20.4.2 operator=()

```
LsodeControlDataInterface& jeod::LsodeControlDataInterface::operator= (
    const LsodeControlDataInterface & ) [delete]
```

8.20.4.3 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LsodeControlDataInterface::p (
    jeod ,
    LsodeControlDataInterface ) [private]
```

8.20.5 Field Documentation

8.20.5.1 abs_tolerance_error_control

```
double* jeod::LsodeControlDataInterface::abs_tolerance_error_control {}
```

Was ATOL.

Vector of the absolute error tolerances. `trick_units(-)`

Definition at line 178 of file `lsode_control_data_interface.hh`.

Referenced by `allocate_arrays()`, `destroy_allocated_arrays()`, `jeod::LsodeFirstOrderODEIntegrator::load_ew_↔values()`, `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2()`, and `set_abs_tol()`.

8.20.5.2 abs_tolerance_error_control_vec

```
std::vector<double> jeod::LsodeControlDataInterface::abs_tolerance_error_control_vec
```

Temporary pre-initialized place to store loaded error values.

`trick_units(-)`

Definition at line 166 of file `lsode_control_data_interface.hh`.

Referenced by `allocate_arrays()`, `check_interface_data()`, `LsodeControlDataInterface()`, and `set_abs_tol()`.

8.20.5.3 corrector_method

`CorrectorMethod` jeod::LsodeControlDataInterface::corrector_method {FunctionalIteration}

Was MITER, in DLS001 common block.

trick_units(-)

Definition at line 201 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up(), jeod::LsodeFirstOrderODEIntegrator::linear_chord_iteration(), and jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1().

8.20.5.4 error_control_indicator

`ErrorControlIndicator` jeod::LsodeControlDataInterface::error_control_indicator {CommonAbsCommonRel}

Was ITOL.

trick_units(-)

Definition at line 161 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), jeod::LsodeFirstOrderODEIntegrator::load_ew_values(), and jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2().

8.20.5.5 error_control_vector_copied_over

`bool` jeod::LsodeControlDataInterface::error_control_vector_copied_over {}

Definition at line 173 of file lsode_control_data_interface.hh.

Referenced by allocate_arrays(), check_interface_data(), destroy_allocated_arrays(), set_abs_tol(), and set_rel_tol().

8.20.5.6 initial_step_size

`double` jeod::LsodeControlDataInterface::initial_step_size {}

Was H0.

Initial guess at the step size. May be input, will be calculated if not. Note - this is the actual step, not the magnitude of the step. whereas max_step_size and min_step_size are magnitudes. trick_units(-)

Definition at line 221 of file lsode_control_data_interface.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2(), jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start(), and jeod::LsodeFirstOrderODEIntegrator::reset_integrator().

8.20.5.7 integration_method

```
IntegrationMethod jeod::LsodeControlDataInterface::integration_method {ImplicitAdamsNonStiff}
```

Was METH, in DLS001 common block.

trick_units(-)

Definition at line 196 of file lsode_control_data_interface.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::calculate_integration_coefficients(), check_interface_↔data(), jeod::LsodeFirstOrderODEIntegrator::integrator_core(), and jeod::LsodeFirstOrderODEIntegrator::update_↔_control_data().

8.20.5.8 max_correction_iters

```
unsigned int jeod::LsodeControlDataInterface::max_correction_iters {3}
```

Was MAXCOR, in DLS001 common block Maximum number of corrector-iterations to attempt on any step.

trick_units(-)

Definition at line 237 of file lsode_control_data_interface.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration().

8.20.5.9 max_num_conv_failure

```
unsigned int jeod::LsodeControlDataInterface::max_num_conv_failure {10}
```

Was MXNCF, in DLS001 common block.

Maximum number of convergence failures on one step. trick_units(-)

Definition at line 247 of file lsode_control_data_interface.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2().

8.20.5.10 max_num_small_step_warnings

```
unsigned int jeod::LsodeControlDataInterface::max_num_small_step_warnings {10}
```

Was MXHNILI, in DLS001 common block.

Populated from IWORK[7] Maximum number of small-step warnings that may be printed. trick_units(-)

Definition at line 232 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), and jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_↔part2().

8.20.5.11 max_num_steps

```
unsigned int jeod::LsodeControlDataInterface::max_num_steps {500}
```

Was MXSTEP, in DLS001 common block.

Maximum number of steps that the integrator may take. Default = 500. trick_units(-)

Definition at line 252 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), and jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_
part1().

8.20.5.12 max_num_steps_jacobian

```
unsigned int jeod::LsodeControlDataInterface::max_num_steps_jacobian {20}
```

Was MSBP, in DLS001 common block.

Populated from IWORK[6] Maximum number of steps for which the same Jacobian can be used. trick_units(-)

Definition at line 242 of file lsode_control_data_interface.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_predict().

8.20.5.13 max_order

```
unsigned int jeod::LsodeControlDataInterface::max_order {12}
```

Was MAXORD, in DLS001 common block.

Populated from IWORK[5] Maximum order allowable. trick_units(-)

Definition at line 227 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), and jeod::LsodeFirstOrderODEIntegrator::update_control_data().

8.20.5.14 max_step_size

```
double jeod::LsodeControlDataInterface::max_step_size {}
```

was HMAX.

RWORK[6] Maximum absolute value of step size allowable. Default to 0.0, interpreted as infinity. user-specified otherwise. trick_units(-)

Definition at line 214 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), and jeod::LsodeFirstOrderODEIntegrator::update_control_data().

8.20.5.15 min_step_size

```
double jeod::LsodeControlDataInterface::min_step_size {}
```

was HMIN, in DLS001 common block.

Minimum absolute value of step size allowable. Default to 0.0, user-specified otherwise. trick_units(-)

Definition at line 208 of file lsode_control_data_interface.hh.

Referenced by check_interface_data(), jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs(), jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order(), and jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part3().

8.20.5.16 num_odes

```
unsigned int jeod::LsodeControlDataInterface::num_odes {3}
```

Was N, in DLS001 common block.

Number of ODEs to be solved at next step. In this implementation, num_odes = num_equations. In original implementation, num_odes (N) was set to NEQ at the start, the some subset could be identified, NYH and solved for. trick_units(-)

Definition at line 191 of file lsode_control_data_interface.hh.

Referenced by allocate_arrays(), check_interface_data(), jeod::LsodeFirstOrderODEIntegrator::gauss_elim_factor(), jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order(), jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part1(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh(), jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup(), jeod::LsodeFirstOrderODEIntegrator::interpolate_y(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up(), jeod::LsodeFirstOrderODEIntegrator::linear_chord_iteration(), jeod::LsodeFirstOrderODEIntegrator::linear_solver(), jeod::LsodeFirstOrderODEIntegrator::load_derivatives(), jeod::LsodeFirstOrderODEIntegrator::load_new_values(), jeod::LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator(), jeod::LsodeFirstOrderODEIntegrator::magnitude_of_weighted_array(), jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1(), jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2(), jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1(), jeod::LsodeFirstOrderODEIntegrator::process_entry_point_cycle_start(), and jeod::LsodeFirstOrderODEIntegrator::reset_integrator().

8.20.5.17 num_odes_at_alloc

```
unsigned int jeod::LsodeControlDataInterface::num_odes_at_alloc {}
```

Definition at line 174 of file lsode_control_data_interface.hh.

Referenced by allocate_arrays(), set_abs_tol(), and set_rel_tol().

8.20.5.18 rel_tolerance_error_control

```
double* jeod::LsodeControlDataInterface::rel_tolerance_error_control {}
```

Was RTOL.

Vector of the relative error tolerances. `trick_units(-)`

Definition at line 182 of file `lsode_control_data_interface.hh`.

Referenced by `allocate_arrays()`, `destroy_allocated_arrays()`, `jeod::LsodeFirstOrderODEIntegrator::load_ew_↔values()`, `jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part2()`, and `set_rel_tol()`.

8.20.5.19 rel_tolerance_error_control_vec

```
std::vector<double> jeod::LsodeControlDataInterface::rel_tolerance_error_control_vec
```

Temporary pre-initialized place to store loaded error values.

`trick_units(-)`

Definition at line 171 of file `lsode_control_data_interface.hh`.

Referenced by `allocate_arrays()`, `check_interface_data()`, `LsodeControlDataInterface()`, and `set_rel_tol()`.

The documentation for this class was generated from the following files:

- [lsode_control_data_interface.hh](#)
- [lsode_control_data_interface.cc](#)

8.21 jeod::LsodeDataArrays Class Reference

The data arrays.

```
#include <lsode_data_classes.hh>
```

Public Member Functions

- [LsodeDataArrays](#) ()=default
- virtual [~LsodeDataArrays](#) ()
- [LsodeDataArrays](#) & `operator=` (const [LsodeDataArrays](#) &)=delete
- [LsodeDataArrays](#) (const [LsodeDataArrays](#) &)=delete
- void [allocate_arrays](#) (unsigned int `num_odes`, [LsodeControlDataInterface::CorrectorMethod](#) `corrector_↔method`)
Allocates memory for the variable size arrays.
- void [destroy_allocated_arrays](#) ()
Allows for refactoring and reallocation of newly sized arrays.

Data Fields

- int * [pivots](#) {}
Was IWM(21) or IPVT.
- double ** [history](#) {}
was RWORK[LYH:LYH+NYH(MAXORD+1)-1].*
- double [lin_alg_1](#) {}
- double [lin_alg_2](#) {}
- double ** [lin_alg](#) {}
was RWORK[LWM:LWM+LENWM-1].
- double * [error_weight](#) {}
was RWORK[LEWT:LEWT+N-1].
- double * [save](#) {}
was RWORK[LSAVF:LSAVF+N-1].
- double * [accum_correction](#) {}
was RWORK[LACOR:LACOR+N-1].
- unsigned int [lin_alg_index1](#) {}
Number of record, this is the value used for data allocation.
- unsigned int [num_odes](#) {3}
Number of record, this is the value used for data allocation.
- bool [allocated](#) {}
Indicator of whether the arrays have been allocated.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) ([jeod](#), [LsodeDataArrays](#))

8.21.1 Detailed Description

The data arrays.

Definition at line 107 of file `lsode_data_classes.hh`.

8.21.2 Constructor & Destructor Documentation

8.21.2.1 LsodeDataArrays() [1/2]

```
jeod::LsodeDataArrays::LsodeDataArrays ( ) [default]
```

8.21.2.2 ~LsodeDataArrays()

```
virtual jeod::LsodeDataArrays::~~LsodeDataArrays ( ) [inline], [virtual]
```

Definition at line 112 of file `lsode_data_classes.hh`.

8.21.2.3 LsodeDataArrays() [2/2]

```
jeod::LsodeDataArrays::LsodeDataArrays (
    const LsodeDataArrays & ) [delete]
```

8.21.3 Member Function Documentation

8.21.3.1 operator=()

```
LsodeDataArrays& jeod::LsodeDataArrays::operator= (
    const LsodeDataArrays & ) [delete]
```

8.21.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LsodeDataArrays::p (
    jeod ,
    LsodeDataArrays ) [private]
```

8.21.4 Field Documentation

8.21.4.1 accum_correction

```
double* jeod::LsodeDataArrays::accum_correction {}
```

was RWORK[LACOR:LACOR+N-1].

LACOR = LSAVF + N acum_correction[i] = rwork[lacor+i]. trick_units(-)

Definition at line 173 of file lsode_data_classes.hh.

Referenced by `allocate_arrays()`, `destroy_allocated_arrays()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_iteration_loop_part2()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_wrapup()`, and `jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop()`.

8.21.4.2 allocated

```
bool jeod::LsodeDataArrays::allocated {}
```

Indicator of whether the arrays have been allocated.

trick_units(-)

Definition at line 187 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), and destroy_allocated_arrays().

8.21.4.3 error_weight

```
double* jeod::LsodeDataArrays::error_weight {}
```

was RWORK[LEWT:LEWT+N-1].

LEWT = LWM + LENWM error_weight[i] = rwork[lewt+i]. trick_units(-)

Definition at line 163 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), destroy_allocated_arrays(), jeod::LsodeFirstOrderODEIntegrator::jacobian_↵
prep_init(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop(), jeod::LsodeFirstOrderODEIntegrator_↵
::jacobian_prep_wrap_up(), jeod::LsodeFirstOrderODEIntegrator::load_ew_values(), jeod::LsodeFirstOrder_↵
ODEIntegrator::magnitude_of_weighted_array(), jeod::LsodeFirstOrderODEIntegrator::manager_initialize_↵
calculation_part2(), and jeod::LsodeFirstOrderODEIntegrator::manager_integration_loop_part1().

8.21.4.4 history

```
double** jeod::LsodeDataArrays::history {}
```

was RWORK[LYH:LYH+NYH*(MAXORD+1)-1].

LYH = 21 First index is to "i" in y_i, second index is to history order. history[i,j] = rwork[LYH + j*nyh + i], with lyh = 21 typically. trick_units(-)

Definition at line 142 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), destroy_allocated_arrays(), jeod::LsodeFirstOrderODEIntegrator::integrator_↵
_compute_new_order(), jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep(), jeod_↵
::LsodeFirstOrderODEIntegrator::integrator_corrector_converged(), jeod::LsodeFirstOrderODEIntegrator_↵
::integrator_corrector_failed_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration(), jeod_↵
::LsodeFirstOrderODEIntegrator::integrator_error_test_failed(), jeod::LsodeFirstOrderODEIntegrator::integrator_↵
_fail_reset_order_1_part1(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2(), jeod_↵
::LsodeFirstOrderODEIntegrator::integrator_predict(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_↵
_iteration_loop_part1(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh(), jeod::LsodeFirstOrder_↵
ODEIntegrator::interpolate_y(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrder_↵
ODEIntegrator::jacobian_prep_wrap_up(), jeod::LsodeFirstOrderODEIntegrator::load_ew_values(), jeod::Lsode_↵
FirstOrderODEIntegrator::manager_initialize_calculation_part2(), jeod::LsodeFirstOrderODEIntegrator::manager_↵
_integration_loop_part2(), and jeod::LsodeFirstOrderODEIntegrator::reset_integrator().

8.21.4.5 lin_alg

```
double** jeod::LsodeDataArrays::lin_alg {}
```

was RWORK[LWM:LWM+LENWM-1].

$LWM = LYH + (NYH * (MAXORD + 1))$ lin_alg_1 = rwork[lwm] lin_alg_2 = rwork[lwm + 1] lin_alg[i,j] = rwork[lwm + j*n + i + 2]. The first two elements are treated differently, then it goes to an array that is sized based on the correction_↵_method. The array sizes are as follows, ordered by value of correction_method: 0: 0 1,2: n x n 3: 1 x n 4,5: (2*ml+mu+1) x n. trick_units(−)

Definition at line 158 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), destroy_allocated_arrays(), jeod::LsodeFirstOrderODEIntegrator::gauss_↵elim_factor(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrderODEIntegrator↵::jacobian_prep_loop(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up(), jeod::LsodeFirstOrder↵ODEIntegrator::linear_chord_iteration(), and jeod::LsodeFirstOrderODEIntegrator::linear_solver().

8.21.4.6 lin_alg_1

```
double jeod::LsodeDataArrays::lin_alg_1 {}
```

Definition at line 143 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrderODEIntegrator↵::jacobian_prep_loop(), and jeod::LsodeFirstOrderODEIntegrator::manager_initialize_calculation_part1().

8.21.4.7 lin_alg_2

```
double jeod::LsodeDataArrays::lin_alg_2 {}
```

Definition at line 144 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), and jeod::LsodeFirstOrderODEIntegrator↵::linear_chord_iteration().

8.21.4.8 lin_alg_index1

```
unsigned int jeod::LsodeDataArrays::lin_alg_index1 {}
```

Number of record, this is the value used for data allocation.

trick_units(−)

Definition at line 178 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), and destroy_allocated_arrays().

8.21.4.9 num_odes

```
unsigned int jeod::LsodeDataArrays::num_odes {3}
```

Number of record, this is the value used for data allocation.

trick_units(-)

Definition at line 182 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), and destroy_allocated_arrays().

8.21.4.10 pivots

```
int* jeod::LsodeDataArrays::pivots {}
```

Was IWM(21) or IPVT.

Pivot vector generated in dgefa, and used in dgesl. trick_units(-)

Definition at line 127 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), destroy_allocated_arrays(), jeod::LsodeFirstOrderODEIntegrator::gauss_elim_factor(), and jeod::LsodeFirstOrderODEIntegrator::linear_solver().

8.21.4.11 save

```
double* jeod::LsodeDataArrays::save {}
```

was RWORK[LSAVF:LSAVF+N-1].

LSAVF = LEWT + N save[i] = rwork[lsavf+i]. trick_units(-)

Definition at line 168 of file lsode_data_classes.hh.

Referenced by allocate_arrays(), destroy_allocated_arrays(), jeod::LsodeFirstOrderODEIntegrator::integrate(), jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep(), jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_iteration(), jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_loop(), and jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up().

The documentation for this class was generated from the following files:

- [lsode_data_classes.hh](#)
- [lsode_data_classes.cc](#)

8.22 jeod::LsodeDataJacobianPrep Class Reference

Data associated with the method DPREPJ.

```
#include <lsode_data_classes.hh>
```

Public Member Functions

- [LsodeDataJacobianPrep](#) ()=default
- virtual [~LsodeDataJacobianPrep](#) ()=default
- [LsodeDataJacobianPrep](#) & [operator=](#) (const [LsodeDataJacobianPrep](#) &)=delete
- [LsodeDataJacobianPrep](#) (const [LsodeDataJacobianPrep](#) &)=delete

Data Fields

- double [fac](#) {}
- double [hl0](#) {}
- int [index](#) {}
- int [index_max](#) {}
- double [r0](#) {}
- double [yj](#) {}

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [LsodeDataJacobianPrep](#))

8.22.1 Detailed Description

Data associated with the method DPREPJ.

Definition at line 86 of file `lsode_data_classes.hh`.

8.22.2 Constructor & Destructor Documentation

8.22.2.1 LsodeDataJacobianPrep() [1/2]

```
jeod::LsodeDataJacobianPrep::LsodeDataJacobianPrep ( ) [default]
```

8.22.2.2 ~LsodeDataJacobianPrep()

```
virtual jeod::LsodeDataJacobianPrep::~~LsodeDataJacobianPrep ( ) [virtual], [default]
```

8.22.2.3 LsodeDataJacobianPrep() [2/2]

```
jeod::LsodeDataJacobianPrep::LsodeDataJacobianPrep (
    const LsodeDataJacobianPrep & ) [delete]
```

8.22.3 Member Function Documentation

8.22.3.1 operator=()

```
LsodeDataJacobianPrep& jeod::LsodeDataJacobianPrep::operator= (
    const LsodeDataJacobianPrep & ) [delete]
```

8.22.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LsodeDataJacobianPrep::p (
    jeod ,
    LsodeDataJacobianPrep ) [private]
```

8.22.4 Field Documentation

8.22.4.1 fac

```
double jeod::LsodeDataJacobianPrep::fac {}
```

Definition at line 96 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), and jeod::LsodeFirstOrderODEIntegrator↵
::jacobian_prep_loop().

8.22.4.2 hl0

```
double jeod::LsodeDataJacobianPrep::hl0 {}
```

Definition at line 97 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), jeod::LsodeFirstOrderODEIntegrator↵
::jacobian_prep_loop(), and jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_wrap_up().

8.22.4.3 index

```
int jeod::LsodeDataJacobianPrep::index {}
```

Definition at line 98 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), and jeod::LsodeFirstOrderODEIntegrator↵
::jacobian_prep_loop().

8.22.4.4 index_max

```
int jeod::LsodeDataJacobianPrep::index_max {}
```

Definition at line 99 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), and jeod::LsodeFirstOrderODEIntegrator↵
::jacobian_prep_loop().

8.22.4.5 r0

```
double jeod::LsodeDataJacobianPrep::r0 {}
```

Definition at line 100 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), and jeod::LsodeFirstOrderODEIntegrator↵
::jacobian_prep_loop().

8.22.4.6 yj

```
double jeod::LsodeDataJacobianPrep::yj {}
```

Definition at line 101 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::jacobian_prep_init(), and jeod::LsodeFirstOrderODEIntegrator↵
::jacobian_prep_loop().

The documentation for this class was generated from the following file:

- [lsode_data_classes.hh](#)

8.23 jeod::LsodeDataStode Class Reference

The data associated with method Dstode.

```
#include <lsode_data_classes.hh>
```

Public Member Functions

- [LsodeDataStode](#) ()=default
- virtual [~LsodeDataStode](#) ()=default
- [LsodeDataStode](#) & [operator=](#) (const [LsodeDataStode](#) &)=delete
- [LsodeDataStode](#) (const [LsodeDataStode](#) &)=delete

Data Fields

- double [step_ratio](#) {}
- double [step_ratio_order_inc](#) {}
- double [told](#) {}
- double [dsm](#) {}
- int [iredo](#) {}
- int [iret](#) {}
- unsigned int [ncf](#) {}
- unsigned int [new_method_order](#) {}

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 [p](#) (jeod, [LsodeDataStode](#))

8.23.1 Detailed Description

The data associated with method Dstode.

Definition at line 193 of file lsode_data_classes.hh.

8.23.2 Constructor & Destructor Documentation

8.23.2.1 [LsodeDataStode](#)() [1/2]

```
jeod::LsodeDataStode::LsodeDataStode ( ) [default]
```

8.23.2.2 [~LsodeDataStode](#)()

```
virtual jeod::LsodeDataStode::~~LsodeDataStode ( ) [virtual], [default]
```

8.23.2.3 LsodeDataStode() [2/2]

```
jeod::LsodeDataStode::LsodeDataStode (
    const LsodeDataStode & ) [delete]
```

8.23.3 Member Function Documentation

8.23.3.1 operator=()

```
LsodeDataStode& jeod::LsodeDataStode::operator= (
    const LsodeDataStode & ) [delete]
```

8.23.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::LsodeDataStode::p (
    jeod ,
    LsodeDataStode ) [private]
```

8.23.4 Field Documentation

8.23.4.1 dsm

```
double jeod::LsodeDataStode::dsm {}
```

Definition at line 207 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order(), and jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged().

8.23.4.2 iredo

```
int jeod::LsodeDataStode::iredo {}
```

Definition at line 208 of file lsode_data_classes.hh.

Referenced by jeod::LsodeFirstOrderODEIntegrator::integrator_core(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_converged(), jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2(), jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed(), jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh(), and jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change().

8.23.4.3 iret

```
int jeod::LsodeDataStode::iret {}
```

Definition at line 209 of file `lsode_data_classes.hh`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part2()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order()`.

8.23.4.4 ncf

```
unsigned int jeod::LsodeDataStode::ncf {}
```

Definition at line 210 of file `lsode_data_classes.hh`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2()`.

8.23.4.5 new_method_order

```
unsigned int jeod::LsodeDataStode::new_method_order {}
```

Definition at line 211 of file `lsode_data_classes.hh`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order()`.

8.23.4.6 step_ratio

```
double jeod::LsodeDataStode::step_ratio {}
```

Definition at line 204 of file `lsode_data_classes.hh`.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_check_step_error()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_fail_reset_order_1_part1()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_method_coeffs()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_reset_yh()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_set_new_order()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_test_stepsize_change()`.

8.23.4.7 step_ratio_order_inc

```
double jeod::LsodeDataStode::step_ratio_order_inc {}
```

Definition at line 205 of file lsode_data_classes.hh.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_compute_new_order_prep()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed()`.

8.23.4.8 told

```
double jeod::LsodeDataStode::told {}
```

Definition at line 206 of file lsode_data_classes.hh.

Referenced by `jeod::LsodeFirstOrderODEIntegrator::integrator_core()`, `jeod::LsodeFirstOrderODEIntegrator::integrator_corrector_failed_part2()`, and `jeod::LsodeFirstOrderODEIntegrator::integrator_error_test_failed()`.

The documentation for this class was generated from the following file:

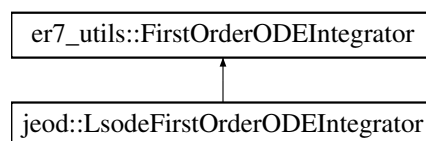
- [lsode_data_classes.hh](#)

8.24 jeod::LsodeFirstOrderODEIntegrator Class Reference

Jeod-compatible version of the Livermore ODE solver, LSODE.

```
#include <lsode_first_order_ode_integrator.hh>
```

Inheritance diagram for `jeod::LsodeFirstOrderODEIntegrator`:



Public Types

- enum `EntryPoint` {
`CycleStartFinish` = 0 , `InitCalc` = 1 , `JacobianPrep` = 2 , `ResetIterLoop` = 3 ,
`IterationLoop` = 4 , `DstodeResetStep` = 5 }
- enum `CalculationTask` {
`Normal` = 1 , `OneStep` = 2 , `CompleteCycle` = 3 , `NormalWithSingularity` = 4 ,
`OneStepWithSingularity` = 5 }

Public Member Functions

- [LsodeFirstOrderODEIntegrator](#) ()
LsodeFirstOrderODEIntegrator default constructor.
- [LsodeFirstOrderODEIntegrator](#) (const [LsodeControlDataInterface](#) &data_in, [er7_utils::IntegrationControls](#) &controls, unsigned int size)
LsodeFirstOrderODEIntegrator non-default constructor.
- [~LsodeFirstOrderODEIntegrator](#) () override
LsodeFirstOrderODEIntegrator destructor.
- [LsodeFirstOrderODEIntegrator](#) & operator= (const [LsodeFirstOrderODEIntegrator](#) &)=delete
- [LsodeFirstOrderODEIntegrator](#) (const [LsodeFirstOrderODEIntegrator](#) &)=delete
- [LsodeFirstOrderODEIntegrator](#) * [create_copy](#) () const override
Create a copy of 'this' *LsodeFirstOrderODEIntegrator* object.
- [EntryPoint](#) [get_re_entry_point](#) ()
Get *re_entry_point* member.
- [er7_utils::IntegratorResult](#) [integrate](#) (double dyn_dt, unsigned int target_stage, const double *[y_dot](#), double *[y](#)) override
Propagate state via *Lsode*'s method.
- void [reset_integrator](#) () override
Resets the integrator when the timestep changes or when identified as needing a reset.
- void [update_control_data](#) ()
Gets the control data from where it can be populated in the constructor and verifies that the input control parameters are not out of sensible range.

Data Fields

- double * [y](#) {}
Was *Y*.
- const double * [y_dot](#) {}
Was *.*
- double [cycle_target_time](#) {}
Was *TOUT*.
- double [convergence_factor](#) {}
was *CONIT*, in *DLS001* common block.
- double [convergence_rate](#) {}
was *CRATE*, in *DLS001* common block.
- unsigned int [order_select_para](#) {}
Was *IALTH*, in *DLS001* common block.
- unsigned int [num_equations](#) {1}
Was *NYH*, in *DLS001* common block.
- unsigned int [num_nordsiek_cols](#) {1}
Was *L*, in *DLS001* common block Number of columns in *Nordsiek* array.
- unsigned int [max_history_size](#) {12}
Was *LMAX*, in *DLS001* common block Maximum allowable number of histories.
- unsigned int [num_predictor_elements](#) {1}
Was *NQNYH*, in *DLS001* common block.
- unsigned int [method_order_current](#) {}
Was *NQ*, in *DLS001* common block.
- double [stage_target_time](#) {}
was *TN*, in *DLS001* common block.
- double [max_step_increase_ratio](#) {1.0}
was *RMAX*, in *DLS001* common block.
- double [max_rel_change_without_jacobian](#) {0.3}
was *CCMAX*, in *DLS001* common block.

Protected Member Functions

- void [process_entry_point_cycle_start](#) ()
The code block from the main integrate routine for re_entry_point=CycleStartFinish.
- void [manager_initialize_calculation_part1](#) ()
Sets the values for the case with calculation_phase = 1.
- void [manager_initialize_calculation_part2](#) ()
- int [manager_check_stop_conditions](#) ()
verifies whether the convergence conditions have been met to end the cycle.
- void [manager_integration_loop_part1](#) ()
The iteration loop for the integration process.
- void [manager_integration_loop_part2](#) ()
- void [manager_integration_loop_part3](#) ()
- void [manager_set_calculation_phase_eq_2_reload](#) ()
- void [integrator_core](#) ()
*integrator_core provides the front-end to all of the integrator_*i methods, which together perform one step of the integration of an initial value problem for a system of ordinary differential equations.*
- void [integrator_reset_method_coeffs](#) ()
Sets/resets the method_coeffs_current array.
- void [integrator_test_stepsize_change](#) ()
Tests h against old h.
- void [integrator_reset_yh](#) ()
Resets history arrays and time-step.
- void [integrator_predict](#) ()
This section computes the predicted values by multiplying the history array by the Pascal Triangle matrix.
- void [integrator_reset_iteration_loop_part1](#) ()
This method resets the iteration loop to the values generated by the integrator_predict method, which populated history[][0].*
- void [integrator_reset_iteration_loop_part2](#) ()
This code follows part 1 after the break-out to get to the external calls, and completes the reset of the integration iteration loop.
- void [integrator_corrector_iteration](#) ()
Keeps looping through the iterations until convergence or failure.
- void [integrator_corrector_failed_part1](#) ()
The corrector iteration failed to converge.
- void [integrator_corrector_failed_part2](#) ()
Retracts the history array in the case that the correction iteration failed to converge with either functional iteration or with an up-to-date Jacobian.
- void [integrator_corrector_converged](#) ()
Starts the processing of a converged iteration.
- void [integrator_error_test_failed](#) ()
Restores the history array following the failure of the corrector for exceeding local error bounds.
- void [integrator_compute_new_order_prep](#) ()
The first steps in computing whether the order of the integrator should be changed.
- void [integrator_compute_new_order](#) ()
Computes the step-size scaling factors that will result once it is determined what happens to the order at the next step.
- void [integrator_compute_new_order_check_step_error](#) ()
- void [integrator_set_new_order](#) ()
Sets the new order and the step-ratio for the next step - or the current step if redoing it.
- void [integrator_fail_reset_order_1_part1](#) ()

- Repopulates the y-array from history, recomputes the original first derivatives, sets the order back to 1, and reduces the step size by a factor of 10.*
- void [integrator_fail_reset_order_1_part2](#) ()
Continue reset, with derivatives now at hand.
- void [integrator_wrapup](#) ()
Wraps up the completion of the integrator.
- void [integrator_terminate](#) ()
this is the only succesful path back from integrator to manager.
- void [calculate_epsilon](#) ()
Identify the smallest double precision value, epsilon, such that the computer can distinguish (1+epsilon) from 1.
- void [calculate_integration_coefficients](#) ()
Modified from original DCFODE subroutine.
- void [interpolate_y](#) ()
Interpolates the zeroth-derivative of y Adapted from subroutine DINTDY, which was a general method for interpolating the K-th derivative of the dependent variable vector, y.
- void [jacobian_prep_init](#) ()
Modified from DPREPJ.
- bool [jacobian_prep_loop](#) ()
- bool [jacobian_prep_wrap_up](#) ()
- void [linear_chord_iteration](#) ()
Modified from DSOLSY.
- void [load_ew_values](#) ()
- double [magnitude_of_weighted_array](#) (const double *v)
returns the RMS value of {V dot W}, where V and W are N-vectors.
- double [magnitude_of_weighted_array](#) (unsigned int ix, double **v)
returns RMS value of v[][index]*
- int [gauss_elim_factor](#) ()
Factors a double array (arrays.lin_alg) by Gaussian elimination.
- void [linear_solver](#) ()
Solves the equation $Y' = A Y$, with $A = \text{arrays.lin_alg}$.
- unsigned int [index_of_max_magnitude](#) (unsigned int num_points, double **mx, int starting_ix)
Modified version of IDAMAX.
- void [load_derivatives](#) (double *derivs)
Load the externally generated derivative values (incoming as y_dot)i into the array derivs.

Protected Attributes

- [LsodeDataJacobianPrep data_prepj](#)
data used exclusively for the DPREPJ method.
- [LsodeDataArrays arrays](#)
data arrays, multiple purposes.
- [LsodeDataStode data_stode](#)
data used exclusively for the DSTODE method.
- unsigned int [num_steps_taken](#) {}
Was NST, in DLS001 common block.
- unsigned int [prior_num_steps](#) {}
Was NSLAST, in DLS001 common block.
- int [step_error](#) {}
Was KFLAG, in DLS001 common block 0: step was successful -1: requested accuracy could not be achieved.
- unsigned int [num_small_step_warnings](#) {}
Was NHNIL, in DLS001 common block.

- unsigned int [num_jacobian_evals](#) {}
Was NJE, in DLS001 common block Number of jacobian evaluations so far for the problem.
- double [iter_delta](#) {}
Was DEL, in DSTODE, local variable.
- double [prev_iter_delta](#) {}
Was DELP, in DSTODE, local variable.
- bool [first_pass](#) {true}
was ISTATE.
- [EntryPoint](#) [re_entry_point](#) {CycleStartFinish}
Indicates where in the integrator to return to following an exit to gether new derivatives.
- bool [initialized](#) {}
was INIT, in DLS001 common block.
- int [internal_state](#) {}
Was JSTART, in DLS001 common block.
- [CalculationTask](#) [calculation_task](#) {Normal}
Was ITASK.
- unsigned int [max_order_internal](#) {}
Was MAXORD.
- [LsodeControlDataInterface](#) [control_data](#)
- [LsodeControlDataInterface::IntegrationMethod](#) [prev_integration_method](#)
Was MEO, in DLS001 common block Integration method used in previous call (see integration_method).
- unsigned int [prev_method_order](#) {}
Was NQU, in DLS001 common block.
- double [method_coeff_first](#) {}
was EL0, in DLS001 common block.
- double [method_coeffs_current](#) [13] {}
was EL, in DLS001 common block.
- double [method_coeffs_complete](#) [13][12] {}
was ELCO, in DLS001 common block.
- double [test_coeffs_complete](#) [3][12] {}
was TESCO, in DLS001 common block.
- double [step_size](#) {}
was H, in DLS001 common block.
- double [prev_step_size](#) {}
was HOLD, in DLS001 common block.
- double [prev_good_step_size](#) {}
was HU, in DLS001 common block.
- double [max_step_size_inv](#) {}
was HMXI.
- bool [jacobian_current](#) {}
Was JCUR, in DLS001 common block Is the jacobian current.
- bool [update_jacobian](#) {true}
Was IPUP, in DLS001 common block.
- unsigned int [step_at_last_jacobian_update](#) {}
Was NSLP, in DLS001 common block.
- unsigned int [convergence_jacobian_flag](#) {}
Was ICF, in DLS001 common block.
- double [rel_change_since_jacobian](#) {}
was RC, in DLS001 common block.
- bool [iteration_matrix_singular](#) {}
Was IERPJ, in DLS001 common block.

- bool `modified_iteration_matrix_singular` {}
Was IERSL, in DLS001 common block.
- unsigned int `iteration_count` {}
- double `epsilon` {1.0E-12}
was UROUND, in DLS001 common block.
- double `sqrt_epsilon` {1.0E-6}
NEW.

Private Member Functions

- `JEOD_CLASS_ESTABLISH_FRIENDS2` (jeod, `LsodeFirstOrderODEIntegrator`)

8.24.1 Detailed Description

Jeod-compatible version of the Livermore ODE solver, LSODE.

Definition at line 96 of file `lsode_first_order_ode_integrator.hh`.

8.24.2 Member Enumeration Documentation

8.24.2.1 CalculationTask

```
enum jeod::LsodeFirstOrderODEIntegrator::CalculationTask
```

Enumerator

Normal	Normal operation. Interpolate to target.
OneStep	Take only one step and return.
CompleteCycle	Stop at first mesh point at or beyond.
NormalWithSingularity	Normal computation, with safeguard on.
OneStepWithSingularity	Take one step without passing t_crit.

Definition at line 127 of file `lsode_first_order_ode_integrator.hh`.

8.24.2.2 EntryPoint

```
enum jeod::LsodeFirstOrderODEIntegrator::EntryPoint
```

Enumerator

CycleStartFinish	Default value. Assumption is that the current.
------------------	--

Enumerator

InitCalc	Reset during initialization. Valid only during.
JacobianPrep	Set at the end of the initialization of the.
ResetIterLoop	Set when the iteration loop (part of DSTODE) has.
IterationLoop	Set during the routine operation of the iteration.
DstodeResetStep	Set in dstode_640 when there have been too many.

Definition at line 104 of file lsode_first_order_ode_integrator.hh.

8.24.3 Constructor & Destructor Documentation

8.24.3.1 LsodeFirstOrderODEIntegrator()

```
jeod::LsodeFirstOrderODEIntegrator::LsodeFirstOrderODEIntegrator (
    const LsodeFirstOrderODEIntegrator & ) [delete]
```

8.24.4 Member Function Documentation

8.24.4.1 get_re_entry_point()

```
EntryPoint jeod::LsodeFirstOrderODEIntegrator::get_re_entry_point ( ) [inline]
```

Get re_entry_point member.

Definition at line 181 of file lsode_first_order_ode_integrator.hh.

Referenced by jeod::LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator().

8.24.4.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::LsodeFirstOrderODEIntegrator::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    LsodeFirstOrderODEIntegrator ) [private]
```

8.24.4.3 operator=()

```
LsodeFirstOrderODEIntegrator& jeod::LsodeFirstOrderODEIntegrator::operator= (
    const LsodeFirstOrderODEIntegrator & ) [delete]
```

8.24.5 Field Documentation

8.24.5.1 arrays

`LsodeDataArrays` jeod::LsodeFirstOrderODEIntegrator::arrays [protected]

data arrays, multiple purposes.

trick_units(-)

Definition at line 393 of file lsode_first_order_ode_integrator.hh.

Referenced by `gauss_elim_factor()`, `integrate()`, `integrator_compute_new_order()`, `integrator_compute_new_order_prep()`, `integrator_core()`, `integrator_corrector_converged()`, `integrator_corrector_failed_part2()`, `integrator_corrector_iteration()`, `integrator_error_test_failed()`, `integrator_fail_reset_order_1_part1()`, `integrator_fail_reset_order_1_part2()`, `integrator_predict()`, `integrator_reset_iteration_loop_part1()`, `integrator_reset_iteration_loop_part2()`, `integrator_reset_yh()`, `integrator_wrapup()`, `interpolate_y()`, `jacobian_prep_init()`, `jacobian_prep_loop()`, `jacobian_prep_wrap_up()`, `linear_chord_iteration()`, `linear_solver()`, `load_ew_values()`, `magnitude_of_weighted_array()`, `manager_initialize_calculation_part1()`, `manager_initialize_calculation_part2()`, `manager_integration_loop_part1()`, `manager_integration_loop_part2()`, `reset_integrator()`, and `~LsodeFirstOrderODEIntegrator()`.

8.24.5.2 calculation_task

`CalculationTask` jeod::LsodeFirstOrderODEIntegrator::calculation_task {Normal} [protected]

Was ITASK.

1: Normal 2:Take one step and return. 3:Stop at first mesh point at or beyond cycle_target_time and return 4:Normal computation, with safeguard on singularity time, t_crit 5:Take one step without passing t_crit. This implementation only allows for calculation_task = 1, so it is a protected variable until such time as it is extended to include additional options.

Only case 1 is supported. trick_units(-)

Definition at line 481 of file lsode_first_order_ode_integrator.hh.

Referenced by `manager_check_stop_conditions()`, and `manager_integration_loop_part3()`.

8.24.5.3 control_data

`LsodeControlDataInterface` jeod::LsodeFirstOrderODEIntegrator::control_data [protected]

Definition at line 488 of file lsode_first_order_ode_integrator.hh.

Referenced by `calculate_integration_coefficients()`, `gauss_elim_factor()`, `integrator_compute_new_order()`, `integrator_compute_new_order_prep()`, `integrator_core()`, `integrator_corrector_converged()`, `integrator_corrector_failed_part1()`, `integrator_corrector_failed_part2()`, `integrator_corrector_iteration()`, `integrator_error_test_failed()`, `integrator_fail_reset_order_1_part1()`, `integrator_fail_reset_order_1_part2()`, `integrator_predict()`, `integrator_reset_iteration_loop_part1()`, `integrator_reset_iteration_loop_part2()`, `integrator_reset_method_coeffs()`, `integrator_reset_yh()`, `integrator_set_new_order()`, `integrator_wrapup()`, `interpolate_y()`, `jacobian_prep_init()`, `jacobian_prep_loop()`, `jacobian_prep_wrap_up()`, `linear_chord_iteration()`, `linear_solver()`, `load_derivatives()`, `load_ew_values()`, `LsodeFirstOrderODEIntegrator()`, `magnitude_of_weighted_array()`, `manager_initialize_calculation_part1()`, `manager_initialize_calculation_part2()`, `manager_integration_loop_part1()`, `manager_integration_loop_part2()`, `manager_integration_loop_part3()`, `process_entry_point_cycle_start()`, `reset_integrator()`, `update_control_data()`, and `~LsodeFirstOrderODEIntegrator()`.

8.24.5.4 convergence_factor

```
double jeod::LsodeFirstOrderODEIntegrator::convergence_factor {}
```

was CONIT, in DLS001 common block.

trick_units(-)

Definition at line 297 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_iteration(), and integrator_reset_method_coeffs().

8.24.5.5 convergence_jacobian_flag

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::convergence_jacobian_flag {} [protected]
```

Was ICF, in DLS001 common block.

0: Solution converged 1: Convergence failed; Jacobian is not current. 2: Convergence failed; Jacobian is current or not needed. trick_units(-)

Definition at line 569 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_failed_part1(), and integrator_corrector_failed_part2().

8.24.5.6 convergence_rate

```
double jeod::LsodeFirstOrderODEIntegrator::convergence_rate {}
```

was CRATE, in DLS001 common block.

trick_units(-)

Definition at line 301 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_iteration(), and jacobian_prep_wrap_up().

8.24.5.7 cycle_target_time

```
double jeod::LsodeFirstOrderODEIntegrator::cycle_target_time {}
```

Was TOUT.

The overall integration target time, reset on each externally-commanded cycle. trick_units(-)

Definition at line 289 of file lsode_first_order_ode_integrator.hh.

Referenced by integrate(), interpolate_y(), manager_check_stop_conditions(), manager_initialize_calculation_part2(), manager_integration_loop_part3(), process_entry_point_cycle_start(), and reset_integrator().

8.24.5.8 data_prepj

`LsodeDataJacobianPrep` `jeod::LsodeFirstOrderODEIntegrator::data_prepj` [protected]

data used exclusively for the DPREPJ method.

`trick_units(-)`

Definition at line 389 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `jacobian_prep_init()`, `jacobian_prep_loop()`, and `jacobian_prep_wrap_up()`.

8.24.5.9 data_stode

`LsodeDataStode` `jeod::LsodeFirstOrderODEIntegrator::data_stode` [protected]

data used exclusively for the DSTODE method.

`trick_units(-)`

Definition at line 397 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_compute_new_order()`, `integrator_compute_new_order_check_step_error()`, `integrator_compute_new_order_prep()`, `integrator_core()`, `integrator_corrector_converged()`, `integrator_corrector_failed_part2()`, `integrator_error_test_failed()`, `integrator_fail_reset_order_1_part1()`, `integrator_fail_reset_order_1_part2()`, `integrator_reset_method_coeffs()`, `integrator_reset_yh()`, `integrator_set_new_order()`, and `integrator_test_stepsize_change()`.

8.24.5.10 epsilon

`double` `jeod::LsodeFirstOrderODEIntegrator::epsilon` {1.0E-12} [protected]

was UROUND, in DLS001 common block.

Small number. `trick_units(-)`

Definition at line 604 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `calculate_epsilon()`, `interpolate_y()`, `jacobian_prep_init()`, `jacobian_prep_wrap_up()`, `manager_initialize_calculation_part2()`, and `manager_integration_loop_part2()`.

8.24.5.11 first_pass

```
bool jeod::LsodeFirstOrderODEIntegrator::first_pass {true} [protected]
```

was ISTATE.

true: was IASTATE = 1: first call for the problem, require initialization. false: was IASTATE = 2: subsequent call, no change to input parameters. not covered: IASTATE = 3: subsequent call, input parameters have changed. `trick_units(-)`

Definition at line 447 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `process_entry_point_cycle_start()`, `reset_integrator()`, and `~LsodeFirstOrderODEIntegrator()`.

8.24.5.12 initialized

```
bool jeod::LsodeFirstOrderODEIntegrator::initialized {} [protected]
```

was INIT, in DLS001 common block.

Flag representing whether the problem has been initialized. `trick_units(-)`

Definition at line 457 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `manager_integration_loop_part3()`, and `process_entry_point_cycle_start()`.

8.24.5.13 internal_state

```
int jeod::LsodeFirstOrderODEIntegrator::internal_state {} [protected]
```

Was JSTART, in DLS001 common block.

0: First step for problem 1: Continue normal calculation -1: Next step has new values of step-size, order, or methods. -2: Undocumented. `trick_units(-)`

Definition at line 465 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_core()`, `integrator_terminate()`, and `manager_initialize_calculation_part1()`.

8.24.5.14 iter_delta

```
double jeod::LsodeFirstOrderODEIntegrator::iter_delta {} [protected]
```

Was DEL, in DSTODE, local variable.

RMS value of {y dot error_weight_data} `trick_units(-)`

Definition at line 431 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_corrector_converged()`, `integrator_corrector_iteration()`, and `integrator_reset_iteration_loop_part2()`.

8.24.5.15 iteration_count

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::iteration_count {} [protected]
```

Definition at line 598 of file lsode_first_order_ode_integrator.hh.

Referenced by `integrator_corrector_converged()`, `integrator_corrector_iteration()`, and `integrator_reset_iteration_↵
loop_part1()`.

8.24.5.16 iteration_matrix_singular

```
bool jeod::LsodeFirstOrderODEIntegrator::iteration_matrix_singular {} [protected]
```

Was IERPJ, in DLS001 common block.

false: Iteration matrix was successfully LU-decomposed (iteration-method = 1,2,4,5) or inverted (iteration-method = 3). true: Matrix is singular. `trick_units(-)`

Definition at line 590 of file lsode_first_order_ode_integrator.hh.

Referenced by `integrator_core()`, `jacobian_prep_init()`, and `jacobian_prep_wrap_up()`.

8.24.5.17 jacobian_current

```
bool jeod::LsodeFirstOrderODEIntegrator::jacobian_current {} [protected]
```

Was JCUR, in DLS001 common block Is the jacobian current.

`trick_units(-)`

Definition at line 552 of file lsode_first_order_ode_integrator.hh.

Referenced by `integrator_core()`, `integrator_corrector_converged()`, `integrator_corrector_failed_part1()`, and `jacobian_prep_init()`.

8.24.5.18 max_history_size

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::max_history_size {12}
```

Was LMAX, in DLS001 common block Maximum allowable number of histories.

`trick_units(-)`

Definition at line 329 of file lsode_first_order_ode_integrator.hh.

Referenced by `integrator_compute_new_order_prep()`, `integrator_core()`, and `integrator_corrector_converged()`.

8.24.5.19 max_order_internal

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::max_order_internal {} [protected]
```

Was MAXORD.

Populated from IWORK[5] Maximum order allowable. trick_units(-)

Definition at line 486 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), and update_control_data().

8.24.5.20 max_rel_change_without_jacobian

```
double jeod::LsodeFirstOrderODEIntegrator::max_rel_change_without_jacobian {0.3}
```

was CCMAX, in DLS001 common block.

Max relative change to (step_size * method_coeff_first) before Jacobian matrix is updated. see also rel_change↔_since_jacobian. // This value was set in DLSODE (line 1385) to 0.3 without any explanation. It can be changed, but takes 0.3 as default. trick_units(-)

Definition at line 365 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_predict().

8.24.5.21 max_step_increase_ratio

```
double jeod::LsodeFirstOrderODEIntegrator::max_step_increase_ratio {1.0}
```

was RMAX, in DLS001 common block.

Max ratio by which step size may be increased. trick_units(-)

Definition at line 355 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_failed_part2(), integrator_error_test_failed(), and integrator↔_reset_yh().

8.24.5.22 max_step_size_inv

```
double jeod::LsodeFirstOrderODEIntegrator::max_step_size_inv {} [protected]
```

was HMXI.

Inverse of maximum absolute step size allowable. Default to 0.0 (i.e. there is no upper bound), calculated from max_step_size if max_step_size is user-specified. trick_units(-)

Definition at line 545 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_reset_yh(), manager_initialize_calculation_part2(), and update_control_data().

8.24.5.23 method_coeff_first

```
double jeod::LsodeFirstOrderODEIntegrator::method_coeff_first {} [protected]
```

was EL0, in DLS001 common block.

method coefficient l_0 for current method and order. trick_units(-)

Definition at line 508 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_reset_method_coeffs(), jacobian_prep_init(), and linear_chord_↔ iteration().

8.24.5.24 method_coeffs_complete

```
double jeod::LsodeFirstOrderODEIntegrator::method_coeffs_complete[13][12] {} [protected]
```

was ELCO, in DLS001 common block.

The array of all of the method coefficients. trick_units(-)

Definition at line 517 of file lsode_first_order_ode_integrator.hh.

Referenced by calculate_integration_coefficients(), integrator_core(), and integrator_reset_method_coeffs().

8.24.5.25 method_coeffs_current

```
double jeod::LsodeFirstOrderODEIntegrator::method_coeffs_current[13] {} [protected]
```

was EL, in DLS001 common block.

trick_units(-)

Definition at line 512 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_compute_new_order(), integrator_core(), integrator_corrector_converged(), integrator_↔ corrector_iteration(), and integrator_reset_method_coeffs().

8.24.5.26 method_order_current

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::method_order_current {}
```

Was NQ, in DLS001 common block.

Method order being tried on this or next step. trick_units(-)

Definition at line 341 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_compute_new_order(), integrator_compute_new_order_prep(), integrator_core(), integrator_corrector_converged(), integrator_corrector_failed_part2(), integrator_corrector_iteration(), integrator_↔ _error_test_failed(), integrator_fail_reset_order_1_part2(), integrator_predict(), integrator_reset_method_coeffs(), integrator_set_new_order(), interpolate_y(), manager_initialize_calculation_part2(), and reset_integrator().

8.24.5.27 modified_iteration_matrix_singular

```
bool jeod::LsodeFirstOrderODEIntegrator::modified_iteration_matrix_singular {} [protected]
```

Was IERSL, in DLS001 common block.

Like iteration_matrix_singular, only applied to the iteration matrix that has been modified to account for the new step for iteration-method 3. trick_units(-)

Definition at line 597 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_iteration(), and linear_chord_iteration().

8.24.5.28 num_equations

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::num_equations {1}
```

Was NYH, in DLS001 common block.

Number of ODEs to be solved in the current problem. In this implementation, num_odes = num_equations. trick_units(-)

Definition at line 317 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_failed_part2(), integrator_error_test_failed(), integrator_predict(), integrator_reset_method_coeffs(), and process_entry_point_cycle_start().

8.24.5.29 num_jacobian_evals

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::num_jacobian_evals {} [protected]
```

Was NJE, in DLS001 common block Number of jacobian evaluations so far for the problem.

trick_units(-)

Definition at line 426 of file lsode_first_order_ode_integrator.hh.

Referenced by jacobian_prep_init().

8.24.5.30 num_nordsiek_cols

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::num_nordsiek_cols {1}
```

Was L, in DLS001 common block Number of columns in Nordsiek array.

This appears to be a variable that is equal to the current order of the integrator + 1. trick_units(-)

Definition at line 324 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_compute_new_order(), integrator_compute_new_order_prep(), integrator_core(), integrator_corrector_converged(), integrator_fail_reset_order_1_part2(), integrator_reset_method_coeffs(), integrator_reset_yh(), integrator_set_new_order(), and interpolate_y().

8.24.5.31 num_predictor_elements

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::num_predictor_elements {1}
```

Was NQNYH, in DLS001 common block.

Number of elements of history array that are changed by predictor. trick_units(-)

Definition at line 334 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), and integrator_reset_method_coeffs().

8.24.5.32 num_small_step_warnings

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::num_small_step_warnings {} [protected]
```

Was NHNIL, in DLS001 common block.

Number of small-step encounters fo the problem so far. trick_units(-)

Definition at line 421 of file lsode_first_order_ode_integrator.hh.

Referenced by manager_integration_loop_part2().

8.24.5.33 num_steps_taken

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::num_steps_taken {} [protected]
```

Was NST, in DLS001 common block.

Number of steps taken for this problem. trick_units(-)

Definition at line 404 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_corrector_converged(), integrator_predict(), jacobian_prep_wrap_up(), manager_check↵_stop_conditions(), and manager_integration_loop_part1().

8.24.5.34 order_select_para

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::order_select_para {}
```

Was IALTH, in DLS001 common block.

0: Select optimal step size and method order 1: If prev_success_order < maximum_order, save vector so that an increase can be considered. >1: Perform neither. trick_units(-)

Definition at line 309 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_compute_new_order(), integrator_compute_new_order_check_step_error(), integrator↵_core(), integrator_corrector_converged(), integrator_fail_reset_order_1_part2(), and integrator_reset_yh().

8.24.5.35 prev_good_step_size

```
double jeod::LsodeFirstOrderODEIntegrator::prev_good_step_size {} [protected]
```

was HU, in DLS001 common block.

The size of the last successful step. `trick_units(-)`

Definition at line 538 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_corrector_converged()`, `interpolate_y()`, and `reset_integrator()`.

8.24.5.36 prev_integration_method

```
LsodeControlDataInterface::IntegrationMethod jeod::LsodeFirstOrderODEIntegrator::prev_integration←
_method [protected]
```

Initial value:

```
{
    LsodeControlDataInterface::ImplicitAdamsNonStiff}
```

Was MEO, in DLS001 common block Integration method used in previous call (see `integration_method`).

`trick_units(-)`

Definition at line 495 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_core()`.

8.24.5.37 prev_iter_delta

```
double jeod::LsodeFirstOrderODEIntegrator::prev_iter_delta {} [protected]
```

Was DELP, in DSTODE, local variable.

Previous value of `iter_delta`, used for comparison to identify rate at which iteration is converging / identifying divergence of iteration. `trick_units(-)`

Definition at line 438 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_corrector_iteration()`, and `integrator_reset_iteration_loop_part2()`.

8.24.5.38 prev_method_order

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::prev_method_order {} [protected]
```

Was NQU, in DLS001 common block.

Method order used in last successful step. `trick_units(-)`

Definition at line 501 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_corrector_converged()`, and `integrator_wrapup()`.

8.24.5.39 prev_step_size

```
double jeod::LsodeFirstOrderODEIntegrator::prev_step_size {} [protected]
```

was HOLD, in DLS001 common block.

`trick_units(-)`

Definition at line 533 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrator_core()`, `integrator_terminate()`, and `integrator_test_stepsize_change()`.

8.24.5.40 prior_num_steps

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::prior_num_steps {} [protected]
```

Was NSLAST, in DLS001 common block.

Number of steps taken for the problem prior to this call to `Lsode`. `trick_units(-)`

Definition at line 409 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `manager_check_stop_conditions()`, and `manager_integration_loop_part1()`.

8.24.5.41 re_entry_point

```
EntryPoint jeod::LsodeFirstOrderODEIntegrator::re_entry_point {CycleStartFinish} [protected]
```

Indicates where in the integrator to return to following an exit to gether new derivatives.

`trick_units(-)`

Definition at line 452 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrate()`, `integrator_corrector_iteration()`, `integrator_fail_reset_order_1_part1()`, `integrator_reset↔_iteration_loop_part1()`, `jacobian_prep_init()`, `manager_initialize_calculation_part1()`, `manager_integration_loop↔_part2()`, and `reset_integrator()`.

8.24.5.42 rel_change_since_jacobian

```
double jeod::LsodeFirstOrderODEIntegrator::rel_change_since_jacobian {} [protected]
```

was RC, in DLS001 common block.

Relative change to (step_size * method_coeff_first) since last update to Jacobian matrix. trick_units(-)

Definition at line 575 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_predict(), integrator_reset_method_coeffs(), integrator_reset_yh(), and jacobian_prep_wrap_up().

8.24.5.43 sqrt_epsilon

```
double jeod::LsodeFirstOrderODEIntegrator::sqrt_epsilon {1.0E-6} [protected]
```

NEW.

square root of epsilon. trick_units(-)

Definition at line 609 of file lsode_first_order_ode_integrator.hh.

Referenced by calculate_epsilon(), and manager_initialize_calculation_part1().

8.24.5.44 stage_target_time

```
double jeod::LsodeFirstOrderODEIntegrator::stage_target_time {}
```

was TN, in DLS001 common block.

Value of the independent variable, typically time, to which the integrator has successfully advanced, or to which it will advance in the next step/stage. trick_units(-)

Definition at line 350 of file lsode_first_order_ode_integrator.hh.

Referenced by integrate(), integrator_core(), integrator_corrector_failed_part2(), integrator_error_test_failed(), integrator_predict(), interpolate_y(), manager_check_stop_conditions(), manager_integration_loop_part1(), manager_integration_loop_part2(), manager_integration_loop_part3(), process_entry_point_cycle_start(), and reset_integrator().

8.24.5.45 step_at_last_jacobian_update

```
unsigned int jeod::LsodeFirstOrderODEIntegrator::step_at_last_jacobian_update {} [protected]
```

Was NSLP, in DLS001 common block.

Step number at last Jacobian update. trick_units(-)

Definition at line 562 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_predict(), and jacobian_prep_wrap_up().

8.24.5.46 step_error

```
int jeod::LsodeFirstOrderODEIntegrator::step_error {} [protected]
```

Was KFLAG, in DLS001 common block 0: step was successful -1: requested accuracy could not be achieved.

<=-2: repeated convergence failures. trick_units(-)

Definition at line 416 of file lsode_first_order_ode_integrator.hh.

Referenced by integrate(), integrator_compute_new_order_check_step_error(), integrator_core(), integrator↵
corrector_converged(), integrator_corrector_failed_part2(), integrator_error_test_failed(), integrator_fail_reset↵
order_1_part1(), and manager_integration_loop_part3().

8.24.5.47 step_size

```
double jeod::LsodeFirstOrderODEIntegrator::step_size {} [protected]
```

was H, in DLS001 common block.

Step size used on this step, or to be attempted on next. trick_units(-)

Definition at line 529 of file lsode_first_order_ode_integrator.hh.

Referenced by integrator_core(), integrator_corrector_converged(), integrator_corrector_failed_part2(), integrator↵
_corrector_iteration(), integrator_error_test_failed(), integrator_fail_reset_order_1_part1(), integrator_fail_reset↵
_order_1_part2(), integrator_predict(), integrator_reset_method_coeffs(), integrator_reset_yh(), integrator_set↵
_new_order(), integrator_terminate(), integrator_test_stepsize_change(), interpolate_y(), jacobian_prep_init(),
jacobian_prep_wrap_up(), linear_chord_iteration(), manager_check_stop_conditions(), manager_initialize↵
calculation_part2(), manager_integration_loop_part2(), and manager_integration_loop_part3().

8.24.5.48 test_coeffs_complete

```
double jeod::LsodeFirstOrderODEIntegrator::test_coeffs_complete[3][12] {} [protected]
```

was TESCO, in DLS001 common block.

The array of all of the test coefficients `trick_units(-)`

Definition at line 522 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `calculate_integration_coefficients()`, `integrator_compute_new_order()`, `integrator_compute_new_order_prep()`, `integrator_core()`, `integrator_corrector_converged()`, `integrator_corrector_iteration()`, and `integrator_wrapup()`.

8.24.5.49 update_jacobian

```
bool jeod::LsodeFirstOrderODEIntegrator::update_jacobian {true} [protected]
```

Was IPUP, in DLS001 common block.

Flag to indicate whether it is necessary to update the Jacobian. `trick_units(-)`

Definition at line 557 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrate()`, `integrator_core()`, `integrator_corrector_failed_part1()`, `integrator_corrector_failed_part2()`, `integrator_fail_reset_order_1_part2()`, `integrator_predict()`, and `jacobian_prep_wrap_up()`.

8.24.5.50 y

```
double* jeod::LsodeFirstOrderODEIntegrator::y {}
```

Was Y.

State vector (zeroth derivative). `trick_units(-)`

Definition at line 277 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrate()`, `integrator_corrector_iteration()`, `integrator_fail_reset_order_1_part1()`, `integrator_reset_iteration_loop_part1()`, `interpolate_y()`, `jacobian_prep_init()`, `jacobian_prep_loop()`, `linear_chord_iteration()`, `linear_solver()`, and `manager_initialize_calculation_part2()`.

8.24.5.51 y_dot

```
const double* jeod::LsodeFirstOrderODEIntegrator::y_dot {}
```

Was .

State vector (first derivative). `trick_units(-)`

Definition at line 282 of file `lsode_first_order_ode_integrator.hh`.

Referenced by `integrate()`, `load_derivatives()`, and `manager_initialize_calculation_part2()`.

The documentation for this class was generated from the following files:

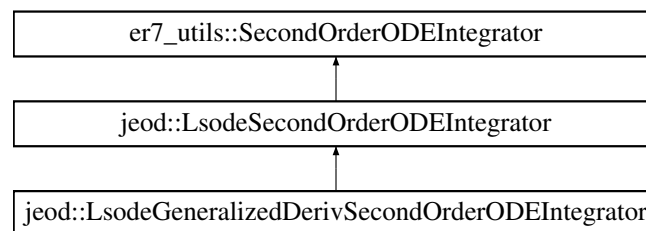
- [lsode_first_order_ode_integrator.hh](#)
- [lsode_first_order_ode_integrator__integrator.cc](#)
- [lsode_first_order_ode_integrator__manager.cc](#)
- [lsode_first_order_ode_integrator__support.cc](#)
- [lsode_first_order_ode_integrator__utility.cc](#)

8.25 jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator Class Reference

JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

```
#include <lsode_generalized_second_order_ode_integrator.hh>
```

Inheritance diagram for `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator`:



Public Member Functions

- `LsodeGeneralizedDerivSecondOrderODEIntegrator ()=default`
LsodeGeneralizedDerivSecondOrderODEIntegrator default constructor.
- `LsodeGeneralizedDerivSecondOrderODEIntegrator (const LsodeGeneralizedDerivSecondOrderODEIntegrator &src)`
LsodeGeneralizedDerivSecondOrderODEIntegrator copy constructor.
- `LsodeGeneralizedDerivSecondOrderODEIntegrator (const LsodeControlDataInterface &data_in, er7_utils::IntegrationControls &controls, const er7_utils::GeneralizedPositionDerivativeFunctions &deriv_funs, unsigned int position_size, unsigned int velocity_size)`
non-default constructor
- `~LsodeGeneralizedDerivSecondOrderODEIntegrator ()` override
LsodeGeneralizedDerivSecondOrderODEIntegrator destructor.
- `LsodeGeneralizedDerivSecondOrderODEIntegrator * create_copy ()` const override
Clone a LsodeGeneralizedDerivSecondOrderODEIntegrator.
- `er7_utils::IntegratorResult integrate (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position)` override
Propagate state via Lsode's method.
- `LsodeGeneralizedDerivSecondOrderODEIntegrator & operator= (const LsodeGeneralizedDerivSecondOrderODEIntegrator &)=delete`

Data Fields

- double * [posdot](#) {}
Stash space for the result of the computation of the derivative of the zeroth-derivative.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [LsodeGeneralizedDerivSecondOrderODEIntegrator](#))

Additional Inherited Members

8.25.1 Detailed Description

JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Definition at line 90 of file `lsode_generalized_second_order_ode_integrator.hh`.

8.25.2 Constructor & Destructor Documentation

8.25.2.1 LsodeGeneralizedDerivSecondOrderODEIntegrator()

```
jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::LsodeGeneralizedDerivSecondOrderODEIntegrator
( ) [default]
```

[LsodeGeneralizedDerivSecondOrderODEIntegrator](#) default constructor.

8.25.3 Member Function Documentation

8.25.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    LsodeGeneralizedDerivSecondOrderODEIntegrator ) [private]
```

8.25.3.2 operator=()

```
LsodeGeneralizedDerivSecondOrderODEIntegrator& jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator←→
::operator= (
    const LsodeGeneralizedDerivSecondOrderODEIntegrator & ) [delete]
```

8.25.4 Field Documentation

8.25.4.1 posdot

```
double* jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::posdot {}
```

Stash space for the result of the computation of the derivative of the zeroth-derivative.

Used with the Generalized derivative form, in which the derivative of the zeroth derivative is not equal to the first-derivative. `trick_units(-)`

Definition at line 149 of file `lsode_generalized_second_order_ode_integrator.hh`.

Referenced by `integrate()`, `LsodeGeneralizedDerivSecondOrderODEIntegrator()`, and `~LsodeGeneralizedDerivSecondOrderODEIntegrator()`.

The documentation for this class was generated from the following files:

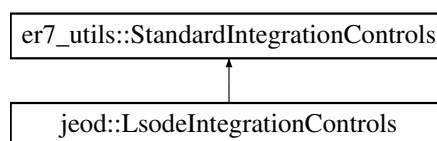
- [lsode_generalized_second_order_ode_integrator.hh](#)
- [lsode_generalized_second_order_ode_integrator.cc](#)

8.26 jeod::LsodeIntegrationControls Class Reference

Contains controls for an LSODE integrator.

```
#include <lsode_integration_controls.hh>
```

Inheritance diagram for `jeod::LsodeIntegrationControls`:



Public Member Functions

- [LsodeIntegrationControls](#) ()=default
- [LsodeIntegrationControls](#) (unsigned int num_stages)
- [~LsodeIntegrationControls](#) () override=default
- [LsodeIntegrationControls](#) & operator= (const [LsodeIntegrationControls](#) &)=delete
- [LsodeIntegrationControls](#) (const [LsodeIntegrationControls](#) &)=delete
- unsigned int [integrate](#) (double start_time, double sim_dt, er7_utils::TimeInterface &time_interface, er7_utils::IntegratorInterface &integ_interface, er7_utils::BaseIntegrationGroup &integ_group) override

Perform one step of the integration process.
- [LsodeIntegrationControls](#) * [create_copy](#) () const override

Create a copy of 'this' StandardIntegrationControls object.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [LsodeIntegrationControls](#))

8.26.1 Detailed Description

Contains controls for an LSODE integrator.

Definition at line 87 of file `lsode_integration_controls.hh`.

8.26.2 Constructor & Destructor Documentation

8.26.2.1 LsodeIntegrationControls() [1/2]

```
jeod::LsodeIntegrationControls::LsodeIntegrationControls ( ) [default]
```

8.26.2.2 ~LsodeIntegrationControls()

```
jeod::LsodeIntegrationControls::~~LsodeIntegrationControls ( ) [override], [default]
```

8.26.2.3 LsodeIntegrationControls() [2/2]

```
jeod::LsodeIntegrationControls::LsodeIntegrationControls (
    const LsodeIntegrationControls & ) [delete]
```

8.26.3 Member Function Documentation

8.26.3.1 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::LsodeIntegrationControls::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    LsodeIntegrationControls ) [private]
```

8.26.3.2 operator=()

```
LsodeIntegrationControls& jeod::LsodeIntegrationControls::operator= (
    const LsodeIntegrationControls & ) [delete]
```

The documentation for this class was generated from the following files:

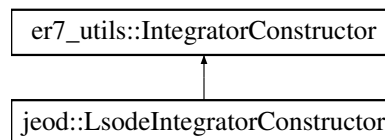
- [lsode_integration_controls.hh](#)
- [lsode_integration_controls.cc](#)

8.27 jeod::LsodeIntegratorConstructor Class Reference

Create state and time integrators that propagate using standard Lsode.

```
#include <lsode_integrator_constructor.hh>
```

Inheritance diagram for jeod::LsodeIntegratorConstructor:



Public Member Functions

- [LsodeIntegratorConstructor](#) ()=default
- [LsodeIntegratorConstructor](#) (const [LsodeIntegratorConstructor](#) &src)
- [LsodeIntegratorConstructor](#) & [operator=](#) (const [LsodeIntegratorConstructor](#) &)=delete
- const char * [get_class_name](#) () const override
Return the class name.
- bool [implements](#) (er7_utils::Integration::ODEProblemType problem_type) const override
Lsode currently does not implement a second order generalized step integrator.
- bool [provides](#) (er7_utils::Integration::ODEProblemType problem_type) const override
Lsode currently does not provide a second order generalized step integrator.
- er7_utils::IntegratorConstructor * [create_copy](#) () const override
Create a duplicate of the constructor.
- er7_utils::IntegrationControls * [create_integration_controls](#) () const override
Create an integration controls that guides the Lsode integration process.
- er7_utils::FirstOrderODEIntegrator * [create_first_order_ode_integrator](#) (unsigned int size, er7_utils::IntegrationControls &controls) const override
Create an Lsode state integrator for a first order ODE.
- er7_utils::SecondOrderODEIntegrator * [create_second_order_ode_integrator](#) (unsigned int size, er7_utils::IntegrationControls &controls) const override
Create an Lsode state integrator for a simple second order ODE.
- er7_utils::SecondOrderODEIntegrator * [create_generalized_deriv_second_order_ode_integrator](#) (unsigned int position_size, unsigned int velocity_size, const er7_utils::GeneralizedPositionDerivativeFunctions &deriv_funcs, er7_utils::IntegrationControls &controls) const override
Create an Lsode state integrator for a generalized second order ODE where generalized position is advanced with the use of the position derivative function.
- unsigned int [get_transition_table_size](#) () const override
Lsode does not use a linear transition table.

Static Public Member Functions

- static er7_utils::IntegratorConstructor * [create_constructor](#) ()
Named constructor; create an [LsodeIntegratorConstructor](#) instance.

Data Fields

- [LsodeControlDataInterface](#) [data_interface](#)

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [LsodeIntegratorConstructor](#))

8.27.1 Detailed Description

Create state and time integrators that propagate using standard Lsode.

Definition at line 96 of file `lsode_integrator_constructor.hh`.

8.27.2 Constructor & Destructor Documentation

8.27.2.1 LsodeIntegratorConstructor()

```
jeod::LsodeIntegratorConstructor::LsodeIntegratorConstructor ( ) [default]
```

8.27.3 Member Function Documentation

8.27.3.1 get_class_name()

```
const char* jeod::LsodeIntegratorConstructor::get_class_name ( ) const [inline], [override]
```

Return the class name.

Definition at line 121 of file `lsode_integrator_constructor.hh`.

8.27.3.2 `get_transition_table_size()`

```
unsigned int jeod::LsodeIntegratorConstructor::get_transition_table_size ( ) const [inline],
[override]
```

Lsode does not use a linear transition table.

Returns

Always returns 0.

Definition at line 201 of file `lsode_integrator_constructor.hh`.

8.27.3.3 `implements()`

```
bool jeod::LsodeIntegratorConstructor::implements (
    er7_utils::Integration::ODEProblemType problem_type ) const [inline], [override]
```

Lsode currently does not implement a second order generalized step integrator.

Definition at line 130 of file `lsode_integrator_constructor.hh`.

8.27.3.4 `JEOD_CLASS_ESTABLISH_FRIENDS2()`

```
jeod::LsodeIntegratorConstructor::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    LsodeIntegratorConstructor ) [private]
```

8.27.3.5 `operator=()`

```
LsodeIntegratorConstructor& jeod::LsodeIntegratorConstructor::operator= (
    const LsodeIntegratorConstructor & ) [delete]
```

8.27.3.6 `provides()`

```
bool jeod::LsodeIntegratorConstructor::provides (
    er7_utils::Integration::ODEProblemType problem_type ) const [inline], [override]
```

Lsode currently does not provide a second order generalized step integrator.

Definition at line 140 of file `lsode_integrator_constructor.hh`.

8.27.4 Field Documentation

8.27.4.1 data_interface

`LsodeControlDataInterface` jeod::LsodeIntegratorConstructor::data_interface

Definition at line 206 of file `lsode_integrator_constructor.hh`.

Referenced by `create_first_order_ode_integrator()`, `create_generalized_deriv_second_order_ode_integrator()`, and `create_second_order_ode_integrator()`.

The documentation for this class was generated from the following files:

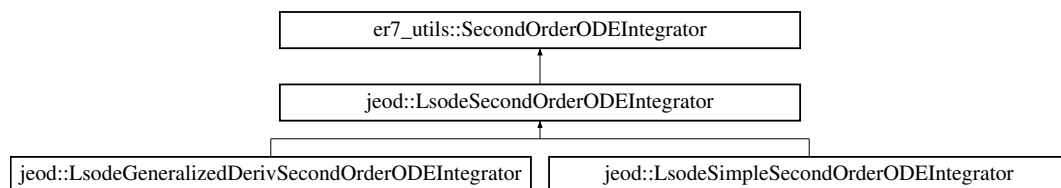
- [lsode_integrator_constructor.hh](#)
- [lsode_integrator_constructor.cc](#)

8.28 jeod::LsodeSecondOrderODEIntegrator Class Reference

JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

```
#include <lsode_second_order_ode_integrator.hh>
```

Inheritance diagram for jeod::LsodeSecondOrderODEIntegrator:



Public Member Functions

- [~LsodeSecondOrderODEIntegrator](#) () override
LsodeSecondOrderODEIntegrator destructor.
- [LsodeSecondOrderODEIntegrator & operator=](#) (const [LsodeSecondOrderODEIntegrator](#) &)=delete
- [LsodeSecondOrderODEIntegrator](#) (const [LsodeSecondOrderODEIntegrator](#) &)=delete
- int [get_re_entry_point](#) ()
Get the integrator's reentry point.

Data Fields

- double * [y](#) {}
State vector (zeroth derivative).
- double * [y_dot](#) {}
State vector (first derivative).
- unsigned int [zeroth_derivative_size](#) {}
- unsigned int [first_derivative_size](#) {}
- [LsodeFirstOrderODEIntegrator](#) [first_order_integrator](#)
- bool [arrays_allocated](#) {}

Protected Member Functions

- [LsodeSecondOrderODEIntegrator](#) ()=default
- [LsodeSecondOrderODEIntegrator](#) (const [LsodeControlDataInterface](#) &data_in, er7_utils::IntegrationControls &controls, unsigned int size)
[LsodeSecondOrderODEIntegrator](#) non-default constructor.
- [LsodeSecondOrderODEIntegrator](#) (const [LsodeControlDataInterface](#) &data_in, er7_utils::IntegrationControls &controls, const er7_utils::GeneralizedPositionDerivativeFunctions &deriv_funs, unsigned int position_size, unsigned int velocity_size)
- void [reset_integrator](#) () override
Reset the integrator.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [LsodeSecondOrderODEIntegrator](#))

8.28.1 Detailed Description

JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Definition at line 90 of file `lsode_second_order_ode_integrator.hh`.

8.28.2 Constructor & Destructor Documentation

8.28.2.1 LsodeSecondOrderODEIntegrator() [1/2]

```
jeod::LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator (
    const LsodeSecondOrderODEIntegrator & ) [delete]
```

References `first_order_integrator`, and `jeod::LsodeFirstOrderODEIntegrator::get_re_entry_point()`.

8.28.2.2 LsodeSecondOrderODEIntegrator() [2/2]

```
jeod::LsodeSecondOrderODEIntegrator::LsodeSecondOrderODEIntegrator ( ) [protected], [default]
```

8.28.3 Member Function Documentation

8.28.3.1 get_re_entry_point()

```
int jeod::LsodeSecondOrderODEIntegrator::get_re_entry_point ( ) [inline]
```

Get the integrator's reentry point.

Definition at line 107 of file lsode_second_order_ode_integrator.hh.

8.28.3.2 JEOD_CLASS_ESTABLISH_FRIENDS2()

```
jeod::LsodeSecondOrderODEIntegrator::JEOD_CLASS_ESTABLISH_FRIENDS2 (
    jeod ,
    LsodeSecondOrderODEIntegrator ) [private]
```

8.28.3.3 operator=()

```
LsodeSecondOrderODEIntegrator& jeod::LsodeSecondOrderODEIntegrator::operator= (
    const LsodeSecondOrderODEIntegrator & ) [delete]
```

8.28.3.4 reset_integrator()

```
void jeod::LsodeSecondOrderODEIntegrator::reset_integrator ( ) [inline], [override], [protected]
```

Reset the integrator.

Definition at line 134 of file lsode_second_order_ode_integrator.hh.

8.28.4 Field Documentation

8.28.4.1 arrays_allocated

```
bool jeod::LsodeSecondOrderODEIntegrator::arrays_allocated {}
```

Definition at line 158 of file lsode_second_order_ode_integrator.hh.

Referenced by LsodeSecondOrderODEIntegrator(), and ~LsodeSecondOrderODEIntegrator().

8.28.4.2 first_derivative_size

```
unsigned int jeod::LsodeSecondOrderODEIntegrator::first_derivative_size {}
```

Definition at line 154 of file lsode_second_order_ode_integrator.hh.

Referenced by `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate()`.

8.28.4.3 first_order_integrator

```
LsodeFirstOrderODEIntegrator jeod::LsodeSecondOrderODEIntegrator::first_order_integrator
```

Definition at line 156 of file lsode_second_order_ode_integrator.hh.

Referenced by `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate()`, `jeod::LsodeSimpleSecondOrderODEIntegrator::integrate()`, and `LsodeSecondOrderODEIntegrator()`.

8.28.4.4 y

```
double* jeod::LsodeSecondOrderODEIntegrator::y {}
```

State vector (zeroth derivative).

`trick_units(-)`

Definition at line 146 of file lsode_second_order_ode_integrator.hh.

Referenced by `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate()`, `jeod::LsodeSimpleSecondOrderODEIntegrator::integrate()`, `LsodeSecondOrderODEIntegrator()`, and `~LsodeSecondOrderODEIntegrator()`.

8.28.4.5 y_dot

```
double* jeod::LsodeSecondOrderODEIntegrator::y_dot {}
```

State vector (first derivative).

`trick_units(-)`

Definition at line 151 of file lsode_second_order_ode_integrator.hh.

Referenced by `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate()`, `jeod::LsodeSimpleSecondOrderODEIntegrator::integrate()`, `LsodeSecondOrderODEIntegrator()`, and `~LsodeSecondOrderODEIntegrator()`.

8.28.4.6 zeroth_derivative_size

```
unsigned int jeod::LsodeSecondOrderODEIntegrator::zeroth_derivative_size {}
```

Definition at line 153 of file `lsode_second_order_ode_integrator.hh`.

Referenced by `jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator::integrate()`, and `jeod::LsodeSimpleSecondOrderODEIntegrator::integrate()`.

The documentation for this class was generated from the following files:

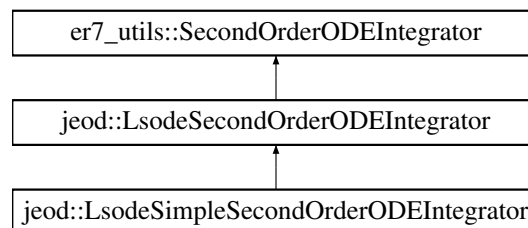
- [lsode_second_order_ode_integrator.hh](#)
- [lsode_second_order_ode_integrator.cc](#)

8.29 jeod::LsodeSimpleSecondOrderODEIntegrator Class Reference

JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

```
#include <lsode_simple_second_order_ode_integrator.hh>
```

Inheritance diagram for `jeod::LsodeSimpleSecondOrderODEIntegrator`:



Public Member Functions

- [LsodeSimpleSecondOrderODEIntegrator](#) ()=default
- [~LsodeSimpleSecondOrderODEIntegrator](#) () override=default
- [LsodeSimpleSecondOrderODEIntegrator](#) & [operator=](#) (const [LsodeSimpleSecondOrderODEIntegrator](#) &)=delete
- [LsodeSimpleSecondOrderODEIntegrator](#) (const [LsodeSimpleSecondOrderODEIntegrator](#) &)=delete
- [LsodeSimpleSecondOrderODEIntegrator](#) (const [LsodeControlDataInterface](#) &data_in, [er7_utils::IntegrationControls](#) &controls, unsigned int size)
[LsodeSimpleSecondOrderODEIntegrator](#) non-default constructor.
- [LsodeSimpleSecondOrderODEIntegrator](#) * [create_copy](#) () const override
- [er7_utils::IntegratorResult](#) [integrate](#) (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position) override
Propagate state via Lsode's method.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2](#) (jeod, [LsodeSimpleSecondOrderODEIntegrator](#))

Additional Inherited Members

8.29.1 Detailed Description

JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Definition at line 93 of file `lsode_simple_second_order_ode_integrator.hh`.

8.29.2 Constructor & Destructor Documentation

8.29.2.1 `LsodeSimpleSecondOrderODEIntegrator()` [1/2]

```
jeod::LsodeSimpleSecondOrderODEIntegrator::LsodeSimpleSecondOrderODEIntegrator ( ) [default]
```

8.29.2.2 `~LsodeSimpleSecondOrderODEIntegrator()`

```
jeod::LsodeSimpleSecondOrderODEIntegrator::~~LsodeSimpleSecondOrderODEIntegrator ( ) [override],  
[default]
```

8.29.2.3 `LsodeSimpleSecondOrderODEIntegrator()` [2/2]

```
jeod::LsodeSimpleSecondOrderODEIntegrator::LsodeSimpleSecondOrderODEIntegrator (   
    const LsodeSimpleSecondOrderODEIntegrator & ) [delete]
```

8.29.3 Member Function Documentation

8.29.3.1 `JEOD_CLASS_ESTABLISH_FRIENDS2()`

```
jeod::LsodeSimpleSecondOrderODEIntegrator::JEOD_CLASS_ESTABLISH_FRIENDS2 (   
    jeod ,   
    LsodeSimpleSecondOrderODEIntegrator ) [private]
```


8.29.3.2 operator=()

```
LsodeSimpleSecondOrderODEIntegrator& jeod::LsodeSimpleSecondOrderODEIntegrator::operator= (
    const LsodeSimpleSecondOrderODEIntegrator & ) [delete]
```

The documentation for this class was generated from the following files:

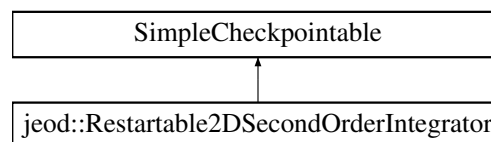
- [lsode_simple_second_order_ode_integrator.hh](#)
- [lsode_simple_second_order_ode_integrator.cc](#)

8.30 jeod::Restartable2DSecondOrderIntegrator Class Reference

Integrates a second order ODE in two dimensional space, $d^2x/dt^2 = a(x,t)$, where x is a two-vector.

```
#include <restartable_2d_second_order_integrator.hh>
```

Inheritance diagram for jeod::Restartable2DSecondOrderIntegrator:



Public Member Functions

- [Restartable2DSecondOrderIntegrator \(\)](#)
Default constructor.
- [~Restartable2DSecondOrderIntegrator \(\)](#) override
Destructor.
- void [create_integrator](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls)
Create the integrator to be managed.
- void [destroy_integrator](#) ()
Destroy the integrator.
- er7_utils::IntegratorResult [integrate](#) (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position)
Propagate state to the specified stage of the integration process for an overall integration time interval of dyn_dt.
- void [reset_integrator](#) ()
Tell the integrator to reset itself.
- void [simple_restore](#) () override
Restore the integrator on restart.
- [Restartable2DSecondOrderIntegrator](#) (const [Restartable2DSecondOrderIntegrator](#) &)=delete
- [Restartable2DSecondOrderIntegrator](#) & [operator=](#) (const [Restartable2DSecondOrderIntegrator](#) &)=delete

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [Restartable2DSecondOrderIntegrator](#))

Private Attributes

- `er7_utils::SecondOrderODEIntegrator * integrator {}`
The pointer to the object that performs integration.
- `RestartableSimpleSecondOrderODEIntegrator< 2 > integrator\_manager`
The object that creates and manages the integrator object.

8.30.1 Detailed Description

Integrates a second order ODE in two dimensional space, $d^2x/dt^2 = a(x,t)$, where x is a two-vector.

Definition at line 82 of file `restartable_2d_second_order_integrator.hh`.

8.30.2 Constructor & Destructor Documentation

8.30.2.1 `Restartable2DSecondOrderIntegrator()` [1/2]

```
jeod::Restartable2DSecondOrderIntegrator::Restartable2DSecondOrderIntegrator ( ) [inline]
```

Default constructor.

Definition at line 88 of file `restartable_2d_second_order_integrator.hh`.

8.30.2.2 `~Restartable2DSecondOrderIntegrator()`

```
jeod::Restartable2DSecondOrderIntegrator::~~Restartable2DSecondOrderIntegrator ( ) [inline],  
[override]
```

Destructor.

Definition at line 97 of file `restartable_2d_second_order_integrator.hh`.

8.30.2.3 `Restartable2DSecondOrderIntegrator()` [2/2]

```
jeod::Restartable2DSecondOrderIntegrator::Restartable2DSecondOrderIntegrator (  
    const Restartable2DSecondOrderIntegrator & ) [delete]
```

8.30.3 Member Function Documentation

8.30.3.1 `create_integrator()`

```
void jeod::Restartable2DSecondOrderIntegrator::create_integrator (  
    const er7_utils::IntegratorConstructor & generator,  
    er7_utils::IntegrationControls & controls ) [inline]
```

Create the integrator to be managed.

Parameters

in	<i>generator</i>	Generator used to create the integrator.
in, out	<i>controls</i>	Controls to be passed to the generator.

Definition at line 107 of file `restartable_2d_second_order_integrator.hh`.

8.30.3.2 destroy_integrator()

```
void jeod::Restartable2DSecondOrderIntegrator::destroy_integrator ( ) [inline]
```

Destroy the integrator.

Definition at line 116 of file `restartable_2d_second_order_integrator.hh`.

8.30.3.3 integrate()

```
er7_utils::IntegratorResult jeod::Restartable2DSecondOrderIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * accel,
    double * velocity,
    double * position ) [inline]
```

Propagate state to the specified stage of the integration process for an overall integration time interval of `dyn_dt`.

Note that this is a pass-through to the encapsulated integrator object. See `er7_utils::SecondOrderODEIntegrator::integrate` for details.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>accel</i>	Time derivative of the generalized velocity.
in, out	<i>velocity</i>	Generalized velocity vector.
in, out	<i>position</i>	Generalized position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 138 of file `restartable_2d_second_order_integrator.hh`.

8.30.3.4 operator=()

```
Restartable2DSecondOrderIntegrator& jeod::Restartable2DSecondOrderIntegrator::operator= (
    const Restartable2DSecondOrderIntegrator & ) [delete]
```

8.30.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::Restartable2DSecondOrderIntegrator::p (
    jeod ,
    Restartable2DSecondOrderIntegrator ) [private]
```

8.30.3.6 reset_integrator()

```
void jeod::Restartable2DSecondOrderIntegrator::reset_integrator ( ) [inline]
```

Tell the integrator to reset itself.

This should be called when the time step or time direction changes or upon a discrete change in state such docking/undocking. Such events invalidate saved state (if any). The integrator needs to restart from scratch when such events occur.

Definition at line 153 of file restartable_2d_second_order_integrator.hh.

8.30.3.7 simple_restore()

```
void jeod::Restartable2DSecondOrderIntegrator::simple_restore ( ) [inline], [override]
```

Restore the integrator on restart.

Definition at line 161 of file restartable_2d_second_order_integrator.hh.

8.30.4 Field Documentation

8.30.4.1 integrator

```
er7_utils::SecondOrderODEIntegrator* jeod::Restartable2DSecondOrderIntegrator::integrator {}
[private]
```

The pointer to the object that performs integration.

This object is created managed by the integrator manager. `trick_units(-)`

Definition at line 176 of file restartable_2d_second_order_integrator.hh.

8.30.4.2 integrator_manager

```
RestartableSimpleSecondOrderODEIntegrator<2> jeod::Restartable2DSecondOrderIntegrator::integrator↔
_manager [private]
```

The object that creates and manages the integrator object.

trick_io(**)

Definition at line 181 of file restartable_2d_second_order_integrator.hh.

The documentation for this class was generated from the following file:

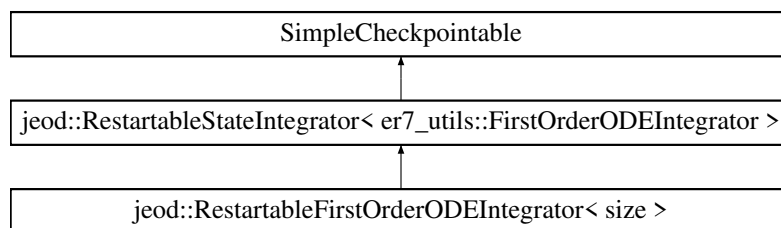
- [restartable_2d_second_order_integrator.hh](#)

8.31 jeod::RestartableFirstOrderODEIntegrator< size > Class Template Reference

A [RestartableFirstOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages an [er7_utils::FirstOrderODEIntegrator](#).

```
#include <restartable_state_integrator_templates.hh>
```

Inheritance diagram for `jeod::RestartableFirstOrderODEIntegrator< size >`:



Public Member Functions

- [RestartableFirstOrderODEIntegrator](#) ()=default
- [~RestartableFirstOrderODEIntegrator](#) () override=default
- [RestartableFirstOrderODEIntegrator](#) (const [RestartableFirstOrderODEIntegrator](#) &)=delete
- [RestartableFirstOrderODEIntegrator](#) & operator= (const [RestartableFirstOrderODEIntegrator](#) &)=delete
- [RestartableFirstOrderODEIntegrator](#) (er7_utils::FirstOrderODEIntegrator *&integ_ref)

Non-default constructor.

Private Member Functions

- er7_utils::FirstOrderODEIntegrator * [create_integrator_internal](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls) override

Create the integrator to be managed.

Additional Inherited Members

8.31.1 Detailed Description

```
template<unsigned int size>
class jeod::RestartableFirstOrderODEIntegrator< size >
```

A [RestartableFirstOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages an [er7_utils::FirstOrderODEIntegrator](#).

Template Parameters

<i>size</i>	Dimensionality of the state vector.
-------------	-------------------------------------

Definition at line 306 of file restartable_state_integrator_templates.hh.

8.31.2 Constructor & Destructor Documentation

8.31.2.1 RestartableFirstOrderODEIntegrator() [1/3]

```
template<unsigned int size>
jeod::RestartableFirstOrderODEIntegrator< size >::RestartableFirstOrderODEIntegrator ( )
[default]
```

8.31.2.2 ~RestartableFirstOrderODEIntegrator()

```
template<unsigned int size>
jeod::RestartableFirstOrderODEIntegrator< size >::~~RestartableFirstOrderODEIntegrator ( )
[override], [default]
```

8.31.2.3 RestartableFirstOrderODEIntegrator() [2/3]

```
template<unsigned int size>
jeod::RestartableFirstOrderODEIntegrator< size >::RestartableFirstOrderODEIntegrator (
    const RestartableFirstOrderODEIntegrator< size > & ) [delete]
```

8.31.2.4 RestartableFirstOrderODEIntegrator() [3/3]

```
template<unsigned int size>
jeod::RestartableFirstOrderODEIntegrator< size >::RestartableFirstOrderODEIntegrator (
    er7_utils::FirstOrderODEIntegrator *& integ_ref ) [inline], [explicit]
```

Non-default constructor.

Parameters

<i>in, out</i>	<i>integ_ref</i>	Reference to the pointer to the integrator that is to be managed.
----------------	------------------	---

Definition at line 319 of file `restartable_state_integrator_templates.hh`.

8.31.3 Member Function Documentation

8.31.3.1 `create_integrator_internal()`

```
template<unsigned int size>
er7_utils::FirstOrderODEIntegrator* jeod::RestartableFirstOrderODEIntegrator< size >::create↵
_integrator_internal (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline], [override], [private],
[virtual]
```

Create the integrator to be managed.

Parameters

<code>in</code>	<code>generator</code>	Integrator constructor used to create the integrator.
<code>in, out</code>	<code>controls</code>	Integration controls to be passed to the generator.

Implements [jeod::RestartableStateIntegrator< er7_utils::FirstOrderODEIntegrator >](#).

Definition at line 332 of file `restartable_state_integrator_templates.hh`.

8.31.3.2 `operator=()`

```
template<unsigned int size>
RestartableFirstOrderODEIntegrator& jeod::RestartableFirstOrderODEIntegrator< size >::operator=
(
    const RestartableFirstOrderODEIntegrator< size > & ) [delete]
```

The documentation for this class was generated from the following file:

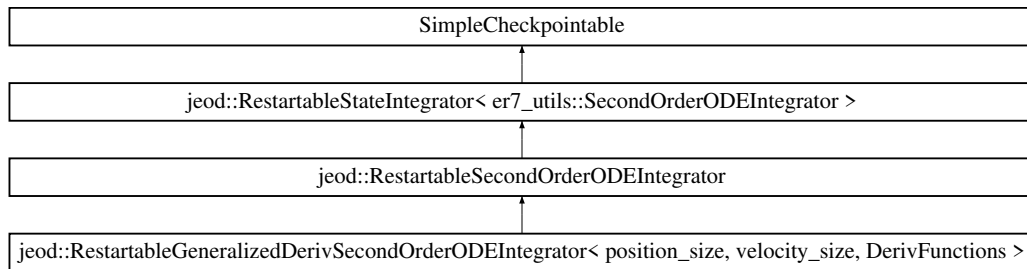
- [restartable_state_integrator_templates.hh](#)

8.32 `jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >` Class Template Reference

A [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

```
#include <restartable_state_integrator_templates.hh>
```

Inheritance diagram for `jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >`:



Public Member Functions

- [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) ()=default
- [~RestartableGeneralizedDerivSecondOrderODEIntegrator](#) () override=default
- [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) (const [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) &)=delete
- [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) & operator= (const [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) &)=delete
- [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) (er7_utils::SecondOrderODEIntegrator *&integ_ref)

Non-default constructor.

Private Member Functions

- er7_utils::SecondOrderODEIntegrator * [create_integrator_internal](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls) override

Create the integrator to be managed.

- void [simple_restore_internal](#) (er7_utils::SecondOrderODEIntegrator *integrator_ptr) override

Perform technique-specific restart actions.

Additional Inherited Members

8.32.1 Detailed Description

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions>
class jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >
```

A [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

Template Parameters

<i>position_size</i>	The dimensionality of the generalized position.
<i>velocity_size</i>	The dimensionality of the generalized velocity.
<i>DerivFunctions</i>	Class that derives from the class er7_utils::GeneralizedPositionDerivativeFunctions.

Definition at line 418 of file restartable_state_integrator_templates.hh.

8.32.2 Constructor & Destructor Documentation

8.32.2.1 RestartableGeneralizedDerivSecondOrderODEIntegrator() [1/3]

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >::RestartableGeneralizedDerivSecondOrderODEIntegrator ( ) [default]
```

8.32.2.2 ~RestartableGeneralizedDerivSecondOrderODEIntegrator()

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >::~~RestartableGeneralizedDerivSecondOrderODEIntegrator ( ) [override], [default]
```

8.32.2.3 RestartableGeneralizedDerivSecondOrderODEIntegrator() [2/3]

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >::RestartableGeneralizedDerivSecondOrderODEIntegrator (
    const RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions > & ) [delete]
```

8.32.2.4 RestartableGeneralizedDerivSecondOrderODEIntegrator() [3/3]

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >::RestartableGeneralizedDerivSecondOrderODEIntegrator (
    er7_utils::SecondOrderODEIntegrator *& integ_ref ) [inline], [explicit]
```

Non-default constructor.

Parameters

in, out	<i>integ_ref</i>	Reference to the pointer to the integrator that is to be managed.
---------	------------------	---

Definition at line 433 of file restartable_state_integrator_templates.hh.

8.32.3 Member Function Documentation

8.32.3.1 `create_integrator_internal()`

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
er7_utils::SecondOrderODEIntegrator* jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator<
position_size, velocity_size, DerivFunctions >::create_integrator_internal (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline], [override], [private],
[virtual]
```

Create the integrator to be managed.

Parameters

<code>in</code>	<i>generator</i>	Integrator constructor used to create the integrator.
<code>in, out</code>	<i>controls</i>	Integration controls to be passed to the generator.

Implements [jeod::RestartableStateIntegrator< er7_utils::SecondOrderODEIntegrator >](#).

Definition at line 446 of file `restartable_state_integrator_templates.hh`.

8.32.3.2 `operator=()`

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
RestartableGeneralizedDerivSecondOrderODEIntegrator& jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator<
position_size, velocity_size, DerivFunctions >::operator= (
    const RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity↔
_size, DerivFunctions > & ) [delete]
```

8.32.3.3 `simple_restore_internal()`

```
template<unsigned int position_size, unsigned int velocity_size, typename DerivFunctions >
void jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size,
DerivFunctions >::simple_restore_internal (
    er7_utils::SecondOrderODEIntegrator * integrator_ptr ) [inline], [override],
[private], [virtual]
```

Perform technique-specific restart actions.

The generalized second order ODE integrators need the pointer to the derivative function to be restored.

Parameters

<code>in, out</code>	<code>integrator_ptr</code>	The base class's integrator data member
----------------------	-----------------------------	---

Reimplemented from [jeod::RestartableStateIntegrator< er7_utils::SecondOrderODEIntegrator >](#).

Definition at line 461 of file `restartable_state_integrator_templates.hh`.

The documentation for this class was generated from the following file:

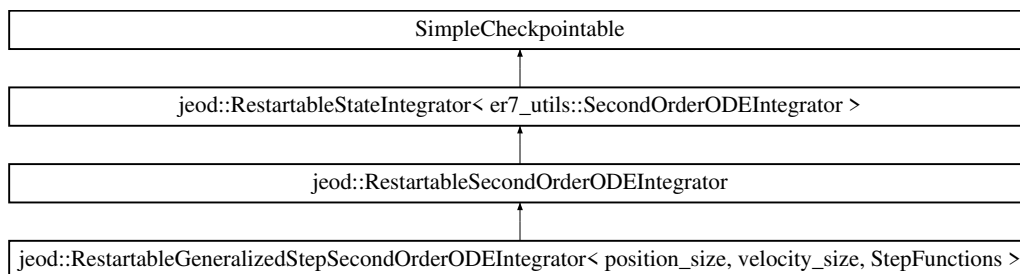
- [restartable_state_integrator_templates.hh](#)

8.33 jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions > Class Template Reference

A [RestartableGeneralizedStepSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

```
#include <restartable_state_integrator_templates.hh>
```

Inheritance diagram for `jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >`:



Public Member Functions

- [RestartableGeneralizedStepSecondOrderODEIntegrator](#) ()=default
- [~RestartableGeneralizedStepSecondOrderODEIntegrator](#) () override=default
- [RestartableGeneralizedStepSecondOrderODEIntegrator](#) (const [RestartableGeneralizedStepSecondOrderODEIntegrator](#) &)=delete
- [RestartableGeneralizedStepSecondOrderODEIntegrator](#) & operator= (const [RestartableGeneralizedStepSecondOrderODEIntegrator](#) &)=delete
- [RestartableGeneralizedStepSecondOrderODEIntegrator](#) (er7_utils::SecondOrderODEIntegrator * &integ_ref)

Non-default constructor.

Private Member Functions

- `er7_utils::SecondOrderODEIntegrator * create_integrator_internal` (const `er7_utils::IntegratorConstructor` &generator, `er7_utils::IntegrationControls` &controls) override
Create the integrator to be managed.
- void `simple_restore_internal` (`er7_utils::SecondOrderODEIntegrator *integrator_ptr`) override
Perform technique-specific restart actions.

Additional Inherited Members

8.33.1 Detailed Description

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions>
class jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >
```

A [RestartableGeneralizedStepSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

Template Parameters

<i>position_size</i>	The dimensionality of the generalized position.
<i>velocity_size</i>	The dimensionality of the generalized velocity.
<i>StepFunctions</i>	Class that derives from <code>er7_utils::GeneralizedPositionStepFunctions</code> .

Definition at line 482 of file `restartable_state_integrator_templates.hh`.

8.33.2 Constructor & Destructor Documentation

8.33.2.1 RestartableGeneralizedStepSecondOrderODEIntegrator() [1/3]

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, Step↵
Functions >::RestartableGeneralizedStepSecondOrderODEIntegrator ( ) [default]
```

8.33.2.2 ~RestartableGeneralizedStepSecondOrderODEIntegrator()

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, Step↵
Functions >::~~RestartableGeneralizedStepSecondOrderODEIntegrator ( ) [override], [default]
```

8.33.2.3 RestartableGeneralizedStepSecondOrderODEIntegrator() [2/3]

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >::RestartableGeneralizedStepSecondOrderODEIntegrator (
    const RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions > & ) [delete]
```

8.33.2.4 RestartableGeneralizedStepSecondOrderODEIntegrator() [3/3]

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >::RestartableGeneralizedStepSecondOrderODEIntegrator (
    er7_utils::SecondOrderODEIntegrator *& integ_ref ) [inline], [explicit]
```

Non-default constructor.

Parameters

<code>in, out</code>	<code>integ_ref</code>	Reference to the pointer to the integrator that is to be managed.
----------------------	------------------------	---

Definition at line 497 of file `restartable_state_integrator_templates.hh`.

8.33.3 Member Function Documentation**8.33.3.1 create_integrator_internal()**

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
er7_utils::SecondOrderODEIntegrator* jeod::RestartableGeneralizedStepSecondOrderODEIntegrator<
position_size, velocity_size, StepFunctions >::create_integrator_internal (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline], [override], [private],
[virtual]
```

Create the integrator to be managed.

Parameters

<code>in</code>	<code>generator</code>	Integrator constructor used to create the integrator.
<code>in, out</code>	<code>controls</code>	Integration controls to be passed to the generator.

Implements `jeod::RestartableStateIntegrator< er7_utils::SecondOrderODEIntegrator >`.

Definition at line 510 of file `restartable_state_integrator_templates.hh`.

8.33.3.2 operator=()

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
RestartableGeneralizedStepSecondOrderODEIntegrator& jeod::RestartableGeneralizedStepSecondOrderODEIntegrator<
position_size, velocity_size, StepFunctions >::operator= (
    const RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_↵
_size, StepFunctions > & ) [delete]
```

8.33.3.3 simple_restore_internal()

```
template<unsigned int position_size, unsigned int velocity_size, typename StepFunctions >
void jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size,
StepFunctions >::simple_restore_internal (
    er7_utils::SecondOrderODEIntegrator * integrator_ptr ) [inline], [override],
[private], [virtual]
```

Perform technique-specific restart actions.

The generalized second order ODE integrators need the pointer to the derivative function to be restored.

Parameters

<i>in, out</i>	<i>integrator_ptr</i>	The base class's integrator data member
----------------	-----------------------	---

Reimplemented from [jeod::RestartableStateIntegrator< er7_utils::SecondOrderODEIntegrator >](#).

Definition at line 525 of file `restartable_state_integrator_templates.hh`.

The documentation for this class was generated from the following file:

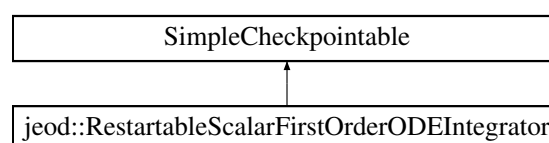
- [restartable_state_integrator_templates.hh](#)

8.34 jeod::RestartableScalarFirstOrderODEIntegrator Class Reference

A [RestartableScalarFirstOrderODEIntegrator](#) integrates a first order ODE, $dx/dt = v(x,t)$, where x is a scalar.

```
#include <restartable_state_integrator.hh>
```

Inheritance diagram for `jeod::RestartableScalarFirstOrderODEIntegrator`:



Public Member Functions

- [RestartableScalarFirstOrderODEIntegrator](#) ()
Default constructor.
- [~RestartableScalarFirstOrderODEIntegrator](#) () override=default
- [RestartableScalarFirstOrderODEIntegrator](#) (const [RestartableScalarFirstOrderODEIntegrator](#) &)=delete
- [RestartableScalarFirstOrderODEIntegrator](#) & operator= (const [RestartableScalarFirstOrderODEIntegrator](#) &)=delete
- void [create_integrator](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls)
Create the integrator to be managed.
- void [destroy_integrator](#) ()
Destroy the integrator.
- er7_utils::IntegratorResult [integrate](#) (double dyn_dt, unsigned int target_stage, double *xdot, double *x)
Propagate state to the specified stage of the integration process for an overall integration time interval of dyn_dt.
- void [reset_integrator](#) ()
Tell the integrator to reset itself.
- void [simple_restore](#) () override
Restore the integrator on restart.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [RestartableScalarFirstOrderODEIntegrator](#))

Private Attributes

- er7_utils::FirstOrderODEIntegrator * [integrator](#) {}
Pointer to the object that performs integration.
- [RestartableFirstOrderODEIntegrator](#)< 1 > [integrator_manager](#)
Object that creates and manages the integrator object.

8.34.1 Detailed Description

A [RestartableScalarFirstOrderODEIntegrator](#) integrates a first order ODE, $dx/dt = v(x,t)$, where x is a scalar.

Definition at line 89 of file `restartable_state_integrator.hh`.

8.34.2 Constructor & Destructor Documentation

8.34.2.1 RestartableScalarFirstOrderODEIntegrator() [1/2]

```
jeod::RestartableScalarFirstOrderODEIntegrator::RestartableScalarFirstOrderODEIntegrator ( )  
[inline]
```

Default constructor.

Definition at line 95 of file `restartable_state_integrator.hh`.

8.34.2.2 ~RestartableScalarFirstOrderODEIntegrator()

```
jeod::RestartableScalarFirstOrderODEIntegrator::~~RestartableScalarFirstOrderODEIntegrator ( )
[override], [default]
```

8.34.2.3 RestartableScalarFirstOrderODEIntegrator() [2/2]

```
jeod::RestartableScalarFirstOrderODEIntegrator::~RestartableScalarFirstOrderODEIntegrator (
    const RestartableScalarFirstOrderODEIntegrator & ) [delete]
```

8.34.3 Member Function Documentation

8.34.3.1 create_integrator()

```
void jeod::RestartableScalarFirstOrderODEIntegrator::create_integrator (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline]
```

Create the integrator to be managed.

Parameters

in	<i>generator</i>	Integrator constructor used to create the integrator.
in, out	<i>controls</i>	Integration controls to be passed to the generator.

Definition at line 110 of file restartable_state_integrator.hh.

8.34.3.2 destroy_integrator()

```
void jeod::RestartableScalarFirstOrderODEIntegrator::destroy_integrator ( ) [inline]
```

Destroy the integrator.

Definition at line 119 of file restartable_state_integrator.hh.

8.34.3.3 integrate()

```
er7_utils::IntegratorResult jeod::RestartableScalarFirstOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    double * xdot,
    double * x ) [inline]
```

Propagate state to the specified stage of the integration process for an overall integration time interval of `dyn_dt`.

Note that this is a pass-through to the encapsulated integrator object. See `er7_utils::FirstOrderODEIntegrator::integrate` for details.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>xdot</i>	Time derivative of <i>x</i> .
in, out	<i>x</i>	Item to be integrated.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 139 of file `restartable_state_integrator.hh`.

8.34.3.4 operator=()

```
RestartableScalarFirstOrderODEIntegrator& jeod::RestartableScalarFirstOrderODEIntegrator<
::operator= (
    const RestartableScalarFirstOrderODEIntegrator & ) [delete]
```

References `jeod::RestartableStateIntegrator< IntegratorType >::create_integrator()`, and `integrator_manager`.

8.34.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RestartableScalarFirstOrderODEIntegrator::p (
    jeod ,
    RestartableScalarFirstOrderODEIntegrator ) [private]
```

8.34.3.6 reset_integrator()

```
void jeod::RestartableScalarFirstOrderODEIntegrator::reset_integrator ( ) [inline]
```

Tell the integrator to reset itself.

This should be called when the time step or time direction changes or upon a discrete change in state such docking/undocking. Such events invalidate saved state (if any). The integrator needs to restart from scratch when such events occur.

Definition at line 153 of file restartable_state_integrator.hh.

8.34.3.7 simple_restore()

```
void jeod::RestartableScalarFirstOrderODEIntegrator::simple_restore ( ) [inline], [override]
```

Restore the integrator on restart.

Definition at line 161 of file restartable_state_integrator.hh.

8.34.4 Field Documentation

8.34.4.1 integrator

```
er7_utils::FirstOrderODEIntegrator* jeod::RestartableScalarFirstOrderODEIntegrator::integrator  
{ } [private]
```

Pointer to the object that performs integration.

The object is created managed by the integrator manager. `trick_units(-)`

Definition at line 172 of file restartable_state_integrator.hh.

8.34.4.2 integrator_manager

```
RestartableFirstOrderODEIntegrator<1> jeod::RestartableScalarFirstOrderODEIntegrator::integrator↔  
_manager [private]
```

Object that creates and manages the integrator object.

`trick_io(**)`

Definition at line 177 of file restartable_state_integrator.hh.

Referenced by operator=().

The documentation for this class was generated from the following file:

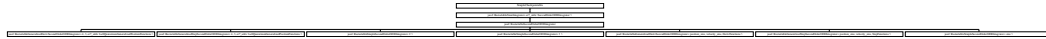
- [restartable_state_integrator.hh](#)

8.35 jeod::RestartableSecondOrderODEIntegrator Class Reference

A [RestartableSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a second order ODE problem.

```
#include <restartable_state_integrator_templates.hh>
```

Inheritance diagram for jeod::RestartableSecondOrderODEIntegrator:



Public Member Functions

- [~RestartableSecondOrderODEIntegrator](#) () override=default
- [RestartableSecondOrderODEIntegrator](#) (const [RestartableSecondOrderODEIntegrator](#) &)=delete
- [RestartableSecondOrderODEIntegrator](#) & operator= (const [RestartableSecondOrderODEIntegrator](#) &)=delete

Protected Member Functions

- [RestartableSecondOrderODEIntegrator](#) ()=default
- [RestartableSecondOrderODEIntegrator](#) (er7_utils::SecondOrderODEIntegrator *&integ_ref)

Non-default constructor.

8.35.1 Detailed Description

A [RestartableSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a second order ODE problem.

Definition at line 344 of file restartable_state_integrator_templates.hh.

8.35.2 Constructor & Destructor Documentation

8.35.2.1 ~RestartableSecondOrderODEIntegrator()

```
jeod::RestartableSecondOrderODEIntegrator::~~RestartableSecondOrderODEIntegrator ( ) [override],
[default]
```

8.35.2.2 RestartableSecondOrderODEIntegrator() [1/3]

```
jeod::RestartableSecondOrderODEIntegrator::RestartableSecondOrderODEIntegrator (
    const RestartableSecondOrderODEIntegrator & ) [delete]
```

8.35.2.3 RestartableSecondOrderODEIntegrator() [2/3]

```
jeod::RestartableSecondOrderODEIntegrator::RestartableSecondOrderODEIntegrator ( ) [protected],
[default]
```

8.35.2.4 RestartableSecondOrderODEIntegrator() [3/3]

```
jeod::RestartableSecondOrderODEIntegrator::RestartableSecondOrderODEIntegrator (
    er7_utils::SecondOrderODEIntegrator *& integ_ref ) [inline], [explicit], [protected]
```

Non-default constructor.

Parameters

<i>in, out</i>	<i>integ_ref</i>	Reference to the pointer to the integrator that is to be managed.
----------------	------------------	---

Definition at line 360 of file restartable_state_integrator_templates.hh.

8.35.3 Member Function Documentation

8.35.3.1 operator=()

```
RestartableSecondOrderODEIntegrator& jeod::RestartableSecondOrderODEIntegrator::operator= (
    const RestartableSecondOrderODEIntegrator & ) [delete]
```

The documentation for this class was generated from the following file:

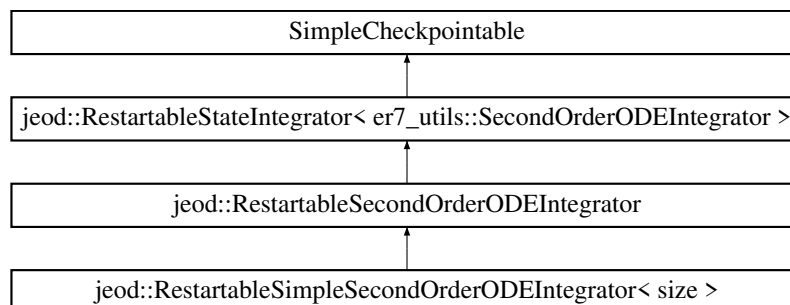
- [restartable_state_integrator_templates.hh](#)

8.36 jeod::RestartableSimpleSecondOrderODEIntegrator< size > Class Template Reference

A [RestartableSimpleSecondOrderODEIntegrator](#) is-a [RestartableSecondOrderODEIntegrator](#) that manages the integrator for a simple second order ODE problem, one in which the time derivative of the generalized position is the generalized velocity.

```
#include <restartable_state_integrator_templates.hh>
```

Inheritance diagram for jeod::RestartableSimpleSecondOrderODEIntegrator< size >:



Public Member Functions

- [RestartableSimpleSecondOrderODEIntegrator](#) ()=default
- [~RestartableSimpleSecondOrderODEIntegrator](#) () override=default
- [RestartableSimpleSecondOrderODEIntegrator](#) (const [RestartableSimpleSecondOrderODEIntegrator](#) &)=delete
- [RestartableSimpleSecondOrderODEIntegrator](#) & operator= (const [RestartableSimpleSecondOrderODEIntegrator](#) &)=delete
- [RestartableSimpleSecondOrderODEIntegrator](#) (er7_utils::SecondOrderODEIntegrator *&integ_ref)

Non-default constructor.

Private Member Functions

- er7_utils::SecondOrderODEIntegrator * [create_integrator_internal](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls) override

Create the integrator to be managed.

Additional Inherited Members

8.36.1 Detailed Description

```
template<unsigned int size>
class jeod::RestartableSimpleSecondOrderODEIntegrator< size >
```

A [RestartableSimpleSecondOrderODEIntegrator](#) is-a [RestartableSecondOrderODEIntegrator](#) that manages the integrator for a simple second order ODE problem, one in which the time derivative of the generalized position is the generalized velocity.

Template Parameters

<i>size</i>	Size of the position vector (and also of velocity).
-------------	---

Definition at line 373 of file restartable_state_integrator_templates.hh.

8.36.2 Constructor & Destructor Documentation

8.36.2.1 RestartableSimpleSecondOrderODEIntegrator() [1/3]

```
template<unsigned int size>
jeod::RestartableSimpleSecondOrderODEIntegrator< size >::RestartableSimpleSecondOrderODEIntegrator
( ) [default]
```

8.36.2.2 ~RestartableSimpleSecondOrderODEIntegrator()

```
template<unsigned int size>
jeod::RestartableSimpleSecondOrderODEIntegrator< size >::~~RestartableSimpleSecondOrderODEIntegrator
( ) [override], [default]
```

8.36.2.3 RestartableSimpleSecondOrderODEIntegrator() [2/3]

```
template<unsigned int size>
jeod::RestartableSimpleSecondOrderODEIntegrator< size >::~RestartableSimpleSecondOrderODEIntegrator
(
    const RestartableSimpleSecondOrderODEIntegrator< size > & ) [delete]
```

8.36.2.4 RestartableSimpleSecondOrderODEIntegrator() [3/3]

```
template<unsigned int size>
jeod::RestartableSimpleSecondOrderODEIntegrator< size >::~RestartableSimpleSecondOrderODEIntegrator
(
    er7_utils::SecondOrderODEIntegrator *& integ_ref ) [inline], [explicit]
```

Non-default constructor.

Parameters

<i>in, out</i>	<i>integ_ref</i>	Reference to the pointer to the integrator that is to be managed.
----------------	------------------	---

Definition at line 386 of file restartable_state_integrator_templates.hh.

8.36.3 Member Function Documentation

8.36.3.1 create_integrator_internal()

```
template<unsigned int size>
er7_utils::SecondOrderODEIntegrator* jeod::RestartableSimpleSecondOrderODEIntegrator< size >↵
::create_integrator_internal (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline], [override], [private],
[virtual]
```

Create the integrator to be managed.

Parameters

in	<i>generator</i>	Integrator constructor used to create the integrator.
in, out	<i>controls</i>	Integration controls to be passed to the generator.

Implements [jeod::RestartableStateIntegrator< er7_utils::SecondOrderODEIntegrator >](#).

Definition at line 399 of file restartable_state_integrator_templates.hh.

8.36.3.2 operator=()

```
template<unsigned int size>
RestartableSimpleSecondOrderODEIntegrator& jeod::RestartableSimpleSecondOrderODEIntegrator<
size >::operator= (
    const RestartableSimpleSecondOrderODEIntegrator< size > & ) [delete]
```

The documentation for this class was generated from the following file:

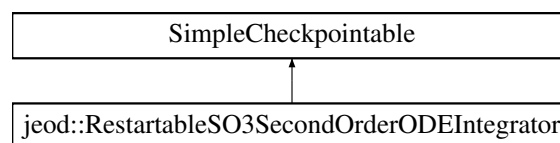
- [restartable_state_integrator_templates.hh](#)

8.37 jeod::RestartableSO3SecondOrderODEIntegrator Class Reference

A [RestartableSO3SecondOrderODEIntegrator](#) integrates a generalized second order ODE that describes rotation in three space.

```
#include <restartable_state_integrator.hh>
```

Inheritance diagram for jeod::RestartableSO3SecondOrderODEIntegrator:



Public Member Functions

- [RestartableSO3SecondOrderODEIntegrator \(\)](#)
Default constructor.
- [~RestartableSO3SecondOrderODEIntegrator \(\)](#) override
Destructor.
- [RestartableSO3SecondOrderODEIntegrator \(const RestartableSO3SecondOrderODEIntegrator &\)=delete](#)
- [RestartableSO3SecondOrderODEIntegrator & operator= \(const RestartableSO3SecondOrderODEIntegrator &\)=delete](#)
- void [create_integrator](#) ([GeneralizedSecondOrderODETechnique::TechniqueType](#) technique_in, const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls)
Create the integrator to be managed.

- void [destroy_integrator](#) ()
Destroy the integrator.
- [er7_utils::IntegratorResult integrate](#) (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position)
Propagate state to the specified stage of the integration process for an overall integration time interval of dyn_dt.
- void [reset_integrator](#) ()
Tell the integrator to reset itself.
- void [simple_restore](#) () override
Restore the integrator on restart.

Private Member Functions

- [JEOD_CLASS_ESTABLISH_FRIENDS2 p](#) (jeod, [RestartableSO3SecondOrderODEIntegrator](#))

Private Attributes

- [GeneralizedSecondOrderODETechnique::TechniqueType technique](#)
Specifies the mechanism for integrating rotational state.
- [er7_utils::SecondOrderODEIntegrator * integrator](#) {}
Pointer to the object that performs integration.
- [RestartableGeneralizedDerivSecondOrderODEIntegrator< 4, 3, er7_utils::LeftQuaternionGeneralizedPositionFunctions > generalized_deriv_integrator_manager](#)
SO3 generalized derivative integrator.
- [RestartableGeneralizedStepSecondOrderODEIntegrator< 4, 3, er7_utils::LeftQuaternionGeneralizedPositionFunctions > generalized_step_integrator_manager](#)
SO3 Lie Group integrator.

8.37.1 Detailed Description

A [RestartableSO3SecondOrderODEIntegrator](#) integrates a generalized second order ODE that describes rotation in three space.

Definition at line 281 of file `restartable_state_integrator.hh`.

8.37.2 Constructor & Destructor Documentation

8.37.2.1 [RestartableSO3SecondOrderODEIntegrator\(\)](#) [1/2]

```
jeod::RestartableSO3SecondOrderODEIntegrator::RestartableSO3SecondOrderODEIntegrator ( ) [inline]
```

Default constructor.

Definition at line 287 of file `restartable_state_integrator.hh`.

8.37.2.2 ~RestartableSO3SecondOrderODEIntegrator()

```
jeod::RestartableSO3SecondOrderODEIntegrator::~~RestartableSO3SecondOrderODEIntegrator ( )
[inline], [override]
```

Destructor.

Definition at line 295 of file restartable_state_integrator.hh.

8.37.2.3 RestartableSO3SecondOrderODEIntegrator() [2/2]

```
jeod::RestartableSO3SecondOrderODEIntegrator::RestartableSO3SecondOrderODEIntegrator (
    const RestartableSO3SecondOrderODEIntegrator & ) [delete]
```

8.37.3 Member Function Documentation

8.37.3.1 create_integrator()

```
void jeod::RestartableSO3SecondOrderODEIntegrator::create_integrator (
    GeneralizedSecondOrderODETechnique::TechniqueType technique_in,
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline]
```

Create the integrator to be managed.

Parameters

in	<i>technique_{in}</i>	Integration technique; generalized step vs deriv.
in	<i>generator</i>	Integrator constructor used to create the integrator.
in, out	<i>controls</i>	Integration controls to be passed to the generator.

Definition at line 309 of file restartable_state_integrator.hh.

8.37.3.2 destroy_integrator()

```
void jeod::RestartableSO3SecondOrderODEIntegrator::destroy_integrator ( ) [inline]
```

Destroy the integrator.

Definition at line 339 of file restartable_state_integrator.hh.

Referenced by operator=().

8.37.3.3 integrate()

```
er7_utils::IntegratorResult jeod::RestartableSO3SecondOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * accel,
    double * velocity,
    double * position ) [inline]
```

Propagate state to the specified stage of the integration process for an overall integration time interval of `dyn_dt`.

Note that this is a pass-through to the encapsulated integrator object. See `er7_utils::SecondOrderODEIntegrator::integrate` for details.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>accel</i>	Time derivative of the generalized velocity.
in, out	<i>velocity</i>	Generalized velocity vector.
in, out	<i>position</i>	Generalized position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 363 of file `restartable_state_integrator.hh`.

8.37.3.4 operator=()

```
RestartableSO3SecondOrderODEIntegrator& jeod::RestartableSO3SecondOrderODEIntegrator::operator=
(
    const RestartableSO3SecondOrderODEIntegrator & ) [delete]
```

References `destroy_integrator()`.

8.37.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RestartableSO3SecondOrderODEIntegrator::p (
    jeod ,
    RestartableSO3SecondOrderODEIntegrator ) [private]
```

8.37.3.6 reset_integrator()

```
void jeod::RestartableSO3SecondOrderODEIntegrator::reset_integrator ( ) [inline]
```

Tell the integrator to reset itself.

This should be called when the time step or time direction changes or upon a discrete change in state such docking/undocking. Such events invalidate saved state (if any). The integrator needs to restart from scratch when such events occur.

Definition at line 378 of file restartable_state_integrator.hh.

8.37.3.7 simple_restore()

```
void jeod::RestartableSO3SecondOrderODEIntegrator::simple_restore ( ) [inline], [override]
```

Restore the integrator on restart.

Definition at line 386 of file restartable_state_integrator.hh.

8.37.4 Field Documentation

8.37.4.1 generalized_deriv_integrator_manager

```
RestartableGeneralizedDerivSecondOrderODEIntegrator<4, 3, er7_utils::LeftQuaternionGeneralized↵  
PositionFunctions> jeod::RestartableSO3SecondOrderODEIntegrator::generalized_deriv_integrator↵  
_manager [private]
```

SO3 generalized derivative integrator.

trick_io(**)

Definition at line 423 of file restartable_state_integrator.hh.

8.37.4.2 generalized_step_integrator_manager

```
RestartableGeneralizedStepSecondOrderODEIntegrator<4, 3, er7_utils::LeftQuaternionGeneralized↵  
PositionFunctions> jeod::RestartableSO3SecondOrderODEIntegrator::generalized_step_integrator↵  
_manager [private]
```

SO3 Lie Group integrator.

trick_io(**)

Definition at line 429 of file restartable_state_integrator.hh.

8.37.4.3 integrator

```
er7_utils::SecondOrderODEIntegrator* jeod::RestartableSO3SecondOrderODEIntegrator::integrator
{} [private]
```

Pointer to the object that performs integration.

This object is created and managed by one of the integrator managers defined below. The technique dictates which of the two is used. `trick_units(-)`

Definition at line 417 of file `restartable_state_integrator.hh`.

8.37.4.4 technique

```
GeneralizedSecondOrderODETechnique::TechniqueType jeod::RestartableSO3SecondOrderODEIntegrator↔
::technique [private]
```

Initial value:

```
{
    GeneralizedSecondOrderODETechnique::Unspecified}
```

Specifies the mechanism for integrating rotational state.

`trick_units(-)`

Definition at line 409 of file `restartable_state_integrator.hh`.

The documentation for this class was generated from the following file:

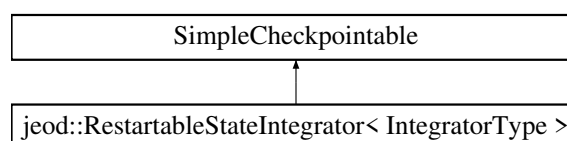
- [restartable_state_integrator.hh](#)

8.38 jeod::RestartableStateIntegrator< IntegratorType > Class Template Reference

A [RestartableStateIntegrator](#) establishes the basic capabilities needed to make a state integrator a managed resource.

```
#include <restartable_state_integrator_templates.hh>
```

Inheritance diagram for `jeod::RestartableStateIntegrator< IntegratorType >`:



Public Member Functions

- [~RestartableStateIntegrator](#) () override
Destructor.
- void [create_integrator](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls)
Create the integrator to be managed.
- void [destroy_integrator](#) ()
Destroy the integrator.
- void [clear_integrator_reference](#) ()
Clear the pointer to the external integrator object.
- void [set_integrator_reference](#) (IntegratorType *&integ_ptr)
Set the pointer to the external integrator object.
- void [simple_restore](#) () override
Restore the integrator on restart.
- [RestartableStateIntegrator](#) (const [RestartableStateIntegrator](#) &)=delete
- [RestartableStateIntegrator](#) & operator= (const [RestartableStateIntegrator](#) &)=delete

Protected Member Functions

- [RestartableStateIntegrator](#) ()=default
- [RestartableStateIntegrator](#) (IntegratorType *&integ_ref)
Non-default constructor.

Private Member Functions

- virtual IntegratorType * [create_integrator_internal](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls)=0
Create the integrator to be managed.
- virtual void [simple_restore_internal](#) (IntegratorType *integrator_ptr)
Perform technique-specific restart actions.

Private Attributes

- IntegratorType ** [integrator_handle](#) {}
Pointer to the containing object's integrator pointer.

8.38.1 Detailed Description

```
template<typename IntegratorType>
class jeod::RestartableStateIntegrator< IntegratorType >
```

A [RestartableStateIntegrator](#) establishes the basic capabilities needed to make a state integrator a managed resource.

This includes

- Creating the integrator object,
- Restoring the integrator's derivative function on restart, and
- Destroying the integrator when the [RestartableStateIntegrator](#) instance goes out of scope.

Template Parameters

<i>IntegratorType</i>	The type of integrator to be managed, either <code>er7_utils::FirstOrderODEIntegrator</code> or <code>er7_utils::SecondOrderODEIntegrator</code> .
-----------------------	--

Usage: This base class template is not directly usable. One must instead use one of the three class templates that derive from this class template:

- [RestartableFirstOrderODEIntegrator](#) to manage an `er7_utils::FirstOrderODEIntegrator` pointer;
- [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) to manage an `er7_utils::SecondOrderODEIntegrator` pointer for a generalized second order ODE, one in which the time derivative of generalized position is a function of generalized position and generalized velocity; and
- [RestartableGeneralizedStepSecondOrderODEIntegrator](#) to manage an `er7_utils::SecondOrderODEIntegrator` pointer for a generalized second order ODE, one in which the time derivative of generalized position is a function of generalized position and generalized velocity; and
- [RestartableSimpleSecondOrderODEIntegrator](#) to manage an `er7_utils::SecondOrderODEIntegrator` pointer for a simple second order ODE, one in which the time derivative of generalized position is the generalized velocity.

Each state integrator to be used in some class needs a pair of data members declared in the definition of that class. The first of these pairs is a pointer to the appropriate state integrator type (either `er7_utils::FirstOrderODEIntegrator` or `er7_utils::SecondOrderODEIntegrator`). The second of the pairs of data members is an instance of the appropriate derived class of [RestartableStateIntegrator](#) that will manage the pointer. An example:

```
class MyClass {
...
    er7_utils::SecondOrderODEIntegrator * integrator;
    RestartableSimpleSecondOrderODEIntegrator<3> integ_manager;
...
};
```

The pointer itself must be exposed to Trick for checkpoint and restart. The [RestartableStateIntegrator](#)-derived object should be hidden from Trick.

The connection between the pointer and the manager for that pointer is made in the initializer lists of the constructors for the class. Use the [RestartableStateIntegrator](#) non-default constructor to tie the [RestartableStateIntegrator](#) object with the pointer it is to manage:

```
MyClass::MyClass ()
:
...
    integrator(nullptr),
    integ_manager(integrator),
...
{
    // Body of MyClass constructor
}
```

Note that there the [RestartableStateIntegrator](#) default constructors exist only for the sake of the simulation engine. The default constructors do not create viable instances. One must use the non-default constructor to initialize [RestartableStateIntegrator](#) instances.

The integrator itself is not created at construction time. Creating the integrator is the job of the `create_integrator` method. This method should be called at initialization time after having created the integration constructor and the integration controls objects.

The counterpart to the `create_integrator` method is `destroy_integrator`. This method must be called prior to calling `create_integrator` if the integrator has already been created via a previous call to `create_integrator`. (For example, switching to a different integration technique). The `destroy_integrator` can be called at shutdown or destruction time, but this call is not essential. The [RestartableStateIntegrator](#) object will call this method internally when it goes out of scope.

Class Design This class template uses the non-virtual interface (NVI) design pattern, aka the template method design pattern (no relation to C++ class templates). The public `create_integrator` and `simple_restore` member functions use the private virtual `create_integrator_internal` and `simple_restore_internal` functions to create and restore the integrators.

Definition at line 171 of file `restartable_state_integrator_templates.hh`.

8.38.2 Constructor & Destructor Documentation

8.38.2.1 ~RestartableStateIntegrator()

```
template<typename IntegratorType >
jeod::RestartableStateIntegrator< IntegratorType >::~~RestartableStateIntegrator ( ) [inline],
[override]
```

Destructor.

Definition at line 179 of file `restartable_state_integrator_templates.hh`.

References `jeod::RestartableStateIntegrator< IntegratorType >::destroy_integrator()`.

8.38.2.2 RestartableStateIntegrator() [1/3]

```
template<typename IntegratorType >
jeod::RestartableStateIntegrator< IntegratorType >::~RestartableStateIntegrator (
    const RestartableStateIntegrator< IntegratorType > & ) [delete]
```

8.38.2.3 RestartableStateIntegrator() [2/3]

```
template<typename IntegratorType >
jeod::RestartableStateIntegrator< IntegratorType >::~RestartableStateIntegrator ( ) [protected],
[default]
```

8.38.2.4 RestartableStateIntegrator() [3/3]

```
template<typename IntegratorType >
jeod::RestartableStateIntegrator< IntegratorType >::~RestartableStateIntegrator (
    IntegratorType *& integ_ref ) [inline], [explicit], [protected]
```

Non-default constructor.

Parameters

<i>in, out</i>	<i>integ_ref</i>	Reference to the pointer to the integrator that is to be managed.
----------------	------------------	---

Definition at line 268 of file restartable_state_integrator_templates.hh.

8.38.3 Member Function Documentation

8.38.3.1 clear_integrator_reference()

```
template<typename IntegratorType >
void jeod::RestartableStateIntegrator< IntegratorType >::clear_integrator_reference ( ) [inline]
```

Clear the pointer to the external integrator object.

This currently (pre-Trick 13.0) needs to be called on restart because both pointers point to invalid objects.

Definition at line 225 of file restartable_state_integrator_templates.hh.

References jeod::RestartableStateIntegrator< IntegratorType >::integrator_handle.

8.38.3.2 create_integrator()

```
template<typename IntegratorType >
void jeod::RestartableStateIntegrator< IntegratorType >::create_integrator (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline]
```

Create the integrator to be managed.

Parameters

<i>in</i>	<i>generator</i>	Integrator constructor used to create the integrator.
<i>in, out</i>	<i>controls</i>	Integration controls to be passed to the generator.

Definition at line 189 of file restartable_state_integrator_templates.hh.

References jeod::RestartableStateIntegrator< IntegratorType >::create_integrator_internal(), jeod::RestartableStateIntegrator< IntegratorType >::destroy_integrator(), jeod::RestartableStateIntegrator< IntegratorType >::integrator_handle, and jeod::IntegrationMessages::invalid_request.

Referenced by jeod::RestartableScalarFirstOrderODEIntegrator::operator=().

8.38.3.3 create_integrator_internal()

```
template<typename IntegratorType >
virtual IntegratorType* jeod::RestartableStateIntegrator< IntegratorType >::create_integrator↵
_internal (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [private], [pure virtual]
```

Create the integrator to be managed.

Parameters

in	<i>generator</i>	Integrator constructor used to create the integrator.
in, out	<i>controls</i>	Integration controls to be passed to the generator.

Returns

The constructed integrator.

Implemented in [jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >](#), [jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< 4, 3, er7_utils::LeftQuaternionGeneralizedPositionFunctions >](#), [jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >](#), [jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< 4, 3, er7_utils::LeftQuaternionGeneralizedPositionFunctions >](#), [jeod::RestartableSimpleSecondOrderODEIntegrator< size >](#), [jeod::RestartableSimpleSecondOrderODEIntegrator< 2 >](#), [jeod::RestartableSimpleSecondOrderODEIntegrator< 3 >](#), [jeod::RestartableFirstOrderODEIntegrator< size >](#), and [jeod::RestartableFirstOrderODEIntegrator< 1 >](#).

Referenced by [jeod::RestartableStateIntegrator< IntegratorType >::create_integrator\(\)](#).

8.38.3.4 destroy_integrator()

```
template<typename IntegratorType >
void jeod::RestartableStateIntegrator< IntegratorType >::destroy_integrator ( ) [inline]
```

Destroy the integrator.

Definition at line 212 of file `restartable_state_integrator_templates.hh`.

References [jeod::RestartableStateIntegrator< IntegratorType >::integrator_handle](#).

Referenced by [jeod::RestartableStateIntegrator< IntegratorType >::create_integrator\(\)](#), [jeod::RestartableT3↵SecondOrderODEIntegrator::destroy_integrator\(\)](#), and [jeod::RestartableStateIntegrator< IntegratorType >::~↵RestartableStateIntegrator\(\)](#).

8.38.3.5 operator=()

```
template<typename IntegratorType >
RestartableStateIntegrator& jeod::RestartableStateIntegrator< IntegratorType >::operator= (
    const RestartableStateIntegrator< IntegratorType > & ) [delete]
```

8.38.3.6 set_integrator_reference()

```
template<typename IntegratorType >
void jeod::RestartableStateIntegrator< IntegratorType >::set_integrator_reference (
    IntegratorType *& integ_ptr ) [inline]
```

Set the pointer to the external integrator object.

This currently (pre-Trick 13.0) needs to be called on restart because the `integrator_handle` is not properly restored by checkpoint.

Parameters

<code>in, out</code>	<code>integ_ptr</code>	Reference to the external integrator object.
----------------------	------------------------	--

Definition at line 236 of file `restartable_state_integrator_templates.hh`.

References `jeod::RestartableStateIntegrator< IntegratorType >::integrator_handle`.

Referenced by `jeod::RestartableT3SecondOrderODEIntegrator::simple_restore()`.

8.38.3.7 simple_restore()

```
template<typename IntegratorType >
void jeod::RestartableStateIntegrator< IntegratorType >::simple_restore ( ) [inline], [override]
```

Restore the integrator on restart.

This currently (pre-Trick 13.0) needs to be called after calling `set_integrator_reference`.

Definition at line 246 of file `restartable_state_integrator_templates.hh`.

References `jeod::RestartableStateIntegrator< IntegratorType >::integrator_handle`, and `jeod::RestartableStateIntegrator< IntegratorType >::simple_restore_internal()`.

Referenced by `jeod::RestartableT3SecondOrderODEIntegrator::simple_restore()`.

8.38.3.8 simple_restore_internal()

```
template<typename IntegratorType >
virtual void jeod::RestartableStateIntegrator< IntegratorType >::simple_restore_internal (
    IntegratorType * integ_ptr ) [inline], [private], [virtual]
```

Perform technique-specific restart actions.

The default is to do nothing.

Parameters

<code>in, out</code>	<code>integrator_ptr</code>	The integrator object to be restored
----------------------	-----------------------------	--------------------------------------

Reimplemented in [jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< position_size, velocity_size, StepFunctions >](#), [jeod::RestartableGeneralizedStepSecondOrderODEIntegrator< 4, 3, er7_utils::LeftQuaternionGeneralizedPositionFunctions >](#), [jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< position_size, velocity_size, DerivFunctions >](#), and [jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator< 4, 3, er7_utils::LeftQuaternionGeneralizedPositionFunctions >](#).

Definition at line 290 of file `restartable_state_integrator_templates.hh`.

Referenced by [jeod::RestartableStateIntegrator< IntegratorType >::simple_restore\(\)](#).

8.38.4 Field Documentation

8.38.4.1 integrator_handle

```
template<typename IntegratorType >
IntegratorType** jeod::RestartableStateIntegrator< IntegratorType >::integrator_handle {}
[private]
```

Pointer to the containing object's integrator pointer.

`trick_io(**)`

Definition at line 297 of file `restartable_state_integrator_templates.hh`.

Referenced by [jeod::RestartableStateIntegrator< IntegratorType >::clear_integrator_reference\(\)](#), [jeod::RestartableStateIntegrator< IntegratorType >::create_integrator\(\)](#), [jeod::RestartableStateIntegrator< IntegratorType >::destroy_integrator\(\)](#), [jeod::RestartableStateIntegrator< IntegratorType >::set_integrator_reference\(\)](#), and [jeod::RestartableStateIntegrator< IntegratorType >::simple_restore\(\)](#).

The documentation for this class was generated from the following file:

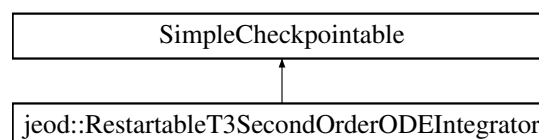
- [restartable_state_integrator_templates.hh](#)

8.39 jeod::RestartableT3SecondOrderODEIntegrator Class Reference

A [RestartableT3SecondOrderODEIntegrator](#) integrates a second order ODE in three space, $d^2x/dt^2 = a(x,t)$, where x is a three-vector.

```
#include <restartable_state_integrator.hh>
```

Inheritance diagram for [jeod::RestartableT3SecondOrderODEIntegrator](#):



Public Member Functions

- [RestartableT3SecondOrderODEIntegrator](#) ()
Default constructor.
- [~RestartableT3SecondOrderODEIntegrator](#) () override=default
- [RestartableT3SecondOrderODEIntegrator](#) (const [RestartableT3SecondOrderODEIntegrator](#) &)=delete
- [RestartableT3SecondOrderODEIntegrator](#) & operator= (const [RestartableT3SecondOrderODEIntegrator](#) &)=delete
- void [create_integrator](#) (const er7_utils::IntegratorConstructor &generator, er7_utils::IntegrationControls &controls)
Create the integrator to be managed.
- void [destroy_integrator](#) ()
Destroy the integrator.
- er7_utils::IntegratorResult [integrate](#) (double dyn_dt, unsigned int target_stage, const double *accel, double *velocity, double *position)
Propagate state to the specified stage of the integration process for an overall integration time interval of dyn_dt.
- void [reset_integrator](#) ()
Tell the integrator to reset itself.
- void [simple_restore](#) () override
Restore the integrator on restart.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [RestartableT3SecondOrderODEIntegrator](#))

Private Attributes

- er7_utils::SecondOrderODEIntegrator * [integrator](#) {}
Pointer to the object that performs integration.
- [RestartableSimpleSecondOrderODEIntegrator](#)< 3 > [integrator_manager](#)
Object that creates and manages the integrator object.

8.39.1 Detailed Description

A [RestartableT3SecondOrderODEIntegrator](#) integrates a second order ODE in three space, $d^2x/dt^2 = a(x,t)$, where x is a three-vector.

Definition at line 184 of file restartable_state_integrator.hh.

8.39.2 Constructor & Destructor Documentation

8.39.2.1 RestartableT3SecondOrderODEIntegrator() [1/2]

```
jeod::RestartableT3SecondOrderODEIntegrator::RestartableT3SecondOrderODEIntegrator ( ) [inline]
```

Default constructor.

Definition at line 190 of file restartable_state_integrator.hh.

8.39.2.2 ~RestartableT3SecondOrderODEIntegrator()

```
jeod::RestartableT3SecondOrderODEIntegrator::~~RestartableT3SecondOrderODEIntegrator ( ) [override],
[default]
```

8.39.2.3 RestartableT3SecondOrderODEIntegrator() [2/2]

```
jeod::RestartableT3SecondOrderODEIntegrator::~RestartableT3SecondOrderODEIntegrator (
    const RestartableT3SecondOrderODEIntegrator & ) [delete]
```

8.39.3 Member Function Documentation

8.39.3.1 create_integrator()

```
void jeod::RestartableT3SecondOrderODEIntegrator::create_integrator (
    const er7_utils::IntegratorConstructor & generator,
    er7_utils::IntegrationControls & controls ) [inline]
```

Create the integrator to be managed.

Parameters

in	<i>generator</i>	Integrator constructor used to create the integrator.
in, out	<i>controls</i>	Integration controls to be passed to the generator.

Definition at line 205 of file restartable_state_integrator.hh.

8.39.3.2 destroy_integrator()

```
void jeod::RestartableT3SecondOrderODEIntegrator::destroy_integrator ( ) [inline]
```

Destroy the integrator.

Definition at line 214 of file restartable_state_integrator.hh.

References jeod::RestartableStateIntegrator< IntegratorType >::destroy_integrator(), and integrator_manager.

8.39.3.3 integrate()

```
er7_utils::IntegratorResult jeod::RestartableT3SecondOrderODEIntegrator::integrate (
    double dyn_dt,
    unsigned int target_stage,
    const double * accel,
    double * velocity,
    double * position ) [inline]
```

Propagate state to the specified stage of the integration process for an overall integration time interval of `dyn_dt`.

Note that this is a pass-through to the encapsulated integrator object. See `er7_utils::SecondOrderODEIntegrator::integrate` for details.

Parameters

in	<i>dyn_dt</i>	Dynamic time step, in dynamic time seconds.
in	<i>target_stage</i>	The stage of the integration process that the integrator should try to attain.
in	<i>accel</i>	Time derivative of the generalized velocity.
in, out	<i>velocity</i>	Generalized velocity vector.
in, out	<i>position</i>	Generalized position vector.

Returns

The status (time advance, pass/fail status) of the integration.

Definition at line 235 of file `restartable_state_integrator.hh`.

8.39.3.4 operator=()

```
RestartableT3SecondOrderODEIntegrator& jeod::RestartableT3SecondOrderODEIntegrator::operator=
(
    const RestartableT3SecondOrderODEIntegrator & ) [delete]
```

8.39.3.5 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::RestartableT3SecondOrderODEIntegrator::p (
    jeod ,
    RestartableT3SecondOrderODEIntegrator ) [private]
```

8.39.3.6 reset_integrator()

```
void jeod::RestartableT3SecondOrderODEIntegrator::reset_integrator ( ) [inline]
```

Tell the integrator to reset itself.

This should be called when the time step or time direction changes or upon a discrete change in state such docking/undocking. Such events invalidate saved state (if any). The integrator needs to restart from scratch when such events occur.

Definition at line 250 of file restartable_state_integrator.hh.

8.39.3.7 simple_restore()

```
void jeod::RestartableT3SecondOrderODEIntegrator::simple_restore ( ) [inline], [override]
```

Restore the integrator on restart.

Definition at line 258 of file restartable_state_integrator.hh.

References `integrator`, `integrator_manager`, `jeod::RestartableStateIntegrator< IntegratorType >::set_integrator_↔reference()`, and `jeod::RestartableStateIntegrator< IntegratorType >::simple_restore()`.

8.39.4 Field Documentation

8.39.4.1 integrator

```
er7_utils::SecondOrderODEIntegrator* jeod::RestartableT3SecondOrderODEIntegrator::integrator  
{ } [private]
```

Pointer to the object that performs integration.

The object is created managed by the integrator manager. `trick_units(-)`

Definition at line 269 of file restartable_state_integrator.hh.

Referenced by `simple_restore()`.

8.39.4.2 integrator_manager

```
RestartableSimpleSecondOrderODEIntegrator<3> jeod::RestartableT3SecondOrderODEIntegrator←
::integrator_manager [private]
```

Object that creates and manages the integrator object.

trick_io(**)

Definition at line 274 of file restartable_state_integrator.hh.

Referenced by destroy_integrator(), and simple_restore().

The documentation for this class was generated from the following file:

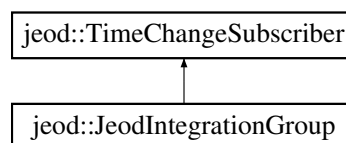
- [restartable_state_integrator.hh](#)

8.40 jeod::TimeChangeSubscriber Class Reference

A [TimeChangeSubscriber](#) is some object that wants to be notified of changes in the nature of time.

```
#include <time_change_subscriber.hh>
```

Inheritance diagram for jeod::TimeChangeSubscriber:



Public Member Functions

- [TimeChangeSubscriber](#) ()=default
- virtual [~TimeChangeSubscriber](#) ()=default
- [TimeChangeSubscriber](#) (const [TimeChangeSubscriber](#) &)=default
- [TimeChangeSubscriber](#) & operator= (const [TimeChangeSubscriber](#) &)=default
- virtual void [respond_to_time_change](#) ()=0

Somehow respond to a change in the nature of time.

Private Member Functions

- JEOD_CLASS_ESTABLISH_FRIENDS2 p (jeod, [TimeChangeSubscriber](#))

8.40.1 Detailed Description

A [TimeChangeSubscriber](#) is some object that wants to be notified of changes in the nature of time.

Definition at line 74 of file time_change_subscriber.hh.

8.40.2 Constructor & Destructor Documentation

8.40.2.1 TimeChangeSubscriber() [1/2]

```
jeod::TimeChangeSubscriber::TimeChangeSubscriber ( ) [default]
```

8.40.2.2 ~TimeChangeSubscriber()

```
virtual jeod::TimeChangeSubscriber::~~TimeChangeSubscriber ( ) [virtual], [default]
```

8.40.2.3 TimeChangeSubscriber() [2/2]

```
jeod::TimeChangeSubscriber::TimeChangeSubscriber (
    const TimeChangeSubscriber & ) [default]
```

8.40.3 Member Function Documentation

8.40.3.1 operator=()

```
TimeChangeSubscriber& jeod::TimeChangeSubscriber::operator= (
    const TimeChangeSubscriber & ) [default]
```

8.40.3.2 p()

```
JEOD_CLASS_ESTABLISH_FRIENDS2 jeod::TimeChangeSubscriber::p (
    jeod ,
    TimeChangeSubscriber ) [private]
```

8.40.3.3 respond_to_time_change()

```
virtual void jeod::TimeChangeSubscriber::respond_to_time_change ( ) [pure virtual]
```

Somehow respond to a change in the nature of time.

Implemented in [jeod::JeodIntegrationGroup](#).

The documentation for this class was generated from the following file:

- [time_change_subscriber.hh](#)

8.41 er7_utils::TwoDArray< T > Class Template Reference

RAII template class that implements a rectangular two dimensional array.

```
#include <two_d_array.hh>
```

Public Member Functions

- [TwoDArray](#) ()=default
Default constructor.
- [TwoDArray](#) (const [TwoDArray](#)< T > &src)
Copy constructor.
- [~TwoDArray](#) ()
Destructor.
- [TwoDArray](#)< T > & [operator=](#) ([TwoDArray](#)< T > src)
Copy and swap assignment constructor.
- const T * [operator\[\]](#) (int N) const
Const overloaded index operator.
- T * [operator\[\]](#) (int N)
Non-const overloaded index operator.
- const T & [operator\(\)](#) (int N, int M) const
Const overloaded function operator.
- T & [operator\(\)](#) (int N, int M)
Non-const overloaded function operator.
- [operator const T *const *](#) () const
Const conversion operator to T const const*.*
- [operator T*const *](#) ()
*Non-const conversion operator to T*const*.*
- const T * [at](#) (int N) const
Range-checked equivalent of const T [operator\[\]](#)(int N) const.*
- T * [at](#) (int N)
Range-checked equivalent of T [operator\[\]](#)(int N).*
- const T & [at](#) (int N, int M) const
Range-checked equivalent of const T& [operator\(\)](#)(int N, int M) const.
- T & [at](#) (int N, int M)
Range-checked equivalent of T& [operator\(\)](#)(int N, int M).
- void [allocate](#) (std::size_t N, std::size_t M)

- *Allocate the array.*
- void [rotate_down](#) (int limit)
Rotate elements 0 to limit downward, with array element 0 moved to array element limit.
- void [rotate_up](#) (int limit)
Rotate elements 0 to limit upward, with array element limit moved to array element 0.
- void [downsample](#) (int limit)
Downsample the array by swapping pointers.
- void [swap](#) ([TwoDArray](#)< T > &other)
*Swap the contents of *this with the other.*

Protected Attributes

- int [n](#) {}
The number of rows in the array.
- int [m](#) {}
The number of columns in the array.
- T * [data_array](#) {}
The array data, as an NxM array of T.
- T ** [row_array](#) {}
The rows in the array.

Private Member Functions

- void [allocate_internal](#) ()
Allocate memory for the array.
- void [deallocate_internal](#) ()
Deallocate memory for the array.

Friends

- void [swap](#) ([TwoDArray](#)< T > &first, [TwoDArray](#)< T > &second)
Swap the contents of the two provided arrays.

8.41.1 Detailed Description

```
template<typename T>
class er7_utils::TwoDArray< T >
```

RAII template class that implements a rectangular two dimensional array.

The implementation provides two special-purpose features that are needed by some of the ER7 utilities and JEOD integrators. The rows of the array can be rotated and downsampled.

Template Parameters

<i>T</i>	Type of each element of the array.
----------	------------------------------------

Definition at line 79 of file two_d_array.hh.

8.41.2 Constructor & Destructor Documentation

8.41.2.1 TwoDArray() [1/2]

```
template<typename T >
er7_utils::TwoDArray< T >::TwoDArray ( ) [default]
```

Default constructor.

8.41.2.2 TwoDArray() [2/2]

```
template<typename T >
er7_utils::TwoDArray< T >::TwoDArray (
    const TwoDArray< T > & src ) [inline]
```

Copy constructor.

Parameters

<i>src</i>	Item to be copied.
------------	--------------------

Definition at line 91 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::allocate()`, `er7_utils::TwoDArray< T >::m`, `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

8.41.2.3 ~TwoDArray()

```
template<typename T >
er7_utils::TwoDArray< T >::~~TwoDArray ( ) [inline]
```

Destructor.

Definition at line 109 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::deallocate_internal()`.

8.41.3 Member Function Documentation

8.41.3.1 allocate()

```
template<typename T >
void er7_utils::TwoDArray< T >::allocate (
    std::size_t N,
    std::size_t M ) [inline]
```

Allocate the array.

Parameters

<i>N</i>	Number of rows in the array.
<i>M</i>	Number of columns in the array.

Exceptions

<i>std::domain_error</i>	N and/or M won't fit in a signed int.
--------------------------	---------------------------------------

Definition at line 252 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::allocate_internal()`, `er7_utils::TwoDArray< T >::data_array`, `er7_utils::TwoDArray< T >::deallocate_internal()`, `er7_utils::TwoDArray< T >::m`, `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::GaussJacksonIntegratorBase()`, and `er7_utils::TwoDArray< T >::TwoDArray()`.

8.41.3.2 allocate_internal()

```
template<typename T >
void er7_utils::TwoDArray< T >::allocate_internal ( ) [inline], [private]
```

Allocate memory for the array.

Definition at line 384 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::data_array`, `er7_utils::TwoDArray< T >::m`, `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

Referenced by `er7_utils::TwoDArray< T >::allocate()`.

8.41.3.3 at() [1/4]

```
template<typename T >
T* er7_utils::TwoDArray< T >::at (
    int N ) [inline]
```

Range-checked equivalent of `T* operator[](int N)`.

Parameters

<i>N</i>	Row index.
----------	------------

Returns

Modifiable pointer to the Nth row in the array.

Exceptions

<i>std::out_of_range</i>	If <i>N</i> is an invalid index.
--------------------------	----------------------------------

Definition at line 205 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

8.41.3.4 at() [2/4]

```
template<typename T >
const T* er7_utils::TwoDArray< T >::at (
    int N ) const [inline]
```

Range-checked equivalent of `const T* operator[](int N) const`.

Parameters

<i>N</i>	Row index.
----------	------------

Returns

Const pointer to the Nth row in the array.

Exceptions

<i>std::out_of_range</i>	If <i>N</i> is an invalid index.
--------------------------	----------------------------------

Definition at line 190 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

8.41.3.5 at() [3/4]

```
template<typename T >
T& er7_utils::TwoDArray< T >::at (
```

```
int N,
int M ) [inline]
```

Range-checked equivalent of T& [operator\(\)\(int N, int M\)](#).

Parameters

<i>N</i>	Row index.
<i>M</i>	Column index.

Returns

Reference to the N,M element of the array.

Exceptions

<i>std::out_of_range</i>	If <i>N</i> or <i>M</i> is an invalid index.
--------------------------	--

Definition at line 237 of file two_d_array.hh.

References [er7_utils::TwoDArray< T >::m](#), [er7_utils::TwoDArray< T >::n](#), and [er7_utils::TwoDArray< T >::row_↵array](#).

8.41.3.6 at() [4/4]

```
template<typename T >
const T& er7\_utils::TwoDArray< T >::at (
    int N,
    int M ) const [inline]
```

Range-checked equivalent of const T& [operator\(\)\(int N, int M\) const](#).

Parameters

<i>N</i>	Row index.
<i>M</i>	Column index.

Returns

Const reference to the N,M element of the array.

Exceptions

<i>std::out_of_range</i>	If <i>N</i> or <i>M</i> is an invalid index.
--------------------------	--

Definition at line 221 of file two_d_array.hh.

References `er7_utils::TwoDArray< T >::m`, `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

8.41.3.7 `deallocate_internal()`

```
template<typename T >
void er7_utils::TwoDArray< T >::deallocate_internal ( ) [inline], [private]
```

Deallocate memory for the array.

Definition at line 393 of file `two_d_array.hh`.

References `er7_utils::TwoDArray< T >::data_array`, and `er7_utils::TwoDArray< T >::row_array`.

Referenced by `er7_utils::TwoDArray< T >::allocate()`, and `er7_utils::TwoDArray< T >::~~TwoDArray()`.

8.41.3.8 `downsample()`

```
template<typename T >
void er7_utils::TwoDArray< T >::downsample (
    int limit ) [inline]
```

Downsample the array by swapping pointers.

Parameters

<i>limit</i>	Number of usable rows after downsample.
--------------	---

Exceptions

<code>std::out_of_range</code>	If <i>limit</i> represents an invalid index.
--------------------------------	--

Definition at line 321 of file `two_d_array.hh`.

References `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::downsample_hist()`.

8.41.3.9 `operator const T *const *()`

```
template<typename T >
er7_utils::TwoDArray< T >::operator const T *const * ( ) const [inline]
```

Const conversion operator to `T const* const*`.

Returns

Non-modifiable pointer to the array.

Definition at line 170 of file two_d_array.hh.

8.41.3.10 operator T*const*()

```
template<typename T >
er7_utils::TwoDArray< T >::operator T*const * ( ) [inline]
```

Non-const conversion operator to T*const*.

Returns

Modifiable pointer to the array.

Definition at line 179 of file two_d_array.hh.

8.41.3.11 operator()() [1/2]

```
template<typename T >
T& er7_utils::TwoDArray< T >::operator() (
    int N,
    int M ) [inline]
```

Non-const overloaded function operator.

Parameters

<i>N</i>	Row index.
<i>M</i>	Column index.

Returns

Reference to the N,M element of the array.

Definition at line 161 of file two_d_array.hh.

References er7_utils::TwoDArray< T >::row_array.

8.41.3.12 operator>() [2/2]

```
template<typename T >
const T& er7_utils::TwoDArray< T >::operator() (
    int N,
    int M ) const [inline]
```

Const overloaded function operator.

Parameters

<i>N</i>	Row index.
<i>M</i>	Column index.

Returns

Const reference to the N,M element of the array.

Definition at line 150 of file two_d_array.hh.

References er7_utils::TwoDArray< T >::row_array.

8.41.3.13 operator=()

```
template<typename T >
TwoDArray<T>& er7_utils::TwoDArray< T >::operator= (
    TwoDArray< T > src ) [inline]
```

Copy and swap assignment constructor.

Parameters

<i>src</i>	Item to be copied.
------------	--------------------

Definition at line 118 of file two_d_array.hh.

References er7_utils::TwoDArray< T >::swap().

8.41.3.14 operator[]() [1/2]

```
template<typename T >
T* er7_utils::TwoDArray< T >::operator[] (
    int N ) [inline]
```

Non-const overloaded index operator.

Parameters

<i>N</i>	Row index.
----------	------------

Returns

Modifiable pointer to the Nth row in the array.

Definition at line 139 of file two_d_array.hh.

References er7_utils::TwoDArray< T >::row_array.

8.41.3.15 operator[]() [2/2]

```
template<typename T >
const T* er7_utils::TwoDArray< T >::operator[] (
    int N ) const [inline]
```

Const overloaded index operator.

Parameters

<i>N</i>	Row index.
----------	------------

Returns

Const pointer to the Nth row in the array.

Definition at line 129 of file two_d_array.hh.

References er7_utils::TwoDArray< T >::row_array.

8.41.3.16 rotate_down()

```
template<typename T >
void er7_utils::TwoDArray< T >::rotate_down (
    int limit ) [inline]
```

Rotate elements 0 to limit downward, with array element 0 moved to array element limit.

Parameters

<i>limit</i>	Index of last element participating in the rotation.
--------------	--

Exceptions

<code>std::out_of_range</code>	If <i>limit</i> is an invalid index.
--------------------------------	--------------------------------------

Definition at line 280 of file `two_d_array.hh`.

References `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

Referenced by `jeod::GaussJacksonIntegratorBase< State, Primer >::rotate_acc_hist()`.

8.41.3.17 rotate_up()

```
template<typename T >
void er7_utils::TwoDArray< T >::rotate_up (
    int limit ) [inline]
```

Rotate elements 0 to limit upward, with array element limit moved to array element 0.

Parameters

<i>limit</i>	Index of last element participating in the rotation.
--------------	--

Exceptions

<code>std::out_of_range</code>	If <i>limit</i> is an invalid index.
--------------------------------	--------------------------------------

Definition at line 301 of file `two_d_array.hh`.

References `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

8.41.3.18 swap()

```
template<typename T >
void er7_utils::TwoDArray< T >::swap (
    TwoDArray< T > & other ) [inline]
```

Swap the contents of `*this` with the other.

Parameters

<i>other</i>	Other array.
--------------	--------------

Definition at line 338 of file `two_d_array.hh`.

References `er7_utils::TwoDArray< T >::data_array`, `er7_utils::TwoDArray< T >::m`, `er7_utils::TwoDArray< T >::n`, and `er7_utils::TwoDArray< T >::row_array`.

Referenced by `er7_utils::TwoDArray< T >::operator=()`, and `jeod::GaussJacksonIntegratorBase< State, Primer >::swap()`.

8.41.4 Friends And Related Function Documentation

8.41.4.1 swap

```
template<typename T >
void swap (
    TwoDArray< T > & first,
    TwoDArray< T > & second ) [friend]
```

Swap the contents of the two provided arrays.

Parameters

<i>first</i>	First array.
<i>second</i>	Second array.

Definition at line 351 of file `two_d_array.hh`.

8.41.5 Field Documentation

8.41.5.1 data_array

```
template<typename T >
T* er7_utils::TwoDArray< T >::data_array {} [protected]
```

The array data, as an NxM array of T.

`trick_units(-)`

Definition at line 373 of file `two_d_array.hh`.

Referenced by `er7_utils::TwoDArray< T >::allocate()`, `er7_utils::TwoDArray< T >::allocate_internal()`, `er7_utils::TwoDArray< T >::deallocate_internal()`, and `er7_utils::TwoDArray< T >::swap()`.

8.41.5.2 m

```
template<typename T >
int er7_utils::TwoDArray< T >::m {} [protected]
```

The number of columns in the array.

trick_units(—)

Definition at line 368 of file two_d_array.hh.

Referenced by `er7_utils::TwoDArray< T >::allocate()`, `er7_utils::TwoDArray< T >::allocate_internal()`, `er7_utils::TwoDArray< T >::at()`, `er7_utils::TwoDArray< T >::swap()`, and `er7_utils::TwoDArray< T >::TwoDArray()`.

8.41.5.3 n

```
template<typename T >
int er7_utils::TwoDArray< T >::n {} [protected]
```

The number of rows in the array.

trick_units(—)

Definition at line 363 of file two_d_array.hh.

Referenced by `er7_utils::TwoDArray< T >::allocate()`, `er7_utils::TwoDArray< T >::allocate_internal()`, `er7_utils::TwoDArray< T >::at()`, `er7_utils::TwoDArray< T >::downsample()`, `er7_utils::TwoDArray< T >::rotate_down()`, `er7_utils::TwoDArray< T >::rotate_up()`, `er7_utils::TwoDArray< T >::swap()`, and `er7_utils::TwoDArray< T >::TwoDArray()`.

8.41.5.4 row_array

```
template<typename T >
T** er7_utils::TwoDArray< T >::row_array {} [protected]
```

The rows in the array.

trick_units(—)

Definition at line 378 of file two_d_array.hh.

Referenced by `er7_utils::TwoDArray< T >::allocate()`, `er7_utils::TwoDArray< T >::allocate_internal()`, `er7_utils::TwoDArray< T >::at()`, `er7_utils::TwoDArray< T >::deallocate_internal()`, `er7_utils::TwoDArray< T >::downsample()`, `er7_utils::TwoDArray< double >::operator const double *const *()`, `er7_utils::TwoDArray< double >::operator double *const *()`, `er7_utils::TwoDArray< T >::operator()()`, `er7_utils::TwoDArray< T >::operator[]()`, `er7_utils::TwoDArray< T >::rotate_down()`, `er7_utils::TwoDArray< T >::rotate_up()`, `er7_utils::TwoDArray< T >::swap()`, and `er7_utils::TwoDArray< T >::TwoDArray()`.

The documentation for this class was generated from the following file:

- [two_d_array.hh](#)

Chapter 9

File Documentation

9.1 gauss_jackson_coefficients_pair.cc File Reference

Defines member functions for the class GaussJacksonCoefficientsPair.

```
#include "../include/gauss_jackson_coefficients_pair.hh"
#include "../include/gauss_jackson_two_state.hh"
#include "er7_utils/interface/include/alloc.hh"
#include <algorithm>
#include <iostream>
```

Namespaces

- [jeod](#)

Namespace jeod.

9.1.1 Detailed Description

Defines member functions for the class GaussJacksonCoefficientsPair.

9.2 gauss_jackson_coefficients_pair.hh File Reference

Defines the class GaussJacksonCoefficientsPair, which contains summed Adams and Gauss-Jackson coefficient pair.

```
#include "gauss_jackson_one_state.hh"
#include "gauss_jackson_two_state.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/interface/include/config.hh"
#include <cstdint>
#include <iosfwd>
```

Data Structures

- class [jeod::GaussJacksonCoefficientsPair](#)
Contains a summed Adams and Gauss-Jackson coefficient pair.

Namespaces

- [jeod](#)
Namespace jeod.

9.2.1 Detailed Description

Defines the class GaussJacksonCoefficientsPair, which contains summed Adams and Gauss-Jackson coefficient pair.

9.3 gauss_jackson_coeffs.cc File Reference

Defines member functions for the class GaussJacksonCoeffs.

```
#include "../include/gauss_jackson_coeffs.hh"
#include "../include/gauss_jackson_coefficients_pair.hh"
#include "../include/gauss_jackson_rational_coeffs.hh"
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/math/include/n_choose_m.hh"
#include <algorithm>
#include <cassert>
#include <iostream>
```

Namespaces

- [jeod](#)
Namespace jeod.

Functions

- std::ostream & [jeod::operator<<](#) (std::ostream &stream, const GaussJacksonCoeffs &coeff)

9.3.1 Detailed Description

Defines member functions for the class GaussJacksonCoeffs.

9.4 gauss_jackson_coefs.hh File Reference

Defines the class GaussJacksonCoeffs, which contains the Gauss-Jackson predictor and corrector coefficients.

```
#include "gauss_jackson_coefficients_pair.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include <iosfwd>
```

Data Structures

- class [jeod::GaussJacksonCoeffs](#)

Contains the Gauss-Jackson predictor and corrector coefficients.

Namespaces

- [jeod](#)

Namespace jeod.

9.4.1 Detailed Description

Defines the class GaussJacksonCoeffs, which contains the Gauss-Jackson predictor and corrector coefficients.

9.5 gauss_jackson_config.cc File Reference

Defines member functions for the class GaussJacksonIntegratorConstructor.

```
#include "utils/math/include/numerical.hh"
#include "../include/gauss_jackson_config.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include <algorithm>
#include <string>
```

Namespaces

- [jeod](#)

Namespace jeod.

Functions

- static GaussJacksonConfig [jeod::set_default_config_values](#) (const GaussJacksonConfig &config)
Swap the negative ones in the supplied config with the default values, some of which are computed.
- static unsigned int [jeod::validate_config](#) (const GaussJacksonConfig &config)
Check for invalid values in the supplied config.

9.5.1 Detailed Description

Defines member functions for the class GaussJacksonIntegratorConstructor.

9.6 gauss_jackson_config.hh File Reference

Defines the class GaussJacksonConfig, which specifies Gauss-Jackson configuration data.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/integration_technique.hh"
```

Data Structures

- class [jeod::GaussJacksonConfig](#)
Contains Gauss-Jackson configuration data.

Namespaces

- [jeod](#)
Namespace jeod.

9.6.1 Detailed Description

Defines the class GaussJacksonConfig, which specifies Gauss-Jackson configuration data.

9.7 gauss_jackson_first_order_ode_integrator.hh File Reference

Defines the class GaussJacksonFirstOrderODEIntegrator, which integrates a first order ODE using the summed Adams technique.

```
#include "gauss_jackson_integrator_base_first.hh"
```

Data Structures

- class [jeod::GaussJacksonFirstOrderODEIntegrator](#)
Integrates a first order ODE using the summed Adams technique that is embedded within the Gauss-Jackson technique.

Namespaces

- [jeod](#)
Namespace jeod.

9.7.1 Detailed Description

Defines the class GaussJacksonFirstOrderODEIntegrator, which integrates a first order ODE using the summed Adams technique.

- Note: This is a header-only implementation. There is no source file that corresponds to this header.

9.8 gauss_jackson_generalized_second_order_ode_integrator.cc File Reference

Defines member functions for the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator.

```
#include "../include/gauss_jackson_generalized_second_order_ode_integrator.↵  
hh"  
#include "er7_utils/interface/include/alloc.hh"  
#include <algorithm>
```

Namespaces

- [jeod](#)
Namespace jeod.

9.8.1 Detailed Description

Defines member functions for the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator.

9.9 gauss_jackson_generalized_second_order_ode_integrator.hh File Reference

Defines the class GaussJacksonGeneralizedDerivSecondOrderODEIntegrator, which integrates a generalized derivative second order ODE using Gauss-Jackson.

```
#include "gauss_jackson_first_order_ode_integrator.hh"  
#include "gauss_jackson_simple_second_order_ode_integrator.hh"  
#include "utils/sim_interface/include/jeod_class.hh"  
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵  
hh"
```

Data Structures

- class [jeod::GaussJacksonGeneralizedDerivSecondOrderODEIntegrator](#)
Integrates a generalized derivative second order ODE using Gauss-Jackson.

Namespaces

- [jeod](#)

Namespace jeod.

9.9.1 Detailed Description

Defines the class `GaussJacksonGeneralizedDerivSecondOrderODEIntegrator`, which integrates a generalized derivative second order ODE using Gauss-Jackson.

9.10 gauss_jackson_integration_controls.cc File Reference

Defines member functions for the class `GaussJacksonIntegrationControls`.

```
#include "utils/math/include/numerical.hh"
#include "../include/gauss_jackson_integration_controls.hh"
#include "er7_utils/integration/core/include/base_integration_group.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include "er7_utils/integration/core/include/integrator_result.hh"
#include "er7_utils/integration/core/include/time_interface.hh"
#include "er7_utils/interface/include/alloc.hh"
#include <cassert>
```

Namespaces

- [jeod](#)

Namespace jeod.

9.10.1 Detailed Description

Defines member functions for the class `GaussJacksonIntegrationControls`.

9.11 gauss_jackson_integration_controls.hh File Reference

Defines the class `GaussJacksonIntegrationControls`, which controls Gauss-Jackson integration process.

```
#include "gauss_jackson_coeffs.hh"
#include "gauss_jackson_config.hh"
#include "gauss_jackson_state_machine.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
```

Data Structures

- class [jeod::GaussJacksonIntegrationControls](#)
IntegrationControls specialized for Gauss-Jackson integration.

Namespaces

- [jeod](#)
Namespace jeod.

9.11.1 Detailed Description

Defines the class GaussJacksonIntegrationControls, which controls Gauss-Jackson integration process.

9.12 gauss_jackson_integrator_base.hh File Reference

Defines the template class GaussJacksonIntegratorBase, which provides the basis for Gauss-Jackson integration.

```
#include "gauss_jackson_coeffs.hh"
#include "gauss_jackson_config.hh"
#include "gauss_jackson_integration_controls.hh"
#include "gauss_jackson_state_machine.hh"
#include "two_d_array.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integrator_result.hh"
#include "er7_utils/interface/include/alloc.hh"
#include <algorithm>
#include <cassert>
```

Data Structures

- class [jeod::GaussJacksonIntegratorBase< State, Primer >](#)
Base template class for integrating state via the Gauss-Jackson technique.

Namespaces

- [jeod](#)
Namespace jeod.

9.12.1 Detailed Description

Defines the template class GaussJacksonIntegratorBase, which provides the basis for Gauss-Jackson integration.

9.13 gauss_jackson_integrator_base_first.hh File Reference

Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE.

```
#include "gauss_jackson_integrator_base.hh"
#include "gauss_jackson_one_state.hh"
#include "er7_utils/integration/core/include/first_order_ode_integrator.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include "er7_utils/interface/include/alloc.hh"
#include <algorithm>
#include <cmath>
```

Namespaces

- [jeod](#)

Namespace jeod.

Typedefs

- using [jeod::GaussJacksonIntegratorBaseFirst](#) = GaussJacksonIntegratorBase< GaussJacksonOneState, er7_utils::FirstOrderODEIntegrator >

Alias for a first order Gauss Jackson integrator.

9.13.1 Detailed Description

Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE.

9.14 gauss_jackson_integrator_base_second.hh File Reference

Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE.

```
#include "gauss_jackson_integrator_base.hh"
#include "gauss_jackson_two_state.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include "er7_utils/interface/include/alloc.hh"
#include <algorithm>
#include <cmath>
```

Namespaces

- [jeod](#)

Namespace jeod.

Typedefs

- using [jeod::GaussJacksonIntegratorBaseSecond](#) = GaussJacksonIntegratorBase< GaussJacksonTwoState, er7_utils::SecondOrderODEIntegrator >

Alias for a second order Gauss Jackson integrator.

9.14.1 Detailed Description

Defines the template specializations of GaussJacksonIntegratorBase for a first order ODE.

9.15 gauss_jackson_integrator_constructor.cc File Reference

Defines member functions for the class GaussJacksonIntegratorConstructor.

```
#include "../include/jeod/jeod_gauss_jackson_integrator_constructor.hh"
#include "../include/jeod/jeod_gauss_jackson_first_order_ode_integrator.hh"
#include "../include/jeod/jeod_gauss_jackson_generalized_second_order_ode_integrator.↵
hh"
#include "../include/jeod/jeod_gauss_jackson_simple_second_order_ode_integrator.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/integration/core/include/integrator_constructor_factory.↵
hh"
#include "er7_utils/integration/core/include/integrator_constructor_utils.↵
hh"
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include <algorithm>
```

Namespaces

- [jeod](#)

Namespace jeod.

Functions

- static GaussJacksonIntegrationControls * [jeod::cast_to_gj_controls](#) (er7_utils::IntegrationControls &controls)

Cast the provided integration controls to a [GaussJacksonIntegrationControls](#).

9.15.1 Detailed Description

Defines member functions for the class GaussJacksonIntegratorConstructor.

9.16 gauss_jackson_integrator_constructor.hh File Reference

Defines the class GaussJacksonIntegratorConstructor, which constructs integrators that use Gauss-Jackson integration.

```
#include "gauss_jackson_config.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include <string>
```

Data Structures

- class [jeod::GaussJacksonIntegratorConstructor](#)
Create state and time integrators that propagate using Gauss-Jackson.

Namespaces

- [jeod](#)
Namespace jeod.

9.16.1 Detailed Description

Defines the class GaussJacksonIntegratorConstructor, which constructs integrators that use Gauss-Jackson integration.

9.17 gauss_jackson_one_state.hh File Reference

Defines the class GaussJacksonOneState, which contains a double* pointer.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include <cstddef>
```

Data Structures

- class [jeod::GaussJacksonOneState](#)
Essentially just a double.*

Namespaces

- [jeod](#)
Namespace jeod.

9.17.1 Detailed Description

Defines the class GaussJacksonOneState, which contains a double* pointer.

9.18 gauss_jackson_rational_coefs.cc File Reference

Defines member functions for the class GaussJacksonRationalCoefficients.

```
#include "../include/ gauss_jackson_rational_coefs.hh"
#include <cassert>
```

Namespaces

- [jeod](#)

Namespace jeod.

9.18.1 Detailed Description

Defines member functions for the class GaussJacksonRationalCoefficients.

9.19 gauss_jackson_rational_coefs.hh File Reference

Defines the class GaussJacksonRationalCoefficients, which contains a set of Adams or Stormer-Cowell coefficients.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/math/include/n_choose_m.hh"
#include "er7_utils/math/include/ratio128.hh"
#include <vector>
```

Data Structures

- class [jeod::GaussJacksonRationalCoefficients](#)

Contains a set of Adams or Stormer-Cowell coefficients.

Namespaces

- [jeod](#)

Namespace jeod.

9.19.1 Detailed Description

Defines the class GaussJacksonRationalCoefficients, which contains a set of Adams or Stormer-Cowell coefficients.

9.20 gauss_jackson_simple_second_order_ode_integrator.hh File Reference

Defines the class GaussJacksonSimpleSecondOrderODEIntegrator, which integrates a simple second order ODE using Gauss-Jackson.

```
#include "gauss_jackson_integrator_base_second.hh"
#include "gauss_jackson_two_state.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
```

Data Structures

- class [jeod::GaussJacksonSimpleSecondOrderODEIntegrator](#)
Integrates a simple second order ODE using the Gauss-Jackson technique.

Namespaces

- [jeod](#)
Namespace jeod.

9.20.1 Detailed Description

Defines the class GaussJacksonSimpleSecondOrderODEIntegrator, which integrates a simple second order ODE using Gauss-Jackson.

9.21 gauss_jackson_state_machine.cc File Reference

Defines member functions for the class GaussJacksonStateMachine.

```
#include "../include/gauss_jackson_state_machine.hh"
#include "../include/gauss_jackson_config.hh"
#include <algorithm>
#include <cassert>
#include <string>
```

Namespaces

- [jeod](#)
Namespace jeod.

9.21.1 Detailed Description

Defines member functions for the class GaussJacksonStateMachine.

9.22 gauss_jackson_state_machine.hh File Reference

Defines the class GaussJacksonStateMachine, which guides the Gauss-Jackson integration process.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include <string>
```

Data Structures

- class [jeod::GaussJacksonStateMachine](#)
Guides the behavior of the Gauss-Jackson integration process via a finite state machine.

Namespaces

- [jeod](#)
Namespace jeod.

9.22.1 Detailed Description

Defines the class GaussJacksonStateMachine, which guides the Gauss-Jackson integration process.

9.23 gauss_jackson_two_state.hh File Reference

Defines the class GaussJacksonTwoState, which contains a pair of double* pointers.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include <cstddef>
```

Data Structures

- class [jeod::GaussJacksonTwoState](#)
Essentially just std::pair<double>.*

Namespaces

- [jeod](#)
Namespace jeod.

9.23.1 Detailed Description

Defines the class GaussJacksonTwoState, which contains a pair of double* pointers.

9.24 generalized_second_order_ode_technique.cc File Reference

Define class GeneralizedSecondOrderODETechnique methods.

```
#include <cstdlib>
#include "utils/message/include/message_handler.hh"
#include "../include/generalized_second_order_ode_technique.hh"
#include "../include/integration_messages.hh"
```

Namespaces

- [jeod](#)

Namespace jeod.

9.24.1 Detailed Description

Define class GeneralizedSecondOrderODETechnique methods.

9.25 generalized_second_order_ode_technique.hh File Reference

Define the static class GeneralizedSecondOrderODETechnique.

```
#include "er7_utils/integration/core/include/integration_technique.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
```

Data Structures

- class [jeod::GeneralizedSecondOrderODETechnique](#)

Enumerates the integration schemes, generalized Cartesian or Lie group integration, and provides simple utilities that work with this.

Namespaces

- [jeod](#)

Namespace jeod.

9.25.1 Detailed Description

Define the static class GeneralizedSecondOrderODETechnique.

9.26 integration_messages.cc File Reference

Implement the class IntegrationMessages.

```
#include "utils/message/include/make_message_code.hh"
#include "../include/integration_messages.hh"
```

Namespaces

- [jeod](#)
Namespace jeod.

Macros

- #define [MAKE_INTEGRATION_MESSAGE_CODE](#)(id) JEOD_MAKE_MESSAGE_CODE(IntegrationMessages, "utils/integration/", id)

9.26.1 Detailed Description

Implement the class IntegrationMessages.

9.26.2 Macro Definition Documentation

9.26.2.1 MAKE_INTEGRATION_MESSAGE_CODE

```
#define MAKE_INTEGRATION_MESSAGE_CODE(  
    id ) JEOD_MAKE_MESSAGE_CODE(IntegrationMessages, "utils/integration/", id)
```

Definition at line 37 of file integration_messages.cc.

9.27 integration_messages.hh File Reference

Define the class IntegrationMessages, the class that specifies the message IDs used in the integration model.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::IntegrationMessages](#)
Declares messages associated with the integration test model.

Namespaces

- [jeod](#)

Namespace jeod.

9.27.1 Detailed Description

Define the class IntegrationMessages, the class that specifies the message IDs used in the integration model.

9.28 jeod_integration_group.cc File Reference

Define JeodIntegrationGroup methods.

```
#include "../include/jeod_integration_group.hh"
#include "../include/integration_messages.hh"
#include "../include/jeod_integration_time.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/integrable_object.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include <algorithm>
#include <cstdint>
```

Namespaces

- [jeod](#)

Namespace jeod.

9.28.1 Detailed Description

Define JeodIntegrationGroup methods.

9.29 jeod_integration_group.hh File Reference

Define the extensible class IntegrationGroup, an instance of which is responsible for integrating the states of a set of DynBody objects.

```
#include "time_change_subscriber.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_integrator_interface.hh"
#include "er7_utils/integration/core/include/base_integration_group.hh"
#include "er7_utils/integration/core/include/integrable_object.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include "er7_utils/integration/core/include/integrator_result_merger_↵
container.hh"
#include <cstdint>
```

Data Structures

- class [jeod::JeodIntegrationGroupOwner](#)
The abstract class `IntegrationGroupOwner` contains an `IntegrationGroup`.
- class [jeod::JeodIntegrationGroup](#)
A `JeodIntegrationGroup` integrates the state of a set of objects over time.

Namespaces

- [jeod](#)
Namespace `jeod`.

9.29.1 Detailed Description

Define the extensible class `IntegrationGroup`, an instance of which is responsible for integrating the states of a set of `DynBody` objects.

9.30 jeod_integration_time.cc File Reference

Define `JeodIntegrationTime` methods.

```
#include <algorithm>
#include <cstdlib>
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "../include/integration_messages.hh"
#include "../include/jeod_integration_time.hh"
#include "../include/time_change_subscriber.hh"
```

Namespaces

- [jeod](#)
Namespace `jeod`.

9.30.1 Detailed Description

Define `JeodIntegrationTime` methods.

9.31 jeod_integration_time.hh File Reference

Define the class `JeodIntegrationTime`.

```
#include "er7_utils/integration/core/include/time_interface.hh"
#include "utils/container/include/pointer_vector.hh"
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::JeodIntegrationTime](#)

The class *JeodIntegrationTime* adds the concepts of timestamping and notification of changes in the nature of time to the ER7 numerical utilities *TimeInterface* class.

Namespaces

- [jeod](#)

Namespace *jeod*.

9.31.1 Detailed Description

Define the class *JeodIntegrationTime*.

9.32 lside_control_data_interface.cc File Reference

Define member functions for the class *LsideControlDataInterface*.

```
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "../include/lside_control_data_interface.hh"
```

9.32.1 Detailed Description

Define member functions for the class *LsideControlDataInterface*.

9.33 lside_control_data_interface.hh File Reference

Define the class *LsideControlDataInterface*.

```
#include <vector>
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::LsideControlDataInterface](#)

Specifies controls for an *LSODE* integrator.

Namespaces

- [jeod](#)

Namespace jeod.

9.33.1 Detailed Description

Define the class LsodeControlDataInterface.

9.34 Isode_data_classes.cc File Reference

Define member functions for the data-grouping classes specified in Isode_data_classes.

```
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "er7_utils/interface/include/alloc.hh"
#include "../include/lsode_data_classes.hh"
```

9.34.1 Detailed Description

Define member functions for the data-grouping classes specified in Isode_data_classes.

9.35 Isode_data_classes.hh File Reference

Define LSODE classes that contain just data members.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "lsode_control_data_interface.hh"
```

Data Structures

- class [jeod::LsodeDataJacobianPrep](#)
Data associated with the method DPREPJ.
- class [jeod::LsodeDataArrays](#)
The data arrays.
- class [jeod::LsodeDataStode](#)
The data associated with method Dstode.

Namespaces

- [jeod](#)

Namespace jeod.

9.35.1 Detailed Description

Define LSODE classes that contain just data members.

9.36 Lsode_first_order_ode_integrator.hh File Reference

Define the class LsodeFirstOrderODEIntegrator, the Jeod-compatible version of the Livermore ODE solver, LSODE.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/first_order_ode_integrator.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integrator_result.hh"
#include "lsode_control_data_interface.hh"
#include "lsode_data_classes.hh"
#include "lsode_integration_controls.hh"
```

Data Structures

- class [jeod::LsodeFirstOrderODEIntegrator](#)
Jeod-compatible version of the Livermore ODE solver, LSODE.

Namespaces

- [jeod](#)
Namespace jeod.

9.36.1 Detailed Description

Define the class LsodeFirstOrderODEIntegrator, the Jeod-compatible version of the Livermore ODE solver, LSODE.

9.37 Lsode_first_order_ode_integrator__integrator.cc File Reference

Define member functions for the class LsodeFirstOrderODEIntegrator.

```
#include <cmath>
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "utils/math/include/numerical.hh"
#include "../include/lsode_first_order_ode_integrator.hh"
```

9.37.1 Detailed Description

Define member functions for the class LsodeFirstOrderODEIntegrator.

9.38 Lsode_first_order_ode_integrator__manager.cc File Reference

Define member functions for the class LsodeFirstOrderODEIntegrator.

```
#include <cmath>
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "utils/math/include/numerical.hh"
#include "../include/lsode_first_order_ode_integrator.hh"
```

9.38.1 Detailed Description

Define member functions for the class LsodeFirstOrderODEIntegrator.

9.39 Lsode_first_order_ode_integrator__support.cc File Reference

Define member functions for the class LsodeFirstOrderODEIntegrator.

```
#include <cmath>
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "utils/math/include/numerical.hh"
#include "../include/lsode_first_order_ode_integrator.hh"
```

9.39.1 Detailed Description

Define member functions for the class LsodeFirstOrderODEIntegrator.

9.40 Lsode_first_order_ode_integrator__utility.cc File Reference

Define member functions for the class LsodeFirstOrderODEIntegrator.

```
#include <cmath>
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "../include/lsode_first_order_ode_integrator.hh"
```

9.40.1 Detailed Description

Define member functions for the class `LsodeFirstOrderODEIntegrator`.

9.41 `lsode_generalized_second_order_ode_integrator.cc` File Reference

Define member functions for the class `LsodeGeneralizedDerivSecondOrderODEIntegrator`.

```
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "../include/lsode_control_data_interface.hh"
#include "../include/lsode_generalized_second_order_ode_integrator.hh"
```

9.41.1 Detailed Description

Define member functions for the class `LsodeGeneralizedDerivSecondOrderODEIntegrator`.

9.42 `lsode_generalized_second_order_ode_integrator.hh` File Reference

Define the class `LsodeGeneralizedDerivSecondOrderODEIntegrator`.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include "lsode_second_order_ode_integrator.hh"
```

Data Structures

- class [jeod::LsodeGeneralizedDerivSecondOrderODEIntegrator](#)
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Namespaces

- [jeod](#)
Namespace jeod.

9.42.1 Detailed Description

Define the class `LsodeGeneralizedDerivSecondOrderODEIntegrator`.

9.43 `lsode_integration_controls.cc` File Reference

Define the methods for the class `LsodeIntegrationControls`.

```
#include "er7_utils/integration/core/include/base_integration_group.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/integration/core/include/integrator_interface.hh"
#include "er7_utils/integration/core/include/time_interface.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "utils/math/include/numerical.hh"
#include "../include/lsode_integration_controls.hh"
```

9.43.1 Detailed Description

Define the methods for the class `LsodeIntegrationControls`.

9.44 `lsode_integration_controls.hh` File Reference

Define the class `LsodeIntegrationControls`.

```
#include "er7_utils/interface/include/alloc.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/generalized_position_derivative.↵
hh"
#include "er7_utils/integration/core/include/standard_integration_controls.↵
hh"
```

Data Structures

- class [jeod::LsodeIntegrationControls](#)
Contains controls for an LSODE integrator.

Namespaces

- [jeod](#)
Namespace jeod.

9.44.1 Detailed Description

Define the class `LsodeIntegrationControls`.

9.45 lside_integrator_constructor.cc File Reference

Define the methods in the class LsodeIntegratorConstructor.

```
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/integration/core/include/integrator_constructor_utils.↵
hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "../include/lside_first_order_ode_integrator.hh"
#include "../include/lside_generalized_second_order_ode_integrator.hh"
#include "../include/lside_integration_controls.hh"
#include "../include/lside_integrator_constructor.hh"
#include "../include/lside_second_order_ode_integrator.hh"
#include "../include/lside_simple_second_order_ode_integrator.hh"
```

9.45.1 Detailed Description

Define the methods in the class LsodeIntegratorConstructor.

9.46 lside_integrator_constructor.hh File Reference

Define the class LsodeIntegratorConstructor, the class that constructs the integration controls and the integrators for the LSODE method.

```
#include <vector>
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include "lsode_control_data_interface.hh"
#include "lsode_first_order_ode_integrator.hh"
#include "lsode_generalized_second_order_ode_integrator.hh"
#include "lsode_second_order_ode_integrator.hh"
#include "lsode_simple_second_order_ode_integrator.hh"
```

Data Structures

- class [jeod::LsodeIntegratorConstructor](#)
Create state and time integrators that propagate using standard Lsode.

Namespaces

- [jeod](#)
Namespace jeod.

9.46.1 Detailed Description

Define the class LsodeIntegratorConstructor, the class that constructs the integration controls and the integrators for the LSODE method.

9.47 `lsode_second_order_ode_integrator.cc` File Reference

Define member functions for the class `LsodeSecondOrderODEIntegrator`.

```
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integ_utils.hh"
#include "er7_utils/integration/core/include/integration_controls.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "../include/lsode_control_data_interface.hh"
#include "../include/lsode_second_order_ode_integrator.hh"
```

9.47.1 Detailed Description

Define member functions for the class `LsodeSecondOrderODEIntegrator`.

9.48 `lsode_second_order_ode_integrator.hh` File Reference

Define the class `LsodeSecondOrderODEIntegrator`.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include "lsode_first_order_ode_integrator.hh"
#include "lsode_integration_controls.hh"
```

Data Structures

- class [jeod::LsodeSecondOrderODEIntegrator](#)
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Namespaces

- [jeod](#)
Namespace jeod.

9.48.1 Detailed Description

Define the class `LsodeSecondOrderODEIntegrator`.

9.49 lside_simple_second_order_ode_integrator.cc File Reference

Define member functions for the class LsideSimpleSecondOrderODEIntegrator.

```
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/integration/core/include/integration_messages.hh"
#include "er7_utils/interface/include/message_handler.hh"
#include "../include/lside_control_data_interface.hh"
#include "../include/lside_simple_second_order_ode_integrator.hh"
```

9.49.1 Detailed Description

Define member functions for the class LsideSimpleSecondOrderODEIntegrator.

9.50 lside_simple_second_order_ode_integrator.hh File Reference

Define the class LsideSimpleSecondOrderODEIntegrator.

```
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include "lside_second_order_ode_integrator.hh"
```

Data Structures

- class [jeod::LsideSimpleSecondOrderODEIntegrator](#)
JEOD-compatible version of the Livermore ODE solver, LSODE, capable of integrating second-order ODEs.

Namespaces

- [jeod](#)
Namespace jeod.

9.50.1 Detailed Description

Define the class LsideSimpleSecondOrderODEIntegrator.

9.51 restartable_2d_second_order_integrator.hh File Reference

Defines the class Restartable2DSecondOrderODEIntegrator.

```
#include "generalized_second_order_ode_technique.hh"
#include "integration_messages.hh"
#include "restartable_state_integrator_templates.hh"
#include "utils/container/include/simple_checkpointable.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/first_order_ode_integrator.hh"
#include "er7_utils/integration/core/include/left_quaternion_functions.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include <cstdint>
```

Data Structures

- class [jeod::Restartable2DSecondOrderIntegrator](#)

Integrates a second order ODE in two dimensional space, $d^2x/dt^2 = a(x,t)$, where x is a two-vector.

Namespaces

- [jeod](#)

Namespace jeod.

9.51.1 Detailed Description

Defines the class Restartable2DSecondOrderODEIntegrator.

9.52 restartable_state_integrator.hh File Reference

Define classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators.

```
#include "generalized_second_order_ode_technique.hh"
#include "integration_messages.hh"
#include "restartable_state_integrator_templates.hh"
#include "utils/container/include/simple_checkpointable.hh"
#include "utils/memory/include/jeod_alloc.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "er7_utils/integration/core/include/first_order_ode_integrator.hh"
#include "er7_utils/integration/core/include/left_quaternion_functions.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include <cstdint>
```

Data Structures

- class [jeod::RestartableScalarFirstOrderODEIntegrator](#)
A [RestartableScalarFirstOrderODEIntegrator](#) integrates a first order ODE, $dx/dt = v(x,t)$, where x is a scalar.
- class [jeod::RestartableT3SecondOrderODEIntegrator](#)
A [RestartableT3SecondOrderODEIntegrator](#) integrates a second order ODE in three space, $d^2x/dt^2 = a(x,t)$, where x is a three-vector.
- class [jeod::RestartableSO3SecondOrderODEIntegrator](#)
A [RestartableSO3SecondOrderODEIntegrator](#) integrates a generalized second order ODE that describes rotation in three space.

Namespaces

- [jeod](#)
Namespace *jeod*.

9.52.1 Detailed Description

Define classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators.

9.53 restartable_state_integrator_templates.hh File Reference

Define template classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators.

```
#include <cstddef>
#include "er7_utils/integration/core/include/first_order_ode_integrator.hh"
#include "er7_utils/integration/core/include/integrator_constructor.hh"
#include "er7_utils/integration/core/include/second_order_ode_integrator.↵
hh"
#include "utils/container/include/simple_checkpointable.hh"
#include "utils/message/include/message_handler.hh"
#include "utils/sim_interface/include/jeod_class.hh"
#include "integration_messages.hh"
```

Data Structures

- class [jeod::RestartableStateIntegrator< IntegratorType >](#)
A [RestartableStateIntegrator](#) establishes the basic capabilities needed to make a state integrator a managed resource.
- class [jeod::RestartableFirstOrderODEIntegrator< size >](#)
A [RestartableFirstOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages an [er7_utils::FirstOrder↵ODEIntegrator](#).
- class [jeod::RestartableSecondOrderODEIntegrator](#)
A [RestartableSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a second order ODE problem.
- class [jeod::RestartableSimpleSecondOrderODEIntegrator< size >](#)

A [RestartableSimpleSecondOrderODEIntegrator](#) is-a [RestartableSecondOrderODEIntegrator](#) that manages the integrator for a simple second order ODE problem, one in which the time derivative of the generalized position is the generalized velocity.

- class [jeod::RestartableGeneralizedDerivSecondOrderODEIntegrator](#)< position_size, velocity_size, DerivFunctions >

A [RestartableGeneralizedDerivSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

- class [jeod::RestartableGeneralizedStepSecondOrderODEIntegrator](#)< position_size, velocity_size, StepFunctions >

A [RestartableGeneralizedStepSecondOrderODEIntegrator](#) is-a [RestartableStateIntegrator](#) that manages the integrator for a generalized second order ODE problem, one in which the time derivative of the generalized position is some function of the generalized position and the generalized velocity.

Namespaces

- [jeod](#)

Namespace *jeod*.

9.53.1 Detailed Description

Define template classes that encapsulate the construction, destruction, checkpointing, and restarting of state integrators.

9.54 time_change_subscriber.hh File Reference

Define the class TimeChangeSubscriber.

```
#include "utils/sim_interface/include/jeod_class.hh"
```

Data Structures

- class [jeod::TimeChangeSubscriber](#)

A [TimeChangeSubscriber](#) is some object that wants to be notified of changes in the nature of time.

Namespaces

- [jeod](#)

Namespace *jeod*.

9.54.1 Detailed Description

Define the class TimeChangeSubscriber.

9.55 two_d_array.hh File Reference

Defines the template class [er7_utils::TwoDArray](#), which implements an RAI rectangular 2D array.

```
#include "er7_utils/interface/include/alloc.hh"
#include "er7_utils/interface/include/er7_class.hh"
#include "utils/sim_interface/include/config.hh"
#include <algorithm>
#include <cstdlib>
#include <cstring>
#include <limits>
#include <stdexcept>
```

Data Structures

- class [er7_utils::TwoDArray< T >](#)
RAII template class that implements a rectangular two dimensional array.
- class [er7_utils::DoubleTwoDArray](#)
2D array, specialized for doubles.

Namespaces

- [er7_utils](#)

9.55.1 Detailed Description

Defines the template class [er7_utils::TwoDArray](#), which implements an RAI rectangular 2D array.

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